

$\text{Span}_2(\mathcal{C}) \rightarrow \overline{\text{Enr}}_{\text{PrL}}$

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Introduction

In this draft, we construct a lax symmetric monoidal $(\infty, 2)$ functor:

$$\text{Span}_2(\mathcal{C})^{\otimes} \rightarrow \overline{\text{Enr}}_{\text{Pr}^L}^{\otimes}$$

where

- $\mathcal{C}$ admits all pullbacks and a final object $*$, and all the morphisms in $\mathcal{C}$ are truncated (i.e, n -truncated for some $n \geq -2$ which may depend on the morphism).
- The left hand side is equipped with a symmetric monoidal $(\infty, 2)$ structure constructed in [7, section 4.3], which can be regarded as a commutative monoid object in $\text{Cat}_{(\infty, 2)}$:

$$\text{Comm}^{\otimes} \rightarrow \text{Cat}_{(\infty, 2)} : [n] \mapsto \text{Span}_2(\mathcal{C}^{\times n})$$

- The underlying category of $\overline{\text{Enr}}_{\text{Pr}^L}^{\otimes}$ in the right hand side is the category of Pr^L -enriched categories in the Heine's sense (see [14, section 3.4]) with small underlying spaces.
- And there is a symmetric monoidal $(\infty, 1)$ structure on Enr_{Pr^L} : consider the symmetric monoidal coCartesian fibration $\text{Enr} \rightarrow \text{Mon}$ with the coCartesian edge corresponding to the transfer of enrichment, and we define the $\widehat{\text{Enr}}_{\text{Pr}^L}^{\otimes}$ as pullback of $\widehat{\text{Enr}}^{\times} \rightarrow \widehat{\text{Mon}}^{\times}$ along the map $\text{Comm}^{\otimes} \rightarrow \widehat{\text{Mon}}^{\times}$ specifying the commutative algebra object $\text{Pr}^L^{\otimes}$. And we define Enr_{Pr^L} to be the full symmetric monoidal subcategory of $\widehat{\text{Enr}}_{\text{Pr}^L}$ consisting of objects with small underlying spaces.
- Likewise, for any symmetric monoidal category $\mathcal{V}$, we can equip $\text{Enr}_{\mathcal{V}}$ with a symmetric monoidal structure.
- And Enr_{Pr^L} is enhanced into a symmetric monoidal $(\infty, 2)$ category $\overline{\text{Enr}}_{\text{Pr}^L}$ as a full symmetric monoidal $(\infty, 2)$ category of $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{\text{cc}}}$ spanned by the image of $\text{Enr}_{\text{Pr}^L} \rightarrow \widehat{\text{Enr}}_{\widehat{\text{Cat}}_{\text{cc}}}$ which is the transfer or enrichment along $\text{Pr}^L \hookrightarrow \widehat{\text{Cat}}_{\text{cc}}$.

The reason why the right hand side involves the $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{\text{cc}}}$ is due to the fact that $\widehat{\text{Cat}}_{\text{cc}}$ is large presentably symmetric monoidal, and thus $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{\text{cc}}}$ can naturally acquire a symmetric monoidal $(\infty, 2)$ structure, more precisely:

- There is a natural way to enhance a presentably left Cat module category M into an $(\infty, 2)$ category by the lax symmetric monoidal functor developed in [32, chapter 4] (also see section 3.3 for a brief review) :

$$\theta : \text{LMod}_{\text{Cat}}(\text{Pr}^{\text{L}}) \rightarrow \widehat{\text{Cat}}_{(\infty, 2)}$$

- Thus by the equivalence $\text{CAlg}(\text{LMod}_{\text{Cat}}(\text{Pr}^{\text{L}})) \cong \text{CAlg}(\text{Pr}^{\text{L}})_{\text{Cat}} /$ (by [22, corollary 3.4.1.7], or see section 1.7 on the left module objects in a coCartesian ∞ -operad), we can specify a colimit-preserving symmetric monoidal functor $\text{Cat} \rightarrow M$ in $\text{CAlg}(\text{Pr}^{\text{L}})$ to acquire a symmetric monoidal $(\infty, 2)$ category $\overline{M} := \theta(M)$.
- And there is a fact that in general, if $\mathcal{V}$ is presentably symmetric monoidal, so is $\text{Enr}_{\mathcal{V}}$ (see [15, proposition 2.60, corollary 4.62] or [9, proposition 4.3.5]). However, if we enlarge the universe, Pr^{L} is not large presentable, instead, Pr^{L} is the full subcategory of large compact objects in $\widehat{\text{Cat}}_{\text{cc}}$ of large categories with all small colimits and functors between them preserving small colimits (see [32, proposition 5.1.4]). Thus we have a large presentably symmetric monoidal structure on $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{\text{cc}}}$, moreover, consider the canonical map $\mathbf{1} : \widehat{\text{Spc}} \rightarrow \widehat{\text{Cat}}_{\text{cc}}$, there is an induced colimit-preserving symmetric monoidal functor between $\widehat{\text{Cat}}$ and $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{\text{cc}}}$, which gives a symmetric monoidal $(\infty, 2)$ structure on $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{\text{cc}}}$.
- And in the enrichment theory developed by Heine, a fully faithful functor $\text{Pr}^{\text{L}} \hookrightarrow \widehat{\text{Cat}}_{\text{cc}}$ induces a fully faithful functor (by [15, proposition 5.1.4])

$$\widehat{\text{Enr}}_{\text{Pr}^{\text{L}}} \rightarrow \widehat{\text{Enr}}_{\widehat{\text{Cat}}_{\text{cc}}}$$

, thus we define the full symmetric monoidal $(\infty, 2)$ category of $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{\text{cc}}}$ spanned by the image of $\widehat{\text{Enr}}_{\text{Pr}^{\text{L}}}$ as $\overline{\widehat{\text{Enr}}_{\text{Pr}^{\text{L}}}}^{\otimes}$, and the underlying $(\infty, 1)$ symmetric monoidal category of $\overline{\widehat{\text{Enr}}_{\text{Pr}^{\text{L}}}}^{\otimes}$, then it's just $\widehat{\text{Enr}}_{\text{Pr}^{\text{L}}}^{\otimes}$ constructed using pullback of $\widehat{\text{Enr}}^{\times} \rightarrow \widehat{\text{Mon}}^{\times}$, thus we can regard our construction of $\overline{\widehat{\text{Enr}}_{\text{Pr}^{\text{L}}}}^{\otimes}$ as an enhancement of $\widehat{\text{Enr}}_{\text{Pr}^{\text{L}}}^{\otimes}$ into an $(\infty, 2)$ category.

- Moreover, we show the mapping category of $\widehat{\text{Enr}}_{\text{Pr}^{\text{L}}}$ for Pr^{L} -enriched categories M, N is exactly the category $\text{LaxLin}_{\text{Pr}^{\text{L}}}(M, N)$ (see section 6.2), under the Heine's model of enriched ∞ -category using weakly left tensored ∞ -categories, we can regard $\text{LaxLin}_{\text{Pr}^{\text{L}}}(M, N)$ as the category of Pr^{L} -enriched functors between M and N . Or equivalently, if we regard M, N as operads over $\mathcal{LM}^{\otimes}$, $\text{LaxLin}_{\text{Pr}^{\text{L}}}(M, N)$ is exactly the full subcategory of $\text{Alg}_{M/\mathcal{LM}}(N)$ spanned by algebra maps between M and N over $\mathcal{LM}^{\otimes}$ whose Ass-part is fixed to be the identity map at Pr^{L} .

And here is an overview on each section:

In section 1:

- We take a brief review of Heine's enrichment theory using the weakly left tensored ∞ -categories.
- And consider the category ωLMod of weakly left tensored ∞ -categories, then ωLMod is actually equivalent to $\text{Op}_{/\mathcal{LM}^{\otimes}}^{\text{HA}}$ (see section 1.3) where Op^{HA} is the ∞ -category of ∞ -(symmetric) operads in the Lurie's sense developed in [22, chapter 2] (but if not specified, all the operads using in this draft are non-symmetric operads).

- So we can transfer many propositions about Op^{HA} to ωLMod , for example, the [construction of envelope](#), and [presheaf of left modules](#).

And in section 2:

- We review the definition of enriched ∞ -category as a weakly left tensored ∞ -category satisfying certain conditions (see [section 2.1](#)).
- And we review the theory of localizations of operads developed in [\[16, section 2.6\]](#), where Heine defines all kinds of localizations of the presentably $\mathcal{P}\text{Env}(\mathcal{V})$ -left tensored ∞ -category $\mathcal{P}\text{LEnv}(\mathcal{M})$ for a weakly left tensored ∞ -category in $\mathcal{V} : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, for example:
 - If we localize $\mathcal{P}\text{LEnv}(\mathcal{M})$ with respect to Pseu , then we get $\mathcal{P}'(\mathcal{M}) := \mathcal{P}\text{LEnv}(\mathcal{M})_{\text{Pseu}}$, which is a presentably $\mathcal{P}(\mathcal{V})$ -left tensored ∞ -category (see [section 2.3](#)).
 - If $\mathcal{V}$ is a monoidal ∞ -category compatible with all small colimits, we localize $\mathcal{P}\text{LEnv}(\mathcal{M})$ with respect to Enr , then we get the $\mathcal{V}$ -enriched presheaf category of $\mathcal{M}$, $\mathcal{V} : \mathcal{M} \hookrightarrow \mathcal{P}_{\mathcal{V}}(\mathcal{M})$ (see [section 2.4](#)).

And these localizations satisfy the universal property:

$$\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \cong \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{V}^* \mathcal{N})$$

for any $\mathcal{N} \in \text{LMod}_{\mathcal{P}(\mathcal{V})}(\widehat{\text{Cat}}_{cc})$ and $\mathcal{V}^* \mathcal{N}$ is regarding $\mathcal{N}$ as a left $\mathcal{V}$ -module through the Yoneda embedding $\mathcal{V} \rightarrow \mathcal{P}(\mathcal{V})$. And if $\mathcal{V}$ is (not necessarily small) monoidal ∞ -category compatible with small colimits, we have:

$$\text{Lin}_{\mathcal{V}}^L(\mathcal{P}_{\mathcal{V}}(\mathcal{M}), \mathcal{N}) \cong \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})$$

for any $\mathcal{N} \in \text{LMod}_{\mathcal{V}}(\widehat{\text{Cat}}_{cc})$.

- We also review the (co)Cartesian fibrations $\text{Enr} \hookrightarrow \text{PLMod}$ over Mon , and the symmetric monoidal version of them in [section 2.7](#) and [section 2.8](#).

And in section 3, we review the basic theory of $(\infty, 2)$ category and 2-coCartesian fibrations to define the symmetric monoidal $(\infty, 2)$ category and lax symmetric monoidal $(\infty, 2)$ functor between them.

And in section 4, we use the universal property of Span_2 developed in [\[7, section 4\]](#) to construct a lax symmetric monoidal $(\infty, 2)$ functor

$$\psi : \text{Span}_2(\mathcal{C}) \rightarrow \overline{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$$

where

- The right hand side is equipped with a symmetric monoidal $(\infty, 2)$ structure through the colimit-preserving symmetric monoidal functor $\text{Cat} \rightarrow \text{LMod}_{\mathcal{V}}(\text{Cat})$ sending K to $K \times \mathcal{V}$.
- ψ sends $c \in \mathcal{C}$ to a $\text{Span}_1(\mathcal{C})$ -left module $\text{Span}_1(\mathcal{C}/_c)$ where the left action is given by $(x, y \rightarrow c) \mapsto x \times y \rightarrow c$.

And in section 5, we show that there is a lax symmetric monoidal $(\infty, 2)$ embedding:

$$R : \overline{\text{LMod}}_{\mathcal{V}}(\text{Cat}) \rightarrow \overline{\text{Enr}}_{\mathcal{P}(\mathcal{V})}$$

by showing it's the right adjoint of a symmetric monoidal $(\infty, 2)$ functor L .

To do this, we first construct a lax symmetric monoidal $(\infty, 2)$ functor from $\overline{\text{LMod}}_{\mathcal{V}}(\text{Cat}) \rightarrow \overline{\text{PLMod}}_{\mathcal{V}}$ where $\overline{\text{PLMod}}_{\mathcal{V}}$ is a full subcategory of $\omega\text{LMod}_{\mathcal{V}}$ spanned by the [pseudo-enriched](#) categories, and then we show $\overline{\text{PLMod}}_{\mathcal{V}}$ is symmetric monoidal equivalent to $\overline{\text{Enr}}_{\mathcal{P}(\mathcal{V})}$ (here we use $\overline{}$ to represent the symmetric monoidal $(\infty, 2)$ enhancement).

Moreover, by taking $\mathcal{V} = \text{Span}_1(\mathcal{C})$, we can show that the composition of $R : \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) \rightarrow \overline{\text{Enr}}_{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ with $\psi : \text{Span}_2(\mathcal{C}) \rightarrow \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$ actually lands in the full symmetric monoidal $(\infty, 2)$ subcategory $\overline{\text{Enr}}_{\text{Span}_1(\mathcal{C})} \subset \overline{\text{Enr}}_{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ induced by the transfer of enrichment along $\mathfrak{y} : \mathcal{V} \hookrightarrow \mathcal{P}(\mathcal{V})$.

And in section 5, we construct a lax symmetric monoidal $(\infty, 2)$ functor from $\overline{\text{Enr}}_{\widehat{\mathcal{P}}(\text{Span}_1(\mathcal{C}))}$ to $\widehat{\overline{\text{Enr}}_{\widehat{\text{Cat}}_{\text{ce}}}}$, and show that by precomposing with $\text{Span}_2(\mathcal{C}) \rightarrow \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\widehat{\text{Cat}}) \rightarrow \overline{\text{Enr}}_{\widehat{\mathcal{P}}(\text{Span}_1(\mathcal{C}))}$, and the image lands in the full symmetric monoidal $(\infty, 2)$ full subcategory $\overline{\text{Enr}}_{\text{PrL}}$, thus there is a factorization

$$\text{Span}_2(\mathcal{C}) \rightarrow \overline{\text{Enr}}_{\text{PrL}}$$

.

1 ∞ -operads

1.1 LM-operad

This part follows [14, section 2 and section 3].

Notation 1.1. Let $\text{Ass} := \Delta^{op}$ be the category of finite non-empty totally ordered sets and order preserving maps. We call a map $[n] \rightarrow [m] \in \text{Ass}$

- inert if it corresponds to a map of Δ in the form $[m] \cong \{i, i+1, \dots, i+m\} \subset [n]$ for some $i \geq 0$.
- standard inert if it is inert and $m = 1$.
- active if it corresponds to a map of Δ , which preserves the minimum and maximum. Noticing that this may not be a quotient map as active map in $\text{Comm}^{\otimes}$.

Remark 1.2. Every morphism in Ass uniquely factors as an inert morphism followed by an active one: if $f : [m] \rightarrow [n]$ is a map in Δ , we can factor f as $[m] \rightarrow \{f(0), \dots, f(m)\} \subset [n]$.

Notation 1.3. Let $\text{BM} := (\Delta_{/[1]})^{op}$, and $\text{LM} \subset \text{BM}$, $\text{RM} \subset \text{BM}$ the full subcategories of maps $[n] \rightarrow [1]$ sending 0 to 0 and at most one object to 1 and sending n to 1 and at most one object to 0.

Remark 1.4. There is a canonical equivalence $\text{Ass} \times [1] \cong \text{LM}$ corresponding to the natural transformation sending $([n], 0)$ to $[n] \star [0]$ and $([n], 1)$ to $[n] \star \emptyset$, and edge $([n], 0) \rightarrow ([n], [1])$ to the natural transformation $[n] \star \emptyset \rightarrow [n] \star [0]$ over $[0] \star [0] \cong [1]$.

Remark 1.5. For every $n \geq 2$ there are n -many standard inert morphisms $[1] \cong \{i-1, i\} \subset [n]$ in Δ for $1 \leq i \leq n$ that are part of a colimit diagram in Δ :

$$(\{0, 1\} \leftarrow \{1\} \rightarrow \{1, 2\} \leftarrow \{2\} \rightarrow \cdots \leftarrow \{n-1\} \rightarrow \{n-1, n\} \rightarrow [n]).$$

And for any $\psi : \mathcal{V} \rightarrow \text{Ass}$ which is coCartesian fibration over the inert edges, for $X \in \mathcal{V}$ lying over $[n]$, we denote the corresponding lifting of the above diagram as $X \rightarrow (X_{0,1} \rightarrow X_1 \leftarrow X_{1,2} \rightarrow X_2 \leftarrow \cdots \rightarrow X_{n-1} \leftarrow X_{n-1,n})$

Remark 1.6. The right fibration $\Delta_{/[1]} \rightarrow \Delta$ is opposite to the left fibration $\text{BM} \rightarrow \text{Ass}$ that restricts to a coCartesian fibration $\text{LM} \rightarrow \text{Ass}$ relative to the collection of inert morphisms with discrete fibers. We call a morphism of LM inert, active if its image in Ass is. Moreover, there exists a unique morphism in LM lying over an inert map.

Notation 1.7. Let $\mathcal{O} \rightarrow \text{Ass}$ be a coCartesian fibration relative to the collection of inert morphisms. We call a morphism of $\mathcal{O}$ (standard) inert if it is a coCartesian edge over (standard) inert edges in Ass . We call a morphism of $\mathcal{O}$ active if its image in Ass is active.

Definition 1.8. Let $\psi : \mathcal{O} \rightarrow \text{Ass}$ be a coCartesian fibration relative to the collection of inert morphisms.

A generalized $\mathcal{O}$ -operad is a coCartesian fibration $\phi : \mathcal{C} \rightarrow \mathcal{O}$ relative to the collection of inert morphisms such that the following conditions are satisfied:

(1) (Segal conditions) For every diagram

$$X \rightarrow (X_{0,1} \rightarrow X_1 \leftarrow X_{1,2} \rightarrow X_2 \leftarrow \cdots \rightarrow X_{n-1} \leftarrow X_{n-1,n})$$

in $\mathcal{O}$, where all the morphisms are ψ -coCartesian lifting the diagram

$$(\{0, 1\} \leftarrow \{1\} \rightarrow \{1, 2\} \leftarrow \{2\} \rightarrow \cdots \leftarrow \{n-1\} \rightarrow \{n-1, n\} \rightarrow [n]).$$

The induced functor

$$\alpha : \mathcal{C}_X \rightarrow \mathcal{C}_{X_{0,1}} \times_{\mathcal{C}_{X_1}} \mathcal{C}_{X_{1,2}} \times_{\mathcal{C}_{X_2}} \cdots \times_{\mathcal{C}_{X_{n-1}}} \mathcal{C}_{X_{n-1,n}}$$

is an equivalence.

(2) For every diagram

$$Y \rightarrow (Y_{0,1} \rightarrow Y_1 \leftarrow Y_{1,2} \rightarrow Y_2 \leftarrow \cdots \rightarrow Y_{n-1} \leftarrow Y_{n-1,n})$$

in $\mathcal{C}$, where all morphisms are ϕ -coCartesian, and for every $Z \in \mathcal{C}$ the square:

$$\begin{array}{ccc} \mathcal{C}(Z, Y) & \xrightarrow{\quad\quad\quad} & \mathcal{C}(Z, Y_{0,1}) \times_{\mathcal{C}(Z, Y_1)} \cdots \times_{\mathcal{C}(Z, Y_{n-1})} \mathcal{C}(Z, Y_{n-1,n}) \\ \downarrow & & \downarrow \\ \mathcal{O}(\phi(Z), \phi(Y)) & \xrightarrow{\quad\quad\quad} & \mathcal{O}(\phi(Z), \phi(Y_{0,1})) \times_{\mathcal{O}(\phi(Z), \phi(Y_1))} \cdots \times_{\mathcal{O}(\phi(Z), \phi(Y_{n-1}))} \mathcal{O}(\phi(Z), \phi(Y_{n-1,n})) \end{array}$$

is a pullback square.

Remark 1.9. condition (2) implies the functor α in (1) is fully faithful: take $\phi(Z) = \phi(Y) = X$, and $* \rightarrow \mathcal{O}(\phi(Z), \phi(Y))$ specifying the identity.

Remark 1.10. condition (2) is equivalent to saying Y is ϕ -limit of the diagram $Y \rightarrow (Y_{0,1} \rightarrow Y_1 \leftarrow Y_{1,2} \rightarrow Y_2 \leftarrow \cdots \rightarrow Y_{n-1} \leftarrow Y_{n-1,n})$.

Example 1.11. For $\mathcal{O} \in \{\text{Ass}, \text{LM}, \text{RM}, \text{BM}\}$ every diagram in $\mathcal{O}$ lifting the diagram:

$$(\{0, 1\} \leftarrow \{1\} \rightarrow \{1, 2\} \leftarrow \{2\} \rightarrow \cdots \leftarrow \{n-1\} \rightarrow \{n-1, n\} \rightarrow [n].$$

is a limit diagram. Therefore the condition(2) of the definition for the generalized $\mathcal{O}$ -operad is equivalent to the condition that for every diagram

$$Y \rightarrow (Y_{0,1} \rightarrow Y_1 \leftarrow Y_{1,2} \rightarrow Y_2 \leftarrow \cdots \rightarrow Y_{n-1} \leftarrow Y_{n-1,n})$$

in $\mathcal{C}$, where all morphisms are ϕ -coCartesian, and every $Z \in \mathcal{C}$ the canonical map

$$\mathcal{C}(Z, Y) \rightarrow \mathcal{C}(Z, Y_{0,1}) \times_{\mathcal{C}(Z, Y_1)} \cdots \times_{\mathcal{C}(Z, Y_{n-1})} \mathcal{C}(Z, Y_{n-1,n})$$

Definition 1.12. Let $\psi : \mathcal{O} \rightarrow \text{Ass}$ be a coCartesian fibration relative to the collection of inert morphisms.

- An $\mathcal{O}$ -operad is a generalized $\mathcal{O}$ -operad $\phi : \mathcal{C} \rightarrow \mathcal{O}$ such that the functor $\mathcal{C}_{[0]} \rightarrow \mathcal{O}_{[0]}$ is an equivalence.
- A $\mathcal{O}$ -monoidal ∞ -category is a $\mathcal{O}$ -operad $\phi : \mathcal{C} \rightarrow \mathcal{O}$ that is a coCartesian fibration.

Notation 1.13. Given two $\mathcal{O}$ -operads $\mathcal{M}, \mathcal{N}$, we denote the full subcategory of $\text{Fun}_{\mathcal{O}}(\mathcal{M}, \mathcal{N})$ consisting of functors preserving inert lifts as $\text{Alg}_{\mathcal{M}}(\mathcal{N})$.

Remark 1.14. When $\mathcal{O} = \text{Ass}$, if $\mathcal{C} \rightarrow \text{Ass}$ is an Ass-operad, then the condition (2) is equivalent to Y being the product of $\prod_{i=0}^{n-1} Y_{i,i+1}$.

Remark 1.15. When $\mathcal{O} = \text{LM}$, LM-operads $\mathcal{C} \rightarrow \text{LM}$ are exactly the LM fibrous objects defined in [22, 2.3.3.28] when presented as an inner fibration (which is automatically a categorical fibration by [21, 2.4.6.5]).

Remark 1.16. Ass-operads are non-symmetric ∞ -operads in the sense of [9, definition 3.13]. We refer to Ass-operads as ∞ -operads.

Notation 1.17. Let $\mathcal{O} \rightarrow \text{Ass}$ be a coCartesian fibration relative to the collection of inert morphisms. Let

- $\text{Cat}_{\infty/\mathcal{O}}^{\text{inert}} \subset \text{Cat}_{\infty/\mathcal{O}}$ be the subcategory of coCartesian fibrations relative to the inert edges and functors over $\mathcal{O}$ preserving coCartesian liftings.
- $\text{Op}_{\infty}^{\mathcal{O}} \subset \text{Op}_{\infty}^{\mathcal{O}, \text{gen}} \subset \text{Cat}_{\infty/\mathcal{O}}^{\text{inert}}$ be the full subcategory of (generalized) $\mathcal{O}$ -operads.
- $\text{Op}_{\infty}^{\mathcal{O}, \text{mon}} \subset \text{Cat}_{\infty/\mathcal{O}}^{\text{inert}}$ be the (non-full) subcategory of $\mathcal{O}$ -monoidal ∞ -categories and $\mathcal{O}$ -monoidal functors.

Recall that in [22, chapter 4], there is also another version of (symmetric) operads Ass and LM encoding the algebra and left module structure, which we denote as $\mathcal{A}\text{ssoc}^\otimes$ and $\mathcal{LM}^\otimes$.

Remark 1.18. There exists morphisms $\text{Cut} : \text{Ass} = \Delta^{op} \rightarrow \mathcal{A}\text{ssoc}^\otimes$ and $\text{LCut} : \text{LM} = (\Delta_{/[1]})^{op} \cong \text{Ass} \times [1] \rightarrow \mathcal{LM}^\otimes$ which are approximations of operads respectively. Moreover, their underlying categories are equivalent.

And we have the following lemmas illustrating their relations.

Lemma 1.19. *The pullback functor along $\text{Cut} : \text{Ass} \rightarrow \mathcal{A}\text{ssoc}^\otimes$ induces an equivalence:*

$$\text{Op}_{/\mathcal{A}\text{ssoc}^\otimes}^{HA} \rightarrow \text{Op}_\infty^{\text{Ass}}$$

where the left hand side is the ∞ slice category of Op^{HA} defined in [22, 2.1.4], and the right hand side is defined as above.

Proof. By [22, theorem 4.3.14]. □

Notation 1.20. We denote $\text{Op}_\infty^{\text{Ass}}$ as Op , and call Ass -(generalized) operads as (generalized) operads.

Lemma 1.21. *The pullback functor along $\text{LCut} : \mathcal{O} \rightarrow \mathcal{C}^\otimes$ induces an equivalence:*

$$\text{Op}_{/\mathcal{LM}^\otimes}^{HA} \rightarrow \text{Op}_\infty^{\text{LM}}$$

where the left hand side is the ∞ slice category of Op^{HA} defined in [22, 2.1.4], and right hand side is defined as above.

Proof. by [18, proposition 2.7.2]. □

1.2 weakly left tensored category

Notation 1.22. We say a morphism in Ass preserves the minimum (maximum) if it corresponds to a map $[m] \rightarrow [n]$ in Δ sending 0 to 0 (sending m to n).

Definition 1.23. Let $\mathcal{V}^\otimes \rightarrow \text{Ass}$ be an ∞ -operad and $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ a map of coCartesian fibrations relative to the inert maps of Ass that preserves the maximum.

We call $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ an ∞ -category weakly left tensored over $\mathcal{V}$ (or an ∞ -category with weak left $\mathcal{V}$ -action) if the following conditions hold:

- (1) for every $n \geq 0$ the map $[0] \cong \{n\} \subset [n]$ in Δ induces an equivalence

$$\theta : \mathcal{M}_{[n]}^\otimes \rightarrow \mathcal{V}_{[n]}^\otimes \times \mathcal{M}_{[0]}^\otimes$$

- (2) for every $X, Y \in \mathcal{M}^\otimes$ lying over $[m], [n] \in \text{Ass}$ the coCartesian lift $Y \rightarrow Y'$ of the map $[0] \cong \{n\} \subset [n]$ in Δ induces a pullback square:

$$\begin{array}{ccc}
\mathcal{M}^{\otimes}(X, Y) & \longrightarrow & \mathcal{M}^{\otimes}(X, Y') \\
\downarrow & & \downarrow \\
\mathcal{V}^{\otimes}(\phi(X), \phi(Y)) & \longrightarrow & \mathcal{V}^{\otimes}(\phi(X), \phi(Y'))
\end{array}$$

Remark 1.24. (2) implies that the functor in (1) is fully faithful by taking the fiber along $\xrightarrow{\text{id}} \mathcal{V}^{\otimes}(\phi(X), \phi(Y))$. So if (2) holds, (1) holds if and only if θ is fully faithful.

Moreover observe that θ is essentially surjective if and only if for every $V \in \mathcal{V}^{\otimes}$ lying over $[n]$, the coCartesian lifting $V \rightarrow *$ of the maps $[0] \cong \{n\} \subset [n]$ induces an essential surjective functor $\mathcal{M}_V^{\otimes} \rightarrow \mathcal{M}_{[0]}^{\otimes}$.

Remark 1.25. The equivalence θ is natural with respect to an inert morphism $a : [n] \rightarrow [m]$ preserving the maximum in Ass:

$$\begin{array}{ccc}
\mathcal{M}_{[n]}^{\otimes} & \xrightarrow{\theta_{[n]}} & \mathcal{V}_{[n]}^{\otimes} \times \mathcal{M}_{[0]}^{\otimes} \\
f \downarrow & & \downarrow g \\
\mathcal{M}_{[m]}^{\otimes} & \xrightarrow{\theta_{[m]}} & \mathcal{V}_{[m]}^{\otimes} \times \mathcal{M}_{[0]}^{\otimes}
\end{array}$$

where f and g are induced by the coCartesian liftings in $\mathcal{M}^{\otimes}$ and $\mathcal{V}^{\otimes}$ of map a .

Notation 1.26. For any ∞ -category $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ with weak left action we call $\mathcal{M} := \mathcal{M}_{[0]}^{\otimes}$ the underlying ∞ -category of ϕ and say that ϕ exhibits $\mathcal{M}$ as weakly left tensored over $\mathcal{V}$.

Remark 1.27. Given a weakly left tensored category $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, and a fully faithful functor $\mathcal{N}^{\otimes} \hookrightarrow \mathcal{M}^{\otimes}$. Assume that the fiber transports of ϕ along the inert morphisms of Ass that preserves the maximum, preserve $\mathcal{N}^{\otimes}$. Also assume that for every coCartesian lift $X \rightarrow Y$ in $\mathcal{M}^{\otimes}$ of the map $[0] \cong \{n\} \subset [n]$, where $Y \in \mathcal{N}^{\otimes}$, the object X belongs to $\mathcal{N}^{\otimes}$. Then the restriction $\mathcal{N}^{\otimes} \hookrightarrow \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ is a weakly left tensored ∞ -category.

Example 1.28. Let $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ be a weakly left tensored ∞ -category and $\mathcal{N} \subset \mathcal{M}$ a full subcategory. Let $\mathcal{N}^{\otimes} \subset \mathcal{M}^{\otimes}$ the full subcategory of $X \in \mathcal{M}^{\otimes}$ lying over $[n] \in \text{Ass}$ corresponding to a family of objects $V_1, \dots, V_n \in \mathcal{V}, Y \in \mathcal{N}$ (that is, filling the Ass-part). By the above remark, the restriction $\mathcal{N}^{\otimes} \hookrightarrow \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ is an ∞ -category with weak left action, where $\mathcal{N}_{[0]}^{\otimes} \cong \mathcal{N}$. And we call $\mathcal{N}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ the full subcategory of $\mathcal{M}^{\otimes}$ with weak left action spanned by $\mathcal{N}$.

Definition 1.29. Let $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ be an ∞ -category with weak left action.

- We say that $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ exhibits $\mathcal{M}$ as left tensored over $\mathcal{V}$ if ϕ is a map of coCartesian fibrations over Ass. In this case, $\mathcal{V}^{\otimes}$ is a monoidal category.
- We say that $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ exhibits $\mathcal{M}$ as left tensored over $\mathcal{V}$ compatible with small colimits if $\mathcal{V}^{\otimes}$ is a monoidal category compatible with small colimits, $\mathcal{M}$ has small colimits and for any $V \in \mathcal{V}, X \in \mathcal{M}$ the functors $(-) \otimes X : \mathcal{V} \rightarrow \mathcal{M}, V \otimes (-) : \mathcal{M} \rightarrow \mathcal{M}$ preserve small colimits.

- A presentable left tensored category is a left tensored category $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ compatible with small colimits such that $\mathcal{V}$ and $\mathcal{M}$ are presentable.

Notation 1.30. (Multi-morphism spaces)

- Let $\mathcal{V}^{\otimes} \rightarrow \text{Ass}$ be an ∞ -operad and $V_1, \dots, V_n, W \in \mathcal{V}$ for some $n \geq 0$. We write

$$\text{Mul}_{\mathcal{V}}(V_1, \dots, V_n; W)$$

for the full subspace of $\mathcal{V}^{\otimes}(V, W)$ spanned by the morphisms $V \rightarrow W$ in $\mathcal{V}^{\otimes}$ lying over the active morphism $[1] \rightarrow [n]$ in Δ , where $V \in \mathcal{V}_{[n]}^{\otimes} \cong \mathcal{V}^{\times n}$ corresponds to $(V_1, \dots, V_n)$.

- Let $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ be an ∞ -category weakly left tensored over ∞ -operad and $V_1, \dots, V_n \in \mathcal{V}, X, Y \in \mathcal{M}$ for $n, m \geq 0$. We write

$$\text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, X; Y)$$

for the full subspace of $\mathcal{M}^{\otimes}(Z, Y)$ spanned by the morphisms $Z \rightarrow Y$ in $\mathcal{M}^{\otimes}$ lying over the maps $[0] \cong \{0\} \subset [n]$ in Δ , where $Z \in \mathcal{M}_{[n]}^{\otimes} \cong \mathcal{V}^{\times n} \times \mathcal{M}$ corresponds to $(V_1, \dots, V_n, X)$.

- There is a decomposition of mapping space in $\mathcal{M}^{\otimes}$ into multimorphism spaces:

$$\text{Map}_{\mathcal{M}}((V_1, \dots, V_n, X), (V'_1, \dots, V'_m, Y)) \cong \text{Map}_{\mathcal{V}^{\otimes}}((V_1, \dots, V_n), (V'_1, \dots, V'_m)) \times \text{Map}_{\mathcal{M}^{\otimes}}((V_1, \dots, V_n, X), Y)$$

by the condition (2) of weakly left tensored ∞ -category, and further we have equivalence:

$$\text{Map}_{\mathcal{M}^{\otimes}}((V_1, \dots, V_n, X), Y) \cong \prod_{i=1, \dots, n} \text{Mul}_{\mathcal{M}}(V_i, \dots, V_n, X; Y)$$

by the fact that any morphism in Ass in the form $([0] \rightarrow [n])^{op}$ sending $\{0\}$ to $\{i\}$ can be decomposed as $([0] \rightarrow \{i, i+1, \dots, n\} \cong [n-i+1] \rightarrow \{1, \dots, n\} = [n])^{op}$ where the last arrow is inert.

Definition 1.31. We say a weakly left tensored ∞ -category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$

- is locally small if $\mathcal{M}^{\otimes}, \mathcal{V}^{\otimes}$ are both locally small.
- is small if $\mathcal{M}_{[0]}^{\otimes} \cong \mathcal{M}$ is small and $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ is locally small.
- is absolute small if $\mathcal{M}^{\otimes}$ and $\mathcal{V}^{\otimes}$ are small.

Now we define morphisms between weakly left tensored ∞ -categories.

Definition 1.32. Let $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}, \phi' : \mathcal{M}'^{\otimes} \rightarrow \mathcal{V}'^{\otimes}$ be ∞ -categories with weak left action. We call a commutative diagram square of ∞ -categories over Ass

$$\begin{array}{ccc} \mathcal{M}^{\otimes} & \xrightarrow{\gamma} & \mathcal{M}'^{\otimes} \\ \phi \downarrow & & \downarrow \phi' \\ \mathcal{V}^{\otimes} & \xrightarrow{\rho} & \mathcal{V}'^{\otimes} \end{array}$$

a lax linear functor between weakly left tensored ∞ -categories if γ preserves coCartesian lifts of inert morphisms of Ass that preserves the maximum. We call a map of weakly left tensored ∞ -categories

- a linear functor if ϕ and ϕ' exhibits $\mathcal{M}$ and $\mathcal{M}'$ as left tensored and γ preserves all coCartesian lifts.
- a lax $\mathcal{V}$ -linear functor if ρ is the identity.
- a $\mathcal{V}$ -linear if it's lax $\mathcal{V}$ -linear and ϕ and ϕ' exhibits $\mathcal{M}, \mathcal{M}'$ as left tensored over $\mathcal{V}$, and γ preserves the coCartesian lifts.
- a linear $\mathcal{V}$ functor preserving colimits if ϕ, ϕ' are left tensored categories with small colimits, and $\gamma_{[0]}, \rho_{[0]}$ preserve small colimits.

Notation 1.33. Let $\text{Cat}_{\infty/\text{Ass}}^{\max} \subset \text{Cat}_{\infty/\text{Ass}}$ be the subcategory of coCartesian fibrations relative to the collection of inert morphisms which preserves the maximum. Let

$$\omega\text{LMod} \subset \text{Op}_{\infty} \times_{\text{Fun}(\{1\}, \text{Cat}_{\infty/\text{Ass}}^{\max})} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\max})$$

be the full subcategories of ∞ -categories with weak left action over some ∞ -operad. Let

$$\text{LMod} \subset \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\text{coCart}})$$

be the full subcategory of ∞ -categories with left action of a monoidal ∞ -category. Let

$$\text{LMod}^{\text{cc}} \subset \widehat{\text{LMod}}$$

be the subcategory of ∞ -categories with left action compatible with small colimits and functors preserving small colimits.

Notation 1.34. Consider the natural functor $\omega\text{LMod} \rightarrow \text{Op}_{\infty}$, given an operad $\mathcal{V}^{\otimes}$, we denote the fiber of ωLMod over $\mathcal{V}^{\otimes}$ as $\omega\text{LMod}_{\mathcal{V}}$. And likewise we denote the fiber of $\text{LMod} \rightarrow \text{Op}_{\infty}^{\text{mon}}$ over $\mathcal{W}^{\otimes}$ as $\text{LMod}_{\mathcal{W}}$.

Notation 1.35. Given two weakly left tensored categories $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}, \phi' : \mathcal{M}'^{\otimes} \rightarrow \mathcal{V}'^{\otimes}$:

- We denote the mapping space $\text{Map}_{\omega\text{LMod}}(\phi, \phi')$ as $\text{LaxLin}(\mathcal{M}, \mathcal{M}')^{\simeq}$.
- When $\mathcal{V}^{\otimes} = \mathcal{V}'^{\otimes}$, we denote the mapping space of the fiber $\text{Map}_{\omega\text{LMod}_{\mathcal{V}}}(\mathcal{M}, \mathcal{M}')$ as $\text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{M}')^{\simeq}$.
- By construction, we have the following pullback diagram:

$$\begin{array}{ccc} \text{Map}_{\omega\text{LMod}_{\mathcal{V}}}(\mathcal{M}, \mathcal{M}') & \xrightarrow{\quad} & \text{Map}_{\omega\text{LMod}}(\mathcal{M}, \mathcal{M}') \\ \downarrow & \lrcorner & \downarrow \\ * & \xrightarrow{\text{id}} & \text{Map}_{\text{Op}_{\infty}}(\mathcal{V}, \mathcal{V}') \end{array}$$

- Denote $\text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{M}')$ as the full subcategory of $\text{Fun}_{\mathcal{V}^{\otimes}}(\mathcal{M}^{\otimes}, \mathcal{M}'^{\otimes})$ consisting of lax $\mathcal{V}$ -linear functors.

Remark 1.36. Recall the construction of ωLMod , the underlying space of ∞ -category $\text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{M}')$ is exactly the mapping space of $\omega\text{LMod}_{\mathcal{V}}$ between objects $\mathcal{M}^{\otimes}$ and $\mathcal{M}'^{\otimes}$. This justifies our notation.

Remark 1.37. There is an equivalent definition of $\text{LaxLin}_V(-, -)$ in [17, definition C.2.1 and remark C.2.4]: if we regard ϕ, ϕ' as operads over $\text{LM}^\otimes$, then $\text{LaxLin}_V(\mathcal{M}, \mathcal{N})$ is $\text{Alg}_{\text{LM}^\otimes}(\mathcal{M}^\otimes, \mathcal{N}^\otimes) \times_{\text{Alg}_{\text{Ass}^\otimes}(\mathcal{V}^\otimes, \mathcal{V}^\otimes)} \{\text{id}\}$

Remark 1.38. A point in the space $\text{LaxLin}(\mathcal{M}, \mathcal{M}')^\simeq$ corresponds to a commutative diagram in the definition of lax linear functor. And the inclusion of (non-full) subcategory $\text{LMod} \rightarrow \omega\text{LMod}$ induces a fully faithful embedding (and thus summand of space) $\text{Lin}(\mathcal{M}, \mathcal{M}')^\simeq \hookrightarrow \text{LaxLin}(\mathcal{M}, \mathcal{M}')^\simeq$ consisting of linear functors.

Proposition 1.39. [14, 3.32] *The canonical functors*

$$\omega\text{LMod} \rightarrow \text{Op}_\infty, \text{LMod} \rightarrow \text{Op}_\infty^{\text{mon}}$$

are biCartesian fibrations and the embeddings

$$\omega\text{LMod} \subset \text{Op}_\infty \times_{\text{Cat}_{\infty/\text{Ass}}^{\text{max}}} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\text{max}})$$

$$\text{LMod} \subset \text{Cat}_{\infty/\text{Ass}}^{\text{coCart}} \times_{\text{Cat}_{\infty/\text{Ass}}^{\text{coCart}}} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\text{coCart}})$$

preserves Cartesian morphisms, and admits left adjoint over Op and Op^{mon} .

Proof. (1). First we notice the following facts:

- $\text{Cat}_{\infty/\text{Ass}}$ admits pullbacks, moreover, given diagram $A \rightarrow B \leftarrow C$ in $\text{Cat}_{\infty/\text{Ass}}^{\text{max}} \hookrightarrow \text{Cat}_{\infty/\text{Ass}}$, the pullback lies in $\text{Cat}_{\infty/\text{Ass}}^{\text{max}}$ too. Thus $\text{Cat}_{\infty/\text{Ass}}^{\text{max}}$ admits pullbacks and can be calculated in $\text{Cat}_{\infty/\text{Ass}}$.
- The evaluation map $ev_{\{1\}} : \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\text{max}}) \rightarrow \text{Cat}_{\infty/\text{Ass}}^{\text{max}}$ is coCartesian fibration (given by precomposition) and Cartesian fibration (given by pullback).
- So the pullback along the natural inclusion $\text{Op}_\infty \hookrightarrow \text{Cat}_{\infty/\text{Ass}}^{\text{max}} : \text{Op}_\infty \times_{\text{Cat}_{\infty/\text{Ass}}^{\text{max}}} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\text{max}}) \rightarrow \text{Op}_\infty$ is again biCartesian fibration.

Moreover, the full subcategory $\omega\text{LMod} \subset \text{Op}_\infty \times_{\text{Cat}_{\infty/\text{Ass}}^{\text{max}}} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\text{max}})$ is closed under the Cartesian lifting: Given an operad map $f : \mathcal{W}^\otimes \rightarrow \mathcal{V}^\otimes$ and $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$, the pullback in $\text{Cat}_{\infty/\text{Ass}}^{\text{max}}$ is:

$$\begin{array}{ccc} \mathcal{N}^\otimes & \xrightarrow{\quad f' \quad} & \mathcal{M}^\otimes \\ \phi' \downarrow \lrcorner & & \downarrow \phi \\ \mathcal{W}^\otimes & \xrightarrow{\quad f \quad} & \mathcal{V}^\otimes \end{array}$$

- It suffices to show $\phi' : \mathcal{N}^\otimes \rightarrow \mathcal{W}^\otimes$ is a weakly left tensored category.
- Since pullback in $\text{Cat}_{\infty/\text{Ass}}^{\text{max}}$ can be computed in $\text{Cat}_{\infty/\text{Ass}}$, the fiber of pullback $\mathcal{N}^\otimes$ is the pullback of fiber calculated in Cat_∞ , especially we have $\mathcal{N}_{[0]}^\otimes \cong \mathcal{M}_{[0]}^\otimes$.
- We have the following commutative diagram by coCartesian liftings over the inert maps that preserve the maximum :

$$\begin{array}{ccccc}
& & \mathcal{N}_{[0]}^{\otimes} & \xrightarrow{\quad} & \mathcal{M}_{[0]}^{\otimes} \\
& & \downarrow & \lrcorner & \downarrow \\
& & *= \mathcal{W}_{[0]}^{\otimes} & \xrightarrow{\quad} & *= \mathcal{V}_{[0]}^{\otimes} \\
& & \downarrow & \lrcorner & \downarrow \\
\mathcal{N}_{[n]}^{\otimes} & \xrightarrow{\quad} & \mathcal{M}_{[n]}^{\otimes} & & \\
\downarrow & \nearrow & \downarrow & \nearrow & \\
\mathcal{W}_{[n]}^{\otimes} & \xrightarrow{\quad} & \mathcal{V}_{[n]}^{\otimes} & &
\end{array}$$

- The front, back and right hand side squares are all pullback, so is the left hand side, thus we have $\mathcal{N}_{[n]}^{\otimes} \cong \mathcal{W}_{[n]}^{\otimes} \times \mathcal{N}_{[0]}^{\otimes}$. This is condition (1) of weakly left tensored category.
- The condition (2) of the weakly left tensored category for $\mathcal{N}^{\otimes}$ is by the following pullback squares and the pasting lemma:

$$\begin{array}{ccccc}
& & \mathcal{M}(f(X), f(Y)) & \xrightarrow{\quad} & \mathcal{M}(f(X), f(Y')) \\
& & \downarrow & \lrcorner & \downarrow \\
& & \mathcal{V}(f\phi'(X), f\phi'(Y)) & \xrightarrow{\quad} & \mathcal{V}(f\phi'(X), f\phi'(Y')) \\
& & \downarrow & \lrcorner & \downarrow \\
\mathcal{N}(X, Y) & \xrightarrow{\quad} & \mathcal{N}(X, Y') & & \\
\downarrow & \nearrow & \downarrow & \nearrow & \\
\mathcal{W}(\phi'(X), \phi'(Y)) & \xrightarrow{\quad} & \mathcal{W}(\phi'(X), \phi'(Y')) & &
\end{array}$$

where the back and left and right squares are pullbacks.

Thus ωLMod restricts to a Cartesian fibration over Op_{∞} .

Next we show the full subcategory embedding $\omega\text{LMod} \rightarrow \text{Op}_{\infty} \times_{\text{Cat}_{\infty/\text{Ass}}^{\text{max}}} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\text{max}})$ admits left adjoint fiberwise: fix an operad $\mathcal{V}^{\otimes}$,

- The fiber at the right hand side is equivalent to $(\text{Cat}_{\infty/\text{Ass}}^{\text{max}})_{/\mathcal{V}^{\otimes}} \cong \text{Cat}_{\infty/\mathcal{V}}^{\text{max}}$ which is the category of coCartesian fibrations over the inert maps in $\mathcal{V}^{\otimes}$ which image in Ass preserves the maximum and functors preserving them.
- Consider the categorical pattern on $\mathcal{V}^{\otimes}$: $\beta = (M, T, \Xi)$ where M is the collection of all inert morphisms in $\mathcal{V}^{\otimes}$ which image in Ass preserves the maximum, and T is the collection of all 2-simplexes, and Ξ is the collection of diagrams $[0]^{\triangleright} \cong [1] \rightarrow \mathcal{V}^{\otimes}$ corresponding to the inert map in $\mathcal{V}^{\otimes}$ lying over $([0] \cong \{n\} \rightarrow [n])^{op}$ in Ass . And another categorical pattern on $\mathcal{V}^{\otimes}$: $\beta' = (M, T, \emptyset)$ where M is the collection of all inert morphisms in $\mathcal{V}^{\otimes}$ which image in Ass preserves the maximum, and T is the collection of all 2-simplexes.

- The weakly left tensored categories are exactly those β -fibered objects defined in [22, B.0.19].
- Then by [22, proof of theorem B.0.20], the category $\text{sSet}_{/\beta}^+$ is the localization of $\text{sSet}_{/\beta'}$ with respect to the generating β -anodyne maps introduced in [22, B.1.1].
- Recall that the underlying category of model category $\text{sSet}_{/\beta}^+$ is $\text{Cat}_{\infty/\mathcal{V}^\otimes}^{\max}$ and the underlying category of the model category $\text{sSet}_{/\beta}$ is (the full subcategory) $\omega\text{LMod}_{\mathcal{V}}$. Thus the embedding $\omega\text{LMod}_{\mathcal{V}} \rightarrow \text{Cat}_{\infty/\mathcal{V}^\otimes}^{\max}$ admits a left adjoint.

Since $\omega\text{LMod} \hookrightarrow \text{Op}_{\infty} \times_{\text{Cat}_{\infty/\text{Ass}}^{\max}} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\max})$ is a fully faithful embedding which preserves Cartesian edges, then by [14, lemma 3.31], the embedding admits a left adjoint over Op_{∞} and ωLMod is coCartesian fibration over Op_{∞} . $\square$

Remark 1.40. Moreover, the Cartesian fibration $\omega\text{LMod} \rightarrow \text{Op}_{\infty}$ restricts to the Cartesian fibration $\text{LMod} \rightarrow \text{Op}_{\infty}^{\text{mon}}$ by [15, proposition 2.62].

1.3 identification of LM-operad and weakly left tensored category

In the previous sections, we have introduced two different models as generalizations of left modules: LM-operad and weakly left tensored category; it turns out that they are equivalent to each other.

Example 1.41. Recall that $\text{LM} \cong [1] \times \text{Ass}$, thus an LM operad can be viewed as a category over $[1] \times \text{Ass}$ that is coCartesian over the inert maps in $[1] \times \text{Ass} \cong \text{LM}$, there is a translation between inert maps in LM and the corresponding maps in $[1] \times \text{Ass}$: a map in $[1] \times \text{Ass}$ is inert in LM under the equivalence if:

- the unique map to the final object in the first component $[1]$, inert map in the second component Ass .
- identity at the initial object $\{0\}$, inert map which preserves the maximum in the second component.

Example 1.42. Consider the map $f : [0] \cong \{i\} \subset [n]$ in Δ , then $\text{id}_{\{0\}} \times f^{op}$ as a morphism in $[1] \times \text{Ass}$ corresponds to the active morphism $([1] \rightarrow [n+1])^{op}$ over $[1]$ preserves the maximum, and sending 0 to i . This corresponds to the left action map letting $a_r, a_{r+1}, \dots, a_n$ acting on m $(a_0, \dots, a_r, \dots, m) \rightarrow (a_0, \dots, a_{r-1}, m)$.

Especially an LM-operad is coCartesian over $[1]$, straightening it gives us an object in $\text{Fun}([1], \text{Cat}_{\infty/\text{Ass}})$, that is, a morphism over Ass : $f : A \rightarrow B$. Moreover, it can be shown that f exhibits A as weakly left tensored over A .

Proposition 1.43. *There exists an equivalence $\text{Cat}_{\infty/\text{LM}}^{\text{inert}} \cong \text{Cat}_{\infty/\text{Ass}}^{\text{inert}} \times_{\text{Fun}(\{1\}, \text{Cat}_{\infty/\text{Ass}}^{\max})} \text{Fun}([1], \text{Cat}_{\infty/\text{Ass}}^{\max})$ that restricts to an equivalence:*

$$\text{Op}_{\infty}^{\text{LM}} \cong \omega\text{LMod}, \text{Op}_{\infty}^{\text{LM, mon}} \cong \text{LMod}$$

Proof. By [14, proposition 3.33].

More precisely, given a functor $\alpha : \mathcal{O} \rightarrow \text{LM} \cong \text{Ass} \times [1]$ that is a coCartesian fibration relative to the collection of inert morphisms, then:

- The functor α is a map of coCartesian fibration over $[1]$.
- The pullback $\mathcal{V}^\otimes := (\text{Ass} \times \{1\}) \times_{\text{Ass} \times [1]} \mathcal{O} \rightarrow \text{Ass}$ is a coCartesian fibration relative to the collection of inert morphisms.
- The pullback $\mathcal{M}^\otimes := (\text{Ass} \times \{0\}) \times_{(\text{Ass} \times [1])} \mathcal{O} \rightarrow \text{Ass}$ is a coCartesian fibration relative to the collection of inert morphisms preserving the maximum and the functor $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ classified by α preserves the coCartesian lifts of inert morphisms preserving the maximum.
- For coCartesian fibrations $\mathcal{O} \rightarrow \text{LM}$, $\mathcal{O}' \rightarrow \text{LM}$ relative to the collection of inert morphisms a functor $\theta : \mathcal{O} \rightarrow \mathcal{O}'$ over LM is a map of coCartesian fibrations relative to the collection of inert morphisms if and only if θ is a map of coCartesian fibrations over $[1]$ classifying a commutative diagram of ∞ -categories:

$$\begin{array}{ccc} \mathcal{M}^\otimes & \xrightarrow{\alpha} & \mathcal{N}^\otimes \\ \phi \downarrow & & \downarrow \phi' \\ \mathcal{V}^\otimes & \xrightarrow{\beta} & \mathcal{W} \end{array}$$

over Ass such that β is a map of coCartesian fibrations relative to the collection of inert morphisms and α is a map of coCartesian fibrations relative to the collection of inert maps preserving the maximum.

□

1.4 envelope

This part follows [15, section 2.6].

Notation 1.44. Let $\text{Min}, \text{Max} \subset \text{Fun}([1], \text{Ass})$ be the full subcategories of morphisms preserving the minimum, maximum, respectively, and $\text{Act} = \text{Min} \cap \text{Max}$ is the intersection.

Remark 1.45. We notice the following facts:

- Every morphism in Ass uniquely factors as an inert morphism followed by an active morphism (which preserves maximum and minimum).
- Every morphism in Ass uniquely factors as an inert morphism preserving the maximum (minimum) followed by a morphism preserving minimum (maximum).
- By [21, lemma 5.2.8.19], the embeddings $\text{Act}, \text{Min}, \text{Max} \subset \text{Fun}([1], \text{Ass})$ admit left adjoint, where a morphism in $\text{Fun}([1], \text{Ass})$ with local target is a local equivalence if and only if its image in Ass under evaluation at the source is inert, is inert and preserves maximum, is inert and preserves minimum, respectively, and its image in Ass under evaluation at the target is an equivalence.

Example 1.46. Given a morphism $f : [n] \rightarrow [m]$ in Ass , consider its factorization $f = g \circ f'$ where $f \in \text{Act}$ (or Min , or Max), then the localization sends f to f' :

$$\begin{array}{ccc} [n] & \xrightarrow{f} & [m] \\ g \downarrow & & \downarrow \cong \\ [n'] & \xrightarrow{f'} & [m] \end{array}$$

The next notation is [15, notation 2.92]:

Notation 1.47. Let $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ be a weakly left tensored ∞ -category. Let

- $\text{Env}(\mathcal{V})^\otimes := \text{Act} \times_{\text{Fun}(\{0\}, \text{Ass})} \mathcal{V}^\otimes \rightarrow \text{Fun}(\{1\}, \text{Ass})$.
- $\text{LEnv}(\mathcal{M})^\otimes := \text{Min} \times_{\text{Fun}(\{0\}, \text{Ass})} \mathcal{M}^\otimes \rightarrow \text{Fun}(\{1\}, \text{Ass})$.

Remark 1.48. • The diagonal embedding $\text{Ass} \subset \text{Act}$, $\text{Ass} \subset \text{Min}$ induces embeddings

$$\mathcal{V}^\otimes \subset \text{Env}(\mathcal{V})^\otimes, \mathcal{M}^\otimes \subset \text{LEnv}(\mathcal{M})^\otimes$$

, which admit left adjoints relative to $\text{Fun}(\{1\}, \text{Ass})$.

- $\text{Env}(\mathcal{V})^\otimes \rightarrow \text{Ass}$ and $\text{LEnv}^\otimes \rightarrow \text{Ass}$ are coCartesian fibrations.

Construction 1.49. There is a map

$$\text{LEnv}(\mathcal{M})^\otimes := \text{Min} \times_{\text{Fun}(\{0\}, \text{Ass})} \mathcal{M}^\otimes \subset \text{Fun}([1], \text{Ass}) \times_{\text{Fun}(\{0\}, \text{Ass})} \mathcal{M}^\otimes \rightarrow \text{Act} \times_{\text{Fun}(\{0\}, \text{Ass})} \mathcal{V}^\otimes = \text{Env}(\mathcal{V})^\otimes$$

of coCartesian fibrations over Ass where the last morphism is induced by the localization functor and the structure map $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$.

Definition 1.50. Let $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ be a weakly left tensored ∞ -category.

- (1) The monoidal envelope of $\mathcal{V}^\otimes \rightarrow \text{Ass}$ is the functor

$$\text{Env}(\mathcal{V})^\otimes \rightarrow \text{Act} \rightarrow \text{Fun}(\{1\}, \text{Ass}) \cong \text{Ass}$$

- (2) The left tensored envelope if the functor

$$\text{LEnv}(\mathcal{M})^\otimes \rightarrow \text{Env}(\mathcal{V})^\otimes$$

Here is a list of properties of monoidal envelope and left tensored envelope:

Proposition 1.51. (1) Let $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ be a weakly left tensored ∞ -category classified by an LM-operad $\mathcal{O} \rightarrow \text{LM}$. Then $\text{LEnv}(\mathcal{M})^\otimes \rightarrow \text{Env}(\mathcal{V})^\otimes$ is a left tensored ∞ -category classified by $\text{Env}_{\text{LM}}(\mathcal{O}) \rightarrow \text{LM}$.

- (2) The embedding $\mathcal{V}^\otimes \rightarrow \text{Env}(\mathcal{V})^\otimes$ is a map of operads. For any monoidal category $\mathcal{W}^\otimes \rightarrow \text{Ass}$ the induced functor $\text{Fun}^\otimes(\text{Env}(\mathcal{V}), \mathcal{W}) \rightarrow \text{Alg}_{\mathcal{V}}(\mathcal{W})$ is an equivalence.

- (3) *The embedding $\mathcal{M}^\otimes \rightarrow \text{LEnv}(\mathcal{M})^\otimes$ is a lax linear functor. For any left tensored ∞ -category $\mathcal{N}^\otimes \rightarrow \text{Env}(\mathcal{V})^\otimes$, the induced map $\text{Lin}_{\text{Env}(\mathcal{V})}(\text{LEnv}(\mathcal{M}), \mathcal{N})^\simeq \rightarrow \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})^\simeq$ is an equivalence, where on the right hand side $\mathcal{N}$ is regarded as $\mathcal{V}$ left tensored category via the pullback along the natural embedding $\mathcal{V}^\otimes \rightarrow \text{Env}(\mathcal{V})^\otimes$.*

Proof. (1) is by [14, proposition 3.111]. (2) is by [15, proposition 2.97(1)].

And recall that in the world of $\text{Op}_\infty^{\text{LM}}$, the forgetful functor $\text{Alg}_{\text{LEnv}(\mathcal{M})}(\mathcal{N}) \rightarrow \text{Alg}_{\mathcal{M}}(\mathcal{N})$ admits a fully faithful left adjoint by [14, proposition 9.30], which restricts to an equivalence between $\text{Fun}^\otimes(\text{LEnv}(\mathcal{M}), \mathcal{N}) \cong \text{Alg}_{\mathcal{M}}(\mathcal{N})$. This implies that by taking summand of $\text{Fun}^\otimes(\text{LEnv}(\mathcal{M}), \mathcal{N})^\simeq \subset \text{Alg}_{\text{LEnv}(\mathcal{M})}(\mathcal{N})^\simeq$ consisting of functors preserving all coCartesian liftings, the functor induced by precomposition $\text{Alg}_{\text{LEnv}(\mathcal{M})}(\mathcal{N})^\simeq \rightarrow \text{Alg}_{\mathcal{M}}(\mathcal{N})^\simeq$ restricts to an equivalence of spaces $\text{Fun}^\otimes(\text{LEnv}(\mathcal{M}), \mathcal{N})^\simeq \cong \text{Alg}_{\mathcal{M}}(\mathcal{N})^\simeq$.

Noticing that the mapping space of $\text{Op}_\infty^{\text{LM}}$ is $\text{Map}_{\text{Op}_\infty^{\text{LM}}}(\mathcal{M}, \mathcal{N}) \cong \text{Alg}_{\mathcal{M}}(\mathcal{N})^\simeq$, and by the equivalence $\omega\text{LMod} \cong \text{Op}_\infty^{\text{LM}}$, we have the equivalence of mapping spaces $\text{Lin}_{\text{Env}(\mathcal{V})}(\text{LEnv}(\mathcal{M}), \mathcal{N})^\simeq \cong \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})^\simeq$ induced by precomposition. $\square$

An object in $\text{LEnv}(\mathcal{M})$ is in the form of $((V_1, \dots, V_n, X), f : [n+1] \rightarrow [0] \in \text{Min})$, and by construction it is the coCartesian lifting over the active map f of $(V_1, \dots, V_n)$ regarded as an object over $[n+1]$ as the image of f . And noticing that the embedding $\mathcal{M}^\otimes \rightarrow \text{LEnv}(\mathcal{M})^\otimes$ is fully faithful, we have the following description of the mapping space in $\text{LEnv}(\mathcal{M})$:

Lemma 1.52. *The embedding $\mathcal{M}^\otimes \rightarrow \text{LEnv}(\mathcal{M})^\otimes$ induces an equivalence of mapping spaces:*

$$\text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, X; Y) \rightarrow \text{Map}_{\text{LEnv}(\mathcal{M})}(((V_1, \dots, V_n, X), f : [n+1] \rightarrow [0]), (Y, \text{id}))$$

Remark 1.53. From another point of view, by definition, $\text{Map}_{\text{LEnv}(\mathcal{M})}(((V_1, \dots, V_n, X), f : [n+1] \rightarrow [0]), (Y, \text{id}))$ is the pullback $\text{Map}_{\text{Min}}(f, \text{id}_{[0]}) \times_{\text{Map}_{\text{Ass}}([n+1], [0])} \text{Map}_{\mathcal{M}^\otimes}((V_1, \dots, V_n, X), Y)$. Noticing that in Ass , there exists only one edge $e : [n+1] \rightarrow [0]$ making the following diagram commute:

$$\begin{array}{ccc} [0] & \xlongequal{\quad} & [0] \\ e \uparrow & & \uparrow g \\ [n+1] & \xrightarrow{\quad f \quad} & [0] \end{array}$$

since there exists only one choice for $g = \text{id} : [0] \rightarrow [0]$, and $e = f$. Thus the pullback $\text{Map}_{\text{Min}}(f, \text{id}_{[0]}) \times_{\text{Map}_{\text{Ass}}([n+1], [0])} \text{Map}_{\mathcal{M}^\otimes}((V_1, \dots, V_n, X), Y)$ is just the fiber of $\text{Map}_{\mathcal{M}^\otimes}((V_1, \dots, V_n, X), Y) \rightarrow \text{Map}_{\text{Ass}}([n+1], [0])$ over the point $f : [n+1] \rightarrow [0]$ corresponding to the map preserving the minimum $[0] \cong \{0\} \rightarrow [n] \in \Delta$, which is exactly Mul .

Moreover, when $\mathcal{V}$ is monoidal, we have the following description of the local equivalences:

Proposition 1.54. *When $\mathcal{V}$ is monoidal ∞ -category, the embedding $\mathcal{V}^\otimes \rightarrow \text{Env}(\mathcal{V})^\otimes$ admits a left adjoint over Ass .*

Proof. This is [14, lemma A.7]. Especially, we have the description of the set of local equivalences: given an object $((V_1, \dots, V_n), f : [n] \rightarrow [1]) \in \text{Act}$ in $\text{Env}(\mathcal{V})$, denote the coCartesian edge

$$\begin{array}{ccc} [n] & \xrightarrow{f} & [1] \\ f \downarrow & & \parallel \\ [1] & \xlongequal{\quad} & [1] \end{array}$$

over the active map $f : [n] \rightarrow [1]$ as c , then the map $(c, \text{the above square})$ in $\text{Env}(\mathcal{V})$ is a local equivalence. □

1.5 symmetric operad

In this subsection, we define the symmetric operad following [22, chapter 2]. Although the objects in $\text{Op}_{\infty}^{\text{LM}, \text{mon}}$ are left modules defined by non-symmetric operad, the symmetric monoidal structure on $\text{Op}_{\infty}^{\text{LM}, \text{mon}}$ itself is defined using symmetric operad.

Definition 1.55. Let $\mathcal{F}in_*$ be the category of pointed finite sets. We denote pointed finite sets as $\langle n \rangle := \{*, 1, \dots, n\}$ for $n \geq 0$, where $*$ is the base point. We call a map α of pointed finite sets $\langle n \rangle \rightarrow \langle m \rangle$

- inert if for every $1 \leq i \leq n$, the fiber of α consists precisely of one element.
- standard inert if $m = 1$.
- active if $\alpha^{-1}(*) = \{*\}$.

For every $n \geq 0$ there are n -many standard inert morphisms $\langle n \rangle \rightarrow \langle 1 \rangle$, where the i -th standard inert morphism $\langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$ sends i to 1.

Definition 1.56. A symmetric ∞ -operad is a coCartesian fibration $\phi : \mathcal{O}^{\otimes} \rightarrow \mathcal{F}in_*$ relative to the collection of inert morphisms such that the following conditions are satisfied:

- (1) for any $\langle n \rangle \in \mathcal{F}in_*$ the standard inert morphisms $\langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$ induce an equivalence $\mathcal{O}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{O}^{\times n}$ where $\mathcal{O} := \mathcal{O}_{\langle n1 \rangle}^{\otimes}$, and $\mathcal{O}_{\langle 0 \rangle}^{\otimes}$ is contractible.
- (2) for any $Y \in \mathcal{O}^{\otimes}$ lying over $\langle n \rangle$ and $Z \in \mathcal{O}^{\otimes}$ lying over $\langle m \rangle$ the coCartesian liftings $Y \rightarrow Y_i$ of the standard inert morphisms induce an equivalence

$$\mathcal{O}^{\otimes}(Z, Y) \rightarrow \mathcal{F}in_*(\langle m \rangle, \langle n \rangle) \times_{\prod_{i=1}^n \mathcal{F}in_*(\langle m \rangle, \langle 1 \rangle)} \prod_{i=1}^n \mathcal{O}^{\otimes}(Z, Y_i)$$

For any symmetric ∞ -operad $\mathcal{O}^{\otimes} \rightarrow \mathcal{F}in_*$ we call $\mathcal{O} := \mathcal{O}_{\langle 1 \rangle}^{\otimes}$ the underlying ∞ -category.

Example 1.57. $\mathcal{F}in_*$ itself is a symmetric operad, and we denote it as $\text{Comm}^{\otimes}$. We also consider maps of symmetric operads:

Definition 1.58. Let $\mathcal{O}^{\otimes} \rightarrow \mathcal{F}in_*$, $\mathcal{O}'^{\otimes} \rightarrow \mathcal{F}in_*$ be symmetric ∞ -operads.

- A map of symmetric ∞ -operads $\phi : \mathcal{O}'^{\otimes} \rightarrow \mathcal{O}^{\otimes}$ or $\mathcal{O}'$ -algebra in $\mathcal{O}$ is a map of coCartesian fibrations $\mathcal{O}'^{\otimes} \rightarrow \mathcal{O}^{\otimes}$ relative to the collection of inert morphisms of $\mathcal{F}in_*$.

- An $\mathcal{O}$ -monoidal ∞ -category is a map of symmetric ∞ -operads $\phi : \mathcal{O}'^{\otimes} \rightarrow \mathcal{O}^{\otimes}$ that is a coCartesian fibration. For $\mathcal{O}^{\otimes} = \mathcal{F}in_*$ we call $\mathcal{O}$ -monoidal ∞ -categories symmetric monoidal ∞ -categories.

Let $\mathcal{O}'^{\otimes} \rightarrow \mathcal{O}^{\otimes}$, $\mathcal{O}''^{\otimes} \rightarrow \mathcal{O}^{\otimes}$ be $\mathcal{O}$ -monoidal ∞ -categories.

- A lax $\mathcal{O}$ -monoidal functor $\mathcal{O}'^{\otimes} \rightarrow \mathcal{O}''^{\otimes}$ is a map of symmetric ∞ -operads $\mathcal{O}'^{\otimes} \rightarrow \mathcal{O}''^{\otimes}$ over $\mathcal{O}^{\otimes}$.
- An $\mathcal{O}$ -monoidal ∞ functor is a map of coCartesian fibrations preserving all coCartesian liftings over $\mathcal{O}^{\otimes}$.

Notation 1.59. Let $\text{Op}^{HA} \subset \text{Cat}_{\infty/\mathcal{F}in_*}$ denote the subcategory consisting of symmetric operads and maps between symmetric operads.

Notation 1.60. Let $\mathcal{O}'^{\otimes} \rightarrow \mathcal{O}^{\otimes}$, $\mathcal{O}''^{\otimes} \rightarrow \mathcal{O}^{\otimes}$ be maps of symmetric ∞ -operads. Let

$$\text{Alg}_{\mathcal{O}'/\mathcal{O}}(\mathcal{O}'') \subset \text{Fun}_{\mathcal{O}^{\otimes}}(\mathcal{O}'^{\otimes}, \mathcal{O}''^{\otimes})$$

be the full subcategory of maps of symmetric ∞ -operads over $\mathcal{O}^{\otimes} \rightarrow \mathcal{F}in_*$. For $\mathcal{O}^{\otimes} = \mathcal{F}in_*$, we set

$$\text{Alg}_{\mathcal{O}'}(\mathcal{O}'') := \text{Alg}_{\mathcal{O}'/\mathcal{O}}(\mathcal{O}'')$$

We usually build a symmetric monoidal ∞ -category from a special kind of symmetric monoidal operad:

Definition 1.61. We call a symmetric monoidal ∞ -category $\mathcal{O}^{\otimes} \rightarrow \mathcal{F}in_*$ Cartesian if the tensor unit $\mathbf{1}$ is a final object of $\mathcal{O}$ and for any $X, Y \in \mathcal{O}$ the canonical maps $X \otimes Y \rightarrow X \otimes \mathbf{1} \cong X$, $X \otimes Y \rightarrow \mathbf{1} \otimes Y \cong Y$ exhibits $X \otimes Y$ as the product of X and Y in $\mathcal{O}$.

Definition 1.62. Dually, we call a symmetric monoidal ∞ -category $\mathcal{O}^{\otimes} \rightarrow \mathcal{F}in_*$ coCartesian if the tensor unit $\mathbf{1}$ is an initial object of $\mathcal{O}$ and for any $X, Y \in \mathcal{O}$ the canonical maps $X \cong \mathbf{1} \otimes X \rightarrow Y \otimes X$, $Y \cong \mathbf{1} \otimes Y \rightarrow X \otimes Y$ exhibits $Y \otimes X$ as the coproduct of X and Y in $\mathcal{O}$.

Definition 1.63. Let $\mathcal{D}$ be an ∞ -category with finite products and $\phi : \mathcal{O}^{\otimes} \rightarrow \mathcal{F}in_*$ a symmetric ∞ -operad. We call a functor $\alpha : \mathcal{O}^{\otimes} \rightarrow \mathcal{D}$ an $\mathcal{O}$ -monoid in $\mathcal{D}$ if for every $X \in \mathcal{O}^{\otimes}$ lying over $\langle n \rangle$ the coCartesian liftings $X \rightarrow X_i$ of the standard inert morphisms induce an equivalence $\alpha(X) \rightarrow \prod_{i=1}^n \alpha(X_i)$.

Example 1.64. Note that $\mathcal{O}$ -monoids in Cat are classified by $\mathcal{O}$ -monoidal categories.

Notation 1.65. Let $\text{Mon}_{\mathcal{O}}(\mathcal{D}) \subset \text{Fun}(\mathcal{O}^{\otimes}, \mathcal{D})$ be the full subcategory of $\mathcal{O}$ -monoids.

By [22, proposition 2.4.1.7] for any ∞ -category $\mathcal{D}$ with finite products there is a Cartesian symmetric monoidal ∞ -category $\mathcal{D}^{\times} \rightarrow \mathcal{F}in_*$ where $\mathcal{D}_{(1)}^{\times} \cong \mathcal{D}$, and there exists a canonical functor $\mathcal{D}^{\times} \rightarrow \mathcal{D}$ such that for any symmetric ∞ -operad $\mathcal{O} \rightarrow \mathcal{F}in_*$ the functor $\text{Alg}_{\mathcal{O}}(\mathcal{D}) \rightarrow \text{Mon}_{\mathcal{O}}(\mathcal{D})$ is an equivalence.

1.6 presheaf of left module

In this section:

- We consider the presheaf of a left tensored category, and the multiplicative structure on it.
- This subsection is just a brief review on the presehaf functor taking presheaf on $\text{Alg}_{\mathcal{LM}}(\text{Cat})$ developed in [22, proposition 4.8.1.10], and transfer it to our version of category of left modules LMod under the [the identification between them](#).

Consider the Cartesian symmetric monoidal structure on $\widehat{\text{Cat}}^\times$, we have the following modification of it:

Notation 1.66. Let $(\text{Cat}^{cc})^\otimes \subset \widehat{\text{Cat}}^\times$ be the symmetric suboperad whose colors are the ∞ -categories having small colimits and whose multi-morphisms $C_1, \dots, C_n \rightarrow D$ correspond to a functor $C_1 \times \dots \times C_n \rightarrow D$ preserving small colimits componentwise. Let $(\text{Pr}^L)^\otimes \subset \text{Cat}_\infty^\otimes$ be the full symmetric suboperad of presentable ∞ -categories.

By [22, remark 4.8.1.8], the inclusion $i : (\text{Cat}^{cc})^\otimes \subset \widehat{\text{Cat}}^\times$ admits a symmetric monoidal left adjoint $\mathcal{P} : \widehat{\text{Cat}}^\times \rightarrow (\text{Cat}^{cc})^\otimes$ where for a small category $\mathcal{C} \in \text{Cat} \subset \widehat{\text{Cat}}$, $\mathcal{P}(\mathcal{C})$ is the presheaf category of a small ∞ -category.

Considering the symmetric operad $\mathcal{LM}$ developed in [22, 4.2.1], by [22, remark 4.19], we have the following adjunction:

$$i : \text{Alg}_{\mathcal{LM}}(\text{Pr}^L) \rightleftarrows \text{Alg}_{\mathcal{LM}}(\widehat{\text{Cat}}) : \mathcal{P}$$

- Remark 1.67.**
- Given an inclusion of symmetric operads $i : \mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$, consider the pullback of the following diagram $\mathcal{C}^\otimes \xrightarrow{i} \mathcal{D}^\otimes \xleftarrow{i} \mathcal{C}^\otimes$ in Op^{HA} , since this pullback can be calculated in $\text{Cat}/\mathcal{F}in_*$, and $\mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$ is an inclusion (that is, a monomorphism in Cat), we have $\mathcal{C}^\otimes$ being the pullback of the above diagram in Op^{HA} .
 - By [22, section 3.2.4], $\text{Alg}_-(\text{--})$ is the underlying category of $\text{Alg}_-(\text{--})^\otimes$ in [22, 3.2.4.1], and $\text{Alg}_-(\text{--})^\otimes$ is the internal Hom of Op^{HA} (with respect to the tensor product between operads), thus preserves limits at the second variable. So for an inclusion $\mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$, $\text{Alg}_{\mathcal{LM}}(\mathcal{C})^\otimes \rightarrow \text{Alg}_{\mathcal{LM}}(\mathcal{D})^\otimes$ is again an inclusion (of ∞ -operads)
 - Especially, consider the underlying category of operads, $i : \text{Alg}_{\mathcal{LM}}(\text{Pr}^L) \rightarrow \text{Alg}_{\mathcal{LM}}(\widehat{\text{Cat}})$ is an inclusion, and can be regarded as a (non-full) subcategory of the latter.
 - Moreover, by the description of coCartesian edges in $\text{Alg}_{(-)}(\text{--})^\otimes$ in [22, proposition 3.2.4.3], $\mathcal{P} : \text{Alg}_{\mathcal{LM}}(\text{Cat})^\otimes \rightarrow \text{Alg}_{\mathcal{LM}}(\text{Pr}^L)^\otimes$ is symmetric monoidal.

Remark 1.68. By the [22, proof of theorem 4.8.5.16], we have the following descriptions on the suboperad inclusion $\text{Alg}_{\mathcal{LM}}^\otimes(\text{Pr}^L) \subset \text{Alg}_{\mathcal{LM}}^\otimes(\widehat{\text{Cat}})^\otimes$, which image (regarded as a non-full subcategory) consists of:

- Object: $\{(C_i, M_i)\}_i$ where (C_i, M_i) is a presentably left tensored category, that is, an object in $\text{Alg}_{\text{LM}}(\text{Pr}^L)$.
- Morphism: a multimorphism over $\alpha : \langle n \rangle \rightarrow \langle m \rangle$ in $\text{Alg}_{\text{LM}}(\widehat{\text{Cat}})^{\otimes=\times}$, $F : \{(C_i, M_i)\}_{i \in \langle n \rangle^\circ} \rightarrow \{(D_j, N_j)\}_{j \in \langle m \rangle^\circ}$ is a multimorphism in $\text{Alg}_{\text{LM}}(\text{Pr}^L)^{\otimes}$ iff for each $j \in \langle m \rangle^\circ$ $\prod_{i: \alpha(i)=j} C_i \rightarrow D_j$ preserves colimits separately in each variable, and $\prod_{i: \alpha(i)=j} M_i \rightarrow N_j$ preserves colimits separately in each variable.

Notation 1.69. Under the [equivalence between LMod and \$\text{Op}_{\infty}^{\text{mon}, \text{LM}}\$](#) , and by the equivalence $\text{Alg}_{\text{LM}}(\text{Cat})^{\otimes} \cong \text{Mon}_{\text{LM}}(\text{Cat}) \cong \text{Op}_{\infty}^{\text{mon}, \text{LM}}$, let

- PrLMod denote the subcategory of $\widehat{\text{LMod}}$ corresponding to $\text{Alg}_{\text{LM}}(\text{Pr}^L)$. Objectwise, this corresponds to [presentable left tensored category in the previous definition](#), and morphism-wise this corresponds to [linear functor preserving colimits](#).
- $\text{PrLMod}^{\otimes}$ denote the suboperad of $\widehat{\text{LMod}}^{\times}$ corresponding to $\text{Alg}_{\text{LM}}(\text{Pr}^L)^{\otimes}$. And especially, given presentable left tensored categories $\{\mathcal{M}_i^{\otimes} \rightarrow \mathcal{V}_i^{\otimes}\}$ and $\mathcal{N}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, we can interpret the multimorphism space of $\text{PrLMod}^{\otimes}$ as a full subspace of multimorphism space of $\widehat{\text{LMod}}^{\times}$ consisting of those linear functors $\prod_{i=1}^n \mathcal{M}_i^{\otimes} \rightarrow \mathcal{N}^{\otimes}$ preserving colimits componentwise.
- Given a small left tensored category $\mathcal{M}^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$, we denote its image under the functor $\mathcal{P}(-) : \widehat{\text{LMod}} \rightarrow \text{PrLMod}$ as $\mathcal{P}(\mathcal{M})^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$, and regard it as the ‘presheaf of left module $\mathcal{M}$ ’. And taking presheaf of a left module is symmetric monoidal, since $\mathcal{P} : \text{Alg}_{\text{LM}}(\text{Cat}) \rightarrow \text{Alg}_{\text{LM}}(\text{Pr}^L)$.
- And the unit map of the above adjunction gives us the Yoneda embedding of left modules

$$\mathcal{M}^{\otimes} \hookrightarrow \mathcal{P}(\mathcal{M})^{\otimes}$$

And especially, we have the following universal property of $\mathcal{P}(-)$:

Proposition 1.70. (1) *Given a left tensored ∞ -category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, and presentably left tensored ∞ -category $\mathcal{N}^{\otimes} \rightarrow \mathcal{W}^{\otimes}$, there is an equivalence of spaces induced by the Yoneda embedding of left modules:*

$$\text{Lin}^L(\mathcal{P}(\mathcal{M}), \mathcal{N})^{\simeq} \rightarrow \text{Lin}(\mathcal{M}, \mathcal{N})^{\simeq}$$

where the left hand side is the space of linear functors preserving colimits.

(2) *Given a left tensored category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, and presentable left tensored category $\mathcal{N}^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$, there is an equivalence of spaces:*

$$\text{Lin}_{\mathcal{P}}^L(\mathcal{P}(\mathcal{M}), \mathcal{Y}^* \mathcal{N})^{\simeq} \rightarrow \text{Lin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})^{\simeq}$$

induced by the Yoneda embedding $\mathcal{M}^{\otimes} \rightarrow \mathcal{P}(\mathcal{M})^{\otimes}$, where $\mathcal{Y}^* \mathcal{N}$ is the pullback of $\mathcal{N}^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$ along the Yoneda embedding $\mathcal{Y} : \mathcal{V}^{\otimes} \hookrightarrow \mathcal{P}(\mathcal{V})^{\otimes}$, which is the Cartesian lifting of $\widehat{\text{LMod}} \rightarrow \text{Mon}$ over the map $\mathcal{Y}$ by [proposition 2.64](#).

- (3) Given a weakly left tensored ∞ -category $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$, and a presentably left tensored ∞ -category $\mathcal{N}^\otimes \rightarrow \mathcal{W}^\otimes$, there is an equivalence induced by the embedding $\mathcal{M}^\otimes \rightarrow \text{LEnv}(\mathcal{M})^\otimes \rightarrow \text{PLEnv}(\mathcal{M})^\otimes$:

$$\text{Lin}^L(\text{PLEnv}(\mathcal{M}), \mathcal{N})^\simeq \rightarrow \text{LaxLin}(\mathcal{M}, \mathcal{N})^\simeq$$

Proof. (1) and (2) is by adjunction. (3) is by the universal property of $\text{LEnv}(-)$ and $\mathcal{P}(-)$. $\square$

Remark 1.71. By [14, corollary 9.32], the equivalence in (1) can be enhanced into an equivalence:

$$\text{LaxLin}^L(\mathcal{P}(\mathcal{M}), \mathcal{N}) \rightarrow \text{LaxLin}(\mathcal{M}, \mathcal{N})$$

where the left hand side is the lax linear functors between $\mathcal{P}(\mathcal{M})^\otimes$ and $\mathcal{N}^\otimes$ which fiberwise preserves colimits. And it restricts to an equivalence:

$$\text{Lin}^L(\mathcal{P}(\mathcal{M}), \mathcal{N}) \rightarrow \text{Lin}(\mathcal{M}, \mathcal{N})$$

.

Likewise we can define the suboperad $\text{PrMon}^\otimes \subset \widehat{\text{Mon}}^\times(\text{Cat})$ consisting of presentably monoidal ∞ -categories and monoidal ∞ functor preserving colimits.

By [22, lemma 4.5.3.2], $\text{LMod} \rightarrow \text{Mon}(\text{Cat})$ is a coCartesian fibration, which is obviously product preserving, thus can be enhanced into a symmetric monoidal functor $\text{LMod}^\times \rightarrow \text{Mon}(\text{Cat})^\times$, and obviously this is again a coCartesian fibration.

By restricting to the (non) full subcategory $\text{PrLMod} \subset \widehat{\text{LMod}}$, $\text{PrLMod} \rightarrow \text{PrMon}$ is also a coCartesian fibration by [22, lemma 4.5.3.6], and can be enhanced into a symmetric monoidal coCartesian fibration between ∞ -operads $\text{PrLMod}^\otimes \rightarrow \text{PrMon}^\otimes$.

Notation 1.72. And given a symmetric monoidal ∞ -category $\mathcal{V}^\otimes$, we can define the symmetric monoidal ∞ -category $\text{LMod}_{\mathcal{V}}(\text{Cat})^\otimes$ as:

$$\begin{array}{ccc} \text{LMod}_{\mathcal{V}}(\text{Cat})^\otimes & \longrightarrow & \text{LMod}^\times \\ \downarrow \lrcorner & & \downarrow \\ \text{Comm}^\otimes & \xrightarrow{\{\mathcal{V}^\otimes\}} & \text{PrMon}^\otimes \end{array}$$

.

Notation 1.73. And given a presentably symmetric monoidal ∞ -category $\mathcal{V}^\otimes$, we can define the symmetric monoidal ∞ -category $\text{LMod}_{\mathcal{V}}(\text{Pr}^L)^\otimes$ as:

$$\begin{array}{ccc} \text{LMod}_{\mathcal{V}}(\text{Pr}^L)^\otimes & \longrightarrow & \text{PrLMod}^\otimes \\ \downarrow \lrcorner & & \downarrow \\ \text{Comm}^\otimes & \xrightarrow{\{\mathcal{V}^\otimes\}} & \text{PrMon}^\otimes \end{array}$$

, which can be identified with [22, construction 5.1.4.6]. And we denote the tensor product in $\mathbf{LMod}_V(\mathbf{Pr}^L)^\otimes$ as $\otimes_V$.

Consider the coCartesian edge $d : \{(C_i, M_i)\}_{i \in \langle n \rangle^\circ} \rightarrow (\otimes_{i=1}^n C_i, \otimes_{i=1}^n M_i)$ in $\mathbf{PrLMod}^\otimes$ over an active map $\alpha : \langle n \rangle \rightarrow \langle 1 \rangle$ (which is in this form since fiberwise the tensor product operation in $\mathbf{Alg}_{\mathbf{LM}}(\mathbf{Pr}^L)^\otimes$ is just tensor product operation in $\mathbf{Pr}^{L,\otimes}$ by [22, 3.2.4.4]).

Consider the image of d under the inclusion $\mathbf{PrLMod}^\otimes \hookrightarrow \widehat{\mathbf{LMod}}^\times$, we have the following factorization in $\widehat{\mathbf{LMod}}^\times$:

$$\begin{array}{ccccc} \{(C_i, M_i)\}_i & \xrightarrow{d_1} & (\prod C_i, \prod M_i) & \xrightarrow{d_2} & (\otimes C_i, \otimes M_i) \\ \downarrow & & \downarrow & & \downarrow \\ \{C_i\}_i & \longrightarrow & \prod C_i & \xrightarrow{q} & \otimes C_i \end{array}$$

where d_1 is the coCartesian lifting in $\widehat{\mathbf{LMod}}^\times$, and fiberwise d_2 and q are the universal componentwise colimit preserving functor $\prod C_i \rightarrow \otimes C_i, \prod M_i \rightarrow \otimes M_i$ developed in [22, 4.8.1].

Proposition 1.74. *Given n presentably left tensored ∞ -categories $M_i^\otimes \rightarrow C_i^\otimes$, and any presentable left tensored ∞ -category $N^\otimes \rightarrow \otimes C_i^\otimes$, there is an equivalence of mapping spaces induced by $d_2 : \prod_{i=1}^n M_i \rightarrow \otimes_{i=1}^n M_i$ constructed as above:*

$$d_2^* : \mathrm{Lin}_{\prod C_i}^{LL}(\prod_{i=1}^n M_i, q^*N)^\simeq \rightarrow \mathrm{Lin}_{\otimes C_i}^L(\otimes_{i=1}^n M_i, N)^\simeq$$

where q^*N is the pullback of N along q , and is again a presentable left tensored ∞ -category by [proposition 2.59](#)

Proof. By the description of the multimorphism space in $\mathbf{PrLMod}^\otimes \subset \widehat{\mathbf{LMod}}^\times$, there is an equivalence of mapping spaces:

$$\mathrm{Lin}^{LL}(\prod M_i, N)^\simeq \cong \mathrm{Lin}^L(\otimes M_i, N)^\simeq$$

And also we have an pullback of mapping spaces by the universal property of Cartesian lifting:

$$\begin{array}{ccc} \mathrm{Lin}(\prod M_i, q^*N)^\simeq & \longrightarrow & \mathrm{Lin}(\prod M_i, N)^\simeq \\ \downarrow & \lrcorner & \downarrow \\ * & \xrightarrow[\{q\}]{} & \mathrm{Map}_{\widehat{\mathbf{Mon}}}(\prod C_i, \otimes C_i) \end{array}$$

And combining the above two things, there is a commutative diagram of spaces:

$$\begin{array}{ccccc}
& & \text{Lin}_{\prod C_i}(\prod M_i, q^*N)^\simeq & \hookrightarrow & \text{Lin}(\prod M_i, N)^\simeq \\
& \nearrow & \downarrow & \lrcorner & \downarrow \\
& & * & \xrightarrow{\{q\}} & \text{Map}_{\widehat{\text{Mon}}}(\prod C_i, \otimes C_i) \\
& \nearrow & \downarrow & \lrcorner & \downarrow \\
\text{Lin}_{\otimes C_i}^L(\otimes M_i, N)^\simeq & \hookrightarrow & \text{Lin}^{LL}(\prod M_i, N)^\simeq & \xrightarrow{\cong} & \text{Lin}^L(\otimes M_i, N)^\simeq \\
\downarrow & \nearrow & \downarrow & \nearrow & \downarrow \\
* & \xrightarrow{\{q\}} & \text{Map}_{\widehat{\text{Mon}}}^{LL}(\prod C_i, \otimes C_i) & \xrightarrow{\cong} & \text{Map}_{\text{PrMon}}(\otimes C_i, \otimes C_i) \\
& \searrow & \downarrow & \searrow & \downarrow \\
& & & & \text{Map}_{\text{PrMon}}(\otimes C_i, \otimes C_i)
\end{array}$$

$\{id\}$

where the back and the front squares are pullbacks.

Consider the top square: all the edges except the left vertical edge are inclusions of components of spaces, so is the left vertical edge: a morphism between spaces is an inclusion of components if it's both Kan fibration and monomorphism by [24, Example 01GE], and both monomorphism and Kan fibration satisfy 2-out-of-3 property.

Thus we know there is an embedding $\text{Lin}_{\otimes C_i}^L(\otimes M_i, N)^\simeq \rightarrow \text{Lin}_{\prod C_i}(\prod M_i, q^*N)^\simeq$, and it remains to check that the image consists of exactly those points in $\text{Lin}_{\prod C_i}(\prod M_i, q^*N)^\simeq$ which preserve colimits componentwise, which is obvious true by the commutative diagram. Thus we have an equivalence between mapping spaces induced by $d_2 : \prod_{i=1}^n M_i \rightarrow \otimes_{i=1}^n M_i$:

$$\text{Lin}_{\prod C_i}^{LL}(\prod_{i=1}^n M_i, q^*N)^\simeq \cong \text{Lin}_{\otimes C_i}^L(\otimes M_i, N)^\simeq$$

□

And we have the following corollaries about the mapping spaces:

Corollary 1.75. *Given left tensored categories $\{(\mathcal{V}_i, \mathcal{M}_i)\}_{i=1}^n$, and $\mathcal{N}^\otimes \rightarrow \otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)^\otimes$ be a pre-sentable left tensored category over $\otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)$, then there is an equivalence of spaces:*

$$\text{Lin}_{\otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)}^L(\otimes_{i=1}^n \mathcal{P}(\mathcal{M}_i), \mathcal{N})^\simeq \rightarrow \text{Lin}_{\prod_{i=1}^n \mathcal{P}(\mathcal{V}_i)}^{LL}(\prod_{i=1}^n \mathcal{P}(\mathcal{M}_i), \mathcal{N})^\simeq \rightarrow \text{Lin}_{\prod_{i=1}^n \mathcal{V}_i}(\mathcal{M}_i, \mathcal{N})^\simeq$$

where $\text{Lin}^{LL}(-)^\simeq$ denote the space of linear functors from $\prod \mathcal{P}(\mathcal{M}_i)$ to $\mathcal{N}$ which preserve colimits separately.

Proof.

Consider the coCartesian edge over the active map α in $\text{PrLMod}^\otimes$ over the active map $\langle n \rangle \rightarrow \langle 1 \rangle$, and its image under the embedding $\text{PrLMod}^\otimes \hookrightarrow \widehat{\text{LMod}}^\times$, we have the following factorization in $\widehat{\text{LMod}}^\times$:

$$\begin{array}{ccccc}
& & \xrightarrow{d} & & \\
\{(\mathcal{P} \mathcal{V}_i, \mathcal{P} \mathcal{M}_i)\}_i & \xrightarrow{d_1} & (\prod \mathcal{P} \mathcal{V}_i, \prod \mathcal{P} \mathcal{M}_i) & \xrightarrow{d_2} & (\otimes \mathcal{P} \mathcal{V}_i, \otimes \mathcal{P} \mathcal{M}_i) \\
\downarrow & & \downarrow & & \downarrow \\
\{\mathcal{P} \mathcal{V}_i\}_i & \longrightarrow & \prod \mathcal{P} \mathcal{V}_i & \xrightarrow{q} & \otimes \mathcal{P} \mathcal{V}_i
\end{array}$$

where d_1 is the coCartesian lifting in $\widehat{\mathbf{LMod}}^\times$, and fiberwise d_2 is the universal componentwise colimit preserving functor $\prod \mathcal{P} \mathcal{V}_i \rightarrow \otimes \mathcal{P} \mathcal{V}_i$, $\prod \mathcal{P} \mathcal{M}_i \rightarrow \otimes \mathcal{P} \mathcal{M}_i$.

So for any presentable left tensored category $\mathcal{N}^\otimes \rightarrow \mathcal{W}^\otimes$, by the description of the multimorphism space described above, we have the following factorization of spaces:

$$\mathrm{Lin}_{\otimes_{i=1}^n \mathcal{P} \mathcal{V}_i}^L(\otimes_{i=1}^n \mathcal{P}(\mathcal{M}_i), \mathcal{N})^\simeq \rightarrow \mathrm{Lin}_{\prod_{i=1}^n \mathcal{P} \mathcal{V}_i}^{LL}(\prod_{i=1}^n \mathcal{P}(\mathcal{M}_i), \mathcal{N})^\simeq \rightarrow \mathrm{Lin}_{\prod_{i=1}^n \mathcal{V}_i}(\mathcal{M}_i, \mathcal{N})^\simeq$$

Since the first arrow and the composition of the first and the second arrow are equivalences, so all arrows are equivalences. $\square$

Corollary 1.76. *Given a presentably symmetric monoidal category $\mathcal{V}^\otimes$, and n presentably left tensored ∞ -categories $\{\mathcal{M}_i^\otimes \rightarrow \mathcal{V}^\otimes\}_{i=1}^n$, and a presentably left tensored ∞ -category $\mathcal{N}^\otimes \rightarrow \mathcal{V}^\otimes$, there is an equivalence of mapping spaces induced by the coCartesian lifting $c : \otimes_{i=1}^n \mathcal{M}_i \rightarrow (\otimes_{\mathcal{V}})_{i=1}^n \mathcal{M}_i$ in $\mathbf{PrLMod}^\otimes$ over the tensor operation $\otimes : \otimes_{i=1}^n \mathcal{V} \rightarrow \mathcal{V} \in \mathbf{PrMon}$:*

$$\mathrm{Lin}_{\mathcal{V}}^L(\otimes_{\mathcal{V}} \mathcal{M}_i, \mathcal{N})^\simeq \xrightarrow{c^*} \mathrm{Lin}_{\otimes_{i=1}^n \mathcal{V}}^L(\otimes \mathcal{M}_i, \otimes^* \mathcal{N})^\simeq \xrightarrow{d_2^*} \mathrm{Lin}_{\prod_{i=1}^n \mathcal{V}}^{LL}(\prod \mathcal{M}_i, q^* \otimes^* \mathcal{N})^\simeq$$

where $q : \prod_{i=1}^n \mathcal{V} \rightarrow \otimes_{i=1}^n \mathcal{V}$ is the universal functor preserving colimits componentwise.

1.7 left module object in a coCartesian symmetric monoidal category

This subsection is an expansion of Tyler Lawson's answer in [1, this post in mathoverflow].

Given a coCartesian symmetric monoidal structure on $\mathcal{C}$, where $\mathcal{C}$ admits coproducts, then an $\mathcal{LM}^\otimes$ algebra object (A, M) in $\mathcal{C}^\mathbb{I}$ should encode a left action map:

$$act : A \coprod M \rightarrow M$$

and this corresponds to two morphisms $A \rightarrow M$ and $M \rightarrow M$, while the latter map is identity since it's equivalent to the composition map $\mathbf{1} \coprod M \cong M \rightarrow A \coprod M \rightarrow M$ where $\mathbf{1}$ is both the initial and the unit object in $\mathcal{C}$. Thus act is determined by a morphism $A \rightarrow M$ in $\mathcal{C}$. And in this subsection, we give a complete proof on this idea:

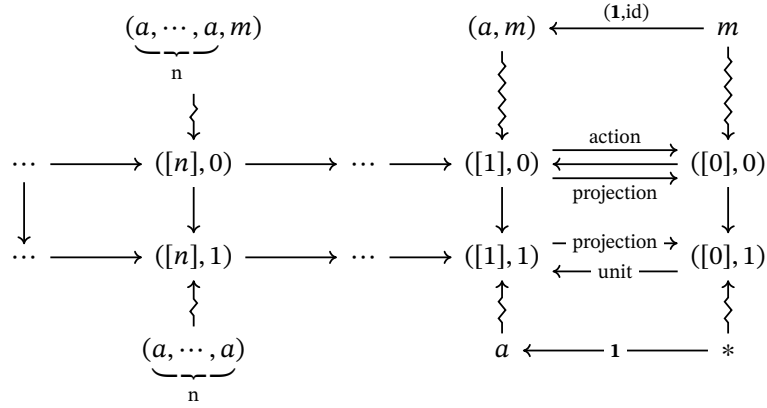
Proposition 1.77. *Given the coCartesian symmetric monoidal structure on $\mathcal{C}$ where $\mathcal{C}$ admits coproducts, then there is an equivalence:*

$$\mathrm{Alg}_{\mathcal{LM}}(\mathcal{C}) \cong \mathrm{Fun}(\Delta^1, \mathcal{C})$$

where the equivalence sends a left module object (A, M) to a morphism $A \rightarrow M$ in $\mathcal{C}$ as illustrated above.

Before the proof, recall that there is an approximation of the ∞ -operad $\gamma : \Delta^{op} \times [1] \rightarrow \mathcal{LM}^\otimes$ (see [22, remark 4.2.2.8]), where if we regard objects in the underlying category $\mathcal{LM}$ as a, m (as in the [22, notation 4.2.1.6]), then:

- γ sends $([n], 0)$ to $(\underbrace{a, \dots, a}_n, m)$ in $\mathcal{LM}^\otimes$ lying over $\langle n+1 \rangle \in \text{Comm}^\otimes$.
- γ sends $([n], 1)$ to $(\underbrace{a, \dots, a}_n, a)$ in $\text{Assoc}^\otimes \hookrightarrow \mathcal{LM}^\otimes$ lying over $\langle n \rangle$.
- Consider the following diagram in $\Delta^{op} \times [1]$:



then all the vertical edges correspond to the inert map $(a, \dots, a, m) \rightarrow (a, \dots, a)$ under the map γ , and the bottom horizontal edges correspond to morphisms in $\text{Assoc}^\otimes$ under the approximation $\text{Cut} : \Delta^{op} \rightarrow \text{Assoc}^\otimes$ (see [22, construction 4.2.1.9]).

- And let M be the collection of morphisms in $\Delta^{op} \times [1]$ in the form of
 - $([n], 0) \rightarrow ([m], 0)$ induced by an inert map $[m] \rightarrow [n] \in \Delta$ preserving the maximum.
 - $([n], 0) \rightarrow ([n], 1)$.
 - and the composition of the morphisms in the above two forms.

where M plays the role of inert maps: we can see that M is the inerse image of inert maps in $\mathcal{LM}^\otimes$ under the approximation $\gamma : \Delta^{op} \times [1] \rightarrow \mathcal{LM}^\otimes$.

$\Delta^{op} \times [1]$ can be regarded as a simplicial set over Fin_* through the composition $\Delta^{op} \times [1] \xrightarrow{\gamma} \mathcal{LM}^\otimes \rightarrow \text{Fin}_*$.

Recall the [22, construction 2.4.3.1] of Γ^* consisting of

- objects: pairs $(\langle n \rangle, i)$ where $i \in \langle n \rangle^\circ$.

- a morphism in Γ^* from $(\langle n \rangle, i)$ to $(\langle m \rangle, j)$ is a map of pointed sets $\alpha : \langle n \rangle \rightarrow \langle m \rangle$ such that $\alpha(i) = j$.

Construction 1.78. And we let $D = (\Delta^{op} \times [1]) \times_{\text{Fin}_*} \Gamma^*$. An object in D is in the form of $([n], 0, i \in \langle n+1 \rangle^\circ)$ or $([n], 1, i \in \langle n \rangle^\circ)$. Especially, let $\mathcal{X}$ be the full subcategory of D consisting of the following objects:

- $([0], 0, 1 \in \langle 1 \rangle^\circ)$ (this is just m).
- $([1], 0, 2 \in \langle 2 \rangle^\circ)$ (this is (a, m) with a mark in m).
- $([1], 0, 1 \in \langle 2 \rangle^\circ)$ (this is (a, m) with a mark in a).
- $([1], 1, 1 \in \langle 1 \rangle^\circ)$ (this is just a).

and we denote the above four objects as X_1, X_2, X_3, X_4 respectively.

Recall that to specify a map f in D , it suffices to specify a tuple of map (f_1, f_2) where $f_1 \in \Delta^{op} \times [1]$, and $f_2 \in \Gamma^*$, with the condition that the image of f_2 in Fin_* under the projection map $\Gamma^* \rightarrow \text{Fin}_*$ coincides with the image of f_1 under the map $\Delta^{op} \times [1] \rightarrow \text{Fin}_*$, thus f is determined by f_1 .

Example 1.79. And here is a full list of morphisms in $\mathcal{X}$:

- the unique map $m \rightarrow ((a, m), \{m\})$:

$$u_0 : ([0], 0, 1) \rightarrow ([1], 0, 1)$$

induced by the unique degeneracy map $[1] \rightarrow [0]$ in Δ .

- the inert map $((a, m), \{a\}) \rightarrow a$:

$$u_1 : ([1], 0, 1) \rightarrow ([1], 1, 1)$$

induced by the unique map in Δ^1 .

- the inert map $((a, m), \{m\}) \rightarrow m$:

$$u_2 : ([1], 0, 2) \rightarrow ([0], 0, 1)$$

induced by the map $[0] \rightarrow [1], 0 \mapsto 1$ in Δ .

- the active map $((a, m), \{m\}) \rightarrow m$:

$$u_3 : ([1], 0, 2) \rightarrow ([0], 0, 1)$$

induced by the map $[0] \rightarrow [1], 0 \mapsto 0$ in Δ .

- the active map $((a, m), \{a\}) \rightarrow m$:

$$u_4 : ([1], 0, 1) \rightarrow ([0], 0, 1)$$

induced by the map $[0] \rightarrow [1], 0 \mapsto 0$ in Δ .

- and the morphisms generated under compositions by the above morphisms, for example, $u_0 \circ u_3$ and $u_0 \circ u_2$

, and we can visualize the category $\mathcal{X}$ as a diagram:

$$\begin{array}{ccccc}
 & & m & & \\
 & & \downarrow & & \\
 & & X_2 & & \\
 & & \uparrow u_3 \quad \uparrow u_2 \quad \downarrow u_0 & & \\
 ((a, m), \{a\}) & \rightsquigarrow & X_3 & \xrightarrow{u_4} & X_1 & \rightsquigarrow & ((a, m), \{m\}) \\
 & & \downarrow u_1 & & & & \\
 & & X_4 & & & & \\
 & & \uparrow & & & & \\
 & & a & & & &
 \end{array}$$

Construction 1.80. For any $Y \in D$, we construct a morphism $g_Y : Y \rightarrow X_{i_Y}$ in D for some $i_Y \in \{1, 2, 3, 4\}$:

- For any $Y \in \mathcal{X}$, we take $g_Y = \text{id}$. And in the following case, we assume $n \geq 1$:
- For $Y = ([n], 0, i)$ where $i \neq n + 1$, then we take g_Y as:

$$([1], 0, 1) \rightarrow ([n], 0, i)$$

which is induced by a map in Δ : $[n] \rightarrow [1], 0, \dots, i-1 \mapsto 0, i, \dots, n \mapsto 1$ and the identity map in $[1]$ (this is the map in $\mathcal{LM}^\otimes$: $(a, m) \rightarrow (\underbrace{a, \dots, a}_n, m)$ where a goes to the i -th place and m goes to the m).

- For $Y = ([n], 0, i)$ where $i = n + 1$, then we take g_Y as:

$$([1], 0, 2) \rightarrow ([n], 0, n + 1)$$

which is induced by a map in Δ : $[n] \rightarrow [1], 0, \dots, n-1 \mapsto 0, n \mapsto 1$ and identity in $[1]$, which corresponds to the map in $\mathcal{LM}^\otimes$: $(a, m) \rightarrow (\underbrace{a, \dots, a}_n, m)$ sending (a, m) to the last two places in $(\underbrace{a, \dots, a}_n, m)$.

- For $Y = ([n], 1, i)$, then we take g_Y as:

$$([1], 1, 1) \rightarrow ([n], 1, i)$$

which is induced by a map $[n] \rightarrow [1]$ in Δ sending $j < i$ to 0 and $j \geq i$ to 1, which corresponds to a map $a \rightarrow (a, \dots, a)$ sending a to the i -th place.

Noticing that all the maps g_Y above admit a left inverse: $f_Y : Y \rightarrow X_{i_Y}$:

- For any $Y \in \mathcal{X}$, we take $f_Y = \text{id}$. And in the following case, we assume $n \geq 1$:

- For $Y = ([n], 0, i)$ where $i \neq n + 1$, we define f_Y as

$$([n], 0, i) \rightarrow ([1], 0, 1)$$

which is induced by a map in Δ : $[1] \rightarrow [n]$, $0 \mapsto i - 1$, $1 \mapsto n$ and the identity map in $[1]$, this is the inert map in $\mathcal{LM}^\otimes$: $(\underbrace{a, \dots, a}_n, m) \rightarrow (a, m)$ projecting the i -th and the last components.

- For $([n], 0, i)$ where $i = n + 1$, then there is a map:

$$([n], 0, n + 1) \rightarrow ([1], 0, 2)$$

which is induced by a map in Δ : $0 \mapsto n - 1$, $1 \mapsto n$ and identity in $[1]$, which corresponds to the inert map in $\mathcal{LM}^\otimes$: $(\underbrace{a, \dots, a}_n, m) \rightarrow (a, m)$ projecting the last two components.

- For $([n], 1, i)$, then there is map:

$$([n], 1, i) \rightarrow ([1], 1, 1)$$

which is induced by a map $[1] \rightarrow [n]$ in Δ sending 0 to $i - 1$ and 1 to i , which corresponds to the inert map ρ^i .

And we have:

Lemma 1.81. *The natural embedding $\mathcal{X} \hookrightarrow D$ admits a right adjoint r . More precisely, given any object $X_i \in \mathcal{X}$, and any object $Y \in D$, the map $g_Y : X_{i_Y} \rightarrow Y$ constructed as above induces an equivalence (bijection) between the Hom sets:*

$$\text{Hom}_D(X_i, X_{i_Y}) \cong \text{Hom}_D(X_i, Y)$$

. Thus $\mathcal{X} \hookrightarrow D$ is a colocalization with respect to the set of morphisms g_Y .

Proof. Both $\mathcal{X}$ and D are ordinary categories, thus we can check the isomorphism between Hom sets directly, and it's obvious that the restriction map induces a bijection between Hom sets. $\square$

Lemma 1.82. *Consider the natural embedding $i : \mathcal{X} \rightarrow D$, then the restriction map:*

$$i^* : \text{Fun}(D, \mathcal{C}) \rightarrow \text{Fun}(\mathcal{X}, \mathcal{C})$$

admits a fully faithful left adjoint r^ by pre-composition with the right adjoint $r : D \rightarrow \mathcal{X}$. Moreover, the essential image of r^* consists of exactly those functors $A : D \rightarrow \mathcal{C}$ such that $A(g_Y)$ is an equivalence.*

Proof. • By the adjunction $i : \mathcal{X} \rightleftharpoons D : r$, there is an induced adjunction by [30, proposition 1.3.7 (iv)]:

$$r^* : \text{Fun}(\mathcal{X}, \mathcal{C}) \rightleftharpoons \text{Fun}(D, \mathcal{C}) : i^*$$

- The left adjoint r^* is fully faithful since the unit is an isomorphism.

- And by the dual version of [21, proposition 5.2.7.12], the image of r^* consists of exactly those functors $A : D \rightarrow C$ such that $A(g_Y)$ is equivalence.

□

Now recall the construction of the coCartesian ∞ -operad $\mathcal{C}^{\text{II}}$ in [22, construction 2.4.3.1], which is characterized by the canonical equivalence of sets for any $K \in \text{sSet}_{/\text{Fin}_*}$:

$$\text{Hom}_{\text{Fin}_*}(K, \mathcal{C}^{\text{II}}) \cong \text{Hom}_{\text{sSet}}(K \times_{\text{Fin}_*} \Gamma^*, \mathcal{C})$$

, thus by replace K by $\Delta^n \times K$, there is an equivalence of ∞ -categories:

$$\text{Fun}_{/\text{Fin}_*}(K, \mathcal{C}^{\text{II}}) \cong \text{Fun}(K \times_{\text{Fin}_*} \Gamma^*, \mathcal{C})$$

, thus we have:

$$\text{Fun}_{/\text{Fin}_*}(\Delta^{op} \times [1], \mathcal{C}^{\text{II}}) \cong \text{Fun}(D, \mathcal{C})$$

.

And by [22, 4.2.2.13], there is an equivalence of ∞ -categories induced by $\gamma : \Delta^{op} \times [1] \rightarrow \mathcal{LM}^{\otimes}$:

$$\text{Alg}_{\mathcal{LM}}(\mathcal{O}) \rightarrow \text{Fun}'_{/\text{Fin}_*}(\Delta^{op} \times [1], \mathcal{O}^{\otimes})$$

where $\text{Fun}'_{/\text{Fin}_*}(\Delta^{op} \times [1], \mathcal{O}^{\otimes})$ is the full subcategory of $\text{Fun}(\Delta^{op} \times [1], \mathcal{O}^{\otimes})$ spanned by those functors $A : \Delta^{op} \times [1] \rightarrow \mathcal{O}^{\otimes}$ taking the morphisms in M to the inert maps in $\mathcal{O}^{\otimes}$.

Thus, By taking $\mathcal{O}^{\otimes} = \mathcal{C}^{\text{II}}$:

Lemma 1.83. *There is an equivalence induced by $\gamma : \Delta^{op} \times [1] \rightarrow \mathcal{LM}^{\otimes}$:*

$$\text{Alg}_{\mathcal{LM}}(\mathcal{C}) \rightarrow \text{Fun}'_{/\text{Fin}_*}(\Delta^{op} \times [1], \mathcal{C}^{\text{II}}) \cong \text{Fun}'(D, \mathcal{C})$$

where $\text{Fun}'(D, \mathcal{C})$ is the full subcategory of $\text{Fun}(D, \mathcal{C})$ consisting of functors $A : D \rightarrow \mathcal{C}$ taking the morphisms in D which image in $\Delta^{op} \times [1]$ lands in M to equivalences in $\mathcal{C}$.

Let $\text{Fun}'(\mathcal{X}, \mathcal{C})$ be the full subcategory of $\text{Fun}(D, \mathcal{C})$ consisting of functors $A : \mathcal{X} \rightarrow \mathcal{C}$ taking the morphisms landing in M to equivalences in $\mathcal{C}$.

More precisely, $\text{Fun}'(\mathcal{X}, \mathcal{C}) \subset \text{Fun}(\mathcal{X}, \mathcal{C})$ consists of those functors $A : \mathcal{X} \rightarrow \mathcal{C}$ taking map u_1, u_2 to equivalences. Noticing that u_0 is the right inverse of u_1 , and u_3 is the left inverse of u_0 , thus the full subcategory $\text{Fun}'(\mathcal{X}, \mathcal{C}) \subset \text{Fun}(\mathcal{X}, \mathcal{C})$ can be characterized as those functors $A : \mathcal{X} \rightarrow \mathcal{C}$ satisfying $A(u_i)$ is equivalence for $i = 0, 1, 2, 3$.

Lemma 1.84. *The adjunction:*

$$r^* : \text{Fun}(\mathcal{X}, \mathcal{C}) \rightleftarrows \text{Fun}(D, \mathcal{C}) : i^*$$

restricts to an equivalence adjunction:

$$r^* : \text{Fun}'(\mathcal{X}, \mathcal{C}) \cong \text{Fun}'(D, \mathcal{C}) : i^*$$

.

Proof. • It's obvious that i^* restricts to the full subcategories Fun' , and we first show $r^* : \text{Fun}(\mathcal{X}, \mathcal{C}) \rightarrow \text{Fun}(D, \mathcal{C})$ restricts to $\text{Fun}'(\mathcal{X}, \mathcal{C}) \rightarrow \text{Fun}'(D, \mathcal{C})$.

- By the lemma above, the essential image of $r^* : \text{Fun}(\mathcal{X}, \mathcal{C}) \rightarrow \text{Fun}(D, \mathcal{C})$ consists of exactly those functors $A : D \rightarrow \mathcal{C}$ such that $A(g_Y)$ is an equivalence, and equivalently, the essential image can be characterized as those functors A such that $A(f_Y)$ are equivalences for all Y .
- And noticing that for any map $f : Y = ([n], 0, i) \rightarrow Z = ([m], 0, j)$ in M , $n, m \geq 1$, there

is a commutative diagram:

$$\begin{array}{ccc} Y & \xrightarrow{f} & Z \\ & \searrow f_Y & \swarrow f_Z \\ & X_{i_Y} & \end{array} .$$

- And also for any map in D involving the unique map in $[1]$: $q : ([n], 0, i \neq n+1) \rightarrow ([n], 1, i)$, it can be factored as:

$$\begin{array}{ccc} ((\underbrace{a, \dots, a}_n, m), \{a\}_i) & & ((a, m), \{a\}) \\ \downarrow \wr & & \downarrow \wr \\ ([n], 0, i) & \xrightarrow{\text{inert}} & ([1], 0, 1) \\ \downarrow q & & \downarrow u_1 \\ (a, \dots, a) & \xrightarrow{\text{inert}} & a \end{array}$$

- So any map in M can be related with maps in the form of f_Y in a 2-out-of-3 diagram.
- Thus a functor $A : D \rightarrow \mathcal{C}$ takes all the maps in D which image lands in M to equivalences iff A takes all f_Y to equivalences and those maps in $\mathcal{X}$ which land in M to equivalences.
- Thus the essential image of $r^* : \text{Fun}'(\mathcal{X}, \mathcal{C})$ consists of exactly those functors $A : D \rightarrow \mathcal{C}$ taking the maps in M to equivalences, that is, $r^* : \text{Fun}'(\mathcal{X}, \mathcal{C}) \rightarrow \text{Fun}'(D, \mathcal{C})$ is essential surjective, and it's also fully faithful, thus r^* is an equivalence.
- And the adjunction $r^* \dashv i^*$ restricts to the adjunction between $\text{Fun}'(\mathcal{X}, \mathcal{C})$ and $\text{Fun}'(D, \mathcal{C})$, since the left adjoint r^* is an equivalence, so is the right adjoint.

□

Recall the category $\mathcal{X}$:

$$\begin{array}{ccccc}
& & m & & \\
& & \downarrow & & \\
& & X_2 & & \\
& & \uparrow u_3 \quad \uparrow u_2 \quad \downarrow u_0 & & \\
((a, m), \{a\}) & \rightsquigarrow & X_3 & \xrightarrow{u_4} & X_1 & \leftarrow \rightsquigarrow & ((a, m), \{m\}) \\
& & \downarrow u_1 & & \\
& & X_4 & & \\
& & \uparrow & & \\
& & a & &
\end{array}$$

Now we show that a functor A in $\text{Fun}'(\mathcal{X}, \mathcal{C})$ is determined by its restriction to u_4 : since u_0, u_1, u_2, u_3 are all sent to equivalences in $\mathcal{C}$, thus the only nontrivial information is about the image of u_4 .

Moreover, u_4 determines a map $z : A \rightarrow M$ in $\mathcal{C}$ where A, M constitutes the left A module (A, M) determined by the algebra map $\mathcal{LM}^\otimes \rightarrow \mathcal{C}^\Pi$.

Lemma 1.85. *By restriction to $u_4 : X_3 = ([1], 0, 1) \rightarrow X_1 = ([0], 0, 1)$, there is an equivalence:*

$$\text{Fun}'(\mathcal{X}, \mathcal{C}) \rightarrow \text{Fun}(\Delta^1, \mathcal{C})$$

.

Proof. • Noticing that $\mathcal{X}$ can be decomposed as the pushouts

$$\begin{array}{ccc}
\begin{array}{ccc}
\bullet : X_3 & \xRightarrow{\quad} & [1] : X_3 \rightarrow X_1 \\
\Downarrow & & \Downarrow \\
[1] : X_3 \rightarrow X_4 & \xRightarrow{\quad} & X_3 \\
& \swarrow & \searrow \\
& X_4 & X_1
\end{array}
& &
\begin{array}{ccc}
\bullet : X_1 & \xRightarrow{\quad} & X_2 \\
\Downarrow & & \uparrow \uparrow \downarrow \\
X_3 & \xRightarrow{\quad} & X_1 \\
& \swarrow & \searrow \\
& X_4 & X_1
\end{array}
\end{array}$$

• Thus $\text{Fun}(\mathcal{X}, \mathcal{C})$ is equivalent to the fiber products:

$$\text{Fun}([1], \mathcal{C}) \times_{\text{Fun}(*, \mathcal{C})} \text{Fun}([1], \mathcal{C}) \times_{\text{Fun}(*, \mathcal{C})} \text{Fun}\left(\begin{array}{c} X_2 \\ \uparrow \uparrow \downarrow \\ X_1 \end{array}, \mathcal{C}\right)$$

And the full subcategory $\text{Fun}'(\mathcal{X}, \mathcal{C})$ restricts the first and the last components to be equivalent to $\text{Fun}(*, \mathcal{C})$ induced by the degeneracy maps, thus:

$$\text{Fun}'(\mathcal{X}, \mathcal{C}) \cong \text{Fun}(*, \mathcal{C}) \times_{\text{Fun}(*, \mathcal{C})} \text{Fun}([1], \mathcal{C}) \times_{\text{Fun}(*, \mathcal{C})} \text{Fun}(*, \mathcal{C}) \cong \text{Fun}([1], \mathcal{C})$$

□

And if $\mathcal{C}$ admits coproducts and we consider back to the ∞ -operad $\mathcal{C}^{\text{II}}$, $z : A \rightarrow M$ can be embedded into the following diagram:

$$\begin{array}{ccccc} \mathcal{C}^{\text{II}} : & (A, \emptyset) & \xrightarrow{\text{co}} & A & \\ & \downarrow & & \downarrow & \searrow z \\ & (A, M) & \xrightarrow{\text{co}} & A \amalg M & \\ & \uparrow & & \uparrow & \swarrow \text{act} \\ & (\emptyset, M) & \xrightarrow{\text{co}} & M & \xrightarrow[\text{id}]{} M \end{array}$$

$$\text{Fin}_* : \quad \langle 2 \rangle \xrightarrow{\text{act}} \langle 1 \rangle$$

Proof of proposition 1.77. By the lemmas above, we have a series of equivalences:

$$\text{Alg}_{\mathcal{LM}}(\mathcal{C}) \rightarrow \text{Fun}'_{/\text{Fin}_*}(\Delta^{op} \times [1], \mathcal{C}^{\text{II}}) \rightarrow \text{Fun}'(D, \mathcal{C}) \rightarrow \text{Fun}'(\mathcal{X}, \mathcal{C}) \cong \text{Fun}([1], \mathcal{C})$$

□

Corollary 1.86. *Given any ∞ -operad $\mathcal{M}$, if $\mathcal{M}$ is symmetric monoidal, then there are equivalences*

$$\text{Fun}(\Delta^1, \text{CAlg}(\mathcal{M})) \cong \text{Alg}_{\mathcal{LM}}(\text{CAlg}(\mathcal{M})) \cong \text{Alg}_{\mathcal{LM} \otimes \text{Comm}}(\mathcal{M}) \cong \text{CAlg}(\text{Alg}_{\mathcal{LM}}(\mathcal{M}))$$

.

Proof. Notice that $\text{CAlg}(\mathcal{M})$ is a coCartesian ∞ -operad by [22, 3.2.4.10], and if $\mathcal{M}$ is symmetric monoidal, $\text{CAlg}(\mathcal{M})$ admits finite coproducts and is symmetric monoidal. □

Especially in the above case, if we assume $\mathcal{M}$ is symmetric monoidal, then given a morphism $f : A \rightarrow M$ in $\text{CAlg}(\mathcal{M})$, we have the following description of the left action morphism $\text{act} : A \otimes M \rightarrow M$:

$$\text{act} : A \otimes M \xrightarrow{f \otimes \text{id}} M \otimes M \xrightarrow{\otimes_{\mathcal{M}}} M$$

by the following commutative diagram in $\text{CAlg}(\mathcal{M})$:

$$\begin{array}{ccc} A \otimes M & & \\ \downarrow (\text{id} \otimes 1) \otimes (1 \otimes \text{id}) & \searrow \text{id} & \\ (A \otimes M) \otimes (A \otimes M) & \xrightarrow{\otimes_{A \otimes M}} & A \otimes M \\ \downarrow \text{act} \otimes \text{act} & & \downarrow \text{act} \\ M \otimes M & \xrightarrow{\otimes_M} & M \end{array}$$

2 Enrichment

2.1 enrichment and pseudo-enrichment

This part follows [15, section 3.1] and [14, definition 3.121].

Definition 2.1. Let $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ be a weakly left tensored ∞ -category.

- (1) We say that ϕ exhibits $\mathcal{M}$ as (left) pseudo-enriched in $\mathcal{V}$ if $\mathcal{V}^{\otimes} \rightarrow \text{Ass}$ is a monoidal ∞ -category and for every $X, Y \in \mathcal{M}$ and $V_1, \dots, V_n \in \mathcal{V}$, the map

$$\text{Mul}_{\mathcal{M}}(V_1 \otimes \dots \otimes V_n, X; Y) \rightarrow \text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, X; Y)$$

induced by the active morphism $(V_1, \dots, V_n) \rightarrow V_1 \otimes \dots \otimes V_n$ is an equivalence.

- (2) a morphism object of $X, Y \in \mathcal{M}$ is an object $\text{Mor}_{\mathcal{M}}(X, Y) \in \mathcal{V}$ together with a multimorphism $\alpha \in \text{Mul}_{\mathcal{M}}(\text{Mor}_{\mathcal{M}}(X, Y), X; Y)$ that induces for all objects $Z_1, \dots, Z_n \in \mathcal{V}$ an equivalence of spaces:

$$\text{Mul}_{\mathcal{V}}(Z_1, \dots, Z_n; \text{Mor}_{\mathcal{M}}(X, Y)) \cong \text{Mul}_{\mathcal{M}}(Z_1, \dots, Z_n, X; Y)$$

is an equivalence.

- (3) We say that $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ exhibits $\mathcal{M}$ enriched in $\mathcal{V}$ if for every $X, Y \in \mathcal{M}$, there exists a morphism object $\text{Mor}_{\mathcal{M}}(X, Y)$.

Noticing that Heine's enrichment is defined for not only for categories enriched in monoidal category, but also for categories enriched in an operad. And even when $\mathcal{V}$ is a monoidal ∞ -category, the above definition of morphism object is different from [22, definition 4.2.128]. But these two definitions are related by the following [14, lemma 3.123]:

Lemma 2.2. Let $\mathcal{V}^{\otimes}$ be a monoidal ∞ -category, and $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ be a weakly left tensored ∞ -category, and let $X, Y \in \mathcal{M}, T \in \mathcal{V}$, and $\alpha \in \text{Mul}_{\mathcal{M}}(T, X; Y)$ be a multi-morphism, then α exhibits T as morphism object of X, Y if and only the following two conditions are satisfied:

- (1) ϕ exhibits $\mathcal{M}$ as pseudo-enriched in $\mathcal{V}$.
(2) For every $Z \in \mathcal{V}$, the canonical map $\text{Map}_{\mathcal{V}}(Z, T) \rightarrow \text{Mul}_{\mathcal{M}}(Z, X; Y)$ is an equivalence.

Remark 2.3. Thus Heine's definition of enriched ∞ -category is automatically a pseudo-enriched ∞ -category.

Notation 2.4. • Let PLMod denote the non-full subcategory of ωLMod consisting of pseudo-

$$\begin{array}{ccc} \mathcal{M}^{\otimes} & \xrightarrow{f} & \mathcal{N}^{\otimes} \\ \downarrow & & \downarrow \\ \mathcal{V}^{\otimes} & \xrightarrow{\alpha} & \mathcal{W}^{\otimes} \end{array} \quad \text{where } \alpha \text{ is a monoidal } \infty \text{ functor.}$$

- Or equivalently, we can regard PLMod as the pullback:

$$\begin{array}{ccccc}
\text{PLMod} & \longrightarrow & \text{P}\omega\text{LMod} & \hookrightarrow & \omega\text{LMod} \\
\downarrow & \lrcorner & \downarrow & & \\
\text{Mon} & \longrightarrow & \text{Op} & &
\end{array}$$

where $\text{P}\omega\text{LMod}$ is the full subcategory of ωLMod consisting of pseuso-enriched categories.

- Let Enr denote the full subcategory of PLMod consisting of enriched categories.
- Given an operad $\mathcal{V}$, denote the fiber over $\mathcal{V}$ of the fibration $\omega\text{LMod} \rightarrow \text{Op}_\infty$ as $\omega\text{LMod}_\mathcal{V}$,
- And if $\mathcal{V}$ is a monoidal category, denote the fiber over $\mathcal{V}$ of the fibration $\text{PLMod} \rightarrow \text{Mon}$ be $\text{PLMod}_\mathcal{V}$, and so is $\text{Enr}_\mathcal{V}$.
- For a monoidal ∞ -category $\mathcal{V}$, there is a full subcategory inclusion $\text{Enr}_\mathcal{V} \hookrightarrow \text{PLMod}_\mathcal{V} \hookrightarrow \omega\text{LMod}_\mathcal{V}$.

Definition 2.5. Given a pseudo-enriched ∞ category $\psi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$,

- (1) given a regular cardinality κ , we say ψ exhibits $\mathcal{M}$ as **(left) κ -enriched** in $\mathcal{V}$ if $\mathcal{V}$ admits κ small colimits and ψ exhibits $\mathcal{M}$ as (left) pseudo-enriched in $\mathcal{V}$, and give any $x, y \in \mathcal{M}$, the presheaf on $\mathcal{V}$: $\text{Mul}_\mathcal{M}(-, x; y)$ preserves κ limits.
- (2) Let σ denote the strong inaccessible cardinality corresponding to the small universe, we say $\psi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ exhibits $\mathcal{M}$ as **(left) quasi-enriched** in $\mathcal{V}$ if ψ exhibits $\mathcal{M}$ as σ -enriched.

Remark 2.6. A category pseudo-enriched in $\mathcal{V}$ can always be regarded as a category enriched in the presheaf $\mathcal{P}(\mathcal{V})$, under this identification, a category left κ -enriched in $\mathcal{V}$ can be regarded as a category enriched in $\text{Ind}_\kappa(\mathcal{V})$, since presheaf preserving κ limits on a small category $\mathcal{V}$ with κ -small colimits corresponds to a presheaf lying in the subcategory $\text{Ind}_\kappa(\mathcal{V})$ by [21, corollary 5.3.4].

Notation 2.7. (1) We denote the subcategory of $\text{PLMod}_\mathcal{V}$ consisting of all (left) κ -enriched categories as ${}^\kappa\text{Enr}_\mathcal{V}$.

- (2) After enlarging the universe, we denote $\widehat{\sigma\text{Enr}}_\mathcal{V}$ as $\text{QEnr}_\mathcal{V}$, and the subcategory of $\widehat{\sigma\text{Enr}}_\mathcal{V}$ consisting of those (left) quasi-enriched category which underlying category is small as $\text{QEnr}_\mathcal{V}$.

Remark 2.8. By [21, 5.5.2.2], a ‘small presheaf’ on a presentable category $\mathcal{V}$ is representable iff it preserves small limits, thus a left quasi-enriched category is enriched iff it is locally small, that is, the multi-morphism space of $\mathcal{M}^\otimes$ is small.

Example 2.9. Take $\mathcal{V} = *$, then $\widehat{\text{PLMod}}_* \cong \widehat{\text{Cat}}^{\text{Spc}}$, and the subcategory of quasi-enriched part is $\widehat{\sigma\text{Enr}}_* \cong \widehat{\text{Cat}}^{\text{Spc}}$, which is the category of all locally small categories.

Example 2.10. Take $\mathcal{V} = \text{Cat}$, then $\widehat{\sigma\text{Enr}}_{\text{Cat}} \cong \text{Cat}^{\text{Cat}} \subset \widehat{\text{Enr}}_{\text{Cat}} \cong \widehat{\text{Cat}}^{\text{Cat}} \subset \widehat{\sigma\text{Enr}}_{\text{Cat}} \subset \widehat{\text{PLMod}}_{\text{Cat}}$.

2.2 enriched adjunction

This part follows [15, section 2.4], and we supplement some additional lemmas on the case of enriched adjunction arising by localization of operads.

Construction 2.11. [14, lemma 3.64] exists a canonical Cat_∞ action on ωLMod by sending $(K \in \text{Cat}_\infty, \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes)$ to $\mathcal{M}^\otimes \times K \rightarrow \mathcal{V}^\otimes$. More precisely, consider the functor $\text{Cat}_\infty \rightarrow \text{Cat}_{\infty/\mathcal{V}^\otimes} : K \mapsto K \times \mathcal{V}^\otimes$, which image lands in $\omega\text{LMod}_{\mathcal{V}}$. Moreover, it can be enhanced into a monoidal ∞ -functor $\text{Cat}_\infty^\times \rightarrow \omega\text{LMod}_{\mathcal{V}}^\times$. Since any monoidal ∞ -category $\mathcal{W}^\otimes$ can be regarded as a self-left tensored ∞ -category $\mathcal{W}^\otimes \rightarrow \mathcal{W}^\otimes$, there exists a map between coCartesian fibrations $\omega\text{LMod}_{\mathcal{V}}^\otimes \rightarrow \omega\text{LMod}_{\mathcal{V}}^\times$, by pullback along $\text{Cat}_\infty^\times \rightarrow \omega\text{LMod}_{\mathcal{V}}^\times$, we have a left Cat_∞ tensored ∞ -category $\omega\text{LMod}_{\mathcal{V}}^\otimes \rightarrow \text{Cat}_\infty^\times$.

Moreover, the above action is closed: recall in the category $\text{Cat}_{\infty/\mathcal{V}^\otimes}$, for any objects $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes, \mathcal{N}^\otimes \rightarrow \mathcal{V}^\otimes$, and any ∞ -category K , there exists an equivalence:

$$\text{Fun}(K, \text{Fun}_{\mathcal{V}^\otimes}(\mathcal{M}^\otimes, \mathcal{N}^\otimes)) \cong \text{Fun}_{\mathcal{V}^\otimes}(\mathcal{M}^\otimes \times K, \mathcal{N}^\otimes)$$

which restricts to an equivalence:

$$\text{Fun}(K, \text{LaxLin}_{\mathcal{V}}(\mathcal{M}^\otimes, \mathcal{N}^\otimes)) \cong \text{LaxLin}_{\mathcal{V}}(\mathcal{M} \times K, \mathcal{N})$$

Thus $\omega\text{LMod}_{\mathcal{V}}$ can be enhanced into an $(\infty, 2)$ category.

Definition 2.12. Given a lax linear functor $F : \mathcal{M}^\otimes \rightarrow \mathcal{N}^\otimes$ lying over the maps of ∞ -operads $\alpha : \mathcal{V}^\otimes \rightarrow \mathcal{V}'^\otimes$ and $G : \mathcal{N}^\otimes \rightarrow \mathcal{M}^\otimes$ lying over $\beta : \mathcal{V}'^\otimes \rightarrow \mathcal{V}^\otimes$, a natural transformation $\eta : \text{id} \rightarrow G \circ F$ lying over $\eta' : \text{id} \rightarrow \beta \circ \alpha$ exhibits F as an enriched left adjoint of G (or G as an enriched right adjoint of F) if

- η exhibits F as the left adjoint of G .
- η' exhibits α as the left adjoint of β relative to Ass .

Remark 2.13. It can be shown that the above definition of enriched adjunction is exactly the adjunction in the $(\infty, 2)$ category enhanced from ωLMod as above.

Remark 2.14. Since there exists a commutative diagram:

$$\begin{array}{ccc} \mathcal{M}^\otimes & \xrightarrow{\quad} & \mathcal{V}^\otimes \\ & \searrow & \swarrow \\ & \text{Ass} & \end{array}, \text{ and the}$$

natural transformation η lies over η' , and η is relative to Ass , we know η makes $F \dashv G$ is also a relative adjunction over Ass .

Remark 2.15. A lax linear functor $G : \mathcal{N}^\otimes \rightarrow \mathcal{M}^\otimes$ lying over the map of ∞ -operads $\beta : \mathcal{V}'^\otimes \rightarrow \mathcal{V}^\otimes$ is an enriched left adjoint if:

- β admits a left adjoint α relative to Ass .

- for every $X \in \mathcal{M}$, there exists $Y \in \mathcal{N}$, and a morphism $X \rightarrow G(Y)$ in $\mathcal{M}$ such that for any $V_1, \dots, V_n \in \mathcal{V}$ for $n \geq 0$, and $Z \in \mathcal{N}$, the following map is an equivalence:

$$\text{Mul}_{\mathcal{N}}(\alpha(V_1), \dots, \alpha(V_n), Y; Z) \rightarrow \text{Mul}_{\mathcal{M}}(\beta\alpha(V_1), \dots, \beta\alpha(V_n), G(Y); G(Z)) \rightarrow \text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, X; G(Z))$$

is an equivalence.

(Noticing that mapping spaces in $\mathcal{M}^{\otimes}, \mathcal{N}^{\otimes}$ can be decomposed as multimorphism spaces and mapping spaces in $\mathcal{V}^{\otimes}$ and $\mathcal{V}'^{\otimes}$.

Lemma 2.16. *Let $G : \mathcal{M}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$ be a lax linear functor lying over $\alpha : \mathcal{V}^{\otimes} \rightarrow \mathcal{V}'^{\otimes}$ between left tensored ∞ -categories, then G admits an enriched left adjoint if and only if the underlying functor $G_{[0]} : \mathcal{M} \rightarrow \mathcal{N}$ admits a left adjoint and α admits a left adjoint F relative to Ass , and the induced morphism*

$$F(V_1 \otimes \dots \otimes V_n \otimes X) \cong \alpha(V_1) \otimes \dots \otimes \alpha(V_n) \otimes F(X)$$

is an equivalence for any $X \in \mathcal{N}, V_1, \dots, V_n \in \mathcal{V}$

Proof. This is [15, lemma 2.73]. □

Remark 2.17. The morphism

$$F(V_1 \otimes \dots \otimes V_n \otimes X) \rightarrow \alpha(V_1) \otimes \dots \otimes \alpha(V_n) \otimes F(X)$$

is induced by adjunction from:

$$V_1 \otimes \dots \otimes V_n \otimes X \rightarrow V_1 \otimes \dots \otimes V_n \otimes GF(X) \rightarrow G(\alpha(V_1) \otimes \dots \otimes \alpha(V_n) \otimes F(X))$$

Remark 2.18. Especially, when $\alpha = \text{id}$, the above lemma implies the enriched left adjoint is linear, which is a special case of [22, corollary 7.3.2.12] when $\mathcal{O} = \text{LM}^{\otimes}$.

Lemma 2.19. *Let $F : \mathcal{M}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$ be a lax linear functor lying over $\alpha : \mathcal{V}^{\otimes} \rightarrow \mathcal{V}'^{\otimes}$ between left tensored ∞ -categories, then G admits an enriched right adjoint if and only if the underlying functor $F_{[0]} : \mathcal{M} \rightarrow \mathcal{N}$ admits a right adjoint and α admits a right adjoint relative to Ass , and F is linear.*

Proof. This is [15, lemma 2.74]. □

2.3 localization of operad

This part follows [16, section 4].

Heine construct the enriched presheaf as the localization of presheaf on the envelope of a weakly left tensored category.

Definition 2.20. A small localization pair is a pair $(\mathcal{V}^{\otimes} \rightarrow \text{Ass}, S)$, where $\mathcal{V}^{\otimes}$ is a small ∞ -operad, and S is a set of morphisms of $\mathcal{P}\text{Env}(\mathcal{V})$ such that its saturated closure is closed under tensor product operation of $\mathcal{P}\text{Env}(\mathcal{V})$ and for any $V \in \mathcal{V} \subset \mathcal{P}\text{Env}(\mathcal{V})$, the map $\mathcal{P}\text{Env}(f, V)$ is an equivalence for $f \in S$.

Remark 2.21. Given a small localization pair $(\mathcal{V}^{\otimes} \rightarrow \text{Ass}, S)$, the embedding $S^{-1}\mathcal{P}\text{Env}(\mathcal{V})^{\otimes} \rightarrow \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$ of the full suboperad consisting of S -local objects admits a left adjoint. And by the definition of localization pair, the embedding $\mathcal{V}^{\otimes} \rightarrow \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$ lands in $S^{-1}\mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$.

Definition 2.22. Given a small localization pair $(\mathcal{V}^\otimes \rightarrow \text{Ass}, S)$, a weakly left tensored category $\phi : \mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ exhibits $\mathcal{M}$ as (left) S -enriched in $\mathcal{V}$ if for any $f \in S, X, Y \in \mathcal{M}$, the induced map $\mathcal{P}\text{LEnv}(\mathcal{M})(f \otimes X, Y)$ is an equivalence.

Notation 2.23. Let ${}^S\text{Enr}_{\mathcal{V}}$ be the full subcategory of $\omega\text{LMod}_{\mathcal{V}}$ consisting of weakly left tensored ∞ -category that is S -enriched in $\mathcal{V}$.

Notation 2.24. For every small monoidal ∞ -category $\mathcal{V}^\otimes \rightarrow \text{Ass}^\otimes$ let $\text{Pseu}_{\mathcal{V}}$ be the set of morphisms of $\text{Env}(\mathcal{V})$ closed under the tensor product generated by $V_1, \dots, V_n \rightarrow V_1 \otimes \dots \otimes V_n$ for $V_1, \dots, V_n \in \mathcal{V}$ and $n \geq 0$, that is, the set of morphisms in $\text{Env}(\mathcal{V})$ in the form of:

$$V'_1, \dots, V'_l, V_1, \dots, V_n, V''_1, \dots, V''_k \rightarrow V_1, \dots, V'_m, V_1 \otimes \dots \otimes V_n, V''_1, \dots, V''_m$$

And we shall use the following theorem on localization of presentable ∞ -categories frequently in this and next subsections:

Lemma 2.25. Let $\mathcal{C}$ be a presentable ∞ -category, and S is a small set of morphisms in $\mathcal{C}$, let $\bar{S}$ be the saturation of S and $\mathcal{C}' \subset \mathcal{C}$ be the full subcategory of S -local objects. Then we have:

- (1) For any $c \in \mathcal{C}$, there exists a S -local equivalence $s : c \rightarrow c'$ where c' is a S -local object.
- (2) $\mathcal{C}'$ is presentable, and the inclusion $\mathcal{C}\epsilon \subset \mathcal{C}$ admits a left adjoint sending c to c'
- (3) For a morphism $f \in \mathcal{C}$, the following conditions are equivalent:
 - f is a S -local equivalence.
 - $f \in \bar{S}$.
 - $L(f)$ is an equivalence.

Proof. This is [22, proposition 5.4.4.15]. □

Lemma 2.26. $(\mathcal{V}^\otimes \rightarrow \text{Ass}, \text{Pseu}_{\mathcal{V}})$ is a small localization pair when $\mathcal{V}$ is a monoidal category, and it induces a localization of $\mathcal{P}(\mathcal{V})^\otimes \subset \mathcal{P}\text{Env}(\mathcal{V})^\otimes$.

- Proof.*
- If we regard Pseu as a set of morphisms in $\text{Env}(\mathcal{V})$, then there is a localization with respect to Pseu $L : \text{Env}(\mathcal{V}) \rightleftarrows \mathcal{V} : i$, and moreover Pseu is saturated by the [description of the local equivalences](#) and is closed under tensor product. Thus it gives rise to a monoidal localization $L : \text{Env}(\mathcal{V})^\otimes \rightleftarrows \mathcal{V}^\otimes : i$.
 - Consider the monoidal localization functor $\text{Env}(\mathcal{V})^\otimes \rightarrow \mathcal{V}^\otimes$, which gives rise to a monoidal functor $\mathcal{P}L : \mathcal{P}\text{Env}(\mathcal{V})^\otimes \rightarrow \mathcal{P}(\mathcal{V})^\otimes$. And fiberwise it admits a fully faithful right adjoint, denoted as $\mathcal{P}i$.
 - By [22, corollary 7.3.2.7], the above adjunction can be enhanced into a monoidal adjunction $\mathcal{P}L : \mathcal{P}\text{Env}(\mathcal{V})^\otimes \rightleftarrows \mathcal{P}(\mathcal{V})^\otimes : \mathcal{P}i$. And since fiberwise the right adjoint is fully faithful, thus the counit map is fiberwise an isomorphism, thus is an isomorphism over Ass , and the right adjoint $\mathcal{P}i$ is a fully faithful embedding of monoidal ∞ -categories.

- We claim: the saturation of Pseu in $\mathcal{P}\text{Env}(\mathcal{V})$ is closed under tensor product. For any two Pseu-local equivalences $e_1 : v_1 \rightarrow v'_1, e_2 : v_2 \rightarrow v'_2$ in $\mathcal{P}\text{Env}(\mathcal{V})$, consider their tensor product in $\mathcal{P}\text{Env}(\mathcal{V})$:

$$L(e_1 \otimes e_2) : L(v_1 \otimes v'_1) \cong L(v_1) \otimes L(v'_1) \rightarrow L(v_2 \otimes v'_2) \cong L(v_2) \otimes L(v'_2)$$

which is an equivalence in $\mathcal{P}(\mathcal{V})$, thus $e_1 \otimes e_2$ is again a local equivalence. Since the collection of local equivalences contains the saturation, thus the saturation of Pseu is closed under tensor product.

•

□

Lemma 2.27. *Given a weakly left tensored ∞ -category $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, ϕ exhibits $\mathcal{M}$ as pseudo-enriched in $\mathcal{V}$ if and only if ϕ exhibits $\mathcal{M}$ as Pseu $_{\mathcal{V}}$ -enriched in $\mathcal{V}$.*

Proof. By the condition of being Pseu-enriched: for any $X, Y \in \mathcal{M}$, and for any morphism $e = (e_1, e_2)$ in the form of $((V_1, \dots, V_n, X), f : [n+1] \rightarrow [0]) \rightarrow ((V_1 \otimes \dots \otimes V_n, X), g : [1] \rightarrow [0]) \in \text{LEnv}(\mathcal{M})$ where f and g preserves the minimum and e_1 is the coCartesian lifting over the (active) map in Ass corresponds to $[1] \rightarrow [n+1], \{0\} \mapsto \{0\}, \{1\} \mapsto \{n\}$ in Δ , and

$$e_2 = \begin{array}{ccc} [n+1] & \xrightarrow{f} & [0] \\ \downarrow & & \downarrow \\ [1] & \xrightarrow{g} & [0] \end{array}, \text{ the induced morphism } \text{LEnv}(e) \text{ is an equivalence (since the}$$

embedding $\text{LEnv}(\mathcal{M})^{\otimes} \hookrightarrow \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes}$ is fully faithful).

Recall the equivalences:

$$\text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, X; Y) \cong \text{Map}_{\text{LEnv}(\mathcal{M})}((V_1, \dots, V_n, X), f : [n+1] \rightarrow [0]), (Y, \text{id}))$$

$$\text{Mul}_{\mathcal{M}}(V_1 \otimes \dots \otimes V_n, X; Y) \cong \text{Map}_{\text{LEnv}(\mathcal{M})}(((V_1 \otimes \dots \otimes V_n, X), g : [1] \rightarrow [0]), (Y, \text{id}))$$

and thus the above condition of Pseu-enriched translates into the condition of being pseudo-enriched.

On the other hand, if $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ is pseudo-enriched, then there are equivalences of multimorphism spaces:

$$\begin{aligned} & \text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, V_{n+1}, \dots, V_m, V_{m+1}, \dots, V_k, X; Y) \rightarrow \\ & \text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, V_{n+1} \otimes \dots \otimes V_m, V_{m+1}, \dots, V_k, X; Y) \cong \text{Mul}_{\mathcal{M}}(V_1 \otimes \dots \otimes V_k, X; Y) \end{aligned}$$

where the second arrow and the composition of the first and the second arrows are equivalences, so is the first arrow.

For $x, y \in \mathcal{M}$, and $V_1, \dots, V_k \in \mathcal{V}$, consider the morphism $e = (e_1, e_2) : ((V_1, \dots, V_n, V_{n+1}, \dots, V_m, V_{m+1}, \dots, V_k, X), f : [k+1] \rightarrow [1]) \rightarrow ((V_1, \dots, V_n, V_{n+1} \otimes \dots \otimes V_m, V_{m+1}, \dots, V_k, X), g : [k-m+n+2] \rightarrow [1])$ in $\text{LEnv}(\mathcal{M})$ where f and g preserve the minimum and e_1 is the coCartesian lifting over the

map in Ass corresponds to $[k - m + n + 2] \rightarrow [k + 1]$ sending $\{n\} \mapsto \{n\}$, $\{n + 1\} \mapsto \{m\}$, and

$$e_2 = \begin{array}{ccc} [n + 1] & \xrightarrow{f} & [0] \\ \downarrow e_1 & & \downarrow \\ [k - m + n + 2] & \xrightarrow{g} & [0] \end{array}$$

Through the equivalences:

$$\begin{aligned} & \text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, V_{n+1}, \dots, V_m, V_{m+1}, \dots, V_k, X; Y) \cong \\ & \text{Map}_{\text{LEnv}(\mathcal{M})}(((V_1, \dots, V_n, V_{n+1}, \dots, V_m, V_{m+1}, \dots, V_k, X; Y), f : [k + 1] \rightarrow [0]), (Y, \text{id})) \\ & \text{Mul}_{\mathcal{M}}(V_1, \dots, V_n, V_{n+1} \otimes \dots \otimes V_m, V_{m+1}, \dots, V_k, X; Y) \cong \\ & \text{Map}_{\text{LEnv}(\mathcal{M})}(((V_1, \dots, V_n, V_{n+1} \otimes \dots \otimes V_m, V_{m+1}, \dots, V_k, X; Y), g : [k - m + n + 2] \rightarrow [0]), (Y, \text{id})) \end{aligned}$$

The condition of being pseudo-enriched translates back to the condition that $\text{LEnv}(e)$ being equivalences in $\text{LEnv}(\mathcal{M}) \hookrightarrow \mathcal{P}\text{LEnv}(\mathcal{M})$, and is equivalent to $\mathcal{M}$ being Pseu-enriched. $\square$

Notation 2.28. Given a small localization pair $(\mathcal{V}, S)$, and a small weakly left tensored category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$.

- Let

$$\mathcal{P}\text{LEnv}(\mathcal{M})_S^{\otimes} \subset S^{-1}\mathcal{P}\text{Env}(\mathcal{V})^{\otimes} \times_{\mathcal{P}\text{Env}(\mathcal{V})^{\otimes}} \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes}$$

be the full weakly left tensored subcategory spanned by those presheaves on $\text{LEnv}(\mathcal{M})$ that are local with respect to the set of morphisms in the form of $f \otimes X$ for every $f \in S$ and $X \in \mathcal{M}$.

- Denote the collection of morphisms in the form of $\{f \otimes \text{id}_x$ for $f \in S$ and $x \in \mathcal{M}$ as T , then the underlying category $\mathcal{P}\text{LEnv}(\mathcal{M})_S$ is the localization of $\mathcal{P}\text{LEnv}(\mathcal{M})$ with respect to the set of morphism T , thus it admits a left adjoint $L : \mathcal{P}\text{LEnv}(\mathcal{M}) \rightarrow \mathcal{P}\text{LEnv}(\mathcal{M})_S$.
- Let $\mathcal{P}'(\mathcal{M})^{\otimes}$ denote $\mathcal{P}\text{LEnv}(\mathcal{M})_{\text{Pseu}}^{\otimes}$.

Remark 2.29. The natural map $\mathcal{P}\text{LEnv}(\mathcal{M})_S^{\otimes} \rightarrow \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes}$ is a fully faithful embedding lying over the natural embedding $S^{-1}\mathcal{P}\text{Env}(\mathcal{V})^{\otimes} \rightarrow \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$: since $\mathcal{P}\text{LEnv}(\mathcal{M})_S^{\otimes} \subset S^{-1}\mathcal{P}\text{Env}(\mathcal{V})^{\otimes} \times_{\mathcal{P}\text{Env}(\mathcal{V})^{\otimes}} \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes}$ is an embedding, so is the pullback along the embedding $S^{-1}\mathcal{P}\text{Env}(\mathcal{V})^{\otimes} \rightarrow \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$: $S^{-1}\mathcal{P}\text{Env}(\mathcal{V})^{\otimes} \times_{\mathcal{P}\text{Env}(\mathcal{V})^{\otimes}} \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes} \rightarrow \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes}$.

Remark 2.30. By the above lemma, the natural embedding $\mathcal{M}^{\otimes} \hookrightarrow \text{LEnv}(\mathcal{M})^{\otimes} \rightarrow \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes}$ lands in $\mathcal{P}\text{LEnv}(\mathcal{M})_{\text{Pseu}}^{\otimes}$.

Lemma 2.31. Given a small localization pair $(\mathcal{V}^{\otimes} \rightarrow \text{Ass}, S)$, consider the saturation $\bar{S}$ of S inside $\mathcal{P}\text{Env}(\mathcal{V})$, then the collection of morphisms $\{f \otimes \text{id}_m\}$ where $f \in \bar{S}$, $m \in \mathcal{M}$ is contained in $\bar{T}$.

Proof. Consider the collection of morphisms U in $\mathcal{P}\text{Env}(\mathcal{V})$ such that for any $f \in U$, $m \in \mathcal{M}$, $f \otimes \text{id}_m$ is a T -local equivalence. Then obviously U contains S , and

- U is stable under colimits inside $\text{Fun}([1], \mathcal{P}\text{Env}(\mathcal{V}))$: since the left action of $\mathcal{P}\text{Env}(\mathcal{V})$ on $\mathcal{P}\text{LEnv}(\mathcal{M})$ is colimit preserving and the collection of T -local equivalences is strongly saturated.

- U is closed under pushout along any morphism in $\mathcal{P}\text{Env}(\mathcal{V})$: since the left action of $\mathcal{P}\text{Env}(\mathcal{V})$ on $\mathcal{P}\text{LEnv}(\mathcal{M})$ preserves colimits and the collection of T local equivalences is strongly saturated.
- U satisfies 2-out-of-3 property: given $v_1 \xrightarrow{f} v_2 \xrightarrow{g} v_3$ where any two of them lie in U , this implies two of the morphisms in $v_1 \otimes m \xrightarrow{f \otimes \text{id}_m} v_2 \otimes m \xrightarrow{g \otimes \text{id}_m} v_3 \otimes m$ lie in $\bar{T}$, thus the remaining one must lie in $\bar{T}$ since it's strongly saturated.

Thus U is strongly saturated, and it must contain $\bar{S}$. $\square$

Thus we can also regard $\mathcal{P}\text{LEnv}(\mathcal{M})_S$ as localization with respect to the set of morphisms in the form of $f \otimes \text{id}_m$ for $f \in \bar{S}$ and $m \in \mathcal{M}$.

Lemma 2.32. [15, remark 4.16] *Given a small localization pair $(\mathcal{V}^\otimes \rightarrow \text{Ass}, S)$, the fully faithful embedding $\mathcal{P}\text{LEnv}(\mathcal{M})_S \rightarrow \mathcal{P}\text{LEnv}(\mathcal{M})$ admits a left adjoint, and can be enhanced into an enriched adjunction*

$$\mathcal{P}\text{LEnv}(\mathcal{M})_S^\otimes \rightleftarrows \mathcal{P}\text{LEnv}(\mathcal{M})^\otimes$$

where the right adjoint is the fully faithful embedding as remarked above.

Proof. We first show that $\mathcal{P}\text{LEnv}(\mathcal{M})_S$ is a left tensored ∞ -category, and then apply the [enriched adjunction theorem](#).

Recall [22, lemma 2.4.2.11], given a full subcategory $D' \subset D$ where D is coCartesian fibration over C , if fiberwise $D'_c \rightarrow D_c$ admits a left adjoint L_c , and given a morphism $e : c \rightarrow c'$, the associated functor $e_! : D_c \rightarrow D_{c'}$ preserves local equivalences, then $D' \rightarrow C$ is also coCartesian fibration. And we apply the lemma on the full subcategory $\mathcal{P}\text{LEnv}(\mathcal{M})_S^\otimes \subset \mathcal{P}\text{LEnv}(\mathcal{M})^\otimes$ over Ass :

- Fiberwise $(\mathcal{P}\text{LEnv}(\mathcal{M})_S^\otimes)_{[n]} \cong S^{-1} \mathcal{P}\text{Env}(\mathcal{V})_{[n]}^\otimes \times \mathcal{P}\text{LEnv}(\mathcal{M})_S^\otimes \hookrightarrow \mathcal{P}\text{Env}(\mathcal{V})_{[n]}^\otimes \times \mathcal{P}\text{LEnv}(\mathcal{M})^\otimes$ is a localization with respect to the collection of morphisms $S^{\times n} \times \mathcal{Q}$ where $\mathcal{Q}$ is the collection of morphisms in the form of $f \otimes X$ for $f \in \bar{S}, X \in \mathcal{M}$. And obviously we have $\bar{T} = \bar{\mathcal{Q}}$.
- Given an inert morphism preserving the maximum $\alpha : [n] \rightarrow [m]$, then $\alpha_!$ is projection $(\mathcal{P}\text{LEnv}(\mathcal{M})_S^\otimes)_{[n]} \rightarrow (\mathcal{P}\text{LEnv}(\mathcal{M})_S^\otimes)_{[m]}$, thus preserves local equivalences, since local equivalences are precisely those morphisms that are fiberwise local equivalences.
- Given any $m \in \mathcal{P}\text{LEnv}(\mathcal{M})_S$, and S -localization in $\mathcal{P}\text{Env}(\mathcal{V})$: $e_1 : v_1 \rightarrow v'_1$, $\mathcal{Q}$ -localization in $\mathcal{P}\text{LEnv}(\mathcal{M})$: $e_2 : m_1 \rightarrow m'_1$, then noticing that $e_1 \otimes e_2 : v_1 \otimes m_1 \rightarrow v'_1 \otimes m'_1$ is the composition $e_1 \otimes \text{id} : v_1 \otimes m_1 \rightarrow v'_1 \otimes m_1$ and $\text{id} \otimes e_2 : v'_1 \otimes m_1 \rightarrow v'_1 \otimes m'_1$, and both are $\mathcal{Q}$ -local equivalences:
- $e_1 \otimes \text{id}$ is $\mathcal{Q}$ -local equivalence since the left action $\mathcal{P}\text{Env}(\mathcal{V})$ sends $\bar{S}$ to the collection of morphisms $\mathcal{Q}$.
- $\text{id}_{v'_1} \otimes e_2$ is $\mathcal{Q}$ -local equivalence: if e_2 lands in $\mathcal{Q}$, then $\text{id}_{v'_1} \otimes e_2$ is also in $\mathcal{Q}$ since $\text{id}_{v'_1}$ is in $\bar{S}$. And now consider the collection of morphisms V in $\mathcal{P}\text{LEnv}(\mathcal{M})$ such that for any $f \in V$, $\text{id}_{v'_1} \otimes f$ is $\mathcal{Q}$ -equivalence. And obviously V contains $\mathcal{Q}$. Then we can show V

is closed under colimits and pushout along morphism in $\mathcal{P}\text{LEnv}(\mathcal{M})$ and satisfies 2-out-of-3 property by the fact that $\mathcal{P}\text{Env}(\mathcal{V})$ action on $\mathcal{P}\text{LEnv}(\mathcal{M})$ preserves colimits. Thus $\overline{Q} = \overline{T} \subset V$, and for any $e_2 \in \overline{T} = \overline{Q}$, $\text{id}_{m_1} \otimes e_2$ is a $\mathcal{Q}$ -local equivalence.

- Thus $e_1 \otimes e_2$ is $\mathcal{Q}$ -local equivalence. This implies left tensor preserves local equivalences.
- Given a morphism preserving the minimum $\beta : ([0] \rightarrow [n])^{op}$, then $\beta_!$ sends $(V_1, \dots, V_n, X)$ to $V_1 \otimes \dots \otimes V_n \otimes X$, which preserves local equivalences.
- Given any morphism preserving the minimum $\beta : ([m] \rightarrow [n])^{op}$, by the equivalence $\text{Ass} \times [1] \cong (\Delta_{/[1]})^{op} = \text{LM}$, it corresponds to an active map in Ass over $[1]$: $([m] \star [0] \cong [m+1] \rightarrow [n] \star [0] \cong [n+1])^{op}$, and more intuitively in the language of colored operad, such active map can be written as $(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n, \mathbf{m}) \rightarrow (\mathbf{a}_1, \dots, \mathbf{a}_m, \mathbf{m})$ which can only involve one possible left action operation $(\mathbf{a}_r, \dots, \mathbf{a}_n, \mathbf{m}) \rightarrow \mathbf{m}$ and an operation only about the $\mathbf{a}$: $(\mathbf{a}_1, \dots, \mathbf{a}_{r-1}) \rightarrow (\mathbf{a}_1, \dots, \mathbf{a}_m)$, since the localization $S^{-1} \mathcal{P}(\mathcal{V})^{\otimes} \rightarrow \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$ is compatible with the monoidal structure, so $\beta_!$ preserves local equivalences.
- Thus $\mathcal{P}\text{LEnv}(\mathcal{M})_S^{\otimes}$ is coCartesian fibration over Ass by [22, lemma 2.2.4.11], and it's obvious that the projection $\mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes} \rightarrow S^{-1} \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$ preserves the coCartesian liftings, and thus it's a left tensored ∞ -category.
- By [enriched adjunction theorem](#) for a lax linear functor between left tensored ∞ -category, the embedding $\mathcal{P}\text{LEnv}(\mathcal{M})_S \hookrightarrow \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes}$ admits an enriched left adjoint.

□

Remark 2.33. Thus $\mathcal{P}\text{LEnv}(\mathcal{M})_S^{\otimes} \rightarrow S^{-1} \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$ is a presentable left tensored category, that is, an object in $\text{LMod}_{S^{-1} \mathcal{P}\text{Env}(\mathcal{V})}(\text{Pr}^L)$.

Example 2.34. Consider the composition of natural inclusion $\mathcal{M}^{\otimes} \hookrightarrow \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes}$ with the localization $\mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes} \rightarrow \mathcal{P}'(\mathcal{M})^{\otimes}$, this functor $\mathcal{M} \rightarrow \mathcal{P}'(\mathcal{M})$ is a fully faithful embedding if $\mathcal{M}$ is pseudo-enriched by [15, remark 4.17]. And in this case, we denote the embedding by $\mathfrak{y}' : \mathcal{M} \hookrightarrow \mathcal{P}'(\mathcal{M})$

And especially, we have the universal property of $\mathcal{P}'(-)$, which is [15, proposition 4.20]:

Proposition 2.35. *Given a pseudo-enriched category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, then the inclusion $\mathcal{M}^{\otimes} \hookrightarrow \mathcal{P}'(\mathcal{M})^{\otimes}$ induces an equivalence for any locally small left tensored category compatible with small colimits $\mathcal{N}^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$:*

$$\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \rightarrow \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})$$

where on the right hand side, $\mathcal{N}$ is regarded as $\mathcal{V}$ -linear module category through the pullback along $\mathfrak{y} : \mathcal{V} \hookrightarrow \mathcal{P}(\mathcal{V})$.

Proof. Let $\widetilde{\mathcal{N}}^{\otimes} \rightarrow \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$ be the pullback of $\mathcal{N}^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$ along the localization map $\text{loc} : \mathcal{P}\text{Env}(\mathcal{V}) \rightarrow \mathcal{P}(\mathcal{V})$, then by the universal property of enveloping and presheaf, the following functor induced by the embedding $\mathcal{M}^{\otimes} \hookrightarrow \mathcal{P}(\text{Env}(\mathcal{M}))^{\otimes}$ is an equivalence:

$$\text{Lin}_{\mathcal{P}\text{Env}(\mathcal{V})}^L(\mathcal{P}(\text{LEnv}(\mathcal{M})), \widetilde{\mathcal{N}}) \rightarrow \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, c^* \widetilde{\mathcal{N}})$$

where $c^* \widetilde{\mathcal{N}}$ is the pullback of $\widetilde{\mathcal{N}}$ along the natural embedding $c : \mathcal{V}^{\otimes} \hookrightarrow \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$.

On the other hand, $c^*\widetilde{\mathcal{N}}$ is equivalent to $\mathfrak{L}_{\mathcal{V}}^*(\mathcal{N})$ since $c^* \circ loc^* \cong (c \circ loc)^* \cong \mathfrak{L}^*$. It remains to show $\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \cong \text{Lin}_{\mathcal{P}(\text{Env}(\mathcal{V}))}^L(\mathcal{P}(\text{LEnv}(\mathcal{M})), \widetilde{\mathcal{N}})$

By construction, $\mathcal{P}'(\mathcal{M})$ is the localization of $\mathcal{P}(\text{LEnv}(\mathcal{M}))$ with respect to the set of morphisms $f \otimes X$ for $X \in \mathcal{M}$, and $f \in \text{LPseu}$ (regarded as a set of morphisms in $\mathcal{P}(\text{LEnv}(\mathcal{M}))$ through the Yoneda embedding), by [15, 4.16], there exists an adjunction where the right adjoint R is the natural embedding (in the $(\infty, 2)$ category $\omega \text{ LMod}$:

$$L : \mathcal{P}\text{LEnv}^{\otimes}(\mathcal{M}) \rightleftarrows \mathcal{P}\text{LEnv}(\mathcal{M})_{\text{LPseu}}^{\otimes} = \mathcal{P}'(\mathcal{M}) : R$$

lying over another adjunction in the $(\infty, 2)$ category Op : $loc : \mathcal{P}(\text{Env}(\mathcal{V}))^{\otimes} \rightleftarrows \mathcal{P}(\mathcal{V})^{\otimes} : i$.

Then we have the following commutative diagram:

$$\begin{array}{ccc} \text{LaxLin}(\mathcal{P}(\text{LEnv}(\mathcal{M})), \widetilde{\mathcal{N}}) & \xrightarrow{R^*} & \text{LaxLin}(\mathcal{P}'(\mathcal{M}), \widetilde{\mathcal{N}}) \\ \downarrow & & \downarrow \\ \text{Alg}_{\mathcal{P}\text{Env}(\mathcal{V})}(\mathcal{P}\text{Env}(\mathcal{V})) & \xrightarrow{i^*} & \text{Alg}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}\text{Env}(\mathcal{V})) \end{array}$$

where the vertical arrows are the projections.

Moreover, by [14, 9.30 (4)], by passing to the left adjoint of the horizontal arrows, we again have a commutative diagram:

$$\begin{array}{ccc} \text{LaxLin}(\mathcal{P}(\text{LEnv}(\mathcal{M})), \widetilde{\mathcal{N}}) & \xleftarrow{L^*} & \text{LaxLin}(\mathcal{P}'(\mathcal{M}), \widetilde{\mathcal{N}}) \\ \downarrow & & \downarrow \\ \text{Alg}_{\mathcal{P}\text{Env}(\mathcal{V})}(\mathcal{P}\text{Env}(\mathcal{V})) & \xleftarrow{loc^*} & \text{Alg}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}\text{Env}(\mathcal{V})) \end{array}$$

And L^* is fully faithful since the R is an embedding by [14, 9.30(2)]. By taking pullback along the choice of object $\{\text{id}\} : * \rightarrow \text{Alg}_{\mathcal{P}\text{Env}(\mathcal{V})}(\mathcal{P}\text{Env}(\mathcal{V}))$, the above adjunction restricts to the following adjunction:

$$L^* : \text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \rightleftarrows \text{LaxLin}_{\mathcal{P}\text{Env}(\mathcal{V})}(\mathcal{P}(\text{LEnv}(\mathcal{M})), \widetilde{\mathcal{N}}) : R^*$$

where the left adjoint is again fully faithful, and by checking at the level of objects, we know the image of L^* consists of those lax linear functors from $\mathcal{P}(\text{LEnv}(\mathcal{M}))$ to $\widetilde{\mathcal{N}}$ sending the morphisms in the set $\{f \otimes X : X \in \mathcal{M}, f \in \text{Pseu}_{\mathcal{V}}\}$ to the equivalences in $\widetilde{\mathcal{N}}$. Recall the construction of $\widetilde{\mathcal{N}}$ is the pullback of $\mathcal{N}$ along the localization $\mathcal{P}\text{Env}(\mathcal{V}) \rightarrow \mathcal{P}(\mathcal{V})$, and $\mathcal{N}$ is left tensored, thus the above adjunction restricts to an equivalence:

$$\text{Lin}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \cong \text{Lin}_{\mathcal{P}\text{Env}(\mathcal{V})}(\mathcal{P}(\text{LEnv}(\mathcal{M})), \widetilde{\mathcal{N}})$$

And especially we have an equivalence $\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \cong \text{Lin}_{\mathcal{P}\text{Env}(\mathcal{V})}^L(\mathcal{P}(\text{LEnv}(\mathcal{M})), \widetilde{\mathcal{N}})$.

And by composition, we have an equivalence: $\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \rightarrow \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})$ $\square$

Moreover, we have the following corollary:

Corollary 2.36. *The forgetful functor $\text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \rightarrow \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})$ admits a fully faithful left adjoint, and the left adjoint lands in the full subcategory $\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}'(\mathcal{M}), \mathcal{N})$.*

Proof. The left adjoint of the functor $\text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \rightarrow \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})$ factors as the composition:

$$\text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N}) \xrightarrow{\alpha} \text{LaxLin}_{\mathcal{PEnv}(\mathcal{V})}(\mathcal{PEnv}(\mathcal{M}), \widetilde{\mathcal{N}}) \xrightarrow{\beta} \text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}'(\mathcal{M}), \mathcal{N})$$

where α is the fully faithful left adjoint of the restriction functor induced by the embedding $\mathcal{M} \rightarrow \mathcal{PEnv}(\mathcal{M})$, and β is the restriction functor R^* in the proof of the above proposition. Thus α lands in $\text{Lin}_{\mathcal{PEnv}(\mathcal{V})}^L(\mathcal{PEnv}(\mathcal{M}), \widetilde{\mathcal{N}})$, and β takes $\text{Lin}_{\mathcal{PEnv}(\mathcal{V})}^L(\mathcal{PEnv}(\mathcal{M}), \widetilde{\mathcal{N}})$ to $\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}'(\mathcal{M}), \mathcal{N})$ $\square$

Remark 2.37. The factorization in the above proof is due to the following diagram:

$$\begin{array}{ccc} \text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}'(\mathcal{M}), \mathcal{N}) & \xrightarrow{id} & \text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}'(\mathcal{M}), \mathcal{N}) \\ \uparrow \wr & \searrow L^* & \uparrow \beta=R^* \\ \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N}) & \xrightarrow{\alpha=\wr} & \text{LaxLin}_{\mathcal{PEnv}(\mathcal{V})}(\mathcal{PEnv}(\mathcal{M}), \widetilde{\mathcal{N}}) \end{array}$$

2.4 enriched presheaf

This part follows [15, section 4].

Notation 2.38. • Let κ be a small regular cardinal and $\mathcal{V}^{\otimes} \rightarrow \text{Ass}$ a small monoidal ∞ -category compatible with κ -small colimits. Let Enr_{κ} be the set of morphisms

$$V'_1, \dots, V'_l, V_1, \dots, V_n, V''_1, \dots, V''_m \rightarrow V'_1, \dots, V'_l, V_1 \otimes \dots \otimes V_n, V''_1, \dots, V''_m,$$

$$V'_1, \dots, V'_l, \text{colim}(\iota \circ F), V''_1, \dots, V''_m \rightarrow V'_1, \dots, V'_l, \iota(\text{colim}(F)), V''_1, \dots, V''_m$$

for $V'_1, \dots, V'_l, V_1, \dots, V_n, V''_1, \dots, V''_m \in \mathcal{V}$ and $l, n, m \geq 0$, where $\iota : \mathcal{V} \subset \mathcal{PEnv}(\mathcal{V})$ is the embedding, $F : K \rightarrow \mathcal{V}$ is a functor and K is κ -small.

- If we enlarge the universe, and let σ corresponding to the small universe, then we denote Enr_{σ} as Enr .

Lemma 2.39. *For every small ∞ -monoidal category $\mathcal{V}$ compatible with small κ -colimits, $(\mathcal{PEnv}(\mathcal{V})^{\otimes} \rightarrow \text{Ass}, \text{Enr}_{\kappa})$ is a small localization pair where $\text{Enr}_{\kappa}^{-1} \mathcal{PEnv}(\mathcal{V})^{\otimes} \cong \text{Ind}_{\kappa}(\mathcal{V})^{\otimes}$ is a monoidal localization of $\mathcal{PEnv}(\mathcal{V})^{\otimes}$.*

Proof. The adjunction induced by localization with respect to $\text{Pseu}_{\mathcal{V}}$ is preserved under the presheaf functor, so there is an adjunction $\mathcal{P}(i) : \mathcal{P}(\mathcal{V}) \rightleftarrows \mathcal{PEnv}(\mathcal{V}) : \mathcal{P}(loc)$, and $\mathcal{P}(i)$ is again fully faithful embedding by [21, proposition 5.1.6.10]. By the uniqueness of right adjoint, $\mathcal{P}(i) \cong loc^* : \text{Fun}(\mathcal{V}^{op}, \text{Spc}) \rightarrow \text{Fun}(\text{Env}(\mathcal{V})^{op}, \text{Spc})$, and by [24, proposition 6.3.3.6, proposition 6.3.6.13], the essential image of this embedding consists of exactly the presheaf $F : \text{Env}(\mathcal{V})^{op} \rightarrow$

$\mathcal{Spc}$ such that $F(f)$ is an equivalence for local equivalence f , which is equivalent to $F(f)$ being an equivalence for all $f \in \mathcal{Pseu}_{\mathcal{V}}$. Thus the full subcategory of local objects with respect to $\mathcal{Pseu}_{\mathcal{V}}$ is exactly $\mathcal{P}(\mathcal{V}) \subset \mathcal{PEnv}(\mathcal{V})$.

On the other hand, denote the collection of morphism in the form of $\text{colim}(\iota \circ F)$ for $F : K \rightarrow \mathcal{V}$ where K is small as Σ , then the presheaf $F \in \mathcal{P}(\mathcal{V})$ which is local with respect to Σ lies exactly in $\text{Ind}_{\kappa}(\mathcal{V}) \subset \mathcal{P}(\mathcal{V})$ by [21, corollary 5.3.5.4]. This gives rise to an adjunction $\text{Ind}_{\kappa}(\mathcal{V}) \rightleftarrows \mathcal{P}(\mathcal{V})$ and implies the full subcategory of Enr_{κ} -local objects should be contained in $\text{Ind}_{\kappa}(\mathcal{V}) \subset \mathcal{P}(\mathcal{V}) \subset \mathcal{PEnv}(\mathcal{V})$.

And on the other hand, for any $f \in \text{Enr}_{\kappa}$ and $F \in \text{Ind}_{\kappa}(\mathcal{V})$, we have $\text{Map}_{\mathcal{PEnv}(\mathcal{V})}(f, i(F)) \cong \text{Map}_{\text{Ind}_{\kappa}}(Lf, F)$ by adjunctions $i : \text{Ind}_{\kappa} \rightleftarrows \mathcal{P}(\mathcal{V}) \rightleftarrows \mathcal{PEnv}(\mathcal{V}) : L$. Since L is monoidal and tensor product on $\mathcal{V}$ is compatible with κ colimits, $L(f) \in \Sigma$, and $\text{Map}_{\text{Ind}_{\kappa}}(L(f), F)$ is an equivalence, so is $\text{Map}_{\mathcal{PEnv}(\mathcal{V})}(f, i(F))$. Thus $\text{Ind}_{\kappa}(\mathcal{V})$ is contained in the full subcategory of Enr_{κ} -local objects, and $\text{Ind}_{\kappa}(\mathcal{V})$ is exactly the full subcategory of Enr_{κ} -local objects.

And it remains to show the above localizations $i : \text{Ind}_{\kappa} \rightleftarrows \mathcal{P}(\mathcal{V}) \rightleftarrows \mathcal{PEnv}(\mathcal{V}) : L$ is monoidal: by [22, proposition 2.2.1.9], it suffices to show the local equivalences are closed under tensor product.

Given two Enr_{κ} local equivalences $e_1 : v_1 \rightarrow v'_1, e_2 : v_2 \rightarrow v'_2$ (especially v'_1, v'_2 lie in $\text{Ind}_{\kappa}(\mathcal{V})$) consider their tensor product $e_1 \otimes e_2$, then $L(e_1 \otimes e_2) : L(v_1 \otimes v_2) \cong L(v_1) \otimes L(v_2) \cong v'_1 \otimes v'_2 \cong L(v'_1) \otimes L(v'_2) \cong L(v'_1 \otimes v'_2)$ is an equivalence, so is a local equivalence. $\square$

Remark 2.40. • a weakly left tensored ∞ -category $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ exhibits $\mathcal{M}$ as [Enr \$_{\kappa}\$ -enriched](#) iff $\mathcal{M}$ is [\$\kappa\$ -enriched](#).

• a weakly left tensored ∞ -category $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ exhibits $\mathcal{M}$ as [Enr-enriched](#) iff $\mathcal{M}$ is [enriched](#).

• $\text{Enr}_{\emptyset} = \mathcal{Pseu}$.

Construction 2.41. Let $\mathcal{V}^{\otimes} \rightarrow \text{Ass}$ be a monoidal ∞ -category compatible with small colimits (for example, a presentable monoidal ∞ -category), and $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ a quasi-enriched category. Let

$$\widehat{\mathcal{P}Env}_{\sigma}(\mathcal{M})_{\text{Enr}} \subset \mathcal{V}^{\otimes} \times_{\text{ind}_{\sigma}(\mathcal{V})^{\otimes}} \widehat{\mathcal{P}Env}(\mathcal{M})_{\text{Enr}}^{\otimes}$$

be the full left tensored subcategory generated by $\mathcal{M}$ under small colimits and $\mathcal{V}$ -action. And we denote $\widehat{\mathcal{P}Env}_{\sigma}(\mathcal{M})_{\text{Enr}} \rightarrow \mathcal{V}^{\otimes}$ as $\mathcal{P}_{\mathcal{V}}(\mathcal{M})^{\otimes} \rightarrow \mathcal{V}^{\otimes}$.

Definition 2.42. For a small category enriched in a monoidal ∞ -category compatible with small colimits $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, we define $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$ to be the enriched presheaf category of $\mathcal{M}$.

Remark 2.43. In general, $\mathcal{M} \hookrightarrow \mathcal{P}_{\mathcal{V}}(\mathcal{M})$ doesn't preserve tensors.

Here are several properties of the enriched presheaf $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$:

Proposition 2.44. (1) Given a quasi-enriched ∞ -category $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ where $\mathcal{V}$ is monoidal ∞ -category compatible with small colimits, then the embedding $\mathcal{M}^\otimes \rightarrow \mathcal{P}'(\mathcal{M})^\otimes$ induces an equivalence:

$$\mathrm{Lin}_{\mathcal{V}}^L(\mathcal{P}_{\mathcal{V}}(\mathcal{M}), N) \rightarrow \mathrm{LaxLin}_{\mathcal{V}}(\mathcal{M}, N)$$

for left tensored ∞ -category $N^\otimes \rightarrow \mathcal{V}^\otimes$ compatible with small colimits.

- (2) Given a small enriched ∞ -category $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ where $\mathcal{V}$ is presentable monoidal ∞ -category, $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$ is presentable left tensored ∞ -category.
- (3) More precisely, if $\mathcal{V}$ is κ -compactly generated, denote the pullback of $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ along the embedding of full subcategory of κ -compact objects $(\mathcal{V}^\kappa)^\otimes \hookrightarrow \mathcal{V}^\otimes$ as $\mathcal{M}_\kappa^\otimes \rightarrow (\mathcal{V}^\kappa)^\otimes$, then the lax linear functor $\mathcal{M}_\kappa^\otimes \rightarrow \mathcal{P}_{\mathcal{V}}(\mathcal{M})^\otimes$ induces an equivalence

$$\mathcal{P}\mathrm{LEnv}(\mathcal{M}_\kappa)_{\mathrm{Enr}_\kappa}^\otimes \cong \mathcal{P}_{\mathcal{V}}(\mathcal{M})^\otimes$$

- (4) (enriched Yoneda embedding) Given a small enriched category $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$ where $\mathcal{V}^\otimes$ is compatible with small colimits, then there exists a $\mathcal{V}$ -linear equivalence $\psi : \mathcal{P}_{\mathcal{V}}(\mathcal{M}) \rightarrow \mathrm{LaxLin}_{\mathcal{V}}(\mathcal{M}^{op}, \mathcal{V})$ making the following diagram commute:

$$\begin{array}{ccc} \mathcal{P}_{\mathcal{V}}(\mathcal{M}) & \xrightarrow{\psi} & \mathrm{LaxLin}_{\mathcal{V}}(\mathcal{M}^{op}, \mathcal{V}) \\ & \searrow i & \nearrow \mathfrak{z} \\ & \mathcal{M} & \end{array} \quad \text{where } \mathrm{LaxLin}_{\mathcal{V}}(\mathcal{M}^{op}, \mathcal{V}) \text{ is equipped with}$$

$\mathcal{V}$ -action by [14, 3.86], and i is the natural embedding, $\mathfrak{z}$ sends $m \in \mathcal{M}$ to $\mathrm{Mor}_{\mathcal{M}}(-, m) \in \mathrm{LaxLin}_{\mathcal{V}}(\mathcal{M}^{op}, \mathcal{V})$.

Proof. (1) is by [15, theorem 4.41], or by the fact that $\mathcal{P}(-)$ is the left adjoint to the inclusion $\mathrm{LMod}_{\mathcal{V}}(\widehat{\mathrm{Cat}}_{cc}) \rightarrow \widehat{\mathrm{QEnr}}_{\mathcal{V}}$ when restricted to the full subcategory $\mathrm{Enr}_{\mathcal{V}} \subset \widehat{\mathrm{QEnr}}_{\mathcal{V}}$.

(2) is by [16, proposition 2.156].

(3) is by [16, theorem 2.157]. □

Remark 2.45. Noticing that if $\mathcal{V} = \mathrm{Ind}_{\kappa}(\mathcal{V}_0)$, then $\mathcal{V} = \mathrm{Ind}_{\kappa}(\mathcal{V}^{\kappa})$, and $\mathcal{P}\mathrm{LEnv}(\mathcal{M}_{\kappa})_{\mathrm{Enr}_{\kappa}}$ is presentably $\mathrm{Ind}_{\kappa}(\mathcal{V}^{\kappa}) \cong \mathcal{V}$ left module.

Moreover, there is a functor assigning the enriched presheaf, as an enriched generalization of [21, proposition 5.3.6.2]:

Theorem 2.46. For $\mathcal{V}$ a presentable monoidal ∞ -category, the natural inclusion $\mathrm{LMod}_{\mathcal{V}}(\mathrm{Pr}^L) \rightarrow \widehat{\mathrm{QEnr}}_{\mathcal{V}}$ admits a left adjoint which restricts to the full subcategory $\mathrm{Enr}_{\mathcal{V}} \subset \widehat{\mathrm{QEnr}}_{\mathcal{V}}$ sending a small enriched category $\mathcal{M}$ to its enriched presheaf $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$.

$$\mathcal{P}_{\mathcal{V}}(-) : \mathrm{Enr}_{\mathcal{V}} \rightarrow \mathrm{LMod}_{\mathcal{V}}(\mathrm{Pr}^L)$$

Proof. This is [16, corollary 5.1]. □

2.5 fully faithful (enriched) functor

In the context of enrichment, an enriched functor, that is, a lax linear functor between $\mathcal{V}$ -enriched categories $F : \mathcal{M}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$ being fully faithful should imply the following condition:

$$(*) \quad \text{Mor}_{\mathcal{M}}(X, Y) \cong \text{Mor}_{\mathcal{N}}(F(X), F(Y)) \in \mathcal{V}$$

And on the other hand, we have the following version of ‘fully faithful embedding’:

Notation 2.47. [15, 2.50] A lax linear functor

$$\begin{array}{ccc} M^{\otimes} & \xrightarrow{f} & N^{\otimes} \\ \downarrow & & \downarrow \\ V^{\otimes} & \xrightarrow{f'} & W^{\otimes} \end{array}$$

is an **embedding** if f and f' are fully faithful functors.

In this part, we show that their relations and some other properties of fully faithful (enriched) functor.

Lemma 2.48. *A lax $\mathcal{V}$ linear functor between enriched ∞ - $\mathcal{V}$ categories $F : \mathcal{M}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$ is an embedding if and only if it induces an equivalence between morphism objects $\text{Mor}_{\mathcal{M}}(X, Y) \cong \text{Mor}_{\mathcal{N}}(FX, FY)$ for all $X, Y \in \mathcal{M}$.*

Proof. Noticing the following decomposition of mapping spaces in $\mathcal{M}^{\otimes}$:

$$\text{Map}_{\mathcal{M}}((V_1, \dots, V_n, X), (V'_1, \dots, V'_m, Y)) \cong \text{Map}_{\mathcal{V}^{\otimes}}((V_1, \dots, V_n), (V'_1, \dots, V'_m)) \times \left(\prod_{i=1, \dots, n} \text{Mul}_{\mathcal{M}}(V_i, \dots, V_n, X; Y) \right)$$

by the condition (2) of weakly left tensored ∞ -category.

Since $\mathcal{M}, \mathcal{N}$ are enriched thus pseudo-enriched, thus F is fully faithful iff the induced map between multimorphism spaces:

$$\text{Mul}_{\mathcal{M}}(V, X; Y) \rightarrow \text{Mul}_{\mathcal{N}}(V, FX; FY)$$

is an equivalence for any $V \in \mathcal{V}, X, Y \in \mathcal{M}$.

Noticing the following commutative diagram:

$$\begin{array}{ccc} \text{Mul}_{\mathcal{M}}(V, X; Y) & \xrightarrow{\quad} & \text{Mul}_{\mathcal{N}}(V, FX; FY) \\ \cong \uparrow & & \uparrow \cong \\ \text{Map}_{\mathcal{V}}(V, \text{Mor}_{\mathcal{M}}(X, Y)) & \xrightarrow{\quad} & \text{Map}_{\mathcal{V}}(V, \text{Mor}_{\mathcal{N}}(FX, FY)) \end{array}$$

thus fully faithful functor $F : \mathcal{M}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$ induces equivalence between morphism objects. $\square$

Lemma 2.49. *Given a lax linear functor*

$$\begin{array}{ccc} M^{\otimes} & \xrightarrow{f} & N^{\otimes} \\ \downarrow & & \downarrow \\ V^{\otimes} & \xrightarrow{f'} & W^{\otimes} \end{array}$$

, if $f' : \mathcal{V}^{\otimes} \rightarrow \mathcal{W}^{\otimes}$ is a fully faithful embedding. Then the following conditions are equivalent:

$$(1) \text{ the lax linear functor } \begin{array}{ccc} M^{\otimes} & \xrightarrow{\quad f \quad} & N^{\otimes} \\ \downarrow & & \downarrow \\ V^{\otimes} & \xrightarrow{\quad f' \quad} & W^{\otimes} \end{array} \text{ is an embedding.}$$

(2) f induces an equivalence in $\mathcal{W}$:

$$f'(\text{Mor}_{\mathcal{M}}(X, Y)) \cong \text{Mor}_{\mathcal{N}}(fX, fY)$$

for any $X, Y \in \mathcal{M}$

Proof. • By the decomposition

$$\text{Map}_{\mathcal{M}}((V_1, \dots, V_n, X), (V'_1, \dots, V'_m, Y)) \cong \text{Map}_{\mathcal{V}^{\otimes}}((V_1, \dots, V_n), (V'_1, \dots, V'_m)) \times \left(\prod_{i=1, \dots, n} \text{Mul}_{\mathcal{M}}(V_i, \dots, V_n, X; Y) \right)$$

, thus f is a lax linear embedding iff the induced map between multimorphism spaces:

$$\text{Mul}_{\mathcal{M}}(V, X; Y) \rightarrow \text{Mul}_{\mathcal{N}}(V, fX; fY)$$

is an equivalence for any $V \in \mathcal{V}, X, Y \in \mathcal{M}$.

• Noticing the following commutative diagram:

$$\begin{array}{ccc} \text{Mul}_{\mathcal{M}}(V, X; Y) & \xrightarrow{\quad \quad \quad} & \text{Mul}_{\mathcal{N}}(V, fX; fY) \\ \cong \uparrow & & \uparrow \cong \\ \text{Map}_{\mathcal{V}}(V, \text{Mor}_{\mathcal{M}}(X, Y)) & \xrightarrow{\quad \quad \quad} & \text{Map}_{\mathcal{W}}(f'(V), f'(\text{Mor}_{\mathcal{M}}(X, Y))) \end{array}$$

thus f induces an equivalence $f'(\text{Mor}_{\mathcal{M}}(X, Y)) \cong \text{Mor}_{\mathcal{N}}(fX, fY)$ iff f is fully faithful. $\square$

Lemma 2.50. Given a weakly left tensored category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, the pullback of $\mathcal{M}^{\otimes}$ along a fully faithful embedding of operads $\alpha : \mathcal{V}^{\otimes} \rightarrow \mathcal{W}^{\otimes}$ induces an embedding:

$$\begin{array}{ccc} \mathcal{N}^{\otimes} & \xrightarrow{\quad \alpha \quad} & \mathcal{M}^{\otimes} \\ \downarrow & & \downarrow \\ \mathcal{V}^{\otimes} & \xrightarrow{\quad \gamma \quad} & \mathcal{W}^{\otimes} \end{array}$$

Lemma 2.51. Given an embedding $f : \mathcal{M}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$ over $\mathcal{V}^{\otimes}$, it is a monomorphism in $\omega\text{LMod}_{\mathcal{V}}$: the induced morphism

$$\text{Map}_{\omega\text{LMod}_{\mathcal{V}}}(\mathcal{O}, \mathcal{M}) \rightarrow \text{Map}_{\omega\text{LMod}_{\mathcal{V}}}(\mathcal{O}, \mathcal{N})$$

is an inclusion of components for any $\psi : \mathcal{O}^{\otimes} \rightarrow \mathcal{V}^{\otimes} \in \omega\text{LMod}_{\mathcal{V}}$.

Proof. We notice the following facts:

- $\text{Map}_{\omega\text{LMod}_{\mathcal{V}}}(-, -) \cong \text{LaxLin}_{\mathcal{V}}(-, -)^{\simeq}$, where $\text{LaxLin}(-, -)$ is the full subcategory of $\text{Fun}_{\mathcal{V}}(-, -)$ consisting of lax linear functors.
- By [21, lemma 5.5.5.12], there is a pullback diagram of spaces:

$$\begin{array}{ccc} \text{Fun}_{\mathcal{V}}(\mathcal{O}^{\otimes}, \mathcal{M}^{\otimes})^{\simeq} & \longrightarrow & \text{Fun}(\mathcal{O}^{\otimes}, \mathcal{M}^{\otimes})^{\simeq} \\ \downarrow \lrcorner & & \downarrow f \\ * & \xrightarrow{\{\psi\}} & \text{Fun}(\mathcal{O}^{\otimes}, \mathcal{V}^{\otimes})^{\simeq} \end{array}$$

- Pullback of a fully faithful functor is again fully faithful.
- Given a fully faithful functor $F : A \rightarrow B$, the underlying functor $F^{\simeq} : A^{\simeq} \rightarrow B^{\simeq}$ is an inclusion of components by [24, remark 4.6.2.11].
- Given a fully faithful functor $\mathcal{M}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$ over $\mathcal{V}^{\otimes}$, the induced morphism $\text{Fun}(\mathcal{O}^{\otimes}, \mathcal{M}^{\otimes})^{\simeq} \rightarrow \text{Fun}(\mathcal{O}^{\otimes}, \mathcal{N}^{\otimes})^{\simeq}$ is an inclusion of components by [24, corollary 4.6.2.44], or equivalently, a fully faithful embedding between spaces.
- Thus the vertical and the bottom horizontal morphisms in the following diagram are fully faithful, so is the upper horizontal:

$$\begin{array}{ccc} \text{Fun}_{\mathcal{V}^{\otimes}}(\mathcal{O}^{\otimes}, \mathcal{M}^{\otimes}) & \longrightarrow & \text{Fun}_{\mathcal{V}^{\otimes}}(\mathcal{O}^{\otimes}, \mathcal{N}^{\otimes}) \\ \downarrow & & \downarrow \\ \text{Fun}(\mathcal{O}^{\otimes}, \mathcal{M}^{\otimes}) & \hookrightarrow & \text{Fun}(\mathcal{O}^{\otimes}, \mathcal{N}^{\otimes}) \end{array}$$

- And this implies the restriction $\text{Map}_{\omega\text{LMod}}(\mathcal{O}, \mathcal{M}) \rightarrow \text{Map}_{\omega\text{LMod}}(\mathcal{O}, \mathcal{N})$ is an inclusion of components.

□

2.6 atomic object

This part follows [29, section 3.2].

Given $m \in \mathcal{M} \in \text{LMod}_{\mathcal{V}}(\text{Pr}^L)$, consider the free algebra functor $\mathcal{V} \rightarrow \mathcal{M} : 1_{\mathcal{V}} \mapsto m$, which is a morphism in $\text{LMod}_{\mathcal{V}}(\text{Pr}^L)$, thus by the enriched adjunction theorem, it admits a right adjoint $\text{Mor}_{\mathcal{M}}(m, -) : \mathcal{M} \rightarrow \mathcal{V}, n \mapsto \text{Mor}_{\mathcal{M}}(m, n)$, which is automatically lax $\mathcal{V}$ -linear.

Definition 2.52. An object $m \in \mathcal{M}$ where $\mathcal{M} \in \text{LMod}_{\mathcal{V}}(\text{Pr}^L)$ is called $\mathcal{V}$ -atomic if the functor $\text{Mor}_{\mathcal{M}}(m, -) : \mathcal{M} \rightarrow \mathcal{V}$ preserves colimits and is $\mathcal{V}$ -linear.

Lemma 2.53. If $1_{\mathcal{V}}$ is κ -compact in $\mathcal{V}$, then $\mathcal{V}$ -atomic object m is κ -compact, especially, the full subcategory of $\mathcal{M}$ consisting of $\mathcal{V}$ -atomic objects $\mathcal{M}^{at}$ is small.

Proof. This is [29, proposition 3.23].

□

Definition 2.54. If $\mathcal{M}^{at}$ generates $\mathcal{M}$ under colimits and $\mathcal{V}$ -action, we call $\mathcal{M}$ atomically generated.

In combination of the above lemma, $\mathcal{M}$ is atomically generated if there exists a set of atomic objects which generate $\mathcal{M}$ under colimits and $\mathcal{V}$ -tensoring.

There is a generalization of the properties of atomic objects.

Definition 2.55. A morphism $F : \mathcal{M} \rightarrow \mathcal{N}$ in $\mathbf{LMod}_{\mathcal{V}}(\mathbf{Pr}^L)$ is called internally left adjoint if its right adjoint F^R is colimit preserving and $\mathcal{V}$ -linear.

Thus $\mathcal{V} \rightarrow \mathcal{M}, 1_{\mathcal{V}} \mapsto m$ is internally left adjoint if and only if m is atomic.

Remark 2.56. By [6, section 4], internally left adjoints and their right adjoints form the adjunction in the $(\infty, 2)$ category $\mathbf{Pr}^L$.

Since internally left adjoints are closed under composition, an internally left adjoint preserves atomic objects. And we have the converse results when things are atomically generated.

Lemma 2.57. *Given a morphism $F : \mathcal{M} \rightarrow \mathcal{N}$ in $\mathbf{LMod}_{\mathcal{V}}(\mathbf{Pr}^L)$, if $\mathcal{M}$ is atomically generated, the following conditions are equivalent:*

- (1) *F is internally left adjoint.*
- (2) *F preserves $\mathcal{V}$ -atomic objects.*
- (3) *If $\mathcal{M}_0$ is a full subcategory of $\mathcal{M}^{at}$ which generates $\mathcal{M}$ under colimits and tensoring, then $F(\mathcal{M}_0) \subset \mathcal{N}^{at}$.*

Lemma 2.58. *Given $\mathcal{V}, \mathcal{W} \in \mathbf{Alg}_{\mathcal{A}ss}(\mathbf{Pr}^L)$, if $F : \mathcal{M} \rightarrow \mathcal{N}$ in $\mathbf{LMod}_{\mathcal{V}}(\mathbf{Pr}^L)$ is internally left adjoint,*

- (1) *If $G : \mathcal{M}' \rightarrow \mathcal{N}'$ in $\mathbf{LMod}_{\mathcal{W}}(\mathbf{Pr}^L)$ is internally left adjoint, then $F \otimes G$ is internally left adjoint in $\mathbf{LMod}_{\mathcal{V} \otimes \mathcal{W}}(\mathbf{Pr}^L)$ with right adjoint $F^R \otimes G^R$.*
- (2) *Given $\mathcal{B} \in {}_{\mathcal{V}}\mathbf{BiMod}_{\mathcal{W}}$, then $F \otimes_{\mathcal{V}} \mathcal{B} : \mathcal{M} \otimes_{\mathcal{V}} \mathcal{B} \rightarrow \mathcal{N} \otimes_{\mathcal{V}} \mathcal{B}$ is internally left adjoint in $\mathbf{LMod}_{\mathcal{W}}(\mathbf{Pr}^L)$, with right adjoint $\mathcal{B} \otimes F^R$.*

Proof. This is [29, corollary 3.40]. □

And especially we have the following properties of enriched presheaf:

Proposition 2.59. *Given an absolute small weakly left tensored ∞ -category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ and a small localization pair $(\mathcal{V}^{\otimes}, S)$, the essential image of the functor*

$$\mathcal{M} \hookrightarrow \mathcal{P}\mathbf{LEnv}(\mathcal{M}) \xrightarrow{L} \mathcal{P}\mathbf{LEnv}(\mathcal{M})_S$$

consists of $S^{-1}\mathcal{P}\mathbf{Env}(\mathcal{V})$ -atomic objects, where L is the [left adjoint of the embedding](#) $\mathcal{P}\mathbf{LEnv}(\mathcal{M})_S \rightarrow \mathcal{P}\mathbf{LEnv}(\mathcal{M})$.

Proof. This is [15, theorem 4.50]. □

Moreover, since L preserves colimits, and $\mathcal{P}\text{LEnv}(\mathcal{M})$ is generated by the image of $\mathcal{M}$ under colimits and tensoring, thus $\mathcal{P}\text{LEnv}(\mathcal{M})_S$ is atomically generated by $\mathcal{M}$.

And there are some other properties of $\mathcal{P}\text{LEnv}(\mathcal{M})_S$ in view of its atomic generation.

Lemma 2.60. *A morphism $F : \mathcal{M} \rightarrow \mathcal{N}$ in $\text{LMod}_{\mathcal{V}}(\text{Pr}^L)$ is fully faithful (that is, the underlying functor is fully faithful) if and only if for any $m, n \in \mathcal{M}$, the induced morphism $\text{Mor}_{\mathcal{M}}(m, n) \cong \text{Mor}_{\mathcal{N}}(F(m), F(n))$ in $\mathcal{V}$ is an equivalence. And when $\mathcal{M}$ is atomically generated by $\mathcal{M}_0$, and F is internally left adjoint, then it suffices to verify the condition for $m, n \in \mathcal{M}_0$.*

Proof. This is [29, lemma 3.33].

Let $m_0 \in \mathcal{M}_0$ and $m \in \mathcal{M}$, then denote the full subcategory of $\mathcal{M}$ consisting of those m such that $\text{Mor}_{\mathcal{M}}(m_0, m) \cong \text{Mor}_{\mathcal{N}}(F(m_0), F(m))$ as $\widehat{\mathcal{M}}$, and $\mathcal{M}_0 \subset \widehat{\mathcal{M}}$ by assumption.

$\widehat{\mathcal{M}}$ is closed under linear $\mathcal{V}$ -action: since $\text{Mor}_{\mathcal{M}}(m_0, -)$ and $\text{Mor}_{\mathcal{N}}(F(m_0), -)$ and F preserves $\mathcal{V}$ -action.

$\widehat{\mathcal{M}}$ is closed under colimits: since $\text{Mor}_{\mathcal{M}}(m_0, -)$ and $\text{Mor}_{\mathcal{N}}(F(m_0), -)$ and F preserves $\mathcal{V}$ -action.

Thus $\widehat{\mathcal{M}} = \mathcal{M}$. □

Proposition 2.61. *Given a small localization pair $(\mathcal{V}^{\otimes}, S)$, and a small S -enriched category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, and a lax linear functor $\phi : \mathcal{M}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$ where $\mathcal{N}$ a presentably left tensored ∞ -category $\mathcal{N}^{\otimes} \rightarrow S^{-1} \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$, then by universal property, there exists a colimit preserving linear functor $\psi : \mathcal{P}\text{LEnv}(\mathcal{M})_S \rightarrow \mathcal{N}$, and*

- (1) *ψ is a fully faithful embedding if ϕ is fully faithful embedding and $\phi(X)$ is $S^{-1} \mathcal{P}\text{Env}(\mathcal{V})$ -atomic for all $X \in \mathcal{M}$.*
- (2) *ψ is an equivalence if ϕ is fully faithful embedding and $\phi(X)$ is $S^{-1} \mathcal{P}\text{Env}(\mathcal{V})$ -atomic for all $X \in \mathcal{M}$, and $\mathcal{N}$ is generated under colimit and left tensors by the image $\phi(\mathcal{M})$.*

Proof. Since ϕ lands in $\mathcal{N}^{at}$, and $\mathcal{M}$ generates $\mathcal{P}\text{LEnv}(\mathcal{M})_S$, ψ is internally left adjoint in $\text{LMod}_{S^{-1} \mathcal{P}\text{Env}(\mathcal{V})}(\text{Pr}^L)$. And since ϕ is fully faithful, so is $\psi|_{\mathcal{M}}$, and by the lemma above, ψ is fully faithful embedding. (2) is automatically by (1). □

Given a small category enriched in a presentable monoidal ∞ -category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, recall that $\mathcal{P}_{\mathcal{V}}(\mathcal{M}) \cong \mathcal{P}\text{LEnv}(\mathcal{M}_{\sigma})_{\text{Enr}}$, thus we have:

Proposition 2.62. *Given a small ∞ -category enriched in a presentable monoidal ∞ -monoidal category $\mathcal{V}^{\otimes}$, the essential image of the enriched Yoneda embedding $\mathfrak{y}_{\mathcal{V}} : \mathcal{M}^{\otimes} \rightarrow \mathcal{P}_{\mathcal{V}}(\mathcal{M})^{\otimes}$ consists of atomic objects, and $\mathcal{P}_{\mathcal{V}}(\mathcal{M})$ is atomically generated by the image.*

2.7 transfer of enrichment

Recall we have already defined $\text{Enr} \subset \omega\text{LMod}$, $\text{PLMod} \subset \omega\text{LMod}$, and $\omega\text{LMod} \rightarrow \text{Op}_{\infty}$ is biCartesian fibration. And it's a natural question to consider whether its restriction to the non-full subcategory PLMod or Enr is again (co)Cartesian fibration. In this subsection, we gather some results on them. And especially, the coCartesian edge in Enr corresponds to the transfer of enrichment.

Proposition 2.63. *Consider the Cartesian fibration ωLMod over Op , it restricts to a Cartesian fibration PLMod over Mon , and the inclusion $\text{PLMod} \hookrightarrow \omega\text{LMod}$ preserves the Cartesian edges.*

Proof. It suffices to show the Cartesian edge in ωLMod at a point $N \rightarrow B^\otimes \in \text{PLMod}$ remains in PLMod .

By [15, 2.62], the Cartesian edge in ωLMod over a map of operads α is just the pullback of operads: Given two weakly left tensored categories $M^\otimes \rightarrow A^\otimes$ and $N^\otimes \rightarrow B^\otimes$, an morphism of weakly left tensored modules $\alpha^* : M^\otimes \rightarrow N^\otimes$ is Cartesian iff the following diagram is a pullback diagram:

$$\begin{array}{ccc} M^\otimes & \xrightarrow{\alpha^*} & N^\otimes \\ \downarrow & \lrcorner & \downarrow \\ A^\otimes & \xrightarrow{\alpha} & B^\otimes \end{array}$$

, that is, for any $a_1, a_2, \dots, a_n \in A$ and $m, n \in M$, the following map induced is an equivalence:

$$\text{Mul}_M(a_1, \dots, a_n, m; n) \rightarrow \text{Mul}_N(\alpha(a_1), \dots, \alpha(a_n), \alpha^*(m); \alpha^*(n))$$

Thus if $\alpha : A \rightarrow B$ is a monoidal functor, then if N is pseudo-enriched in A , and for any $a_1, a_2, \dots, a_n \in A$ and $m, n \in M$, we have the following commutative diagram:

$$\begin{array}{ccc} \text{Mul}_M(a_1 \otimes \dots \otimes a_n, m; n) & \xrightarrow{\cong} & \text{Mul}_N(\alpha(a_1 \otimes \dots \otimes a_n), \alpha^*(m); \alpha^*(n)) \\ \downarrow & & \downarrow \cong \\ \text{Mul}_M(a_1, \dots, a_n, m; n) & \xrightarrow{\cong} & \text{Mul}_N(\alpha(a_1), \dots, \alpha(a_n), \alpha^*(m); \alpha^*(n)) \end{array}$$

where the vertical arrows are induced by the active map corresponding to the tensor product. Thus the left vertical map is equivalence, and this implies M is pseudo-enriched in A . $\square$

Remark 2.64. The above statement is not true if the morphism $\alpha : A \rightarrow B$ is in Op instead of Mon , for example, by [15, corollary 4.22], the pullback along the lax monoidal functor $\mathcal{V} \rightarrow \text{Env}(\mathcal{V})$ takes a category pseudo-enriched in $\text{Env}(\mathcal{V})$ to a weakly left tensored category over $\mathcal{V}$.

Noticing that pullback along coCartesian fibration is again coCartesian fibration, we have:

Proposition 2.65. *The (non-full) subcategory inclusion $\text{LMod} \rightarrow \text{PLMod}$ preserves Cartesian edges over Mon .*

Proof. We have already known $\text{LMod} \rightarrow \text{Mon}(\text{Cat})$ is a Cartesian fibration by [22, corollary 4.2.3.2], thus it suffices to show:

- Given a left tensored ∞ -category $\mathcal{N}^\otimes \rightarrow \mathcal{W}^\otimes$, and a monoidal functor $f : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, the pullback in ωLMod :

$$\begin{array}{ccc}
\mathcal{M}^{\otimes} & \xrightarrow{f_!} & \mathcal{N}^{\otimes} \\
\downarrow & \lrcorner & \downarrow \\
\mathcal{V}^{\otimes} & \xrightarrow{f} & \mathcal{W}^{\otimes}
\end{array}$$

lies in $\text{LMod} \subset \text{PLMod}$, which is true since pullback of coCartesian fibration is again coCartesian fibration.

- For any $\mathcal{N}' \in \text{LMod}_{\mathcal{V}}(\text{Cat})$, and a linear functor $\mathcal{N}'^{\otimes} \xrightarrow{F} \mathcal{N}^{\otimes}$, there is a unique $\mathcal{V}$ -linear functor $G : \mathcal{N}' \rightarrow \mathcal{M}$ making the following diagram

$$\begin{array}{ccccc}
& & \mathcal{N}'^{\otimes} & & \\
& \searrow & \downarrow & \xrightarrow{F} & \downarrow \\
& & \mathcal{M}^{\otimes} & \xrightarrow{f_!} & \mathcal{N}^{\otimes} \\
& \searrow & \downarrow & & \downarrow \\
& & \mathcal{V}^{\otimes} & \xrightarrow{f} & \mathcal{W}^{\otimes}
\end{array}$$

, which is also true, since pullback in $\text{coC}_{/\text{Ass}}$ can be computed in $\text{Cat}_{/\text{Ass}}$, so there is a unique induced morphism in $\text{coC}_{/\text{Ass}}$ $\mathcal{N}^{\otimes} \rightarrow \mathcal{M}^{\otimes}$ making the diagram commute, thus preserve coCartesian edges, and is a linear functor.

- Thus $f_!$ is locally Cartesian in $\text{LMod} \rightarrow \text{Mon}(\text{Cat})$, and thus Cartesian.

□

By [22, corollary 4.2.3.2], the functor $\text{PrLMod} \rightarrow \text{PrMon}$ is also a Cartesian fibration, and we have:

Proposition 2.66. *The non-full subcategory inclusion $\text{PrLMod} \rightarrow \widehat{\text{LMod}}$ preserves Cartesian edges over $\text{PrMon} \rightarrow \widehat{\text{Mon}}$.*

- Proof.*
- To show a left tensored ∞ -category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ is presentably, it suffices to show that the underlying categories $\mathcal{M}$ and $\mathcal{V}$ are presentably, and for any $V \in \mathcal{V}, X \in \mathcal{M}$, the functors $(-) \otimes X : \mathcal{V} \rightarrow \mathcal{M}, V \otimes (-) : \mathcal{M} \rightarrow \mathcal{M}$ preserve colimits.
 - Thus given a presentably left tensored ∞ -category $\mathcal{N}^{\otimes} \rightarrow \mathcal{W}^{\otimes}$, and a presentably monoidal functor $f : \mathcal{V}^{\otimes} \rightarrow \mathcal{W}^{\otimes}$, the pullback in ωLMod :

$$\begin{array}{ccc}
\mathcal{M}^{\otimes} & \xrightarrow{f_!} & \mathcal{N}^{\otimes} \\
\downarrow & \lrcorner & \downarrow \\
\mathcal{V}^{\otimes} & \xrightarrow{f} & \mathcal{W}^{\otimes}
\end{array}$$

lies in PrLMod , since the underlying category of $\mathcal{M}^\otimes$ is $\mathcal{M} \cong \mathcal{N}$, and for any $V \in \mathcal{V}$, $X \in \mathcal{M}$, $(-) \otimes X : \mathcal{V} \rightarrow \mathcal{M} \cong \mathcal{N}$ factors through $(-) \otimes X : \mathcal{V} \rightarrow \mathcal{W} \rightarrow \mathcal{N} \cong \mathcal{M}$ which is the composition of colimit preserving functor, thus preserves colimits, and $V \otimes (-) \cong f(V) : \otimes(-)\mathcal{M} \cong \mathcal{N} \rightarrow \mathcal{M} \cong \mathcal{N}$ also preserves colimits.

- And it remains to show for any $\mathcal{N}' \in \text{LMod}_{\mathcal{V}}(\text{Pr}^L)$, and a colimit preserving linear functor

$$\begin{array}{ccc} \mathcal{N}'^\otimes & \xrightarrow{F} & \mathcal{N}^\otimes \\ \downarrow & & \downarrow \\ \mathcal{V}^\otimes & \xrightarrow{f} & \mathcal{W}^\otimes \end{array}, \text{ there is a unique colimit preserving } \mathcal{V}\text{-linear functor } G : \mathcal{N}' \rightarrow$$

$\mathcal{M}$ making the following diagram commute:

$$\begin{array}{ccccc} \mathcal{N}'^\otimes & & & & \\ & \searrow F & & & \\ & & \mathcal{M}^\otimes & \xrightarrow{f_!} & \mathcal{N}^\otimes \\ & \searrow G & \downarrow & & \downarrow \\ & & \mathcal{V}^\otimes & \xrightarrow{f} & \mathcal{W}^\otimes \end{array}$$

, which is true, since the inclusion $\text{Pr}^L \hookrightarrow \text{Cat}$ preserves limits by [21, corollary 5.5.3.13], thus consider the fiber over $[0]$, which corresponds to the underlying category of a left tensored ∞ -category, then there is a unique colimit preserving functor $\mathcal{N}' \rightarrow \mathcal{M}$ making

the diagram

$$\begin{array}{ccccc} \mathcal{N}' & & & & \\ & \searrow F_{[0]} & & & \\ & \searrow G_{[0]} & \mathcal{M} & \xrightarrow{\cong} & \mathcal{N} \\ & & \downarrow & & \downarrow \\ & & \mathcal{V} & \xrightarrow{f_{[0]}} & \mathcal{W} \end{array}$$

commute, thus $G_{[0]}$ is also colimit preserving,

and G is a colimit preserving $\mathcal{V}$ -linear functor, thus lies in PrLMod .

- Again the above points show that $f_!$ is a locally Cartesian edge in PrLMod , thus is Cartesian. $\square$

Remark 2.67. $\text{LMod} \rightarrow \text{PLMod}$ in general doesn't preserve coCartesian edges.

Now comes to the coCartesian fibrations.

First of all, we need a description of coCartesian edges in ωLMod :

Proposition 2.68. *Let $f : \mathcal{M}^\otimes \rightarrow \mathcal{N}^\otimes$ be a lax linear functor lying over an ∞ -operad map $\alpha : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, then the following conditions are equivalent:*

- (1) f is coCartesian edge in $\omega\text{LMod} \rightarrow \text{Op}_\infty$.
- (2) The induced linear map $\text{LEnv}(\mathcal{M})^\otimes \rightarrow \text{LEnv}(\mathcal{N})^\otimes$ exhibits $\text{LEnv}(\mathcal{N})$ as:

$$\text{Env}(\mathcal{W}) \otimes_{\text{Env}(\mathcal{V})} \text{LEnv}(\mathcal{M}) \cong \text{LEnv}(\mathcal{N})$$

- (3) The functor $\mathcal{M} \rightarrow \mathcal{N}$ is essential surjective, and the induced linear functor $\mathcal{P}\text{Env}(\mathcal{M}) \rightarrow \mathcal{P}\text{Env}(\mathcal{N})$ exhibits $\mathcal{P}\text{Env}(\mathcal{N})$ as:

$$\mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{Env}(\mathcal{M})$$

- (4) The functor $\mathcal{M} \rightarrow \mathcal{N}$ is essential surjective, and the induced morphism in $\mathcal{P}\text{Env}(\mathcal{W})$:

$$\mathcal{P}\text{Env}(\alpha)(\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{M})}(X, Y)) \cong \text{Mor}_{\mathcal{P}\text{Env}(\mathcal{N})}(f(X), f(Y))$$

for any $X, Y \in \mathcal{M}$.

Remark 2.69. Noticing that $\mathcal{P}\text{Env}(\mathcal{M})^{\otimes} \rightarrow \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$ is presentably left tensored, thus admits morphism object $\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{M})}$, and for $x, y \in \mathcal{M}$, under the embedding $i : \mathcal{M} \rightarrow \mathcal{P}\text{Env}(\mathcal{M})$, we denote $\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{M})}(i(x), i(y))$ as $\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{M})}(x, y)$.

Proof. This is in [15, theorem 4.67].

- We first show (1) $\Rightarrow$ (2), (3) $\Rightarrow$ (1), (3) $\Rightarrow$ (4).
- We show the existence of the morphism lying over α satisfying the condition (3). Then by the uniqueness of coCartesia edge, this implies (1) $\Rightarrow$ (3). This also implies the existence of morphism satisfying condition (4).
- And then we show if f is lax $\mathcal{V}$ linear (that is, f lying over the identity map between ∞ -operads), and f satisfies the condition (4), then f is an equivalence: this implies if the morphism in (4) exists, then the induced factorization of it is the composition of the coCartesian edge and an equivalence, thus it's also coCartesian lifting. This implies (4) $\Rightarrow$ (1).
- And we use the same trick to show (2) $\Rightarrow$ (3).

(1) $\Rightarrow$ (2): Noticing the following commutative diagram:

$$\begin{array}{ccccc}
 & \text{Map}_{\text{LMod}}(\text{LEnv}(\mathcal{M}), \mathcal{Q}) & \xrightarrow{\quad} & \text{Map}_{\text{LMod}}(\text{LEnv}(\mathcal{N}), \mathcal{Q}) & \\
 & \downarrow \lrcorner & & \downarrow & \\
 & \text{Map}_{\text{Mon}}(\text{Env}(\mathcal{V}), \mathcal{U}) & \xrightarrow{\quad} & \text{Map}_{\text{Mon}}(\text{Env}(\mathcal{W}), \mathcal{U}) & \\
 \swarrow \cong & & \searrow \cong & & \\
 \text{Map}_{\omega\text{LMod}}(\mathcal{M}, \mathcal{Q}) & \xrightarrow{\quad} & \text{Map}_{\omega\text{LMod}}(\mathcal{N}, \mathcal{Q}) & & \\
 \downarrow & & \downarrow & & \\
 \text{Map}_{\text{Op}_{\infty}}(\mathcal{V}, \mathcal{U}) & \xrightarrow{\quad} & \text{Map}_{\text{Op}_{\infty}}(\mathcal{W}, \mathcal{U}) & &
 \end{array}$$

where $\mathcal{Q} \rightarrow \mathcal{U}$ is a left tensored ∞ -category. And if the front square is pullback, so is the back square, which implies $\mathcal{L}\text{Env}(\mathcal{N})$ is the coCartesian lifting of $\mathcal{L}\text{Env}(\mathcal{M})$ over α in $\mathcal{L}\text{Mod}$, that is exactly the relative tensor product $\mathcal{L}\text{Env}(\mathcal{M}) \otimes_{\mathcal{L}\text{Env}(\mathcal{V})} \mathcal{L}\text{Env}(\mathcal{W})$.

Now we show (3) $\Rightarrow$ (1): Since $\mathcal{M} \rightarrow \mathcal{N}$ is essential surjective, we have the following pullback square in $\mathcal{S}\text{pc}$ for any weakly left tensored ∞ -category $\mathcal{Q}^\otimes \rightarrow \mathcal{U}^\otimes$:

$$\begin{array}{ccc} \text{Map}_{\omega\mathcal{L}\text{Mod}}(\mathcal{N}, \mathcal{Q}) & \xrightarrow{\quad\quad\quad} & \text{Map}_{\omega\mathcal{L}\text{Mod}}(\mathcal{M}, \mathcal{Q}) \\ \downarrow & \lrcorner & \downarrow \\ \text{Map}_{\omega\mathcal{L}\text{Mod}}(\mathcal{N}, \tilde{\mathcal{P}}\text{Env}(\mathcal{Q})) & \xrightarrow{\quad\quad\quad} & \text{Map}_{\omega\mathcal{L}\text{Mod}}(\mathcal{M}, \tilde{\mathcal{P}}\text{Env}(\mathcal{Q})) \end{array}$$

where $\tilde{\mathcal{P}}\text{Env}(\mathcal{Q})$ denotes the pullback of $\mathcal{P}\text{Env}(\mathcal{Q})^\otimes$ along the natural embedding $\mathcal{U} \rightarrow \mathcal{P}\text{Env}(\mathcal{U})$.

Since the right vertical arrow is a summand of space, so is the pullback to $\text{Map}_{\omega\mathcal{L}\text{Mod}}(\mathcal{N}, \mathcal{P}\text{Env}(\mathcal{Q}))$, and to show the above diagram is pullback, it remains to show the image of this embedding is spanned by those maps $\mathcal{N} \rightarrow \mathcal{P}\text{Env}(\mathcal{Q})$ which actually lands in $\mathcal{Q}$, and this is true since $\mathcal{N} \rightarrow \mathcal{M}$ is essential surjective: any morphism $\mathcal{M} \rightarrow \mathcal{N} \rightarrow \mathcal{P}\text{Env}(\mathcal{Q})$ which composition lands in $\mathcal{Q}$ must satisfy that the second edge also lies in $\mathcal{Q}$. And we have the following commutative diagram:

$$\begin{array}{ccccc} & & \text{Map}_{\omega\mathcal{L}\text{Mod}}(\mathcal{N}, \mathcal{Q}) & \xrightarrow{\quad\quad\quad} & \text{Map}_{\omega\mathcal{L}\text{Mod}}(\mathcal{M}, \mathcal{Q}) \\ & \swarrow & \downarrow & \lrcorner & \swarrow \\ \text{Map}_{\omega\mathcal{L}\text{Mod}}(\mathcal{N}, \tilde{\mathcal{P}}\text{Env}(\mathcal{Q})) & \xrightarrow{\quad\quad\quad} & \text{Map}_{\omega\mathcal{L}\text{Mod}}(\mathcal{M}, \tilde{\mathcal{P}}\text{Env}(\mathcal{Q})) & & \\ \downarrow & & \downarrow & & \downarrow \\ & \swarrow & \text{Map}_{\text{Op}_\infty}(\mathcal{W}, \mathcal{U}) & \xrightarrow{\quad\quad\quad} & \text{Map}_{\text{Op}_\infty}(\mathcal{V}, \mathcal{U}) \\ & \swarrow \text{id} & \downarrow & \lrcorner & \swarrow \text{id} \\ \text{Map}_{\text{Op}_\infty}(\mathcal{W}, \mathcal{U}) & \xrightarrow{\quad\quad\quad} & \text{Map}_{\text{Op}_\infty}(\mathcal{V}, \mathcal{U}) & & \end{array}$$

Thus it suffices to show the back square is pullback, and we can replace $\mathcal{Q}$ by $\tilde{\mathcal{P}}\text{Env}(\mathcal{M})$.

And by the fact that $\tilde{\mathcal{P}}\text{Env}(\mathcal{M})$ is the Cartesian lifting of $\mathcal{P}\text{Env}(\mathcal{Q})$ along the embedding $\mathcal{U} \rightarrow \mathcal{P}\text{Env}(\mathcal{Q})$, we have the following commutative diagram:

$$\begin{array}{ccccc}
\text{Map}_{\omega\text{LMod}}(\mathcal{N}, \tilde{\mathcal{P}}\text{Env}(\mathcal{Q})) & \xrightarrow{\quad} & \text{Map}_{\omega\text{LMod}}(\mathcal{N}, \mathcal{P}\text{Env}(\mathcal{Q})) & & \\
\downarrow & \searrow & \downarrow & \searrow & \\
\text{Map}_{\text{Op}_{\infty}}(\mathcal{W}, \mathcal{U}) & \xrightarrow{\quad} & \text{Map}_{\text{Op}_{\infty}}(\mathcal{W}, \mathcal{P}\text{Env}(\mathcal{U})) & & \\
& \searrow & & \searrow & \\
& \text{Map}_{\omega\text{LMod}}(\mathcal{M}, \tilde{\mathcal{P}}\text{Env}(\mathcal{Q})) & \xrightarrow{\quad} & \text{Map}_{\omega\text{LMod}}(\mathcal{M}, \mathcal{P}\text{Env}(\mathcal{Q})) & \\
& \downarrow & & \downarrow & \\
& \text{Map}_{\text{Op}_{\infty}}(\mathcal{W}, \mathcal{U}) & \xrightarrow{\quad} & \text{Map}_{\text{Op}_{\infty}}(\mathcal{W}, \mathcal{P}\text{Env}(\mathcal{U})) &
\end{array}$$

where the front and the back squares are pullback since $\tilde{(-)}$ is the Cartesian lifting. And to show the left square is pullback, it suffices to show the right square is pullback, which is true by the following commutative diagram:

$$\begin{array}{ccccc}
& \text{Map}_{\text{PrLMod}}(\mathcal{P}\text{Env}(\mathcal{N}), \mathcal{P}\text{Env}(\mathcal{Q})) & \xrightarrow{\quad} & \text{Map}_{\text{PrLMod}}(\mathcal{P}\text{Env}(\mathcal{M}), \mathcal{P}\text{Env}(\mathcal{Q})) & \\
& \downarrow & \lrcorner & \downarrow & \\
& \text{Map}_{\text{PrMon}}(\mathcal{P}\text{Env}(\mathcal{W}), \mathcal{P}\text{Env}(\mathcal{U})) & \xrightarrow{\quad} & \text{Map}_{\text{PrMon}}(\mathcal{P}\text{Env}(\mathcal{V}), \mathcal{P}\text{Env}(\mathcal{U})) & \\
& \uparrow & \nearrow & \uparrow & \\
\text{Map}_{\omega\text{LMod}}(\mathcal{N}, \mathcal{P}\text{Env}(\mathcal{Q})) & \xrightarrow{\quad} & \text{Map}_{\omega\text{LMod}}(\mathcal{M}, \mathcal{P}\text{Env}(\mathcal{Q})) & & \\
\downarrow & \lrcorner & \downarrow & \lrcorner & \\
\text{Map}_{\text{Op}}(\mathcal{W}, \mathcal{P}\text{Env}(\mathcal{U})) & \xrightarrow{\quad} & \text{Map}_{\text{Op}}(\mathcal{V}, \mathcal{P}\text{Env}(\mathcal{U})) & &
\end{array}$$

where the back square is pullback by the fact that $\mathcal{P}\text{Env}(\mathcal{N})$ is the coCartesian lifting in PrLMod over $\mathcal{P}\text{Env}(\mathcal{V}) \rightarrow \mathcal{P}\text{Env}(\mathcal{W})$, that is, the relative tensor product in Pr^L , and the left and right squares are pullback due to the universal property of taking $\mathcal{P}\text{Env}(-)$, and $\mathcal{P}\text{Env}(-)$.

Now we show (3) $\Rightarrow$ (4): Since the image of the embeddings $(-) \hookrightarrow \mathcal{P}\text{Env}(-)$ consists of atomic objects, we have

$$\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{M})}(X, -) : \mathcal{P}\text{Env}(\mathcal{M}) \rightleftharpoons \mathcal{P}\text{Env}(\mathcal{V}) : L_X$$

where L_X is the unique $\mathcal{P}\text{Env}(\mathcal{V})$ -linear colimit preserving functor sending $1_{\mathcal{V}}$ to X .

And:

$$\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{N})}(f(X), -) : \mathcal{P}\text{Env}(\mathcal{N}) \rightleftharpoons \mathcal{P}\text{Env}(\mathcal{W}) : L_{f(X)}$$

And by [29, corollary 3.40], we have another adjunction by tensoring the first adjunction with $\mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V}\mathcal{V})} (-)$:

$$\text{id} \otimes \text{Mor}_{\mathcal{P}\text{Env}(\mathcal{M})}(X, -) : \mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{Env}(\mathcal{M}) \rightleftharpoons \mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{Env}(\mathcal{V}) \cong \mathcal{P}\text{Env}(\mathcal{W}) : \text{id} \otimes L_X$$

And $\text{id} \otimes L_X$ sends $1_{\mathcal{W}} \otimes 1_{\mathcal{V}} \cong 1_{\mathcal{W}}$ to $1_{\mathcal{W}} \otimes X$ in $\mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M})$, which coincides with $\mathcal{P}\text{LEnv}(f)(X) = f(X)$ under the identification $\mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M}) \cong \mathcal{P}\text{LEnv}(\mathcal{N})$.

And since the right adjoint of $L_{f(X)}$ is unique up to isomorphism, thus we have $\mathcal{P}\text{Env}(\alpha)(\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(X, Y)) \cong \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{N})}(f(X), f(Y))$.

Now we show the existence of edge satisfying (3): Consider the embedding $\mathcal{M}^{\otimes} \rightarrow \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes} \rightarrow (\mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes})$, and consider the full weakly left tensored ∞ -category over $\mathcal{W}$ of $\mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes} \rightarrow \mathcal{P}\text{Env}(\mathcal{W})^{\otimes}$ spanned by the essential image of $\mathcal{M}$ as $\widetilde{\mathcal{M}}^{\otimes} \rightarrow \mathcal{P}\text{Env}(\mathcal{W})$: that is, we take pullback of $\mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M})$ along $\mathcal{W} \rightarrow \mathcal{P}\text{Env}(\mathcal{W})$, and regard it as weakly left tensored ∞ -category over $\mathcal{W}$, and we take the full weakly left tensored subcategory spanned by the image of $\mathcal{M}$. And we denote the embedding as $\tau : \widetilde{\mathcal{M}} \rightarrow \mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M})$. And it remains to show that τ induces an equivalence $\mathcal{P}\text{LEnv}(\widetilde{\mathcal{M}}) \cong \mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M})$.

By [the lemma on fully faithful pullback](#), $\widetilde{\mathcal{M}}^{\otimes} \hookrightarrow (\mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M}))^{\otimes} \rightarrow (\mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M}))^{\otimes}$ is fully faithful, thus it suffices to show the image of τ consists of atomic objects by [the lemma on atomic and embedding](#).

Consider the adjunction in the $(\infty, 2)$ -category $\text{LMod}_{\mathcal{V}}(\text{Pr}^L)$:

$$\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(X, -) : \mathcal{P}\text{Env}(\mathcal{V}) \rightleftarrows \mathcal{P}\text{LEnv}(\mathcal{M}) : L_X$$

which gives rise to an adjunction in $\text{LMod}_{\mathcal{W}}(\text{Pr}^L)$:

$$\text{id} \otimes \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(X, -) : \mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{Env}(\mathcal{V}) \rightleftarrows \mathcal{P}\text{Env}(\mathcal{W}) \otimes_{\mathcal{P}\text{Env}(\mathcal{V})} \mathcal{P}\text{LEnv}(\mathcal{M}) : \text{id} \otimes L_X \cong L_{\tau(X)}$$

Thus the right adjoint of the unique linear $\mathcal{W}$ linear functor $L_{\tau(X)}$ sending $1_{\mathcal{W}}$ to $\tau(X)$ is again colimit preserving and $\mathcal{W}$ -linear, and this implies $\tau(X)$ is $\mathcal{W}$ -atomic. □

Here are some more intuitive interpretations on the description (4) above:

Lemma 2.70. *Given a pseudo-enriched ∞ -category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, for $x, y \in \mathcal{M}$, under the natural embedding*

$$\begin{array}{ccc} \mathcal{M}^{\otimes} & \xhookrightarrow{i} & \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes} \\ \downarrow & & \downarrow \\ \mathcal{V}^{\otimes} & \xhookrightarrow{i} & \mathcal{P}\text{Env}(\mathcal{V})^{\otimes} \end{array}$$

, the morphism object $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y) \in \mathcal{P}\text{Env}(\mathcal{V})$ lands in $\mathcal{P}(\mathcal{V}) \hookrightarrow \mathcal{P}\text{Env}(\mathcal{V})$. Moreover, a weakly left tensored category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ where $\mathcal{V}^{\otimes}$ is monoidal is pseudo-enriched iff $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)$ lands in $\mathcal{P}(\mathcal{V}) \subset \mathcal{P}\text{Env}(\mathcal{V})$ for all $x, y \in \mathcal{M}$.

Proof. • Given a localization of small ∞ -categories: $L : C \rightleftarrows D = S^{-1}C : i$ where S is the set of local equivalences, since the right adjoint i is fully faithful, thus $P(i)$ is fully faithful by [21, proposition 5.1.6.10], and we have a localization again:

$$\mathcal{P}(L) : \mathcal{P}(C) \rightleftarrows \mathcal{P}(D) : \mathcal{P}(R)$$

- Especially, $\mathcal{P}(D) = S^{-1}\mathcal{P}(C)$: since for any $F \in \mathcal{P}(C)$, $\text{Map}_{\mathcal{P}(C)}(x \xrightarrow{s} y, F)$ is an equivalence iff $F(s) : F(y) \rightarrow F(x)$ is an equivalence, after taking $(-)^{op}$:

$$\begin{array}{ccc} \mathcal{P}(D) & \xrightarrow{L^*} & \mathcal{P}(C) \\ \downarrow (-)^{op} & & \downarrow (-)^{op} \\ \text{Fun}(D, \text{Spc}^{op}) & \xrightarrow{L^*} & \text{Fun}(C, \text{Spc}^{op}) \end{array}$$

and by [21, proposition 5.2.7.12], we can see $F(s)$ is an equivalences for all $s \in S$ iff F lands in $\mathcal{P}(D) \subset \mathcal{P}(C)$, thus $\mathcal{P}(D)$ is the S -local objects of $\mathcal{P}(C)$, and we have $\mathcal{P}(D) = S^{-1}\mathcal{P}(C)$, although S may not be the whole collection of local equivalences of $\mathcal{P}(L) : \mathcal{P}(C) \rightarrow \mathcal{P}(D)$.

- Back to our context, take $C = \text{Env}(\mathcal{V})$ and $D = \mathcal{V}$, and S is the set of morphisms in the

$$\begin{array}{ccc} [n] & \xrightarrow{f} & [1] \\ \downarrow f & & \parallel \\ [1] & \xrightarrow{=} & [1] \end{array} \quad \text{) where } f \in \text{Act} \text{ and } c \text{ is the coCartesian lifting of } f.$$

- So the condition that $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)$ lands in $\mathcal{P}(\mathcal{V})$, by the above points, is equivalent to condition that for any morphism $s \in S$, the induced morphism between mapping spaces $s : \text{Map}_{\mathcal{P}\text{Env}(\mathcal{V})}(s, \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y))$ is an equivalence.
- Recall that a weakly left tensored ∞ -category is [pseudo-enriched](#) iff [Pseu-enriched](#), thus we have

$$\text{Map}_{\mathcal{P}\text{Env}(\mathcal{V})}(s, \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)) \cong \text{Map}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(s \otimes x, y)$$

where the latter is an equivalence by the definition of Pseu-enrichment. □

Lemma 2.71. *Given a pseudo-enriched ∞ -category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, if for $x, y \in \mathcal{M}$, there exists a morphism object $\text{Mor}_{\mathcal{M}}(x, y) \in \mathcal{V}$, then the natural embedding*

$$\begin{array}{ccc} \mathcal{M}^{\otimes} & \xhookrightarrow{i} & \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes} \\ \downarrow & & \downarrow \\ \mathcal{V}^{\otimes} & \xhookrightarrow{i} & \mathcal{P}\text{Env}(\mathcal{V})^{\otimes} \end{array}$$

preserves the morphism object, that is $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y) \cong \text{Mor}_{\mathcal{M}}(x, y)$.

Proof. • By definition, there is an equivalence of mapping spaces:

$$\text{Map}_{\mathcal{V}}(z, \text{Mor}_{\mathcal{M}}(x, y)) \cong \text{Mul}_{\mathcal{M}}(z, x; y)$$

- For any $z \in \text{Env}(\mathcal{V})$, there are equivalences of mapping spaces:

$$\text{Map}_{\text{Env}(\mathcal{V})}(z, i(\text{Mor}_{\mathcal{M}}(x, y))) \cong \text{Mor}_{\mathcal{V}}(L(z), \text{Mor}_{\mathcal{M}}(x, y)) \cong \text{Mul}_{\mathcal{M}}(L(z), Li(x); y) \cong \text{Mul}_{\text{LEnv}(\mathcal{M})}(z, i(x); i(y))$$

where the first and the last equivalences are by the localizations $L : \text{Env}(\mathcal{V}) \rightleftarrows \mathcal{V} : i$ and $L : \text{LEnv}(\mathcal{M}) \rightleftarrows \mathcal{M} : i$.

- For any $z \in \mathcal{P}\text{Env}(\mathcal{V})$, write z as $\text{colim}_{\alpha} z_{\alpha}$ where $z_{\alpha} \in \text{Env}(\mathcal{V})$, then there are equivalences of mapping spaces:

$$\begin{aligned} \text{Map}_{\mathcal{P}\text{Env}(\mathcal{V})}(z, \text{Mor}_{\mathcal{M}}(x, y)) &\cong \text{colim}_{\alpha} \text{Map}_{\text{Env}(\mathcal{V})}(z_{\alpha}, \text{Mor}_{\mathcal{M}}(x, y)) \cong \\ &\text{colim}_{\alpha} \text{Mul}_{\text{LEnv}(\mathcal{M})}(z_{\alpha} \otimes x, y) \cong \text{Mul}_{\mathcal{P}\text{LEnv}(\mathcal{M})}((\text{colim}_{\alpha} z_{\alpha}) \otimes x, y) \cong \\ &\text{Mul}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(z, i(x); i(y)) \end{aligned}$$

- Thus $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)$ is exactly $\text{Mor}_{\mathcal{M}}(x, y)$ under the embedding $\mathcal{V} \hookrightarrow \mathcal{P}\text{Env}(\mathcal{V})$. $\square$

Lemma 2.72. *Given a pseudo-enriched category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, $\mathcal{M}$ is enriched in $\mathcal{V}$ iff for any $x, y \in \mathcal{M}$, $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)$ lands in $\mathcal{V} \hookrightarrow \mathcal{P}\text{Env}(\mathcal{V})$.*

Proof. By the above lemma, being enriched implies $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)$ lands in $\mathcal{V}$.

And if $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)$ lands in $\mathcal{V} \subset \mathcal{P}\text{Env}(\mathcal{V})$, then we have:

$$\begin{aligned} \text{Map}_{\mathcal{V}}(v, \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)) &\cong \text{Map}_{\mathcal{P}\text{Env}(\mathcal{V})}(v, \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)) \cong \\ &\text{Mul}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(v, i(x); i(y)) \cong \text{Mul}_{\mathcal{M}}(v, x; y) \end{aligned}$$

where the first and the last equivalences are by the fully faithfulness of the embeddings $\mathcal{V}^{\otimes} \hookrightarrow \mathcal{P}\text{Env}(\mathcal{V})^{\otimes}$ and $\mathcal{M}^{\otimes} \hookrightarrow \mathcal{P}\text{LEnv}(\mathcal{M})^{\otimes}$. Thus $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y)$ is the morphism object. $\square$

Thus by description (4) above of the coCartesian edge, we have the following proposition:

Proposition 2.73. *The inclusions of subcategories $\text{Enr} \subset \text{PLMod} \subset \omega\text{LMod}$ preserve coCartesian liftings over the edges in $\text{Mon} \subset \text{Op}_{\infty}$, thus there is a functor between coCartesian fibrations over Mon preserving coCartesian edges*

$$\text{Enr} \rightarrow \text{PLMod}$$

.

Proof. • Given an enriched category $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, and an ∞ -operad map $\alpha : \mathcal{V}^{\otimes} \rightarrow \mathcal{W}^{\otimes}$, consider the coCartesian lifting $f : \mathcal{M} \rightarrow \mathcal{N}$ in ωLMod over α , then for any $x, y \in \mathcal{M}$, the morphism object $\text{Mor}_{\mathcal{N}}(f(x), f(y)) \cong \alpha(\text{Mor}_{\mathcal{M}}(x, y))$ by the lemma above and the description (4) of the coCartesian edges and the following commutative diagram:

$$\begin{array}{ccc} \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(x, y) \in \mathcal{P}\text{Env}(\mathcal{V}) & \xrightarrow{\mathcal{P}\text{Env}(\alpha)} & \mathcal{P}\text{Env}(\mathcal{W}) \ni \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{N})}(\mathcal{P}\text{Env}(\alpha)(x), \mathcal{P}\text{Env}(\alpha)(y)) \\ \uparrow i & & \uparrow i \\ \text{Mor}_{\mathcal{M}}(x, y) \in \mathcal{V} & \xrightarrow{\alpha} & \mathcal{W} \end{array}$$

- And since f is essential surjective, thus for all $n_1, n_2 \in \mathcal{N}$, $\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{N})}(i(n_1), i(n_2))$ lands in $\mathcal{W} \subset \mathcal{P}\text{Env}(\mathcal{W})$, and $\mathcal{N}$ is enriched in $\mathcal{W}$ by the above lemma.
- So the coCartesian lifting of an enriched ∞ -category is again enriched. And $\text{Enr} \rightarrow \text{Mon}$ is a coCartesian fibration inherited from ωLMod .
- Likewise for pseudo-enriched ∞ category $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$, and a monoidal ∞ -functor $\alpha : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, consider the coCartesian lifting $f : \mathcal{M} \rightarrow \mathcal{N}$ in ωLMod over α , $\mathcal{N}$ is pseudo-enrichedL: for any $x, y \in \mathcal{M}$, the morphism object $\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{V})}(x, y)$ lands in $\mathcal{P}(\mathcal{V})$, so $\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{N})}(f(x), f(y))$ lands in $\mathcal{W}$ by the following commutative diagram:

$$\begin{array}{ccc}
\mathcal{P}\text{Env}(\mathcal{V}) & \xrightarrow{\mathcal{P}\text{Env}(\alpha)} & \mathcal{P}\text{Env}(\mathcal{W}) \\
\uparrow & & \uparrow \\
\text{Mor}_{\mathcal{P}\text{Env}(\mathcal{M})}(x, y) \in \mathcal{P}(\mathcal{V}) & \xrightarrow{\mathcal{P}(\alpha)} & \mathcal{P}(\mathcal{W})
\end{array}$$

□

Although we define Enr as a coCartesian fibration over Mon instead of Mon^{lax} , we can actually see from the above proof that the coCartesian lifting of an enriched ∞ -category over an operad map between the monoidal ∞ -categories is again an enriched categories, thus we have the corollary:

Lemma 2.74. *Given a lax monoidal functor $\alpha : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, the functor $\alpha_! : \omega\text{LMod}_{\mathcal{V}} \rightarrow \omega\text{LMod}_{\mathcal{W}}$ restricts to a functor*

$$\alpha_! : \text{Enr}_{\mathcal{V}} \rightarrow \text{Enr}_{\mathcal{W}}$$

.

Moreover, $\mathcal{P}(-) : \text{Mon}(\text{Cat}) \rightarrow \text{Alg}_{\text{Ass}}(\text{Pr}^L)$ can be generalized to lax case: by [14, corollary 9.32], given a monoidal ∞ -category $C \in \text{Mon}(\text{Cat})$ and a presentably monoidal category $D \in \text{Alg}_{\text{Ass}}(\text{Pr}^L)$, then the monoidal embedding $\mathfrak{z} : C \rightarrow \mathcal{P}(C)$ induces a functor

$$\text{Alg}_{\mathcal{P}(C)}(D) \rightarrow \text{Alg}_C(D)$$

which admits a fully faithful left adjoint which image consists of precisely those lax monoidal morphisms $\mathcal{P}(C) \rightarrow D$ which preserves colimits fiberwise.

Especially, given a lax monoidal functor between monoidal ∞ -categories $\alpha : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, by [14, corollary 9.32], there is again an induced lax monoidal functor:

$$\mathcal{P}(\alpha) : \mathcal{P}(\mathcal{V}) \rightarrow \mathcal{P}(\mathcal{W})$$

, and by the universal property, there is a commutative square:

$$\begin{array}{ccc}
\mathcal{P}\text{Env}(\mathcal{V}) & \xrightarrow{\mathcal{P}\text{Env}(\alpha)} & \mathcal{P}\text{Env}(\mathcal{W}) \\
\uparrow & & \uparrow \\
\mathcal{P}(\mathcal{V}) & \xrightarrow{\mathcal{P}(\alpha)} & \mathcal{P}(\mathcal{W}) \\
\uparrow \mathfrak{z} & & \uparrow \mathfrak{z} \\
\mathcal{V} & \xrightarrow{\alpha} & \mathcal{W}
\end{array}$$

This implies the coCartesian lifting of a pseudo-enriched ∞ -category over a lax monoidal functor $\alpha : \mathcal{V} \rightarrow \mathcal{W}$ is again a pseudo-enriched category. And especially, we have:

Corollary 2.75. *Given a lax monoidal functor $\alpha : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, the functor $\alpha_! : \omega\text{LMod}_{\mathcal{V}} \rightarrow \omega\text{LMod}_{\mathcal{W}}$ restricts to a functor*

$$\alpha_! : \text{PLMod}_{\mathcal{V}} \rightarrow \text{PLMod}_{\mathcal{W}}$$

By pullback along the biCartesian fibration $\omega\text{LMod} \rightarrow \text{Op}$, any lax monoidal functor between monoidal ∞ -categories $\alpha : \mathcal{V} \rightarrow \mathcal{W}$ induces an adjunction

$$\alpha_! : \omega\text{LMod}_{\mathcal{V}} \rightleftarrows \omega\text{LMod}_{\mathcal{W}} : \alpha^*$$

Especially, when α is monoidal, then the above adjunction restricts to an adjunction between pseudo-enriched parts:

$$\alpha_! : \text{PLMod}_{\mathcal{V}} \rightleftarrows \text{PLMod}_{\mathcal{W}} : \alpha^*$$

since $\text{PLMod} \hookrightarrow \omega\text{LMod}$ preserves both the Cartesian and coCartesian edges over the morphism in $\text{Mon} \subset \text{Op}$.

However, even if α is monoidal, the right adjoint α^* in general doesn't preserve enriched categories. But:

Remark 2.76. If α is a monoidal functor and furthermore admits a lax monoidal right adjoint α^R , then by [15, proposition 4.70], there is a natural isomorphism $(\alpha^R)_! \cong \alpha^*$, and thus there is an adjunction:

$$\alpha_! : \omega\text{LMod}_{\mathcal{V}} \rightleftarrows \omega\text{LMod}_{\mathcal{W}} : \alpha^*$$

which by the above corollaries restricts to the adjunctions:

$$\alpha_! : \text{PLMod}_{\mathcal{V}} \rightleftarrows \text{PLMod}_{\mathcal{W}} : (\alpha^R)_!$$

and

$$\alpha_! : \text{Enr}_{\mathcal{V}} \rightleftarrows \text{Enr}_{\mathcal{W}} : (\alpha^R)_!$$

2.8 the symmetric monoidal structure on enrichment

In this subsection, we endow $\text{PLMod}_{\mathcal{V}}$ for a symmetric monoidal ∞ -category $\mathcal{V}^\otimes$ with a symmetric monoidal structure.

We have already seen $\text{Enr} \subset \text{PLMod}$ are embeddings of coCartesian fibrations over Mon , and we have the following commutative diagram of symmetric monoidal ∞ -categories:

$$\begin{array}{ccc} \text{Enr}^\times & \xrightarrow{\quad} & \text{PLMod}^\times \\ & \searrow p & \swarrow q \\ & \text{Mon}^\times & \end{array}$$

And to show p, q are also coCartesian fibration, it suffices to show the coCartesian liftings in $\text{Enr} \subset \text{PLMod} \rightarrow \text{Mon}$ are closed under product:

Lemma 2.77. *Given two pseudo-enriched ∞ -categories $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes, \mathcal{N}^\otimes \rightarrow \mathcal{W}^\otimes$, and given two monoidal ∞ -functors $\alpha : \mathcal{V}^\otimes \rightarrow \mathcal{V}'^\otimes$ and $\beta : \mathcal{W}^\otimes \rightarrow \mathcal{W}'^\otimes$, denote the coCartesian liftings over α and β as f and g , then the product of coCartesian liftings is the coCartesian lifting over $\alpha \times \beta$:*

$$\begin{array}{ccc} \mathcal{M}^\otimes \times_{\text{Ass}} \mathcal{N}^\otimes & \xrightarrow{f \times g} & \alpha_!(\mathcal{M}^\otimes) \times_{\text{Ass}} \beta_!(\mathcal{N}^\otimes) \\ \downarrow & & \downarrow \\ \mathcal{V}^\otimes \times_{\text{Ass}} \mathcal{W}^\otimes & \xrightarrow{\alpha \times \beta} & \mathcal{V}'^\otimes \times_{\text{Ass}} \mathcal{W}'^\otimes \end{array}$$

Proof. This is [16, proposition 4.26].

For any $v \in \mathcal{V}, w \in \mathcal{W}$, and $m_1, m_2 \in \mathcal{M}, n_1, n_2 \in \mathcal{N}$, there is a commutative diagram consisting of natural equivalences:

$$\begin{array}{ccc} \text{Map}_{\mathcal{P}(\mathcal{V} \times \mathcal{W})}((v, w), \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M} \times \mathcal{N})}((m_1, m_2), (n_1, n_2))) & & \\ \cong \downarrow & & \\ \text{Map}_{\mathcal{P}\text{Env}(\mathcal{V} \times \mathcal{W})}(v, \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(m_1, m_2)) & \xrightarrow{\cong} & \text{Mul}_{\mathcal{M} \times \mathcal{N}}((v, w), (m_1, n_1); (m_2, n_2)) \\ \cong \downarrow & & \downarrow \cong \\ \text{Map}_{\mathcal{P}\text{Env}(\mathcal{V})}(v, \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(m_1, m_2)) \times \text{Map}_{\mathcal{P}\text{Env}(\mathcal{W})}(w, \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{N})}(n_1, n_2)) & \xrightarrow{\cong} & \text{Mul}_{\mathcal{M}}(v, m_1; m_2) \times \text{Mul}_{\mathcal{N}}(w, n_1; n_2) \\ \uparrow \cong & & \\ \text{Map}_{\mathcal{P}(\mathcal{V})}(v, \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(m_1, m_2)) \times \text{Map}_{\mathcal{P}(\mathcal{W})}(w, \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{N})}(n_1, n_2)) & & \end{array}$$

Especially, the left vertical arrows implies $\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M} \times \mathcal{N})}((m_1, m_2), (n_1, n_2)) \in \mathcal{P}(\mathcal{V} \otimes \mathcal{W})$ is the image of $(\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M})}(m_1, m_2), \text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{N})}(n_1, n_2)) \in \mathcal{P}(\mathcal{V}) \times \mathcal{P}(\mathcal{W})$ under the universal componentwise colimit preserving functor $\mathcal{P}(\mathcal{V}) \times \mathcal{P}(\mathcal{W}) \rightarrow \mathcal{P}(\mathcal{V} \times \mathcal{W})$ sending $F \times G$ to $(v, w) \mapsto F(v) \times G(w)$. Thus by the following commutative diagram, we have $\alpha(\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M} \times \mathcal{W})}((m_1, m_2), (n_1, n_2))) \cong \text{Mor}_{\mathcal{P}\text{LEnv}(\alpha_! \mathcal{M} \times \beta_! \mathcal{N})}((f(m_1), f(m_2)), (g(n_1), g(n_2)))$:

$$\begin{array}{ccc} \mathcal{P}(\mathcal{V}) \times \mathcal{P}(\mathcal{W}) & \xrightarrow{\mathcal{P}(\alpha) \times \mathcal{P}(\beta)} & \mathcal{P}(\mathcal{V}') \times \mathcal{P}(\mathcal{W}') \\ \downarrow & & \downarrow \\ \mathcal{P}(\mathcal{V} \times \mathcal{W}) & \xrightarrow{\mathcal{P}(\alpha \times \beta)} & \mathcal{P}(\mathcal{V}' \times \mathcal{W}') \end{array}$$

□

And likewise for the enriched case:

Lemma 2.78. *Given two enriched ∞ -categories $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes, \mathcal{N}^\otimes \rightarrow \mathcal{W}^\otimes$, and given two monoidal ∞ -functors $\alpha : \mathcal{V}^\otimes \rightarrow \mathcal{V}'^\otimes$ and $\beta : \mathcal{W}^\otimes \rightarrow \mathcal{W}'^\otimes$, denote the coCartesian liftings over α and β as f and g , then the product of coCartesian liftings is the coCartesian lifting over $\alpha \times \beta$:*

$$\begin{array}{ccc}
\mathcal{M}^{\otimes} \times_{\text{Ass}} \mathcal{N}^{\otimes} & \xrightarrow{f \times g} & \alpha_!(\mathcal{M}^{\otimes}) \times_{\text{Ass}} \beta_!(\mathcal{N}^{\otimes}) \\
\downarrow & & \downarrow \\
\mathcal{V}^{\otimes} \times_{\text{Ass}} \mathcal{W}^{\otimes} & \xrightarrow{\alpha \times \beta} & \mathcal{V}'^{\otimes} \times_{\text{Ass}} \mathcal{W}'^{\otimes}
\end{array}$$

Proof. Given $m_1, m_2 \in \mathcal{M}, n_1, n_2 \in \mathcal{N}$, we have:

$$\text{Mor}_{\mathcal{P}\text{LEnv}(\mathcal{M} \times \mathcal{N})}((m_1, m_2), (n_1, n_2)) \cong \text{Mor}_{\mathcal{M} \times \mathcal{N}}((m_1, m_2), (n_1, n_2)) \cong \text{Mor}_{\mathcal{M}}(m_1, m_2) \times \text{Mor}_{\mathcal{N}}(n_1, n_2)$$

Thus $(\alpha \times \beta)(\text{Mor}_{\mathcal{M} \times \mathcal{N}}((m_1, m_2), (n_1, n_2))) = \text{Mor}_{\mathcal{M}}(m_1, m_2) \times \text{Mor}_{\mathcal{N}}(n_1, n_2) \cong \text{Mor}_{\alpha_! \mathcal{M} \times \beta_! \mathcal{N}}((m_1, m_2), (n_1, n_2))$. And by the description (4) of coCartesian edge, $f \times g$ is again a coCartesian lifting. $\square$

Notation 2.79. Given a small symmetric monoidal category $\mathcal{V}^{\otimes}$, let $\text{Enr}_{\mathcal{V}}^{\otimes}$ be the pullback

$$\begin{array}{ccc}
\text{Enr}_{\mathcal{V}}^{\otimes} & \xrightarrow{\quad} & \text{Enr}^{\times} \\
\downarrow & \lrcorner & \downarrow \\
\text{Comm}^{\otimes} & \xrightarrow{\{\mathcal{V}^{\otimes}\}} & \text{Mon}^{\times}
\end{array}$$

Let $\text{PLMod}_{\mathcal{V}}^{\otimes}$ be the pullback

$$\begin{array}{ccc}
\text{PLMod}_{\mathcal{V}}^{\otimes} & \xrightarrow{\quad} & \text{PLMod}^{\times} \\
\downarrow & \lrcorner & \downarrow \\
\text{Comm}^{\otimes} & \xrightarrow{\{\mathcal{V}^{\otimes}\}} & \text{Mon}^{\times}
\end{array}$$

And the embedding $\text{Enr}^{\times} \hookrightarrow \text{PLMod}^{\times}$ induces a natural symmetric monoidal embedding $\text{Enr}_{\mathcal{V}}^{\otimes} \hookrightarrow \text{PLMod}_{\mathcal{V}}^{\otimes}$. And we denote the tensor product of $\text{Enr}_{\mathcal{V}} \subset \text{PLMod}_{\mathcal{V}}$ as $\boxtimes_{\mathcal{V}}$.

If we enlarge the universe and consider the enrichment in a presentably monoidal category $\mathcal{V}$: For a presentably symmetric monoidal ∞ -category $\mathcal{V}$, if we take $v = \mathbf{1}_{\mathcal{V}}$, then $\text{Map}_{\mathcal{V}}(\mathbf{1}_{\mathcal{V}}, \text{Mor}_{\mathcal{M}}(x, y)) \cong \text{Map}_{\mathcal{M}}(x, y)$ is small, thus a $\mathcal{V}$ -enriched ∞ -category has small underlying category iff its underlying space is small iff it's a small enriched ∞ -category, although not absolute small.

By the construction of $\widehat{\text{Enr}}_{\mathcal{V}}^{\otimes}$, given any two $\mathcal{V}$ -enriched ∞ -categories $\mathcal{M}, \mathcal{N}$, then the tensor product operation in $\widehat{\text{Enr}}_{\mathcal{V}}^{\otimes} \boxtimes_{\mathcal{V}} : (\mathcal{M}, \mathcal{N}) \rightarrow \mathcal{M} \boxtimes_{\mathcal{V}} \mathcal{N}$ factors through the tensor product operation in $\widehat{\text{Enr}}^{\times}(\mathcal{M}, \mathcal{N}) \rightarrow \mathcal{M} \times \mathcal{N} \rightarrow \mathcal{M} \boxtimes_{\mathcal{V}} \mathcal{N}$, and the last arrow $\mathcal{M} \times \mathcal{N} \rightarrow \mathcal{M} \boxtimes_{\mathcal{V}} \mathcal{N}$ is the coCartesian lifting in $\widehat{\text{Enr}}$ over the tensor product morphism $\otimes : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$.

Especially, the coCartesian lifting is essential surjective, thus if $\mathcal{M}, \mathcal{N}$ are with small underlying spaces, so is $\mathcal{M} \times \mathcal{N}$ and $\mathcal{M} \boxtimes_{\mathcal{V}} \mathcal{N}$.

Thus the full subcategory of $\widehat{\text{Enr}}_{\mathcal{V}}$ consisting of objects with small underlying categories is closed under tensor product in $\widehat{\text{Enr}}_{\mathcal{V}}$, and we have the following notation:

Notation 2.80. Given a presentably symmetric monoidal ∞ -category $\mathcal{V}^\otimes$, let $\text{Enr}_{\mathcal{V}}^\otimes$ be the full symmetric monoidal subcategory of $\widehat{\text{Enr}}_{\mathcal{V}}^\otimes$ consisting of $\mathcal{V}$ -enriched categories with small underlying category.

There is also a symmetric monoidal structure on $\text{QEnr}_{\mathcal{V}}$ as defined in [16, Notation 4.25]:

Notation 2.81. Let ${}^{cc}\text{Mon}^\otimes$ be the category $\text{Alg}_{\text{Ass}}(\widehat{\text{Cat}}_{cc})^\otimes$ of large monoidal ∞ -categories compatible with small colimits and small colimit preserving monoidal functor between them. Then let $\widehat{\text{QEnr}}^\otimes$ be the full symmetric suboperad of Q consisting of quasi-enriched categories:

$$\begin{array}{ccccc} \widehat{\text{QEnr}}^\otimes & \hookrightarrow & Q & \xrightarrow{\quad} & \widehat{\text{PLMod}}^\times \\ & & \downarrow & \lrcorner & \downarrow \\ & & {}^{cc}\text{Mon}^\otimes & \hookrightarrow & \widehat{\text{Mon}}^\times \end{array}$$

Lemma 2.82. $\widehat{\text{QEnr}}^\otimes \rightarrow {}^{cc}\text{Mon}^\otimes$ is a coCartesian fibration of ∞ -operads

Proof. This is [16, proposition 4.26]. □

Notation 2.83. Given a presentably symmetric monoidal ∞ -category, let $\widehat{\text{QEnr}}_{\mathcal{V}}^\otimes$ be the pullback:

$$\begin{array}{ccc} \widehat{\text{QEnr}}_{\mathcal{V}}^\otimes & \xrightarrow{\quad} & \widehat{\text{QEnr}}^\otimes \\ \downarrow & \lrcorner & \downarrow \\ \text{Comm}^\otimes & \xrightarrow[\{\mathcal{V}^\otimes\}]{} & {}^{cc}\text{Mon}^\otimes \end{array}$$

Remark 2.84. There is a symmetric monoidal localization functor $\widehat{\text{PLMod}}_{\mathcal{V}}^\otimes \rightarrow \widehat{\text{QEnr}}_{\mathcal{V}}^\otimes$ where the inclusion $\widehat{\text{QEnr}}_{\mathcal{V}}^\otimes \hookrightarrow \widehat{\text{PLMod}}_{\mathcal{V}}^\otimes$ is its lax symmetric monoidal right adjoint, which is not symmetric monoidal, since quasi-enriched category corresponds to enriched in the $\widehat{\text{Ind}}_\sigma(\mathcal{V})$, and $\widehat{\text{Ind}}_\sigma(\mathcal{V}) \hookrightarrow \widehat{\mathcal{P}}(\mathcal{V})$ is in general not symmetric monoidal. On the other hand, $\mathcal{V} \hookrightarrow \widehat{\text{Ind}}_\sigma(\mathcal{V})$ is always symmetric monoidal, and this corresponds to the fact that $\widehat{\text{Enr}}_{\mathcal{V}}^\otimes \hookrightarrow \widehat{\text{QEnr}}_{\mathcal{V}}^\otimes$ is symmetric monoidal by [16, theorem 4.43 (3)].

And we have the following symmetric monoidal version of enriched Yoneda embedding:

Theorem 2.85. Let $\mathcal{V}$ be a presentably symmetric monoidal ∞ -category, then there is a symmetric monoidal adjunction:

$$\mathcal{P}_{\mathcal{V}} : \widehat{\text{QEnr}}_{\mathcal{V}}^\otimes \rightleftarrows \text{LMod}_{\mathcal{V}}(\text{Pr}^{\text{L}})^\otimes : i$$

where i is the natural inclusion and $\mathcal{P}$ sends a small $\mathcal{V}$ -enriched ∞ -category $\mathcal{M}$ to its enriched presheaf $\mathcal{P}_{\mathcal{V}}(\mathcal{M})^\otimes \rightarrow \mathcal{V}^\otimes$, and the left adjoint restricts to a symmetric monoidal functor:

$$\mathcal{P}_{\mathcal{V}} : \text{Enr}_{\mathcal{V}}^\otimes \rightarrow \text{LMod}_{\mathcal{V}}(\text{Pr}^{\text{L}})^\otimes$$

Proof. This is [16, corollary 5.1]. □

3 $\text{Cat}_{(\infty,2)}$

In this part:

- In the first subsection, we review the Segal space model of $\text{Cat}_{(\infty,2)}$, and define the full $(\infty, 2)$ subcategory spanned by a subspace of objects under this model.
- In the second subsection, we define the symmetric monoidal $(\infty, 2)$ category as the commutative algebra object in $\text{Cat}_{(\infty,2)}$, or equivalently, a functor $\text{Span}_1(\text{Fin}) \rightarrow \text{Cat}_{(\infty,2)}$ satisfying certain product conditions. And also we can describe symmetric monoidal $(\infty, 2)$ category as 2-coCartesian fibrations over Fin_* using the (un)straightening for $un : \text{Fun}(\text{Comm}^\otimes, \text{Cat}_{(\infty,2)}) \cong \text{Cat}_{(\infty,2)/\text{Fin}_*} : st$. Under the unstraightening, we define the lax symmetric monoidal $(\infty, 2)$ functor similar to the $(\infty, 1)$ case.
- Especially, the right adjoint of a symmetric monoidal $(\infty, 2)$ functor is lax symmetric monoidal $(\infty, 2)$ functor.
- In the third subsection, we review a lax symmetric monoidal functor developed by Stefanich:

$$\theta : \text{LMod}_{\text{Cat}}(\text{Pr}^{\text{L}}) \rightarrow \widehat{\text{Cat}}_{(\infty,2)}$$

, which produces a symmetric monoidal $(\infty, 2)$ category by specifying an object in $\text{CAlg}(\text{LMod}_{\text{Cat}}(\text{Pr}^{\text{L}})) \cong \text{CAlg}(\text{Pr}^{\text{L}})_{\text{Cat}} /$.

- Especially, given a morphism $F : \mathcal{M} \rightarrow \mathcal{N} \in \text{CAlg}(\text{LMod}_{\text{Cat}}(\text{Pr}^{\text{L}}))$, the underlying $(\infty, 1)$ right adjoint $F^R : \mathcal{N} \rightarrow \mathcal{M}$ can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor.

3.1 $\text{Seg}(\text{Cat}_\infty)$

This part follows [10, section 7].

Definition 3.1. Suppose $\mathcal{C}$ is an ∞ -category with limits, a category object in $\mathcal{C}$ is a simplicial object $F : \Delta^{op} \rightarrow \mathcal{C}$ such that for each n the map

$$F([n]) \rightarrow F([1]) \times_{F([0])} \cdots \times_{F([0])} F([1])$$

induced by $\{i, i+1\} \subset [n], \{i\} \subset [n]$ is an equivalence.

Definition 3.2. A Segal space in Cat_∞ is a category object in $X : \Delta^{op} \rightarrow \text{Cat}_\infty$ such that $X[0] \in \text{Spc}$, we denote the full subcategory of $\text{Cat}(\text{Cat}_\infty)$ consisting of Segal spaces as $\text{Seg}(\text{Cat}_\infty)$.

Remark 3.3. This definition of $\text{Seg}(\text{Cat}_\infty)$ is the same as $\text{SS}_{\text{Spc} \subseteq \text{Cat}_\infty}$ in [20, definition 1.2.7], where we have a pullback diagram of ∞ -categories by [20, remark 1.2.8]:

$$\begin{array}{ccc} \text{Seg}(\text{Cat}_\infty) & \longrightarrow & \text{Cat}(\text{Cat}_\infty) \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{S} & \longrightarrow & \text{Cat}_\infty \end{array}$$

Proposition 3.4. $i^* : \text{Seg}(\text{Cat}_\infty) \rightarrow \text{Spc}$ induced by $i : [0] \rightarrow \Delta^{op}$ is a Cartesian fibration.

Remark 3.5. The functor i^* admits a right adjoint by right Kan extension $i_* : \mathbf{Spc} \rightarrow \mathbf{Cat} \cong \mathbf{Fun}(*, \mathbf{Cat}) \rightarrow \mathbf{Fun}(\Delta^{op}, \mathbf{Cat})$, which sends a space C to its Čech nerve $C^{\cdot+1}$.

The above proposition is a restricted version of [11, lemma 4.15], based on the following lemmas:

First we shall introduce a criterion determining whether a categorical fibration is (co)Cartesian fibration if it admits an adjoint:

Lemma 3.6. [12, corollary 4.5.2] Suppose $p : \mathcal{E} \rightarrow \mathcal{C}$ is a categorical fibration between ∞ -categories, and that p has a left adjoint $F : \mathcal{C} \rightarrow \mathcal{E}$. Then the followings are equivalent:

(1) p is a coCartesian fibration.

(2) For every $e \in \mathcal{E}$ and every morphism $\phi : p(e) \rightarrow x$ in $\mathcal{C}$, there is a pushout square in $\mathcal{E}$

$$\begin{array}{ccc} Fp(e) & \xrightarrow{F(\phi)} & F(x) \\ \text{counit} \downarrow & & \downarrow v \\ e & \xrightarrow[\bar{\phi}]{} & \bar{x} \end{array}$$

such that the composite $x \xrightarrow{\text{unit}} pF(x) \xrightarrow{p(v)} p(\bar{x})$ is an equivalence.

sketch of the proof. Given a morphism $\phi : p(e) \rightarrow x \in \mathcal{C}$, we construct the p -coCartesian lift of ϕ from $\bar{\phi}$ by composing with the lift of the inverse of the equivalence between x and $p(\bar{x})$. Such lifting exists because p is an isofibration. $\square$

Lemma 3.7. [12, corollary 4.5.2] Suppose $p : \mathcal{E} \rightarrow \mathcal{C}$ is a categorical fibration between ∞ -categories, and that p has a right adjoint $F : \mathcal{C} \rightarrow \mathcal{E}$. Then the followings are equivalent:

(1) p is a Cartesian fibration.

(2) For every $e \in \mathcal{E}$ and every morphism $\phi : x \rightarrow p(e)$ in $\mathcal{C}$, there is a pullback square in $\mathcal{E}$:

$$\begin{array}{ccc} \bar{x} & \xrightarrow{v} & F(x) \\ \downarrow & \lrcorner & \downarrow F(\phi) \\ e & \xrightarrow{\text{unit}} & Fp(e) \end{array}$$

such that the composite $p(\bar{x}) \xrightarrow{p(v)} pF(x) \xrightarrow{\text{counit}} x$ is an equivalence.

Proof of $i^ : \mathbf{Seg}(\mathbf{Cat}_\infty) \rightarrow \mathbf{Spc}$ being a Cartesian fibration.* This is just the pullback of the Cartesian fibration $\mathbf{Cat}(\mathbf{Cat}_\infty) \rightarrow \mathbf{Cat}_\infty$ along the inclusion $\mathbf{Spc} \hookrightarrow \mathbf{Cat}_\infty$. Moreover, by the above lemmas, we obtain the exact description of the Cartesian lifting of a morphism $\phi : X \rightarrow Y[0]$ in $\mathbf{Spc}$, for $Y \in \mathbf{Seg}(\mathbf{Cat}_\infty)$: this corresponds to the pullback $\phi^*Y \cong i_*X \times_{i_*(Y[0])} Y \rightarrow Y$. Since the limits in $\mathbf{Seg}(\mathbf{Cat}_\infty) \subset \mathbf{Cat}(\mathbf{Cat}_\infty) \subset \mathbf{Fun}(\Delta^{op}, \mathbf{Cat}_\infty)$ are computed objectwise, we have:

$$\begin{array}{ccc} \phi^*Y[n] & \longrightarrow & Y[n] \\ \downarrow & \lrcorner & \downarrow \\ X^{n+1} & \xrightarrow{\phi^{n+1}} & Y[0]^{n+1} \end{array}$$

□

The category $\text{Seg}(\text{Cat}_\infty)$ serves as a prototype of $\text{Cat}_{(\infty,2)}$: we want the equivalence in this category to be (enriched) fully faithful and essential surjective.

The definition below comes from [11, section 7]

Definition 3.8. Write $\text{Gpd}(\text{Spc})$ for the full subcategory of $\text{Seg}(\text{Spc})$ consisting of groupoid objects, that is, a simplicial object $F : \Delta^{op} \rightarrow \text{Spc}$ such that for any partition $[n] = S \cup S'$, where $S \cap S'$ consists of only one object, and the following diagram is a pullback diagram in Spc

$$\begin{array}{ccc} F([n]) & \longrightarrow & F(S) \\ \downarrow \lrcorner & & \downarrow \\ F(S') & \longrightarrow & F(S \cap S') \end{array}$$

Notice that the inclusion $i : \text{Spc} \hookrightarrow \text{Cat}_\infty$ admits a right adjoint $(-)^{\simeq}$ by taking underlying space, which induces an adjunction:

$$\text{Seg}(i) : \text{Seg}(\text{Spc}) \rightleftarrows \text{Seg}(\text{Cat}_\infty) : ()^{1,\simeq} = \text{Seg}((-)^{\simeq})$$

And the inclusion $\text{Gpd}(\text{Spc}) \hookrightarrow \text{Seg}(\text{Spc})$ admits a right adjoint by [20, proposition 1.1.14]. Thus there is a right adjoint to the inclusion $j : \text{Gpd}(\text{Spc}) \hookrightarrow \text{Seg}(\text{Cat}_\infty)$:

$$j : \text{Gpd}(\text{Spc}) \rightleftarrows \text{Seg}(\text{Cat}_\infty) : \tau$$

Definition 3.9. An object $F \in \text{Seg}(\text{Cat}_\infty)$ is complete (in the Segal sense) if $\tau(F)$ is a constant groupoid object. We denote the full subcategory of $\text{Seg}(\text{Cat}_\infty)$ consisting of complete objects as $\text{Cat}_{(\infty,2)}^{\text{Seg}}$.

Remark 3.10. Let E_n denote the right Kan extension along $\{0\} \hookrightarrow \Delta^{op}$ of the functor $* \rightarrow \text{Spc}$ specifying the discrete space $\{0, \dots, n\}$, it's obvious that E_n is a Segal space, and by [11, proposition 7.16], a Segal space in Cat_∞ is complete iff it is local with respect to the morphism $E^1 \rightarrow E^0$.

Especially, the above localization gives rise to an adjunction:

$$L : \text{Seg}(\text{Cat}) \rightleftarrows \text{Cat}_{(\infty,2)} : I$$

where L sends a Segal category $\mathbf{D}$ to the localization $L(\mathbf{D})$, which is a complete Segal space in Cat .

Lemma 3.11. When $\mathcal{V}$ is a complete object in $\text{Seg}(\text{Cat}_\infty)$, given a morphism $p : S \rightarrow \mathcal{V}[0]$, consider the Cartesian lifting $Y = p^* \mathcal{V} \rightarrow \mathcal{V}$ in $\text{Seg}(\text{Cat}_\infty)$ over p in $\mathcal{S}$, Y is also a complete object iff p is a monomorphism.

Proof. The Cartesian lifting is given by $Y \cong i_* S \times_{i_* \mathcal{V}[0]} \mathcal{V}$, by [11, lemma 7.2.1], this is equivalent to showing that the underlying Segal space of Y is a complete Segal space, under the functor $(-)^{1,\simeq} = \text{Seg}((-)^{\simeq}) : \text{Seg}(\text{Cat}_\infty) \rightarrow \text{Seg}(\text{Spc})$ induced by the core functor $(-)^{\simeq} : \text{Cat}_\infty \rightarrow \text{Spc}$ dropping all the non-invertible edges. Here we can just regard $\mathcal{V}^{1,\simeq}$ as the underlying $(\infty, 1)$ -category of $\mathcal{V}$, in the model of 1-fold Segal space.

By our condition, $\mathcal{V}$ is complete, thus $\mathcal{V}^{1,\simeq}$ is always a complete Segal space, and by [11, theorem 3.6], this is equivalent to saying:

$$\begin{array}{ccccc}
\mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}(E_1, \mathcal{V}^{1,\simeq}) & \xrightarrow{[1] \rightarrow E_1, \cong} & \mathcal{V}_{1,eq}^{1,\simeq} \cong (\mathcal{V}_{1,eq})^{\simeq} & \xleftarrow{\text{degeneracy}, \cong} & \mathcal{V}[0] \\
& \searrow [1] \rightarrow E_1 & \parallel & & \\
& & \mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}([1], \mathcal{V}^{1,\simeq})_{eq} & & \\
& & \downarrow & & \\
& & \mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}([1], \mathcal{V}^{1,\simeq}) = \mathcal{V}_1^{1,\simeq} & &
\end{array}$$

In the above diagram, $\mathcal{V}_{1,eq}^{1,\simeq}$ denotes the summand of the space $\mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}([1], \mathcal{V}^{1,\simeq})$, spanned by the image of the map $\mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}(E_1, \mathcal{V}^{1,\simeq}) \rightarrow \mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}([1], \mathcal{V}^{1,\simeq})$ induced by the embedding $[1] \rightarrow E_1$, thus it's equivalent to $\mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}(E_1, \mathcal{V}^{1,\simeq})$.

Especially, by [11, theorem 3.6], there is an equivalent condition for a Segal space $X_\bullet$ to be complete: the morphism induced by degenerate map $X_0 \rightarrow X_{1,eq}$ is an equivalence, where $X_{1,eq}$ is the subspace of ‘equivalences’ of X_1 defined as above.

This space can be understood as the subspace of ‘equivalences’ in the space of 1-morphisms $\mathcal{V}_1^{1,\simeq}$. The completeness conditions just says this space of equivalences corresponds exactly to the space of objects, that is, the ‘equivalence’ derived from the space of morphisms is the same as the ‘equivalence’ specified in the space of objects.

Then it remains to show the underlying Segal space $Y^{1,\simeq}$ also satisfies the completeness condition. By the pullback $Y \cong i_* S \times_{i^* \mathcal{V}_0} \mathcal{V}$, we get again a pullback $Y^{1,\simeq} \cong i_* S \times_{i^* \mathcal{V}_0} \mathcal{V}^{1,\simeq}$ since the functor $(-)^{1,\simeq}$ is the right adjoint of the embedding $\mathrm{Seg}(\mathrm{Spc}) \rightarrow \mathrm{Seg}(\mathrm{Cat}_\infty)$, thus preserving pullback, and we have the following pullback diagram of mapping spaces:

$$\begin{array}{ccc}
\mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}(E_1, Y^{1,\simeq}) & \longrightarrow & \mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}(E_1, \mathcal{V}^{1,\simeq}) \\
\downarrow & \lrcorner & \downarrow \\
\mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}(E_1, i_* S) & \xrightarrow{p} & \mathrm{Map}_{\mathrm{Seg}(\mathrm{Spc})}(E_1, i_* \mathcal{V}_0)
\end{array}$$

which can be assembled into a bigger diagram:

$$\begin{array}{ccc}
Y_0 = S & \xrightarrow{p} & \mathcal{V}_0 \\
\text{degenerate} \downarrow & & \cong \downarrow \\
Y_{1,eq}^{1,\simeq} & \xrightarrow{\quad} & \text{Map}_{\text{Seg}(\text{Spc})}([1], \mathcal{V}_{1,eq}^{1,\simeq}) \cong \mathcal{V}_{1,eq}^{1,\simeq} \\
\cong, [1] \rightarrow E_1 \uparrow & & \cong \uparrow \\
\text{Map}_{\text{Seg}(\text{Spc})}(E_1, Y_{1,eq}^{1,\simeq}) & \xrightarrow{\quad} & \text{Map}_{\text{Seg}(\text{Spc})}(E_1, \mathcal{V}_{1,eq}^{1,\simeq}) \\
\downarrow \lrcorner & & \downarrow \\
\text{Map}_{\text{Seg}(\text{Spc})}(E_1 = i_*\{*, *\}, i_*S) \cong S \times S & \xrightarrow{p \times p} & \text{Map}_{\text{Seg}(\text{Spc})}(E_1, i^*\mathcal{V}_0) \\
& & \downarrow \text{ev}_0, \cong, \text{adjunction} \\
& & \text{Map}_{\text{Spc}}(* \coprod *, \mathcal{V}_0) \cong \mathcal{V}_0 \times \mathcal{V}_0
\end{array}$$

The target is to show the upper square is also a pullback, if so, then by the fact that the pullback of an equivalence is again an equivalence, and we shall get the morphism $Y_0 \rightarrow Y_{1,eq}$ induced by the degenerate map also being equivalence, and thus Y is complete.

Since the middle square and the bottom squares are all pullback squares, it suffices to show the big outside diagram is a pullback square by the pasting lemma, which is just the following diagram:

$$\begin{array}{ccc}
S & \xrightarrow{p} & \mathcal{V}_0 \\
\downarrow \lrcorner & & \downarrow \text{diagonal id} \times \text{id} \\
S \times S \cong S & \xrightarrow{p \times p} & \mathcal{V}_0 \times \mathcal{V}_0
\end{array}$$

This is obviously a pullback square iff p is a monomorphism. □

And we can use the above Cartesian fibration to define the full $(\infty, 2)$ subcategory:

Definition 3.12. Given an $(\infty, 2)$ category $\mathbf{D}$, and a full subspace of objects $j : C \hookrightarrow \mathbf{D}^\simeq$, then consider the Cartesian lifting $f : j^*\mathbf{D} \rightarrow \mathbf{D}$ over j of $\text{Cat}_{(\infty, 2)} \rightarrow \text{Spc}$, then we define the full $(\infty, 2)$ subcategory of $\mathbf{D}$ spanned by the subspace $j : C \subset \mathbf{D}^\simeq$ as $\overline{C} := j^*\mathbf{D}$.

Recall the definition of monomorphism in $\text{Cat}_{(\infty, 2)}$:

Lemma 3.13. Given a morphism $F : \mathbf{C} \rightarrow \mathbf{D}$ in $\text{Cat}_{(\infty, 2)}$, the following conditions are equivalent:

- (1) F is a monomorphism, that is, the induced morphism $\text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C}) \rightarrow \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{D})$ is a fully faithful embedding of spaces for any $\mathbf{E} \in \text{Cat}_{(\infty, 2)}$.
- (2) The induced diagonal morphism in $\text{Cat}_{(\infty, 2)}$: $\Delta_F : \mathbf{C} \rightarrow \mathbf{C} \times_{\mathbf{D}} \mathbf{C}$ is an isomorphism.

Proof. (1) $\Rightarrow$ (2): Given any $\mathbf{E} \in \text{Cat}_{(\infty, 2)}$, the diagonal morphism induces

$$\text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C}) \rightarrow \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C} \times_{\mathbf{D}} \mathbf{C}) \cong \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C}) \times_{\text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{D})} \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C})$$

which is an equivalence by [Tag 04W5](#): since $\text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C}) \rightarrow \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{D})$ is a monomorphism in Spc . Then by the Yoneda embedding, $\mathbf{C} \cong \mathbf{C} \times_{\mathbf{D}} \mathbf{C}$.

(2) $\Rightarrow$ (1): If (2) holds, then the morphism

$$\text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C}) \rightarrow \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C} \times_{\mathbf{D}} \mathbf{C}) \cong \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C}) \times_{\text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{D})} \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C})$$

is again an equivalence, thus $\text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{C}) \rightarrow \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{D})$ is a monomorphism by [Tag 04W5](#). $\square$

Especially, a full $(\infty, 2)$ subcategory embedding is a monomorphism:

Lemma 3.14. *The full $(\infty, 2)$ subcategory $\overline{\mathbf{C}} \rightarrow \mathbf{D}$ spanned by the full subspace of objects $\mathbf{C} \hookrightarrow \mathbf{D}^{\simeq} = \mathbf{D}[0]$ is a monomorphism in $\text{Cat}_{(\infty, 2)}$, that is, the induced morphism between mapping spaces:*

$$\text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \overline{\mathbf{C}}) \rightarrow \text{Map}_{\text{Cat}_{(\infty, 2)}}(\mathbf{E}, \mathbf{D})$$

is a fully faithful embedding of mapping spaces for any $(\infty, 2)$ category $\mathbf{E}$.

Proof. This is the universal property of Cartesian liftings. $\square$

And we have the following equivalent conditions for full $(\infty, 2)$ subcategory embedding:

Lemma 3.15. *Given a morphism $F : \mathbf{C} \rightarrow \mathbf{D}$ in $\text{Cat}_{(\infty, 2)}$, the following conditions are equivalent:*

- (1) $F : \mathbf{C} \rightarrow \mathbf{D}$ is a full $(\infty, 2)$ subcategory spanned by the subspace $\mathbf{C}^{\simeq} \hookrightarrow \mathbf{D}^{\simeq}$.
- (2) F is a monomorphism, and for any $x, y \in \mathbf{C}$, the induced morphism $F_{x, y} : \mathbf{C}(x, y) \rightarrow \mathbf{D}(F(x), F(y))$ is an equivalence of categories.

Proof.

(1) $\Rightarrow$ (2): By the above lemma, F is a monomorphism.

And the equivalence between the mapping categories is by the following pullback diagrams:

$$\begin{array}{ccccc} \mathbf{C}(x, y) \cong j^* \mathbf{D}(x, y) \cong \mathbf{D}(Fx, Fy) & \longrightarrow & j^* \mathbf{D}[1] & \longrightarrow & \mathbf{D}[1] \\ \downarrow & \lrcorner & \downarrow d_1 \times d_0 & & \downarrow d_1 \times d_0 \\ * & \xrightarrow{\{x, y\}} & \mathbf{C}[0]^{\times 2} & \xrightarrow{F[0]^{\times 2}} & \mathbf{D}[0]^{\times 2} \end{array}$$

(2) $\Rightarrow$ (1): Since the functor sending an $(\infty, 2)$ category to its underlying space is the right adjoint of the inclusion $\mathbf{Spc} \hookrightarrow \mathbf{Cat}_{(\infty, 2)}$, thus preserves limits, so $j : \mathbf{C}^\simeq \hookrightarrow \mathbf{D}^\simeq$ is a fully faithful embedding of spaces, and we consider the full $(\infty, 2)$ subcategory $j^*\mathbf{D} = i_*\mathbf{C}^\simeq \times_{i_*\mathbf{D}^\simeq} \mathbf{D}$ where i_* is the right Kan extension along $i : * \rightarrow \Delta^{op}, * \mapsto \{[0]\}$: $i_* : \mathbf{Spc} \subset \mathbf{Cat} \cong \mathbf{Fun}(*, \mathbf{Cat}) \rightarrow \mathbf{Fun}(\Delta^{op}, \mathbf{Cat})$, and the induced morphism $j^*\mathbf{D} \rightarrow \mathbf{D}$ is the Cartesian edge over $j : \mathbf{C}^\simeq \hookrightarrow \mathbf{D}^\simeq$.

By the universal property of Cartesian edges, there is an induced morphism $s : \mathbf{C} \rightarrow i_*\mathbf{C}[0] \times_{i_*\mathbf{D}[0]} \mathbf{D}$, and it remains to show s is an isomorphism.

Since this is a morphism in $\mathbf{Seg}(\mathbf{Cat}) \subset \mathbf{Fun}(\Delta^{op}, \mathbf{Cat})$, it suffices to show the poinwise induced morphism:

$$s[n] : \mathbf{C}([n]) \rightarrow (i_*\mathbf{C}[0] \times_{i_*\mathbf{D}[0]} \mathbf{D})([n])$$

is an equivalence for all n . And since both sides satisfy the Segal condition, thus it suffices to show the case when $n = 1$.

Notice that the condition $\mathbf{C}(x, y) \rightarrow \mathbf{D}(Fx, Fy)$ is an equivalence for any points $x, y \in \mathbf{C}[0]$ is equivalent to that the pullback of $s[1] : \mathbf{C}[1] \rightarrow (i_*\mathbf{C}[0] \times_{i_*\mathbf{D}[0]} \mathbf{D})[1]$ along the any point $(x, y) : * \rightarrow \mathbf{C}[0]^{\times 2}$ is an equivalence:

$$\begin{array}{ccccc}
 \mathbf{C}(x, y) & \xrightarrow{\quad} & \mathbf{C}[1] & \xrightarrow{\quad F[1] \quad} & \mathbf{D}[1] \\
 \searrow \cong & & \searrow s[1] & & \downarrow d_1 \times d_0 \\
 & & \mathbf{D}(Fx, Fy) & \xrightarrow{\quad} & j^*\mathbf{D}[1] & \xrightarrow{\quad} & \mathbf{D}[1] \\
 & & \downarrow & \lrcorner & \downarrow d_1 \times d_0 & & \downarrow d_1 \times d_0 \\
 & & * & \xrightarrow{\{x, y\}} & \mathbf{C}[0]^{\times 2} & \xrightarrow{F[0]^{\times 2}} & \mathbf{D}[0]^{\times 2}
 \end{array}$$

By [Tag 023K](#), this implies $s[1]$ is an equivalence. □

3.2 symmetric monoidal $(\infty, 2)$ category

Definition 3.16. A symmetric monoidal $(\infty, 2)$ -category is a commutative algebra object in $\mathbf{Cat}_{(\infty, 2)}^\times$.

Definition 3.17. A symmetric monoidal $(\infty, 2)$ functor is a morphism in $\mathbf{CAlg}(\mathbf{Cat}_{(\infty, 2)})$.

And we want a description of symmetric monoidal $(\infty, 2)$ -category in terms of coCartesian fibration over $\mathbf{Comm}^\otimes = \mathbf{Fin}_*$:

Definition 3.18. Given an $(\infty, 2)$ functor $F : \mathbf{S} \rightarrow \mathbf{T}$, we say a 1-morphism $e : s_1 \rightarrow s_2 \in \mathbf{S}$ is **(2-)coCartesian** over $\mathbf{T}$, if for any $s \in \mathbf{S}$, there is an equivalence induced by e :

$$\mathbf{S}(s_2, s) \cong \mathbf{S}(s_1, s) \times_{\mathbf{T}(F(s_1), F(s))} \mathbf{T}(F(s_2), F(s))$$

Definition 3.19. [8, chapter 11, 1.1.2] An $(\infty, 2)$ functor $F : \mathbf{S} \rightarrow \mathbf{T}$ is 2-coCartesian fibration if the following conditions hold:

- (1) For any 1-morphism $d : F(s) \rightarrow t$ in $\mathbf{T}$, there exists a coCartesian 1-morphism $e : s \rightarrow s'$ lying over d .
- (2) For any $s, s' \in \mathbf{S}$, the induced functor $\mathbf{S}(s, s') \rightarrow \mathbf{T}(F(s), F(s'))$ is a Cartesian fibration of $(\infty, 1)$ -categories, and we call the Cartesian edges as Cartesian 2-morphisms.
- (3) And for any 1-morphism $g : s' \rightarrow s'', h : s_0 \rightarrow s$ in $\mathbf{S}$, the induced functor $\mathbf{S}(s, s') \rightarrow \mathbf{S}(s, s'')$ and $\mathbf{S}(s, s') \rightarrow \mathbf{S}(s_0, s')$ preserves Cartesian arrows.

Denote the full 1-subcategory of the $(\infty, 2)$ -category $\mathbf{Cat}_{(\infty, 2)/\mathbf{T}}$ spanned by 2-coCartesian fibrations as $2\text{coC}/\mathbf{T}$ and morphisms between 2-coCartesian fibrations which preserve both coCartesian 1-morphisms and Cartesian 2-morphisms.

Theorem 3.20. [26, theorem 6.1] *There is a natural straightening-unstraightening equivalence:*

$$2\text{coC}/\mathbf{T} \cong \text{Fun}(\mathbf{T}, \text{Cat}_{(\infty, 2)})$$

Thus a symmetric monoidal $(\infty, 2)$ category $\mathcal{C} : \text{Fin}_* \rightarrow \text{Cat}_{(\infty, 2)}$ can be unstraightened to a 2-coCartesian fibration $\mathcal{C}^\otimes$ over Fin_* , and correspondingly a symmetric monoidal $(\infty, 2)$ functor between $\mathcal{C}$ and $\mathcal{D}$ is a morphism between $\mathcal{C}^\otimes$ and $\mathcal{D}^\otimes$ preserving coCartesian 1-morphisms and Cartesian 2-morphisms.

And we have the following definition for lax symmetric monoidal $(\infty, 2)$ functor:

Definition 3.21. Given two symmetric monoidal $(\infty, 2)$ categories $\mathcal{C}, \mathcal{D} \in \text{CAlg}(\text{Cat}_{(\infty, 2)})$, which after unstraightening can be regarded as 2-coCartesian fibrations $\mathcal{C}^\otimes, \mathcal{D}^\otimes$ in 2coC , and an $(\infty, 2)$ functor $F : \mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$ over Fin_* is a lax symmetric monoidal $(\infty, 2)$ functor if F preserves the coCartesian 1-morphisms over the inert maps in Fin_* and all Cartesian 2-morphisms.

Here is a way to produce lax symmetric monoidal $(\infty, 2)$ functors:

Proposition 3.22. *Given a symmetric monoidal $(\infty, 2)$ functor $F : \mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$, if the underlying $(\infty, 2)$ functor $F_{\langle 1 \rangle} : \mathcal{C} := \mathcal{C}_{\langle 1 \rangle}^\otimes \rightarrow \mathcal{D} := \mathcal{D}_{\langle 1 \rangle}^\otimes$ admits a right adjoint g in $\text{Cat}_{(\infty, 2)}$, then there exists a right adjoint G of F relative to Fin_* and G is a lax symmetric monoidal $(\infty, 2)$ functor, and $G_{\langle 1 \rangle} \cong g$.*

Remark 3.23. In this case, we say the $(\infty, 2)$ functor g can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor.

Proof. By [2, Theorem A], there is more general version of straightening-unstraightening, which is an $(\infty, 2)$ equivalence:

$$\text{Fun}(\text{Comm}^\otimes, \text{Cat}_{(\infty, 2)})^{\text{lax}} \cong \mathbf{2coC}_{/\text{Comm}^\otimes}^{\text{lax}}$$

where the left hand side is the internal Hom with respect to the gray tensor product in $\text{Cat}_{(\infty, 2)}$, consisting of $(\infty, 2)$ functors from $\text{Comm}^\otimes$ to $\text{Cat}_{(\infty, 2)}$ and lax natural transformations between them, and the right hand side is the $(\infty, 2)$ category of 2-coCartesian fibrations over $\text{Comm}^\otimes$ and morphisms preserving Cartesian 2-morphisms.

And by [2, Theorem D], a morphism F in $\text{Fun}(\text{Comm}, \text{Cat}_{(\infty, 2)})$ admits a right adjoint iff it admits a right adjoint pointwise. For a symmetric monoidal functor F , at $\langle n \rangle \in \text{Comm}^\otimes$, $F_{\langle n \rangle} = F_{\langle 1 \rangle}^\times : \mathcal{C}_{\langle n \rangle}^\otimes \rightarrow \mathcal{D}_{\langle n \rangle}^\otimes$ admits a right adjoint $g^{\times n}$, thus under unstraightening, the functor $F : \mathcal{C}^\otimes \rightarrow \mathcal{D}^\otimes$ admits a right adjoint G in $\mathbf{2coC}_{/\text{Comm}^\otimes}$, thus it preserves Cartesian 2-morphisms. Moreover, it preserves coCartesian 1-morphisms over the inert maps by the description of $G_{\langle n \rangle} = g^{\times n}$. $\square$

And here is another approach of describing symmetric monoidal $(\infty, 2)$ categories using $\text{Span}_1(\text{Fin})$.

The following part is from [4, Appendix C.1]

Recall there is an equivalence $\text{Fin}_* \cong \text{Span}_1(\text{Fin}, \text{inj}, \text{all})$ by sending

- object: $\langle n \rangle \mapsto [n]$ where $[n]$ is the finite set of n objects, that is, a subset of $\langle n \rangle$ excluding the $*$.
- morphism: $f : \langle n \rangle \rightarrow \langle m \rangle \mapsto$

In particular, we can identify Fin_* with a wide subcategory of $\text{Span}_1(\text{Fin})$ where $\text{Fin}^{op} \subset \text{Span}_1(\text{Fin})$ plays the role of inert maps.

Proposition 3.24. *The restriction functor*

$$\text{Fun}(\text{Span}_1(\text{Fin}), \text{Cat}_{(\infty, 2)}) \rightarrow \text{Fun}(\text{Fin}_*, \text{Cat}_{(\infty, 2)})$$

induces an equivalence between

- (1) *functors $F : \text{Span}_1(\text{Fin}) \rightarrow \text{Cat}_{(\infty, 2)}$ such that the restriction of F to Fin^{op} is product preserving.*
- (2) *the commutative monoids in $\text{Cat}_{(\infty, 2)}$.*

Proof. This is [4, proposition C.1]. $\square$

Thus a symmetric monoidal $(\infty, 2)$ category can equivalently be defined by specifying a functor $F : \text{Span}_1(\text{Fin}) \rightarrow \text{Cat}_{(\infty, 2)}$ under certain product condition, and after unstraightening, this corresponds to a 2-coCartesian fibration $\mathcal{C}'$ over $\text{Span}_1(\text{Fin})$. Since straightening-unstraightening is natural, we can recover $\mathcal{C}^\otimes \rightarrow \text{Fin}_*$ by pullback of $\mathcal{C}'$ along the inclusion $\text{Fin}_* \rightarrow \text{Span}(\text{Fin})$.

Remark 3.25. And there is another way of defining lax symmetric monoidal $(\infty, 2)$ functor using $\text{Span}_1(\text{Fin})$ by [7, definition 4.27]: Given two symmetric monoidal $(\infty, 2)$ categories $\mathcal{C}, \mathcal{D} \in \text{CAlg}(\text{Cat}_{(\infty, 2)})$, under the equivalence $\text{Fun}(\text{Fin}_*, \text{Cat}_{(\infty, 2)}) \cong \text{Fun}(\text{Span}_1(\text{Fin}), \text{Cat}_{(\infty, 2)}) \cong \mathbf{2coC}_{/\text{Span}_1(\text{Fin})}$,

denote their unstraightenings over $\text{Span}_1(\text{Fin})$ as $\mathcal{C}'$ and $\mathcal{D}'$, then a functor

is lax symmetric monoidal $(\infty, 2)$ if it preserves coCartesian 1-morphisms over $\text{Fin}^{op} \hookrightarrow \text{Span}_1(\text{Fin})$ and all Cartesian 2-morphisms.

Since the inert maps in Fin_* correspond exactly to maps lying in $\text{Fin}^{op} \hookrightarrow \text{Span}_1(\text{Fin})$, and thus a morphism $F' : \mathcal{C}' \rightarrow \mathcal{D}'$ preserving coCartesian 1-morphism over Fin^{op} corresponds exactly to a functor $F : \mathcal{C}'^{\otimes} \rightarrow \mathcal{D}'^{\otimes}$ preserving coCartesian 1-morphisms over inert maps under pullback along $\text{Fin}_* \rightarrow \text{Span}_1(\text{Fin})$. Moreover, pullback of a functor preserving Cartesian 2-morphism also preserves Cartesian 2-morphisms, thus the above two versions of definitions of lax symmetric monoidal $(\infty, 2)$ functor coincide after pullback along the wide subcategory inclusion $\text{Fin}_* \rightarrow \text{Span}_1(\text{Fin})$.

As analogy to [22, section 2.2.1], we can define a symmetric monoidal $(\infty, 2)$ subcategory $\mathcal{C}^{\otimes}$ of $\mathcal{D}^{\otimes}$ through a full $(\infty, 2)$ subcategory $\mathcal{C} \subset \mathcal{D}$ where the embedding is closed under tensor product in $\mathcal{D}$.

Construction 3.26. Given a symmetric monoidal $(\infty, 2)$ category $F : \text{Comm}^{\otimes} \rightarrow \text{Cat}_{(\infty, 2)}$ such that $F(\langle 1 \rangle) \cong \mathbf{D}$ and a full $(\infty, 2)$ subcategory $\mathcal{C}$ of $\mathbf{D}$ which is closed under tensor product operation, that is, the image of the $(\infty, 2)$ functor $\mathcal{C} \times \mathcal{C} \hookrightarrow \mathbf{D} \times \mathbf{D} \xrightarrow{\otimes := F(a)} \mathbf{D}$ lands in $\mathcal{C} \hookrightarrow \mathbf{D}$ where a is the active map $\langle 2 \rangle \rightarrow \langle 1 \rangle$, then by [28, proposition A.1], there exists a subfunctor G of F :

$$\alpha : G \Rightarrow F : \text{Comm}^{\otimes} \rightarrow \text{Cat}_{(\infty, 2)}$$

such that $G(\langle n \rangle) \cong \mathcal{C}^{\times n} \hookrightarrow \mathbf{D}^{\times n} \cong F(\langle n \rangle)$. And obviously, F also satisfies the monoid condition, thus we get an $(\infty, 2)$ symmetric monoidal structure on the full subcategory $\mathcal{C}$ of $\mathbf{D}$.

Notation 3.27. In the form of 2-coCartesian fibrations, given a symmetric monoidal $(\infty, 2)$ category $\mathbf{D}^{\otimes}$ and a full $(\infty, 2)$ subcategory $i : \mathcal{C} \hookrightarrow \mathbf{D}$ which is closed under tensor product, we denote the corresponding full symmetric monoidal $(\infty, 2)$ subcategory spanned by $\mathcal{C}$ as:

$$i : \mathcal{C}^{\otimes} \hookrightarrow \mathbf{D}^{\otimes}$$

which is again a full $(\infty, 2)$ subcategory, since this is a morphism in $2\text{coC}/_{\text{Comm}^{\otimes}} \cong \text{Fun}(\text{Comm}^{\otimes}, \text{Cat}_{(\infty, 2)})$, thus fiberwise fully faithfulness implies fully faithfulness as a whole.

Especially, given a lax symmetric monoidal $(\infty, 2)$ functor $G : \mathbf{B}^{\otimes} \rightarrow \mathbf{D}^{\otimes}$, if the underlying $(\infty, 2)$ functor $G_{\langle 1 \rangle} : \mathbf{B} \rightarrow \mathbf{D}$ lands in the full $(\infty, 2)$ subcategory $\mathcal{C} \hookrightarrow \mathbf{D}$ where $\mathcal{C}$ is closed under tensor product in $\mathbf{D}$, then there is a factorization:

$$G : \mathbf{B}^{\otimes} \rightarrow \mathcal{C}^{\otimes} \hookrightarrow \mathbf{D}^{\otimes}$$

Here is a way to produce full symmetric monoidal $(\infty, 2)$ subcategory from a symmetric monoidal $(\infty, 1)$ full subcategory.

Example 3.28. Given a symmetric monoidal $(\infty, 2)$ category $\mathcal{C}^{\otimes}$, if we are given a symmetric monoidal $(\infty, 1)$ full subcategory $i : \mathcal{B}^{\otimes} \rightarrow \mathcal{C}^{\otimes}$, then we can take the full $(\infty, 2)$ subcategory $\mathcal{B}$ of $\mathcal{C}$ spanned by the full subspace $\mathcal{B}^{\simeq} \hookrightarrow \mathcal{C}^{\simeq}$, then $\mathcal{B} \hookrightarrow \mathcal{C}$ is closed under tensor product since the underlying $(\infty, 1)$ subcategory $\mathcal{B} \hookrightarrow \mathcal{C}$ is, thus the symmetric monoidal $(\infty, 1)$ category $\mathcal{B}$ is enhanced into a symmetric monoidal $(\infty, 2)$ category $\mathcal{B}^{\otimes}$.

3.3 $\theta : \text{LMod}_{\text{Cat}}(\text{Pr}^L) \rightarrow \widehat{\text{Cat}}_{(\infty,2)}$

This part follows from [33, chapter 4].

Proposition 3.29. *There exists a functor $\theta : \text{LMod}_{\text{Cat}}(\text{Pr}^L) \rightarrow \widehat{\text{Cat}}_{(\infty,2)}$ such that:*

- (1) *For any $\mathcal{M} \in \text{LMod}_{\text{Cat}}(\text{Pr}^L)$, $\theta(\mathcal{M})$ has $\mathcal{M}$ as its underlying $(\infty, 1)$ category, and for any $x, y \in \mathcal{M}$, there is an equivalence between $\text{Fun}_{\theta(\mathcal{M})}(x, y)$ and the internal Hom $\text{Mor}_{\mathcal{M}}(x, y)$ under Cat-action.*
- (2) *θ can be enhanced into a lax symmetric monoidal functor where the left hand side is equipped with the relative tensor product symmetric monoidal structure, and the right hand side is equipped with the Cartesian symmetric monoidal structure.*
- (3) *There exists a commutative diagram consisting of lax symmetric monoidal functors:*

$$\begin{array}{ccc}
 \widehat{\text{Cat}} & \xleftarrow{(-)} & \widehat{\text{Cat}}_{(\infty,2)} \\
 \uparrow i & & \uparrow \theta \\
 \text{Pr}^L & \xleftarrow{F} & \text{LMod}_{\text{Cat}}(\text{Pr}^L)
 \end{array}$$

where $(-)$ is the functor taking the underlying $(\infty, 1)$ category of an $(\infty, 2)$ category, and F is the forgetful functor (or can be regarded as the restriction of scalars $\text{LMod}_{\text{Cat}}(\text{Pr}^L) \rightarrow \text{LMod}_{\text{Spc}}(\text{Cat})$), and i is the natural embedding.

Proof. This is [33, proposition 3.2.3, corollary 3.2.14, proposition 3.2.15]. $\square$

Moreover, the above functor θ also provides us with a way to produce adjunction in $\text{Cat}_{(\infty,2)}$:

Proposition 3.30. *Let F be a morphism in $\text{LMod}_{\text{Cat}}(\text{Pr}^L)$, then the $(\infty, 2)$ functor $\theta(F)$ admits a right adjoint $\theta(F)^R$ in $\widehat{\text{Cat}}_{(\infty,2)}$.*

Proof. This is [33, proposition 4.4.1]. $\square$

Remark 3.31. • Regardless of what kind of model of $\text{Cat}_{(\infty,2)}$, the adjunction $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$ in $\text{Cat}_{(\infty,2)}$ can always be defined using the natural transformations $\eta : \text{id} \Rightarrow GF : \mathcal{C} \times [1] \rightarrow \mathcal{C}$ and $\epsilon : FG \Rightarrow \text{id} : \mathcal{D} \times [1] \Rightarrow \mathcal{D}$ satisfying the triangle identity: $F = F \circ \text{id} \xrightarrow{\eta} FGF \xrightarrow{\epsilon} \text{id} \circ F$ and $G = \text{id} \circ G \xrightarrow{\eta} GFG \xrightarrow{\epsilon} G$ are identities.

- Especially, consider the functor taking the underlying $(\infty, 1)$ category of a $(\infty, 2)$ category: $(-) : \text{Cat}_{(\infty,2)} \rightarrow \text{Cat}$, the natural transformations become $\eta : \underline{\mathcal{C}} \times [1] \rightarrow \underline{\mathcal{C}}$ and $\epsilon : \underline{\mathcal{D}} \times [1] \rightarrow \underline{\mathcal{D}}$, and the triangle identities remain the same. Thus $(-)$ preserves the adjunction, or

this can be seen by the fact that there is a symmetric monoidal functor $\text{Cat} \hookrightarrow \text{Cat}_{(\infty,2)} \xrightarrow{(-)} \text{Cat}$, thus $(-)$ is Cat linear, and preserves the adjunction.

- Thus if F is a morphism in $\mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L)$, then F^R is the underlying $(\infty, 1)$ functor of the $(\infty, 2)$ functor $\theta(F)^R$ by the uniqueness of right adjoint. This implies the right adjoint F^R can be enhanced into an $(\infty, 2)$ right adjoint.

Here is a way to produce lax symmetric monoidal $(\infty, 2)$ functor:

Example 3.32. Given a diagram consisting of symmetric monoidal colimit preserving functors between presentably symmetric monoidal ∞ -categories:

$$\begin{array}{ccc} A^\otimes & \xrightarrow{F} & C^\otimes \\ & \nwarrow \quad \nearrow & \\ & \mathbf{Cat}^\times & \end{array}$$

then it corresponds to a morphism $[1] \rightarrow \mathbf{CAlg}(\mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L)) \subset \mathbf{Fun}([1], \mathbf{CAlg}(\mathbf{Cat}))$, thus under the lax symmetric monoidal functor $\theta : \mathbf{CAlg}(\mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L)) \rightarrow \mathbf{CAlg}(\widehat{\mathbf{Cat}}_{(\infty, 2)})$, there is a symmetric monoidal $(\infty, 2)$ functor $\theta(F) : \theta(A) \rightarrow \theta(C)$ which underlying $(\infty, 1)$ functor is the symmetric monoidal functor F by applying $\mathbf{CAlg}(-)$ on the diagram in [proposition 3.30\(3\)](#).

Moreover, since F is colimit preserving, thus it admits a right adjoint between the underlying $(\infty, 1)$ functors $G : C \rightarrow A$, and G can be enhanced into an $(\infty, 2)$ functor $\bar{G} : \theta(C) \rightarrow \theta(A)$, and thus can be further enhanced into a lax symmetric monoidal $(\infty, 2)$ functor $\bar{G}^\otimes : \theta(C)^\otimes \rightarrow \theta(A)^\otimes$ as the right adjoint of a symmetric monoidal $(\infty, 2)$ functor. Especially, by the fact that the right adjoint is unique, the underlying $(\infty, 1)$ functor of $\bar{G}^\otimes : \theta(C) \rightarrow \theta(A)$ coincides with the lax symmetric monoidal structure on G as the right adjoint of a symmetric monoidal functor by [\[22, corollary 7.3.2.7\]](#).

4 $\mathbf{Span}_2(\mathcal{C}) \rightarrow \mathbf{Span}_1(\mathcal{C})\text{-mod}$

4.1 $\mathcal{C}^{op} \rightarrow \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$

In this subsection, we construct a functor $\mathcal{C}^{op} \rightarrow \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$ assigning $x \in \mathcal{C}$ with $(\mathbf{Span}_1(\mathcal{C}), \mathbf{Span}_1(\mathcal{C}_{/x}))$, where the left $\mathbf{Span}_1(\mathcal{C})$ module structure on $\mathbf{Span}_1(\mathcal{C}_{/x})$ is induced by the symmetric monoidal monoidal functor $F : \mathbf{Span}_1(\mathcal{C}) \rightarrow \mathbf{Span}_1(\mathcal{C}_{/x}), c \mapsto c \times x$ and the left

action map is induced by $act : \mathbf{Span}_1(\mathcal{C}) \times \mathbf{Span}_1(\mathcal{C}_{/x}) \rightarrow \mathbf{Span}_1(\mathcal{C}_{/x}) \times \mathbf{Span}_1(\mathcal{C}_{/x}) \xrightarrow{\otimes_{\mathbf{Span}_1(\mathcal{C}_{/x})}} \mathbf{Span}_1(\mathcal{C}_{/x})$. Moreover, we identify the symmetric monoidal structure on $\mathbf{Span}_1(\mathcal{C})$ induced from the lax symmetric monoidal functor $\mathbf{Span}_1(-) : \mathbf{Cat}^{pb} \rightarrow \mathbf{Cat}$ with the symmetric monoidal structure $\mathbf{Span}_1(\mathcal{C})^\otimes$ constructed in [\[17, definition 2.3.2\]](#).

Assume $\mathcal{C}$ admits all **pullbacks**, and denote the final object in $\mathcal{C}$ as $*$.

Remark 4.1. We will consider the Cartesian symmetric monoidal structure on $\mathcal{C}$, thus the tensor product commutes with all pullbacks: $x \times (-)$ preserves pullbacks: we can take $x = x \times_x x$, then use the fact that limit commutes with limit.

Consider arrow category $ev_1 : \mathbf{Arr}(\mathcal{C}) \rightarrow \mathcal{C}$, which is a Cartesian fibration classifying the functor

$\mathcal{C}^{op} \rightarrow \text{Cat}, x \mapsto \mathcal{C}_{/x}$ where the transition map is the pullback functor. Moreover, it preserves products since products in $\text{Arr}(\mathcal{C}) = \text{Fun}([1], \mathcal{C})$ is computed pointwise.

Step 1: a lax symmetric monoidal functor $\Psi : \mathcal{C}^{op} \rightarrow \text{Cat}, x \mapsto \mathcal{C}_{/x}$.

The functor we want is the (contravariant) straightening associated to the Cartesian fibration $ev_1 : \text{Arr}(\mathcal{C}) \rightarrow \mathcal{C}$, this functor is also the (covariant) straightening associated to the conjugate coCartesian fibration of $ev_1 : ev_1^\dagger : \text{Arr}(\mathcal{C})^\dagger \rightarrow \mathcal{C}^{op}$, which can be factored into two steps by [24, Section 04C0]:

- First we take the op , and there is a coCartesian fibration $ev_1^{op} : \text{Arr}(\mathcal{C})^{op} \rightarrow \mathcal{C}^{op}$, classifying the functor:

$$\mathcal{C}^{op} \rightarrow \text{Cat}, c \mapsto \mathcal{C}_{/c}^{op}$$

which is the composition of the functor classified by the Cartesian fibration ev_1 with the automorphism $\text{Cat} \xrightarrow{op} \text{Cat}$.

- Then we take the dual coCartesian fibration of ev_1^{op} , and there is a coCartesian fibration $ev_1^{op, \vee} : \text{Arr}(\mathcal{C})^{op, \vee} \rightarrow \mathcal{C}^{op}$ which is the composition of the functor classified by the coCartesian fibration $ev_1^{op, \vee}$ with the automorphism $\text{Cat} \xrightarrow{op} \text{Cat}$.
- In fact, we don't really construct the dual coCartesian fibration $ev_1^{op, \vee}$, instead, we just construct the op : $\text{Arr}(\mathcal{C})^{op} \rightarrow \mathcal{C}^{op}$, and consider its multiplicative structure, and show its straightening is a lax symmetric monoidal functor to Cat , then we directly compose this straightening with the automorphism $\text{Cat} \xrightarrow{op} \text{Cat}$, which corresponds to the dual coCartesian fibration of ev_1^{op} , and recovers the functor $\mathcal{C}^{op} \rightarrow \text{Cat}$ classified by the Cartesian fibration $ev_1 : \text{Arr}(\mathcal{C}) \rightarrow \mathcal{C}$, and especially, we get a lax symmetric monoidal structure on this functor.

Now consider the op : $ev_1^{op} : \text{Arr}(\mathcal{C})^{op} \rightarrow \mathcal{C}^{op}$, which is a coCartesian fibration sending $x \in \mathcal{C}$ to $(\mathcal{C}_{/x})^{op}$, moreover, it preserves coproducts, so we have a symmetric monoidal ∞ -functor $\text{Arr}^{op, \Pi} \rightarrow \mathcal{C}^{op, \Pi}$, and over each fiber $\langle n \rangle$, the induced functor is a coCartesian fibration, moreover, we have:

Lemma 4.2. *The above functor $\text{Arr}(\mathcal{C})^{op, \Pi} \rightarrow \mathcal{C}^{op, \Pi}$ is a coCartesian fibration of coCartesian symmetric monoidal ∞ categories.*

Proof. • Since both $\mathcal{C}$ and $\text{Arr}(\mathcal{C})$ admit finite products, and $ev_1 : \text{Arr}(\mathcal{C}) \rightarrow \mathcal{C}$ preserves finite products, thus $\mathcal{C}^{op}$ and $\text{Arr}(\mathcal{C})^{op}$ admit coproducts, and by [22, remark 2.4.3.4], $\mathcal{C}^{op, \Pi}$ and $\text{Arr}(\mathcal{C})^{op, \Pi}$ are symmetric monoidal coCartesian ∞ categories and the coproduct preserving functor induces a symmetric monoidal ∞ -functor.

- And the tensor product in these opposite categories (which is just coproduct) can be computed as product back in the original categories.
- By [27, lemma 1.10], to show the ∞ -operad map is again coCartesian fibration, it suffices to show the fiberwise coCartesian lifting is preserved under the operations in $\text{Comm}^\otimes$.

- Given an operation $\alpha : \langle m \rangle \rightarrow \langle n \rangle$, if α is inert, then α_i is just projection, and coCartesian lifting is preserved.

If α is active, first assume that $n = 1$, then consider the diagram:

$$\begin{array}{ccc} \text{Arr}^{op, \Pi}(\mathcal{C})_{\langle m \rangle} & \xrightarrow{\alpha_i} & \text{Arr}^{op, \Pi}(\mathcal{C})_{\langle 1 \rangle} \\ \downarrow & & \downarrow \\ \mathcal{C}_{\langle m \rangle}^{op, \Pi} & \xrightarrow{\alpha_i} & \mathcal{C}_{\langle 1 \rangle}^{op, \Pi} \end{array}$$

then given a morphism $e : (x \leftarrow y)^{op} = ((x_1, \dots, x_m) \leftarrow (y_1, \dots, y_m))^{op}$ in $\mathcal{C}_{\langle m \rangle}^{op, \Pi}$, and a point over x in $\text{Arr}(\mathcal{C})^{op}$ is in the form of $(c_1 \rightarrow x_1, \dots, c_m \rightarrow x_m)$, and the coCartesian lifting over e is $(c_1 \times_{x_1} y_1 \rightarrow y_1, \dots, c_n \times_{x_n} y_n \rightarrow y_n)$.

- Noticing that limit commutes with limit, we have a pullback diagram in the opposite category of $\mathcal{C}_{\langle 1 \rangle}^{op, \Pi}$:

$$\begin{array}{ccc} c_1 \times \dots \times c_n & \xleftarrow{\quad} & \prod_{i=1}^n c_i \times_{x_i} y_i \\ \downarrow & & \downarrow \\ x_1 \times \dots \times x_n & \xleftarrow{\prod e} & y_1 \times \dots \times y_n \end{array}$$

and this implies that the n -ary tensor product $\prod_{i=1}^n c_i \times_{x_i} y_i$ is the coCartesian lifting over $\prod(e)^{op} : x_1 \times \dots \times x_n \rightarrow y_1 \times \dots \times y_n$.

- As for the general case of an active map $\alpha : \langle m \rangle \rightarrow \langle n \rangle$, with source and target written as m -ary products and n -ary products, α_i can be described as a combination of the projection and an i -ary tensor operation for some $0 \leq i \leq m$, and coCartesian lifting in the product is componentwise coCartesian lifting, thus it also preserves coCartesian lifting. □

By [27, theorem 2.1], the above symmetric monoidal coCartesian fibration under straightening gives rise to a lax symmetric monoidal functor:

$$\Phi : \mathcal{C}^{op, \Pi} \rightarrow \text{Cat}^\times$$

However, to get the functor $\mathcal{C}^{op} \rightarrow \text{Cat}$ assigning $c \in \mathcal{C}^{op}$ with $\mathcal{C}_{/c}$, we need to take a further opposite: recall that there is an auto-equivalence of Cat :

$$(-)^{op} : \text{Cat} \rightarrow \text{Cat}$$

sending a category to its opposite and a functor $F : C \rightarrow D$ to $F^{op} : C^{op} \rightarrow D^{op}$. And especially, $(-)^{op}$ preserves limits, thus induces an equivalence between ∞ -operads:

$$\text{Cat}^\times \xrightarrow{(-)^{op}} \text{Cat}^\times$$

Thus we have the composition of ∞ -operad maps:

$$\Psi : \mathcal{C}^{op, \Pi} \rightarrow \text{Cat}^{\times} \xrightarrow{(-)^{op}} \text{Cat}^{\times}, x \mapsto (\mathcal{C}_{/x})^{op} \mapsto (\mathcal{C}_{/x})^{\otimes}$$

which gives rise to a functor:

$$\text{CAlg}(\Psi) : \mathcal{C}^{op} \cong \text{CAlg}(\mathcal{C}^{op}) \rightarrow \text{CAlg}(\text{Cat}), x \mapsto (\mathcal{C}_{/x})^{\otimes}, x \mapsto (\mathcal{C}_{/x})^{\otimes}$$

In the following remarks, we show the symmetric monoidal structure on $(\mathcal{C}_{/x})^{\otimes}$ is Cartesian.

Remark 4.3. We can at the beginning take the conjugate (or dual) fibration of the Cartesian fibration $\text{Arr}(\mathcal{C}) \xrightarrow{ev_1} \mathcal{C}$ instead of taking opposite fibration, where the former is a coCartesian fibration classifying the same functor as the corresponding Cartesian fibration as shown in [5, section 2]. However, we don't know how to show the dual fibration admits coproducts and the fibration preserves it.

Remark 4.4. Especially, by [27, lemma 3.12], we have the following description on the symmetric monoidal structure on the image of $x \in \mathcal{C}^{op} \cong \text{CAlg}(\mathcal{C}^{op})$ under the lax symmetric functor from the straightening of $\text{Arr}(\mathcal{C})^{op} \rightarrow \mathcal{C}^{op}$: consider the ∞ operad map corresponding to x : $\text{Comm}^{\otimes} \rightarrow \mathcal{C}^{op, \Pi}$, then the pullback along x is again a coCartesian fibration, corresponding to a symmetric monoidal structure on $\mathcal{C}_{/x}$:

$$\begin{array}{ccc} ((\mathcal{C}_{/x})^{op})^{\otimes} & \longrightarrow & \text{Arr}(\mathcal{C})^{op, \Pi} \\ \downarrow \lrcorner & & \downarrow \\ \text{Comm}^{\otimes} & \xrightarrow{x} & \mathcal{C}^{op, \Pi} \end{array}$$

And as a property to check, we can see from that the symmetric monoidal structure on $(\mathcal{C}_{/x})^{op}$ is the coCartesian symmetric monoidal structure. Or we can show it by noticing that the pullback of coCartesian symmetric monoidal ∞ -categories is again coCartesian symmetric monoidal:

- Recall the construction of the coCartesian symmetric monoidal structure in [22, Construction 2.4.3.1], given an ∞ category C , C^{Π} is defined as

$$\text{Hom}_{\text{N}(\text{Fin})_*}(K, C^{\Pi}) \cong \text{Hom}_{\text{sSet}}(K \times_{\text{N}(\text{Fin})_*} \Gamma^*, C)$$

Thus given the fiber product $C \times_A B$ in Cat , there is pullback in $\text{Cat}_{/\text{Comm}^{\otimes}}: C^{\Pi} \times_{A^{\Pi}} B^{\Pi} \cong (C \times_A B)^{\Pi}$, thus the pullback of coCartesian symmetric monoidal ∞ -categories is again coCartesian symmetric monoidal.

- Consider the following pullback diagram in Cat :

$$\begin{array}{ccc} (\mathcal{C}_{/x})^{op} & \longrightarrow & \text{Arr}(\mathcal{C})^{op} \\ \downarrow \lrcorner & & \downarrow \\ * & \xrightarrow{\{x\}} & \mathcal{C}^{op} \end{array}$$

- By taking $(-)^{\Pi}$, we can see the induced symmetric monoidal structure on $(\mathcal{C}_{/x})^{op}$ is again coCartesian symmetric monoidal.

Thus we know the tensor product structure on $(\mathcal{C}_{/x})^{op, \otimes}$ just corresponds to the product in $\mathcal{C}_{/x}$.

Remark 4.5. And after composition with $(-)^{op}$, the functor $\mathcal{C}^{op} \cong \text{CAlg}(\mathcal{C}^{op}) \rightarrow \text{CAlg}(\text{Cat}), x \mapsto (\mathcal{C}_{/x})^{op, \otimes}$ becomes:

$$\mathcal{C}^{op} \cong \text{CAlg}(\mathcal{C}^{op}) \rightarrow \text{CAlg}(\text{Cat}) \xrightarrow{(-)^{op}} \text{CAlg}(\text{Cat})$$

And by [22, remark 2.4.3.7], the last morphism in the above composition corresponds to the operation sending a symmetric monoidal category to its opposite symmetric monoidal category.

And obviously, the opposite of the coCartesian symmetric monoidal category C is a Cartesian symmetric monoidal category:

- Noticing that by restriction to the map $[1] \rightarrow \text{Comm}^{\otimes}$ corresponding to the active map $\langle 2 \rangle$, we have the tensor product functor $\otimes : C \times C \rightarrow C$.
- The functor $(-)^{op}$ induces the following correspondence:

$$\begin{array}{ccc}
 (x, 1) & & x \amalg 1 \\
 \downarrow & \lrcorner & \downarrow \\
 (1, y) \longrightarrow (x, y) & & 1 \amalg y \longrightarrow x \amalg y \\
 \\
 (x, 1) & & x \otimes 1 \\
 \uparrow & \llcorner & \uparrow \\
 (1, y) \longleftarrow (x, y) & & * \otimes 1 \longleftarrow x \otimes y
 \end{array}$$

where the right upper corner diagram is a coproduct since it's coCartesian, thus the image under $(-)^{op}$ is a product.

- And obviously the unit object changes from initial object to final object, thus after taking $(-)^{op} : \text{CAlg}(\text{Cat}) \rightarrow \text{CAlg}(\text{Cat})$ sends a coCartesian symmetric monoidal category to Cartesian symmetric monoidal category.

Remark 4.6. Thus we denote the image of $x \in \mathcal{C}$ under the functor $\text{CAlg}(\Phi) : \mathcal{C}^{op} \cong \text{CAlg}(\mathcal{C}^{op}) \rightarrow \text{CAlg}(\text{Cat}), x \mapsto (\mathcal{C}_{/x})^{\otimes}$ as $\mathcal{C}_{/x}^{\times}$, and given a morphism $f : x \rightarrow y$ in $\mathcal{C}$, the image of f^{op} gives rise to a symmetric monoidal functor $f^* : \mathcal{C}_{/y}^{\times} \rightarrow \mathcal{C}_{/x}^{\times}$, which underlying functor is just the product-preserving functor $\mathcal{C}_{/y} \rightarrow \mathcal{C}_{/x}, z \mapsto z \times_y x$ (classified by the Cartesian fibration of $\text{Arr}(\mathcal{C}) \rightarrow \mathcal{C}$).

Step 2: a lax symmetric monoidal functor $\psi : \mathcal{C}^{op} \rightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$.

And we take several steps to construct ψ :

- First we show the lax symmetric monoidal functor $\Psi : \mathcal{C}^{op} \rightarrow \text{Cat}$ factors through the (non-full) symmetric monoidal subcategory $\text{Cat}^{pb} \hookrightarrow \text{Cat}$.
- And then consider the lax symmetric monoidal functor $\text{Span}_1(-) : \text{Cat}^{pb} \rightarrow \text{Cat}$ developed in [17, proposition 2.2.9 and lemma 2.2.11], by composition with Ψ , there is a lax symmetric monoidal functor $\mathcal{C}^{op} \rightarrow \text{Cat}, x \mapsto \text{Span}_1(\mathcal{C}/_x)$.
- And from above there is a functor $\mathcal{C}^{op} \rightarrow \text{CAlg}(\text{Cat})_{\text{Span}_1(\mathcal{C})} \cong \text{CAlg}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}))$ by [22, corollary 3.4.1.7 and corollary 5.4.1.10]. And especially, there is an ∞ -operad map $\mathcal{C}^{op, \Pi} \cong \text{CAlg}(\mathcal{C}^{op})^\Pi \rightarrow \text{CAlg}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}))^\Pi \xrightarrow{ev_{(1)}} \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^\otimes$.

Notation 4.7. Consider the subcategory of Cat consisting of category with pullbacks and functors between them preserving pullbacks, denoted as Cat^{pb} , then the inclusion $\text{Cat}^{pb} \hookrightarrow \text{Cat}$ preserves products, thus there is a symmetric monoidal functor $\text{Cat}^{pb, \times} \rightarrow \text{Cat}^\times$.

Lemma 4.8. *The functor induced above $\text{Cat}^{pb, \times} \rightarrow \text{Cat}^\times$ is a subcategory inclusion.*

Proof. Since $\text{Cat}^{pb} \rightarrow \text{Cat}$ is a subcategory inclusion, thus $\text{Cat}^{pb, \times} \rightarrow \text{Cat}^\times$ is fiberwise subcategory inclusion. If we regard them as Comm -monoids in $\widehat{\text{Cat}}$, then the above ∞ -operad map can be regarded as a map $[1] \rightarrow \text{Mon}_{\text{Comm}}(\widehat{\text{Cat}}) \cong \text{Cat}^\otimes \subset \text{Fun}(\text{Comm}^\otimes, \widehat{\text{Cat}}) \cong \text{coC}_{/\text{Comm}}^\otimes$ where $\text{Cat}^\otimes \subset \text{Op}_\infty$ denotes the (non-full) subcategory consisting of symmetric monoidal ∞ -categories and symmetric monoidal ∞ -functors.

To show that $\text{Cat}^{pb, \times} \rightarrow \text{Cat}^\times$ is a subcategory inclusion, by [24, Tag 04W5], it suffices to show the following commutative diagram is a pullback diagram in $\widehat{\text{Cat}}$:

$$\begin{array}{ccc} \text{Cat}^{pb, \times} & \xrightarrow{\text{id}} & \text{Cat}^{pb, \times} \\ \text{id} \downarrow \lrcorner & & \downarrow \\ \text{Cat}^{pb, \times} & \longrightarrow & \text{Cat}^\times \end{array}$$

And such a pullback diagram can be checked in the slice category $\text{Cat}_{/\text{Comm}^\otimes}$, and also can be checked in $\text{coC}_{/\text{Comm}^\otimes}$, thus it suffices to show the above diagram is pullback in $\text{coC}_{/\text{Comm}^\otimes} \cong \text{Fun}(\text{Comm}^\otimes, \widehat{\text{Cat}})$, which is true: since fiberwise $\text{Cat}^{pb} \rightarrow \text{Cat}$ is a subcategory inclusion, so the following diagram is pullback in $\widehat{\text{Cat}} \cong \text{Fun}(\{n\}, \widehat{\text{Cat}})$:

$$\begin{array}{ccc} \text{Cat}^{pb, \times n} & \xrightarrow{\text{id}} & \text{Cat}^{pb, \times n} \\ \text{id} \downarrow \lrcorner & & \downarrow \\ \text{Cat}^{pb, \times n} & \longrightarrow & \text{Cat}^{\times n} \end{array}$$

□

Especially, as a subcategory, $\text{Cat}^{pb, \times} \subset \text{Cat}^\times$ is spanned by the objects in the form $(X_1, \dots, X_n)$ where $X_i \in \text{Cat}^{pb}$, and within each fiber the morphisms which are componentwise pullback

preserving, and contains all the coCartesian liftings between these objects from $\text{Cat}^{pb,\times}$ inside $\text{Cat}^\times$ since $\text{Cat}^{pb,\times} \rightarrow \text{Cat}^\times$ is coCartesian preserving.

Given $f : x \rightarrow y \in \mathcal{C}$, the induced pullback functor $\mathcal{C}_{/y} \rightarrow \mathcal{C}_{/x}$ is pullback preserving: for any $z, w \in \mathcal{C}_{/y}$,

$$(z \times_y x) \times_x (w \times_y x) \cong (z \times_y w) \times_y x \cong z \times_y w \times_y x$$

Thus the underlying functor $\mathcal{C}^{op} \rightarrow \text{Cat}$ factors through the (non)-full subcategory $\text{Cat}^{pb} \rightarrow \text{Cat}$.

Lemma 4.9. *The functor $\Psi : \mathcal{C}^{op,\Pi} \rightarrow \text{Cat}^\times$ factors through the (non) full subcategory $\text{Cat}^{pb,\times}$, and gives rise to an ∞ -operad map $\mathcal{C}^{op,\Pi} \rightarrow \text{Cat}^{pb,\times}$.*

Proof. • Fiberwise, $\mathcal{C}_{\langle n \rangle}^{op,\Pi} \cong \mathcal{C}^{op,\times n} \rightarrow \text{Cat}_{\langle n \rangle}^\times \cong \text{Cat}^{\times n}$ factors through $\text{Cat}^{pb,\times n}$.

- It suffices to show each coCartesian edge e in $\mathcal{C}^{op,\Pi}$ lands in $\text{Cat}^{pb,\times} \rightarrow \text{Cat}^\times$: since edges between different fibers can be factored as a coCartesian edge and a morphism within the same fiber.
- If e is inert, then it's just projection, and its image under the operad map $\mathcal{C}^{op,\Pi} \rightarrow \text{Cat}^\times$ lands in $\text{Cat}^{pb,\times} \hookrightarrow \text{Cat}^\times$.
- If e is coCartesian lifting over an active map, then it suffices to assume that $e : (c_1, \dots, c_n) \rightarrow c_1 \times \dots \times c_n$ lies over $\langle n \rangle \rightarrow \langle 1 \rangle$ and corresponds to the n -ary tensor product.
- In the above case, the image of e under the functor $\mathcal{C}^{op,\Pi} \rightarrow \text{Cat}^{\times n}$ can be factored as:

$$(\mathcal{C}_{/c_1}, \dots, \mathcal{C}_{/c_n}) \rightarrow \mathcal{C}_{/c_1} \times \dots \times \mathcal{C}_{/c_n} \rightarrow \mathcal{C}_{/c_1 \times \dots \times c_n}, (x_1 \rightarrow c_1, \dots, x_n \rightarrow c_n) \mapsto x_1 \times \dots \times x_n \rightarrow c_1 \times \dots \times c_n$$

where the first edge lands in $\text{Cat}^{pb,\times}$ since $\text{Cat}^{pb,\times} \rightarrow \text{Cat}^\times$ preserves coCartesian edges, and the second edge is pullback preserving since limit commutes with limit, thus also lands in $\text{Cat}^{pb,\times}$.

- Thus $\mathcal{C}^{op,\Pi} \rightarrow \text{Cat}^\times$ factors through $\text{Cat}^{pb,\times}$ and gives rises to an ∞ -operad map

$$\mathcal{C}^{op,\Pi} \rightarrow \text{Cat}^{pb,\times}$$

□

Recall in [17, chapter 2], there is a subcategory of $\text{Fun}([1], \text{Cat})$ called GeomSetup consisting of

- objects: $C_0 \rightarrow C$ is a wide subcategory and C_0 admits pullbacks along morphisms in C and the inclusion $C_0 \rightarrow C$ preserves fiber product.
- morphisms: $(f, f_0) : (C, C_0) \rightarrow (D, D_0)$ such that f_0 preserves pullback in the form $x \rightarrow y \leftarrow z$ where the arrow $x \rightarrow y$ lies in C_0 .

Induced by $[1] \rightarrow [0]$, there is a functor $\text{Cat}^{pb} \rightarrow \text{GeomSetup}$, $C \mapsto (C, C)$, which obviously preserves products. And there is a functor assigning each geometric set up with a span category in [17, section[2.2]:

$$\text{Span}_1(-) : \text{GeomSetup} \rightarrow \text{Cat}$$

which is limit-preserving by [17, lemma 2.2.11].

Now by precomposition, consider the product preserving functor $\text{Span}_1(-) : \text{Cat}^{pb} \rightarrow \text{Cat}$, which gives rise to a symmetric monoidal ∞ -functor $\text{Span}_1(-) : \text{Cat}^{pb, \times} \rightarrow \text{Cat}^\times$, and there is an ∞ -operad map by composition with Ψ :

$$\Phi : \mathcal{C}^{op, \Pi} \xrightarrow{\Psi} \text{Cat}^{pb, \times} \xrightarrow{\text{Span}_1(-)} \text{Cat}^\times$$

And apply the construction $\text{Alg}_{\text{Comm}}(-)^\otimes$ developed in [22, chapter 3.2.4], we have

$$\text{CAlg}(\mathcal{C}^{op, \Pi}) \xrightarrow{\text{CAlg}(\Phi)} \text{CAlg}(\text{Cat})^\Pi$$

here we use the fact that the ∞ -operad structure on $\text{Alg}_{\text{Comm}}(-)^\otimes$ is coCartesian by [22, theorem 3.4.2.7].

We take the underlying category, then there is a functor $\text{CAlg}(\Phi) : \mathcal{C}^{op} \cong \text{CAlg}(\mathcal{C}^{op}) \rightarrow \text{CAlg}(\text{Cat})$ where the first equivalence is by [22, proposition 2.4.3.8]. Since $*$ is the initial object in $\mathcal{C}^{op}$, there is a factorization to the slice category

$$\psi : \mathcal{C}^{op} \rightarrow \text{CAlg}(\text{Cat})_{\text{Span}_1(\mathcal{C})/} \cong \text{CAlg}(\text{Mod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})) \cong \text{CAlg}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}))$$

, where the last equivalence is by [22, corollary 3.4.1.7, corollary 5.1.4.11].

Since all the $\text{CAlg}(-)$ s above can be equipped with a coCartesian symmetric monoidal structure, and by the equivalence $\text{Alg}_C(D) \cong \text{Fun}(C, D)$ for coCartesian symmetric monoidal categories C, D by [22, proposition 2.4.3.8], the above functor ψ corresponds to an ∞ -operad map:

$$\psi : \mathcal{C}^{op, \Pi} \rightarrow \text{CAlg}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}))^{\otimes = \Pi} \xrightarrow{e_{V(1)}} \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^\otimes$$

Remark 4.10. Here are some descriptions on the left module structure on $\psi(x) = (\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/x}))$:

- By the previous description on the symmetric monoidal structure on $\mathcal{C}_{/x}$, which is the Cartesian symmetric monoidal structure, and the symmetric monoidal functor $F : \mathcal{C}_{/*} \cong \mathcal{C} \rightarrow \mathcal{C}_{/x}$ is induced by the pullback functor along $x \rightarrow *$.
- After applying the symmetric monoidal functor $\text{Span}_1(-)$, there is a symmetric monoidal functor induced by F

$$\text{Span}_1(F) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x}), c \mapsto c \times x$$

- And this symmetric monoidal functor gives rise to a left $\text{Span}_1(\mathcal{C})$ module structure on $\text{Span}_1(\mathcal{C}_{/x})$, and the concrete $\text{Span}_1(\mathcal{C})$ action on $\text{Span}_1(\mathcal{C}_{/x})$ can be described as:

$$\begin{aligned} \text{Span}_1(\mathcal{C}) \times \text{Span}_1(\mathcal{C}_{/x}) &\xrightarrow{\text{Span}_1(F) \times \text{id}} \text{Span}_1(\mathcal{C}_{/x}) \times \text{Span}_1(\mathcal{C}_{/x}) \xrightarrow{\otimes_{\text{Span}_1(\mathcal{C}_{/x})}} \\ \text{Span}_1(\mathcal{C}_{/x}), (c, d) &\mapsto F(c) \times d \cong c = (c \times x) \times_x d \cong c \times d \in \mathcal{C}_{/x} \end{aligned}$$

Step 3: identify the symmetric monoidal structures on $\text{Span}_1(\mathcal{C})$

The tensor product of the symmetric monoidal category $\mathcal{C}$ is compatible with the pullback, thus $\mathcal{C}$ is also a commutative algebra object in Cat^{pb} , through the lax symmetric monoidal functor $\text{Span}_1(-) : \text{Cat}^{pb} \rightarrow \text{Cat}$, there is a symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$.

On the other hand, in [17, section 2.3], there is a symmetric monoidal structure constructed on a geometric set up (C, C_0) :

$$\text{Span}_1(C, C_0)^\otimes := \text{Span}_1(C^{op, \Pi})^{op}, (C_0^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \rightarrow \text{Comm}^\otimes$$

which is an ∞ -operad, and is symmetric monoidal ∞ -category when C admits finite products. And by [17, example 2.3.6], when a morphism between geometric set up $(f, f_0) : (C, C_0) \rightarrow (D, D_0)$ satisfies that f preserves finite products, then the induced morphism $\text{Span}_1(C, C_0)^\otimes \rightarrow \text{Span}_1(D, D_0)^\otimes$ is symmetric monoidal.

Remark 4.11. • In [17, proposition 2.3.5], the functor $\text{Span}_1(-)^\otimes : \text{GeomSetup} \rightarrow \text{Op}$ is constructed in the following way:

$$\begin{array}{ccc} \text{GeomSetup} & \xrightarrow{C \mapsto ((C^{op, \Pi})^{op})} & \text{GeomSetup}_{/(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, op})} \\ & & \downarrow \text{pullback along } \text{Comm}^\otimes \rightarrow \text{Span}_1(\text{Comm}^{\otimes, op}) \\ & & \text{Cat}_{/\text{Span}_1(\text{Comm}^{\otimes, op})} \xrightarrow{\text{Span}_1(-)} \text{Cat}_{/\text{Comm}^\otimes} \end{array}$$

- Notice that when $C \rightarrow D$ preserves pullback, given $e : d \rightarrow c \in C$, there is a commutative diagram:

$$\begin{array}{ccc} C/c & \longrightarrow & D/F(c) \\ e^* \downarrow & & \downarrow F(e)^* \\ C/d & \longrightarrow & D/F(d) \end{array}$$

- And we apply the above observation to the pullback-preserving functor $\text{Span}_1(-)$, and $e : (\langle 1 \rangle, \langle 1 \rangle) \rightarrow (\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, op})$, then there is a commutative diagram:

$$\begin{array}{ccccc} \text{GeomSetup} & & \text{Span}_1(-)^\otimes & & \\ \downarrow (-)^{op, \Pi, op} & & \searrow & & \\ \text{GeomSetup}_{/(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, op})} & \xrightarrow{\text{Span}_1(-)} & \text{Cat}_{/\text{Span}_1(\text{Comm}^{\otimes, op})} & \longrightarrow & \text{Op} \subset \text{Cat}_{/\text{Comm}^\otimes} \\ \downarrow e^* & & \downarrow \text{Span}_1(e)^* & & \nearrow \\ \text{GeomSetup}_{/*=\langle \{1\}, \{1\} \rangle} & \xrightarrow{\text{Span}_1(-)} & \text{Cat}_{/*=\{\{1\}\}} & & \end{array}$$

- Thus we have a commutative diagram:

$$\begin{array}{ccccc}
\text{Cat}^{pb} & \xrightarrow{\text{Span}(-)^\otimes} & \text{Cat}^\otimes & \hookrightarrow & \text{Op}^{HA} \\
& \searrow \text{Span}(-) & & \searrow & \downarrow G \\
& & & & \text{Cat}
\end{array}$$

where:

- G is induced by pullback along $* \cong \{ \langle 1 \rangle \} \rightarrow \text{Comm}^\otimes$, and sends an ∞ -operad to its underlying category.
- Moreover, after precomposition with $\text{Cat}^{pb} \rightarrow \text{GeomSetup}$, the functor assigning the symmetric monoidal structure $\text{Span}_1(-)^\otimes$:

$$\text{Span}(-)^\otimes : \text{GeomSetup} \rightarrow \text{Op}^{HA}$$

factors through the non full subcategory of Op^{HA} consisting of symmetric monoidal ∞ -categories and symmetric monoidal ∞ -functors between them.

- Thus, there is a factorization

$$\text{Span}_1(-)^\otimes : \text{Cat}^{pb} \rightarrow \text{Cat}^\otimes \subset \text{Op}^{HA}$$

.

Moreover there is:

Lemma 4.12. *The above functor $\text{Cat}^{pb} \rightarrow \text{GeomSetup} \xrightarrow{\text{Span}_1(-)^\otimes} \text{Cat}^\otimes$, $C \mapsto (C, C) \mapsto \text{Span}_1(C)^\otimes$ preserves products.*

Proof. • Recall the construction of the coCartesian symmetric monoidal structure in [22, Construction 2.4.3.1], given an ∞ category C , C^Π is defined as

$$\text{Hom}_{\text{N}(\text{Fin})_*}(K, C^\Pi) \cong \text{Hom}_{\text{sSet}}(K \times_{\text{N}(\text{Fin})_*} \Gamma^*, C)$$

and it's obvious that for $C \times D$, the representator in the left hand side should be $C^\Pi \times D^\Pi$, which implies $(C \times D)^\Pi \cong (C^\Pi \times_{\text{Comm}^\otimes} D^\Pi)$, where the second product takes place in $\text{Cat}_{/\text{N}(\text{Fin})_*}$, so it's the fiber product over $\text{N}(\text{Fin})_*$.

- So there are equivalences in Cat :

$$((C \times D)^{op, \Pi})^{op} \cong (C^{op, \Pi} \times_{\text{Comm}^\otimes} D^{op, \Pi})^{op} \cong (C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} (D^{op, \Pi})^{op}$$

and so is

$$((C \times D)^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \cong ((C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} (D^{op, \Pi})^{op}) \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}$$

and

$$\begin{aligned}
& ((C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} (D^{op, \Pi})^{op}) \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \cong \\
& ((C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) \times_{\text{Comm}^{\otimes, op}} ((D^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq})
\end{aligned}$$

by the fact that $\text{Comm}^{\otimes, \simeq} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq} \cong \text{Comm}^{\otimes, \simeq}$ and limit commutes with limit.

- Thus we have a fiber product in GeomSetup :

$$\begin{array}{ccc}
 (((C \times D)^{op, \Pi})^{op}, ((C \times D)^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) & \longrightarrow & ((C^{op, \Pi})^{op}, (C^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) \\
 \downarrow \lrcorner & & \downarrow \\
 ((D^{op, \Pi})^{op}, (D^{op, \Pi})^{op} \times_{\text{Comm}^{\otimes, op}} \text{Comm}^{\otimes, \simeq}) & \longrightarrow & (\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, \simeq})
 \end{array}$$

•

- Since $\text{Span}_1(-) : \text{GeomSetup} \rightarrow \text{Cat}$ preserves finite limits, thus there is a pullback diagram in Cat :

$$\begin{array}{ccc}
 \text{Span}_1(C \times D)^{\otimes} & \longrightarrow & \text{Span}_1(C)^{\otimes} \\
 \downarrow \lrcorner & & \downarrow \\
 \text{Span}_1(D)^{\otimes} & \longrightarrow & \text{Comm}^{\otimes}
 \end{array}$$

here we identify $\text{Span}_1(\text{Comm}^{\otimes, op}, \text{Comm}^{\otimes, \simeq}) \cong \text{Comm}^{\otimes}$ by [17, example 2.2.13].

- This implies we have a product in $\text{Cat}^{\otimes} \subset \text{Op}_{\infty}$:

$$\text{Span}_1(C \times D)^{\otimes} \cong \text{Span}_1(C)^{\otimes} \times_{\text{Comm}^{\otimes}} \text{Span}_1(D)^{\otimes}$$

- Thus $\text{Span}_1(-)^{\otimes} : \text{Cat}^{pb} \rightarrow \text{Cat}^{\otimes}$ preserves products.

□

Thus $\text{Cat}^{pb} \xrightarrow{\text{Span}_1(-)^{\otimes}} \text{Cat}^{\otimes}$ preserves products, and so is the functor G by [22, remark 2.1.4.10], thus there is a commutative diagram:

$$\begin{array}{ccc}
 \text{Cat}^{pb, \times} & \xrightarrow{\text{Span}_1(-)^{\otimes}} & (\text{Cat}^{\otimes})^{\times} \cong \text{Mon}_{\text{Comm}}(\text{Cat})^{\times} \\
 & \searrow \text{Span}_1(-) & \downarrow G \\
 & & \text{Cat}^{\times}
 \end{array}$$

which induces the following commutative diagram:

$$\begin{array}{ccccc}
 \text{Cat}^{pb} & & \xrightarrow{\text{Span}_1(-)^{\otimes}} & & \\
 \uparrow ev_{(1)} & & & & \\
 \text{CAlg}(\text{Cat}^{pb}) & \xrightarrow{\text{CAlg}(\text{Span}_1(-)^{\otimes})} & \text{CAlg}(\text{Mon}_{\text{Comm}}(\text{Cat})) \cong \text{CAlg}(\text{CAlg}(\text{Cat})) & \xrightarrow{ev_{(1)}} & \text{Mon}_{\text{Comm}}(\text{Cat}) \\
 & \searrow \text{CAlg}(\text{Span}_1(-)) & \downarrow \cong, \text{CAlg}(G) & & \downarrow G \\
 & & \text{CAlg}(\text{Cat}) & \xrightarrow{ev_{(1)}} & \text{Cat}
 \end{array}$$

Thus, given a commutative algebra object $\mathcal{C}$ in Cat^{pb} , then the induced commutative algebra object $* \xrightarrow{\mathcal{C}} \text{CAlg}(\text{Cat}^{pb}) \xrightarrow{\text{Span}_1(-)} \text{CAlg}(\text{Cat})$ corresponds to one symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$, on the other hand, forget the commutative algebra structure on $\mathcal{C}$, and regarded as an object in Cat^{pb} , by the construction $\text{Span}_1(-)^\otimes$, there is another symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$, these symmetric monoidal structures coincide since $ev_{\langle 1 \rangle} : \text{CAlg}(\text{CAlg}(\text{Cat})) \rightarrow \text{CAlg}(\text{Cat})$ is an equivalence.

From now on, we denote this symmetric monoidal structure on $\text{Span}_1(\mathcal{C})$ as $\text{Span}_1(\mathcal{C})^\otimes$, and recall the six functor formalism, which is a lax symmetric monoidal ∞ -functor:

$$(*) \quad \text{Shv} : \text{Span}_1(\mathcal{C})^\otimes \rightarrow \text{Cat}^\times$$

Especially, consider the lax symmetric monoidal functor $\mathcal{C}^{op} \xrightarrow{\Psi} \text{Cat}^{pb} \xrightarrow{\text{Span}_1(-)} \text{Cat}$, after applying $\text{CAlg}(-)$, it sends the unit $*$ to the symmetric monoidal category $\mathcal{C}^\times$ (which is Cartesian by [remarks 4.4](#)), and then sends $\mathcal{C}^\times$ to $\text{Span}_1(\mathcal{C})^\otimes$, which corresponds to the symmetric monoidal structure on $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^\otimes$:

$$\begin{array}{ccc} \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^\otimes & \longrightarrow & \text{LMod}(\text{Cat})^\times \\ \downarrow & \lrcorner & \downarrow \\ \text{Comm}^\otimes & \xrightarrow{\{\text{Span}_1(\mathcal{C})^\otimes\}} & \text{Mon}^\times \end{array}$$

4.2 biadjointness

This part follows [7, shaper 2].

Definition 4.13. Let $\mathbb{D}$ be an $(\infty, 2)$ -category, consider the following commutative diagram in $\mathbb{D}$:

$$\begin{array}{ccc} X' & \xrightarrow{h^*} & X \\ f^* \downarrow & & \downarrow k^* \\ Y' & \xrightarrow{g^*} & Y \end{array}$$

- (1) The square is vertically left adjointable if f^* and k^* admit left adjoints $f_\#$ and $k_\#$ respectively, and the Beck-Chevalley map:

$$k_\# \circ g^* \xrightarrow{\eta} k_\# \circ g^* \circ f^* \circ f_\# \rightarrow k_\# \circ k^* \circ h^* \circ f_\# \xrightarrow{\epsilon} h^* \circ f_\#$$

is invertible.

- (2) Dually, we say the square is vertically right adjointable if f^* and k^* admit right adjoints f_* and k_* respectively, and the Beck-Chevalley map: $h^* \circ f_* \rightarrow k_* \circ g^*$ is invertible.
- (3) The notion of horizontally left / right adjointable are defined analogously.

- (4) The square is biadjointable if it is horizontally left adjointable, and the resulting square:

$$\begin{array}{ccc} X' & \xleftarrow{h_{\sharp}} & X \\ f^* \downarrow & & \downarrow k^* \\ Y' & \xleftarrow{g_{\sharp}} & Y \end{array}$$

is vertically right adjointable, we refer the associated Beck-Chevalley map $h_{\sharp} \circ k_* \rightarrow f_* \circ g_{\sharp}$ as double Beck-Chevalley map.

Definition 4.14. Let C be a category, and $I, P \subset C$ be two wide subcategories closed under base change, then consider a functor $F : C^{op} \rightarrow \mathbb{D}$, for a morphism $f \in C$, we write f^* as $F(f^{op})$,

- (1) We say F is left I adjointable if for every pullback square:

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ f \downarrow & & \downarrow k \\ Y' & \xrightarrow{g} & Y \end{array}$$

where $g \in I$, the associated square:

$$\begin{array}{ccc} F(Y) & \xrightarrow{g^*} & F(Y') \\ k^* \downarrow & & \downarrow f^* \\ F(X) & \xrightarrow{h^*} & F(X') \end{array}$$

in $\mathbb{D}$ is horizontally left adjointable.

- (2) Dually, We say F is right P adjointable if for every pullback square:

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ f \downarrow & & \downarrow k \\ Y' & \xrightarrow{g} & Y \end{array}$$

where $k \in P$, the associated square:

$$\begin{array}{ccc} F(Y) & \xrightarrow{g^*} & F(Y') \\ k^* \downarrow & & \downarrow f^* \\ F(X) & \xrightarrow{h^*} & F(X') \end{array}$$

in $\mathbb{D}$ is vertically right adjointable.

- (3) We say F is (I, P) -biadjointable if F is left I adjointable and right P adjointable, and the square:

$$\begin{array}{ccc}
F(Y) & \xrightarrow{g^*} & F(Y') \\
k^* \downarrow & & \downarrow f^* \\
F(X) & \xrightarrow{h^*} & F(X')
\end{array}$$

is biajointable.

4.3 The functor ψ is $(\mathcal{C}, \mathcal{C})$ -biadjointable

Recall we have constructed a lax symmetric monoidal ∞ -functor $\psi : \mathcal{C}^{op} \rightarrow \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$ sending x to left $\mathbf{Span}_1(\mathcal{C})$ -module $(\mathbf{Span}_1(\mathcal{C}), \mathbf{Span}_1(\mathcal{C}_{/x}))$ where the action is induced by the symmetric monoidal ∞ -functor $\mathbf{Span}_1(\mathcal{C}) \rightarrow \mathbf{Span}_1(\mathcal{C}_{/x})$. And in this subsection:

- We enhance $\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$ into an $(\infty, 2)$ category $\overline{\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})}$.
- And by [17, proposition 2.5.7], $\mathbf{Span}_1(-) : \mathbf{Cat}^{pb} \rightarrow \mathbf{Cat}$ can be enhanced into an $(\infty, 2)$ functor, thus given $e : y \rightarrow x \in \mathcal{C}$, there is a biadjunction

$$e_* : \mathbf{Span}_1(\mathcal{C}_{/y}) \rightleftarrows \mathbf{Span}_1(\mathcal{C}_{/x}) : e^*$$

where $e_* \dashv e^*$ and $e^* \dashv e_*$.

- And we show that $\psi : \mathcal{C}^{op} \rightarrow \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat}) \hookrightarrow \overline{\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})}$ is a $(\mathcal{C}, \mathcal{C})$ -biadjointable functor.

First of all we enhance $\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$ into an $(\infty, 2)$ category $\overline{\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})}$.

Construction 4.15. Consider the symmetric monoidal functor $\mathbf{1}_{\mathbf{Span}_1(\mathcal{C})} : * \rightarrow \mathbf{Span}_1(\mathcal{C})$, then pullback along $\mathbf{1}_{\mathbf{Span}_1(\mathcal{C})}$ of the coCartesian fibration of ∞ -operads $\mathbf{LMod}^\times \rightarrow \mathbf{Mon}^\times$ gives us a symmetric monoidal functor $F : \mathbf{Cat}^\times \rightarrow \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})^\otimes$. Recall that $\mathbf{LMod} \rightarrow \mathbf{Mon}$ is biCartesian fibration, thus the above functor is the left adjoint of the forgetful functor $G : \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat}) \rightarrow \mathbf{Cat}$.

Given $K \in \mathbf{Cat}$, and $M \in \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$, the left action $\mathbf{Cat} \times \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat}) \rightarrow \mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$, $(K, M) \mapsto K \cdot M$ can be computed as the coCartesian lifting at (K, M) of the coCartesian fibration $\mathbf{LMod}^\times \rightarrow \mathbf{Mon}^\times$ over the map $\alpha : (*, \mathbf{Span}_1(\mathcal{C})) \xrightarrow{(\mathbf{1}, \text{id})} (\mathbf{Span}_1(\mathcal{C}), \mathbf{Span}_1(\mathcal{C})) \xrightarrow{\otimes} \mathbf{Span}_1(\mathcal{C})$ in $\mathbf{Mon}^\times$. Moreover, α can be factored as $(*, \mathbf{Span}_1(\mathcal{C})) \xrightarrow{\times} * \times_{\mathbf{Span}_1(\mathcal{C})} \mathbf{Span}_1(\mathcal{C}) \cong \mathbf{Span}_1(\mathcal{C})$ where $\times$ is the tensor product in the Cartesian symmetric monoidal ∞ -category $\mathbf{Mon}^\times$, thus $K \cdot M$ is precisely $K \times \mathcal{M}^\otimes \cong (K \times \mathbf{Ass}) \times_{\mathbf{Ass}} \mathcal{M}^\otimes \rightarrow \mathbf{Ass} \times_{\mathbf{Ass}} \mathbf{Span}_1(\mathcal{C})^\otimes \cong \mathbf{Span}_1(\mathcal{C})^\otimes$, which is the product of an ∞ category K with the left tensored ∞ -category $\mathcal{M}^\otimes \rightarrow \mathbf{Span}_1(\mathcal{C})^\otimes$ (here we are using [weakly left tensored](#) ∞ -category model for $\mathbf{LMod}$).

Moreover, $\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$ lies in $\mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L)$, thus we can enhance $\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$ to an $(\infty, 2)$ category through the functor $\theta : \mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L) \rightarrow \widehat{\mathbf{Cat}}_{(\infty, 2)}$, denoted as $\overline{\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})} := \theta(\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat}))$.

Recall that under the functor $\theta : \mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L) \rightarrow \mathbf{Cat}_{(\infty,2)}$, the mapping category of $\theta(M)$ is the internal Hom under the left-Cat action, and for the Cat action on $\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$, it induces an internal Hom:

$$\mathbf{Map}_{\mathbf{Cat}}(K, \underline{\mathbf{Hom}}_{\mathbf{Span}_1(\mathcal{C})}(C, D)) \cong \mathbf{Map}_{\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})}(K \times C, D)$$

Especially by taking $K = [1]$, the above equivalence implies that the 2-morphisms in $\overline{\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})}$ are precisely those natural transformations over $\mathbf{Span}_1(\mathcal{C})^{\otimes}$ if we regard objects in $\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$ as left tensored ∞ -categories in the form of $\mathcal{M}^{\otimes} \rightarrow \mathbf{Span}_1(\mathcal{C})^{\otimes} \rightarrow \mathbf{Ass}$, and thus left adjoints in $\overline{\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})}$ are those morphisms between two left $\mathcal{C}$ -modules which are fiberwise left adjoints and the associated right adjoints (which are automatically lax linear) preserve the left $\mathcal{C}$ -action (thus are linear), which is a special case of [enriched adjunction between left tensored \$\infty\$ -categories](#).

Remark 4.16. Equivalently, under the [identification of two models of left modules](#), we can also regard objects M, N in $\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$ as coCartesian fibrations over LM, and then in the $(\infty, 2)$ structure constructed as above, 2-morphisms are the natural transformations relative to LM, and the adjunctions are the adjunctions relative to LM as defined in [22, definition 7.3.2.2].

Given a pullback square in $\mathcal{C}$:

$$\begin{array}{ccc} y' & \xrightarrow{f'} & y \\ g' \downarrow & & \downarrow g \\ x' & \xrightarrow{f} & x \end{array}$$

, the associated square under ψ is:

$$\begin{array}{ccc} \mathbf{Span}_1(\mathcal{C}_{/x}) & \xrightarrow{f^*} & \mathbf{Span}_1(\mathcal{C}_{/x'}) \\ g^* \downarrow & & \downarrow g'^* \\ \mathbf{Span}_1(\mathcal{C}_{/y}) & \xrightarrow{f'^*} & \mathbf{Span}_1(\mathcal{C}_{/y'}) \end{array}$$

noticing that the above diagram is in $\mathbf{LMod}_{\mathbf{Span}_1(\mathcal{C})}(\mathbf{Cat})$, and we only denote the underlying categories of the left modules.

First we show that at the level of underlying categories, the above diagram is biadjointable:

Proposition 4.17. *Given a morphism $e : y \rightarrow x \in \mathcal{C}$, consider the adjunction in Cat:*

$$e_* : \mathcal{C}_{/y} \rightleftarrows \mathcal{C}_{/x} : e^*$$

where e_* is by precomposition, and right adjoint e^* is pullback along e , and there is an induced adjunction in Cat:

$$e_* : \mathbf{Span}_1(\mathcal{C}_{/y}) \rightleftarrows \mathbf{Span}_1(\mathcal{C}_{/x}) : e^*$$

, and moreover, in the latter adjunction, the pair $e_* \dashv e^*$ is a biadjunction, that is, e_* is the left and the right adjoint of e^* .

Before the proof, let's first enhance the functor:

$$\text{Span}_1(-) : \text{GeomSetup} \rightarrow \text{Cat}$$

into an $(\infty, 2)$ functor following [17, section 2.5].

- Given two geometric setups (C, C_0) and (D, D_0) , define $\text{Fun}^1((C, C_0), (D, D_0))$ as the subcategory of $\text{Fun}(C, D)$ consisting of objects which are morphisms between geometric setups, and morphisms are natural transformations $\alpha : F \rightarrow F'$ which satisfies the following

$$\text{condition: for any morphism } f : x \rightarrow y \text{ in } C_0, \text{ the square } \begin{array}{ccc} F(x) & \xrightarrow{\alpha_x} & F'(x) \\ F(f) \downarrow & & \downarrow F'(f) \\ F(y) & \xrightarrow{\alpha_y} & F'(y) \end{array} \text{ is}$$

a pullback diagram in D .

- We denote $\text{Fun}^2((C, C_0), (D, D_0))$ as the wide subcategory of $\text{Fun}^1((C, C_0), (D, D_0))$ with morphisms satisfying the additional conditions: for any $y \in C$, α_y lies in D_0 , and for any f in C , the above diagram is pullback in D .
- Especially, when we precompose with $\text{Cat}^{pb} \rightarrow \text{GeomSetup}$ sending $(\mathcal{C})$ to $(\mathcal{C}, \mathcal{C})$, $\text{Fun}^1((\mathcal{C}, \mathcal{C}), (\mathcal{D}, \mathcal{D}))$ get simplified into a subcategory of $\text{Fun}(\mathcal{C}, \mathcal{D})$ consisting of functors preserving pullback diagrams and natural transformations $\alpha : F \rightarrow F'$ such that for any $f : x \rightarrow y$ in $\mathcal{C}$, the

$$\text{square } \begin{array}{ccc} F(x) & \xrightarrow{\alpha_x} & F'(x) \\ F(f) \downarrow & & \downarrow F'(f) \\ F(y) & \xrightarrow{\alpha_y} & F'(y) \end{array} \text{ is a pullback diagram in } \mathcal{D}. \text{ And } \text{Fun}^2((\mathcal{C}, \mathcal{C}), (\mathcal{D}, \mathcal{D})) \cong$$

$\text{Fun}^1((\mathcal{C}, \mathcal{C}), (\mathcal{D}, \mathcal{D}))$. This can also be regarded as a version of internal Hom of Cat^{pb} .

- The pair $(\text{Fun}^1((C, C_0), (D, D_0)), (\text{Fun}^2((C, C_0), (D, D_0))))$ is a geometric setup if $((C, C_0)$ and (D, D_0) are geometric setups.
- By [17, proposition 2.5.3], GeomSetup is a closed Cartesian symmetric monoidal ∞ -category with internal Hom : $(\text{Fun}^1((C, C_0), (D, D_0)), (\text{Fun}^2((C, C_0), (D, D_0))))$, denoted as $\text{Fun}_{\text{GeomSetup}}((C, C_0), (D, D_0))$, and thus GeomSetup is self-enriched.
- By applying the transfer of enrichment along $\text{Span}_1(-) : \text{GeomSetup} \rightarrow \text{Cat}$, GeomSetup can be enhanced into a Cat -enriched ∞ -category, thus can be regarded as an $(\infty, 2)$ -category, denoted as $\overline{\text{GeomSetup}}$.
- The functor $\text{Span}_1(-)$ can be enhanced into an $(\infty, 2)$ functor

$$\text{Span}_1(-) : \overline{\text{GeomSetup}} \rightarrow \overline{\text{Cat}}$$

by [17, theorem 2.5.7].

Concretely, the $(\infty, 2)$ functor sends

- A geometric setup (C, C_0) to $\text{Span}_1(C, C_0)$.
- A morphism between geometric setups $(f, f_0) : (C, C_0) \rightarrow (D, D_0)$ to

$$\text{Span}_1(f) : \text{Span}_1(C, C_0) \rightarrow \text{Span}_1(D, D_0)$$

- A morphism in $\text{Span}_1(\text{Fun}_{\text{GeomSetup}}((C, C_0), (D, D_0)))$, in the shape of $F' \leftarrow F \rightarrow F''$ to the natural transformation $\text{Span}_1(F') \Rightarrow \text{Span}_1(F'')$ where objectwise it is $F'(x) \leftarrow F(x) \rightarrow F''(x)$.

Now we apply the above $(\infty, 2)$ functor to our context. Consider the adjunction

$$e_* : \mathcal{C}_{/y} \rightleftarrows \mathcal{C}_{/x} : e^*$$

:

- The adjunction is induced by pullback along $[1] \rightarrow \mathcal{C}$ specifying the edge e of the biCartesian fibration $\text{Arr}(\mathcal{C}) \rightarrow \mathcal{C}$.
- The left adjoint $e_* : \mathcal{C}_{/y} \rightarrow \mathcal{C}_{/x} : (z \rightarrow y) \mapsto (z \rightarrow y \rightarrow x)$ preserves pullbacks: since $\mathcal{C}_{/y} \cong (\mathcal{C}_{/x})_{/e}$, and pullback in the slice category can be computed in the original category.
- The right adjoint e^* preserves limits, thus also preserves pullbacks.
- Denote the unit and counit map of the above adjunction as η and ϵ .
- Given an object $c \in \mathcal{C}_{/y}$, $\eta_c : c \rightarrow e^*e_*(c) = c \times_x y$ is induced by the following diagram:

$$\begin{array}{ccc} c & \xrightarrow{\eta_c} & e^*e_*(c) = c \times_x y \longrightarrow c \\ \downarrow & \searrow e_*(c) & \downarrow \quad \quad \downarrow \\ y & \xrightarrow{f} & x \quad \quad y \xrightarrow{f} x \end{array}$$

recall that the right adjoint e^* corresponds to the Cartesian lifting in $\text{Arr}(\mathcal{C})$, so the unit map is just an edge in $\text{Arr}(\mathcal{C})$ induced by the universal property of pullback.

- And given an object $d \in \mathcal{C}_{/x}$, the counit map $\epsilon_d : e_*e^*(d) = d \times_x y \rightarrow d$ is encoded in the following pullback diagram:

$$\begin{array}{ccc} d \times_x y & \xrightarrow{\epsilon_d} & d \\ \downarrow \lrcorner & & \downarrow \\ y & \xrightarrow{f} & x \end{array}$$

Lemma 4.18. (η, ϵ) exhibits e_* as the left adjoint of e^* in $\overline{\text{GeomSetup}}$.

Proof. • $\eta : \text{id} \Rightarrow e^* e_*$ as a morphism in $\text{Fun}(\mathcal{C}_y, \mathcal{C}_y)$, satisfies the following condition:
for any morphism $f : c \rightarrow d$ in $\mathcal{C}_y$, the square

$$\begin{array}{ccc} c & \xrightarrow{\eta_c} & e^* e_* c = c \times_x y \\ f \downarrow & & \downarrow e^* e_*(f) \\ d & \xrightarrow{\eta_d} & e^* e_* d = d \times_x y \end{array}$$

is a pullback square by the descriptions of η_c and η_d and that limit commutes with

$$\begin{array}{ccc} c \cong c \times_x x & \xrightarrow{\quad} & c \times_x y \\ \text{limit: } \downarrow & \lrcorner & \downarrow \\ d \cong d \times_x x & \xrightarrow{\quad} & d \times_x y \end{array}.$$

- So η is a morphism in $\text{Fun}^1((\mathcal{C}_y, \mathcal{C}_y), (\mathcal{C}_y, \mathcal{C}_y))$, especially, under the embedding:

$$\text{Fun}^1((\mathcal{C}_y, \mathcal{C}_y), (\mathcal{C}_y, \mathcal{C}_y)) \rightarrow \text{Span}_1(\text{Fun}^1((\mathcal{C}_y, \mathcal{C}_y), (\mathcal{C}_y, \mathcal{C}_y)), \text{Fun}^{1=2}((\mathcal{C}_y, \mathcal{C}_y), (\mathcal{C}_y, \mathcal{C}_y)))$$

regarded as forward spans, η is a 2-morphism in $\overline{\text{GeomSetup}}$. Likewise, the counit map ϵ also satisfies the same property.

- Without abuse of notation, we again denote these 2-morphisms as η and ϵ :

$$\begin{array}{ccc} & \text{id} & \\ \swarrow & & \searrow \eta \\ \text{id} & & e^* e_* \end{array} \qquad \begin{array}{ccc} & e_* e^* & \\ \swarrow & & \searrow \epsilon \\ e_* e^* & & \text{id} \end{array}$$

- The above η and ϵ exhibit $e_* : (\mathcal{C}_y, \mathcal{C}_y) \rightarrow (\mathcal{C}_x, \mathcal{C}_x)$ as the left adjoint of $e^* : (\mathcal{C}_x, \mathcal{C}_x) \rightarrow (\mathcal{C}_y, \mathcal{C}_y)$ in $\overline{\text{GeomSetup}}$: an invertible morphism in $\text{Span}_1(\text{Fun}_{\text{GeomSetup}}((\mathcal{C}_x, \mathcal{C}_x), (\mathcal{C}_x, \mathcal{C}_x)))$

is in the shape of $\begin{array}{ccc} & F & \\ \swarrow \alpha & & \searrow \beta \\ F' & & F'' \end{array}$ where both legs are invertible by [32, proposition

3.1.11]. So the composition of 2-morphisms in $\overline{\text{GeomSetup}}$: $e_* \Rightarrow e_*(e^* e_*) \Rightarrow (e_* e^*) e_* \Rightarrow e_*$ (as composition of spans) is an invertible 2-morphism in $\overline{\text{GeomSetup}}$. So is $e^* \Rightarrow (e^* e_*) e^* \Rightarrow e^*(e_* e^*) \Rightarrow e^*$. □

On the other hand, consider the following 2-morphisms in $\overline{\text{GeomSetup}}$:

$$\begin{array}{ccc} \tilde{\epsilon} : & \begin{array}{ccc} & \text{id} & \\ \swarrow \eta & & \searrow \\ e^* e_* & & \text{id} \end{array} & \tilde{\eta} : \begin{array}{ccc} & e_* e^* & \\ \swarrow \epsilon & & \searrow \\ \text{id} & & e_* e^* \end{array} \end{array}$$

Likewise, we can show the following lemma:

Lemma 4.19. $(\tilde{\eta}, \tilde{\epsilon})$ exhibits e_* as the right adjoint of e^* .

proof of proposition on biadjointness. Since $\text{Span}_1(-)$ is an $(\infty, 2)$ functor, thus preserves adjunction, thus the biadjunction $e_* \dashv e^*$ is sent to an biadjunction in Cat :

$$e_* : \text{Span}_1(\mathcal{C}_{/y}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/x}) : e^*$$

□

Remark 4.20. By the description of the $(\infty, 2)$ functor $\text{Span}_1(-)$, we have:

- For the adjunction:

$$e_* : \text{Span}_1(\mathcal{C}_{/y}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/x}) : e^*$$

for $c \in \text{Span}_1(\mathcal{C}_{/y})$, the unit map of the above adjunction at c is

$$\begin{array}{ccc} & c & \\ \parallel & \searrow \eta_c & \\ c & & c \times_x y \end{array}$$

, and for $d \in \mathcal{C}_{/x}$, the counit map is

$$\begin{array}{ccc} & d \times_x y & \\ \parallel & \searrow \epsilon_d & \\ d \times_x y & & d \end{array}$$

where η_c and ϵ_d are the unit and counit maps in slice adjunction $e_* \dashv e^*$.

- And for the adjunction

$$e^* : \text{Span}_1(\mathcal{C}_{/x}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/y}) : e_*$$

for $d \in \mathcal{C}_{/x}$, the unit map at d is

$$\begin{array}{ccc} & d \times_x y & \\ \swarrow \epsilon & \parallel & \searrow \\ d & & d \times_x y \end{array}$$

and for $c \in \mathcal{C}_{/y}$, the counit map at c is

$$\begin{array}{ccc} & c & \\ \swarrow \eta_c & \parallel & \searrow \\ c \times_x y & & c \end{array}$$

Now we enhance the biadjointness to the $\overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$:

Construction 4.21. Consider the commutative diagrams of symmetric monoidal ∞ categories with edges also symmetric monoidal:

$$\begin{array}{ccc}
 \mathcal{C} & \xrightarrow{(-) \times x} & \mathcal{C}_{/x} \\
 \searrow & & \downarrow (-) \times_x y \\
 & & \mathcal{C}_{/y}
 \end{array}
 \quad \xRightarrow{\text{Span}_1(-)} \quad
 \begin{array}{ccc}
 \text{Span}_1(\mathcal{C}) & \xrightarrow{(-) \times x} & \text{Span}_1(\mathcal{C}_{/x}) \\
 \searrow & & \downarrow (-) \times_x y \\
 & & \text{Span}_1(\mathcal{C}_{/y})
 \end{array}$$

which gives rise to a morphism $[1] \rightarrow \text{Fun}([1], \text{CAlg}(\text{Cat})_{\text{Span}_1(\mathcal{C})}) \cong \text{CAlg}(\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})) \xrightarrow{\text{forget}} \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$. Thus we have a $\text{Span}_1(\mathcal{C})$ -linear functor $\bar{e}^* : (\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/x})) \rightarrow (\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/y}))$.

And we can show:

Lemma 4.22. Consider the adjunction $e_* : \text{Span}_1(\mathcal{C}_{/y}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/x}) : e^*$, then e_* can be enhanced into a $\text{Span}_1(\mathcal{C})$ -linear functor which is the left adjoint of $\bar{e}^*$ in $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$:

$$\bar{e}_* : (\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/y})) \rightarrow (\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/x})) : \bar{e}^*$$

.

Proof. • During the proof, the unit and counit maps refer to the adjunction:

$$e_* : \text{Span}_1(\mathcal{C}_{/y}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/x}) : e^*$$

for $c \in \text{Span}_1(\mathcal{C}_{/y})$, the unit map of the above adjunction at c is

$$\begin{array}{ccc}
 & c & \\
 \parallel & \searrow \eta_c & \\
 c & & c \times_x y
 \end{array}$$

, and for $d \in \mathcal{C}_{/x}$, the counit map is

$$\begin{array}{ccc}
 & d \times_x y & \\
 \parallel & \searrow \epsilon_d & \\
 d \times_x y & & d
 \end{array}$$

where η_c and ϵ_d are the unit and counit maps in [slice adjunction \$e_* \dashv e^*\$](#)

- Since fiberwise $\bar{e}^*$ admits left adjoint, thus by [enriched adjunction theorem](#) (or relative adjunction theorem [22, corollary 7.3.2.12]), it suffices to show the following map in $\text{Span}_1(\mathcal{C}_{/x})$ is an equivalence:

$$e_*(c_1 \otimes \cdots \otimes c_n \otimes z) \rightarrow c_1 \otimes \cdots \otimes c_n \otimes e_*(z)$$

which is induced by:

$$\theta : c_1 \otimes \cdots \otimes c_n \otimes z \rightarrow e^*(c_1 \otimes \cdots \otimes c_n \otimes e_*(z)) = c_1 \otimes \cdots \otimes c_n \otimes e^*e_*(z)$$

- The map $c_1 \otimes \cdots \otimes c_n \otimes z \rightarrow c_1 \otimes \cdots \otimes c_n \otimes e^* e_*(z)$ (which is induced by unit map) is in the shape of:

$$\begin{array}{ccc}
 & c_1 \times \cdots \times c_n \times z & \\
 \swarrow & & \searrow \eta_z \\
 c_1 \times \cdots \times c_n \times z & & c_1 \times \cdots \times c_n \times (z \times_x y)
 \end{array}$$

- Recall that given an a map $a \rightarrow e^*(b)$, the corresponding map under adjunction is $e_* \dashv e^*$ is $e_*(a) \rightarrow e_* e^*(b) \rightarrow b$ where the last edge is the counit map.
- Thus $\theta : e_*(c_1 \otimes \cdots \otimes c_n \otimes z) \rightarrow c_1 \otimes \cdots \otimes c_n \otimes e_*(z)$ is the composition

$$e_*(c_1 \otimes \cdots \otimes c_n \otimes z) \rightarrow e_* e^*(c_1 \otimes \cdots \otimes c_n \otimes e_*(z)) \rightarrow c_1 \otimes \cdots \otimes c_n \otimes e_*(z)$$

$$\begin{array}{ccccc}
 & & c_1 \times \cdots \times c_n \times z & & \\
 & \swarrow & & \searrow & \\
 c_1 \times \cdots \times c_n \times z & & & & c_1 \times \cdots \times c_n \times (z \times_x y) \\
 \swarrow & \eta_z & & \swarrow & \searrow \epsilon_z \\
 c_1 \times \cdots \times c_n \times z & & c_1 \times \cdots \times c_n \times (z \times_x y) & & c_1 \times \cdots \times c_n \times z
 \end{array}$$

where the left leg of the composition is identity, and the right leg is equivalence by triangle identity of the slice adjunction $e_* : \mathcal{C}_{/y} \rightleftarrows \mathcal{C}_{/x} : e^*$, exhibited in the following diagram:

$$\begin{array}{ccccc}
 (c_1 \times \cdots \times c_n) \times z & \xrightarrow{\text{id}} & (c_1 \times \cdots \times c_n) \times z & & \\
 \eta_z \searrow & & \downarrow & \xrightarrow{\epsilon_z} & \downarrow \\
 (c_1 \times \cdots \times c_n) \times (z \times_x y) & & y & \xrightarrow{\quad} & x
 \end{array}$$

□

And likewise, we can show $\bar{e}^*$ admits a $\mathcal{C}$ -linear right adjoint:

Lemma 4.23. *Consider the adjunction $e^* : \text{Span}_1(\mathcal{C}_{/x}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/y}) : e_*$, then e_* can be enhanced into a $\text{Span}_1(\mathcal{C})$ -linear functor $\bar{e}_*$, which is the right adjoint of $\bar{e}^*$ in $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$:*

$$\bar{e}^* : (\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/x})) \rightarrow (\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/y})) : \bar{e}_*$$

.

Proof. • During this proof, we use the model of coCartesian fibrations over LM for left modules for simplicity.

- And all the unit and counit maps during this proof refer to the adjunction

$$e^* : \text{Span}_1(\mathcal{C}_{/x}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/y}) : e_*$$

, and for $d \in \mathcal{C}_{/x}$, the unit map at d is

$$\begin{array}{ccc} & d \times_x y & \\ \swarrow \epsilon & & \searrow \\ d & & d \times_x y \end{array}$$

and for $c \in \mathcal{C}_{/y}$, the counit map at c is

$$\begin{array}{ccc} & c & \\ \swarrow \eta_c & & \searrow \\ c \times_x y & & c \end{array}$$

- Since $\bar{e}^*$ is $\mathcal{C}$ -linear and fiberwise admits right adjoint, thus regarded as coCartesian fibrations over LM, it admits a relative right adjoint $\tilde{e}_*$ by [22, corollary 7.3.2.7], and it remains to show $\tilde{e}_*$ is $\mathcal{C}$ -linear.
- Consider the coCartesian lifting over an active map in LM corresponding to the linear action $a : (c_1, \dots, c_n, z) \rightarrow c_1 \otimes \dots \otimes c_n \otimes z$ in the left module $(\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/y}))$, then under the lax linear functor $\tilde{e}_*$, there is a factorization diagram in $(\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/x}))$:

$$\begin{array}{ccc} \tilde{e}_*(c_1, \dots, c_n, z) & \xrightarrow{\tilde{e}_*(a)} & e_*(c_1 \otimes \dots \otimes c_n \otimes z) \\ \parallel & \searrow \text{active map} & \uparrow q \\ (c_1, \dots, c_n, e_*(z)) & & c_1 \otimes \dots \otimes c_n \otimes e_*(z) \end{array}$$

, and it suffices to show the induced map $q : c_1 \otimes \dots \otimes c_n \otimes e_*(z) \rightarrow e_*(c_1 \otimes \dots \otimes c_n \otimes z)$ is an equivalence by [22, corollary 7.3.2.12].

- After applying the functor $\bar{e}^*$ on the above diagram, there is a new commutative diagram in $(\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/y}))$:

$$\begin{array}{ccccc} (c_1, \dots, c_n, z) & \xrightarrow{a} & & & c_1 \otimes \dots \otimes c_n \otimes z \\ \uparrow \text{counit} & & & & \uparrow \\ \bar{e}^* \tilde{e}_*(c_1, \dots, c_n, z) & \xrightarrow{\bar{e}^* \tilde{e}_*(a)} & \bar{e}^* \tilde{e}_*(c_1 \otimes \dots \otimes c_n \otimes z) & \xrightarrow{\text{counit}} & c_1 \otimes \dots \otimes c_n \otimes z \\ \parallel & & \nearrow \bar{e}^*(q) & & \nearrow \\ (c_1, \dots, c_n, e^* e_*(z)) & \xrightarrow{\text{active}} & c_1 \otimes \dots \otimes c_n \otimes e^* e_*(z) & \xrightarrow{\text{id} \otimes \text{counit}} & c_1 \otimes \dots \otimes c_n \otimes z \end{array}$$

- Also we notice that the morphism $\tilde{e}_*(a)$ under adjunction corresponds to a morphism in $(\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/y}))$:

$$\tilde{e}^* \tilde{e}_*(c_1, \dots, c_n, z) \xrightarrow{\tilde{e}^* \tilde{e}_*(a)} \tilde{e}^* \tilde{e}_*(c_1 \otimes \dots \otimes c_n \otimes z) \xrightarrow{\text{counit}} c_1 \otimes \dots \otimes c_n \otimes z$$

thus the above diagram implies $q : c_1 \otimes \dots \otimes c_n \otimes e_*(z) \rightarrow e_*(c_1 \otimes \dots \otimes c_n \otimes z)$ corresponds to the map

$$\text{counit} \circ \tilde{e}^*(q) = \text{id} \otimes \text{counit} : c_1 \otimes \dots \otimes c_n \otimes e^* e_*(z) \rightarrow c_1 \otimes \dots \otimes c_n \otimes z$$

under the adjunction $e^* \dashv e_*$, and we can recover q by taking the image of $\text{id} \otimes \text{counit}$ under e_* and then precompose it with the unit map.

- Thus q can be translated into the following composition of morphisms in the underlying category of the left module $\text{Span}_1(\mathcal{C}_{/x})$:

$$\begin{array}{ccccc}
 & & c_1 \times \dots \times c_n \times z & & \\
 & \swarrow & & \searrow & \\
 (c_1 \times \dots \times c_n \times z) \times_x y & & & & c_1 \times \dots \times c_n \times z \\
 \swarrow \text{id} \times \epsilon_z & & \searrow & & \searrow \\
 c_1 \times \dots \times c_n \times z & & (c_1 \times \dots \times c_n \times z) \times_x y & & c_1 \times \dots \times c_n \times z \\
 & & \parallel & & \\
 & & c_1 \times \dots \times c_n \times (z \times_x y) & & \\
 & \swarrow \text{id} \times \eta_z & & \searrow & \\
 & & & & c_1 \times \dots \times c_n \times z
 \end{array}$$

where the right leg of the composition is identity, and the left leg is equivalence by triangle identity of the slice adjunction $e_* : \mathcal{C}_{/y} \rightleftarrows \mathcal{C}_{/x} : e^*$, exhibited in the following diagram:

$$\begin{array}{ccccc}
 (c_1 \times \dots \times c_n) \times z & \xrightarrow{\text{id}} & (c_1 \times \dots \times c_n) \times z & & \\
 \eta_z \searrow & & \downarrow \epsilon_z & & \downarrow \\
 (c_1 \times \dots \times c_n) \times (z \times_x y) & \xrightarrow{\epsilon_z} & (c_1 \times \dots \times c_n) \times z & & \\
 \downarrow & & \downarrow & & \\
 y & \xrightarrow{\quad} & x & &
 \end{array}$$

□

By far we have seen that, given a pullback diagram in $\mathcal{C}$:

$$\begin{array}{ccc}
 y' & \xrightarrow{f'} & y \\
 g' \downarrow & & \downarrow g \\
 x' & \xrightarrow{f} & x
 \end{array}$$

all the edges in its image

under ψ in $\overline{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$ admit left and right adjoints:

$$\begin{array}{ccc}
\text{Span}_1(\mathcal{C}_{/x}) & \xrightarrow{f^*} & \text{Span}_1(\mathcal{C}_{/x'}) \\
g^* \downarrow & & \downarrow g'^* \\
\text{Span}_1(\mathcal{C}_{/y}) & \xrightarrow{f'^*} & \text{Span}_1(\mathcal{C}_{/y'})
\end{array}$$

And it remains to show the above diagram is biadjointable:

Lemma 4.24. *Under the functor by taking the underlying category $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) \rightarrow \text{Cat}$, the above diagram is biadjointable in $\overline{\text{Cat}}$.*

Proof. • It's vertically left adjointable, the Beck-Chevalley map sends

$$\begin{array}{ccc}
z \times_y y' & \xrightarrow{\quad} & z \\
\downarrow & \lrcorner & \downarrow \\
y' & \xrightarrow{f'} & y \\
\downarrow & & \\
x' & &
\end{array} \text{ to }$$

$$\begin{array}{ccc}
z \times_x x' & \xrightarrow{\quad} & z \\
\downarrow & \lrcorner & \downarrow \\
y' & \xrightarrow{f'} & y \\
g' \downarrow & \lrcorner & \downarrow g \\
x' & \xrightarrow{f} & x
\end{array} \text{ which is equivalence by pasting lemma.}$$

- Dually, it's horizontally right adjointable: the Beck-Chevalley map is the same as the previous case but change the leg.
- And it's biadjointable: after taking the horizontal left adjoints, there is a commutative diagram:

$$\begin{array}{ccc}
\text{Span}_1(\mathcal{C}_{/x}) & \xleftarrow{f_*} & \text{Span}_1(\mathcal{C}_{/x'}) \\
\downarrow g_* & & \downarrow g'_* \\
\text{Span}_1(\mathcal{C}_{/y}) & \xleftarrow{f'_*} & \text{Span}_1(\mathcal{C}_{/y'})
\end{array}$$

and we further take vertical right adjoints, then the associated Beck-Chevalley map is:

$$f_* g'_* \Rightarrow g_* g^* f_* g'_* \cong g_* f'_* g'^* g'_* \Rightarrow g_* f'_*$$

, for any $z \in \mathcal{C}_{/y'}$, the above natural transformation at z becomes a morphism where the left leg is:

$$\begin{array}{ccc}
F(X) \otimes F(Y) & \xrightarrow{F(f) \otimes \text{id}} & F(X') \otimes F(Y) \\
\downarrow & & \downarrow \\
F(X \otimes Y) & \xrightarrow{F(f \otimes \text{id})} & F(X' \otimes Y)
\end{array}$$

is horizontally left adjointable and horizontally right adjointable.

And we define a symmetric monoidal $(\infty, 2)$ category $\text{Span}_2(\mathcal{C})$ for an $(\infty, 1)$ symmetric monoidal category $\mathcal{C}$ with all pullbacks and the tensor product structure is compatible with the pullbacks:

Construction 4.27. [7, section 4.3] Given a symmetric monoidal $(\infty, 1)$ category $\mathcal{C}^\otimes$ such that the tensor product is compatible with all pullbacks, by [4, proposition C.1], $\mathcal{C}^\otimes$ is equivalent to a functor:

$$\mathcal{C}^\otimes : \text{Span}_1(\text{Fin}) \rightarrow \text{Cat}$$

that its restriction to Fin^{op} is product preserving. And by our condition that tensor product commutes with pullbacks, thus $\mathcal{C}^\otimes$ admits a factorization through $\text{Cat}^{pb} \rightarrow \text{Cat}$, thus we have:

$$\mathcal{C}^\otimes : \text{Span}_1(\text{Fin}) \rightarrow \text{Cat}^{pb}$$

, then by composition with the product-preserving functor $\text{Span}_2 : \text{Cat}^{pb} \rightarrow \text{Cat}$, there is a functor:

$$\text{Span}_2(\mathcal{C})^\otimes : \text{Span}_1(\text{Fin}) \rightarrow \text{Cat}_{(\infty, 2)}$$

where $\text{Span}_2(\mathcal{C})^\otimes([n]) \cong \text{Span}_2(\mathcal{C}^{\times n})$ and its restriction to Fin^{op} is product-preserving, thus corresponds to a commutative monoid in $\text{Cat}_{(\infty, 2)}$, and especially, this gives rise to a symmetric monoidal $(\infty, 2)$ structure on $\text{Span}_2(\mathcal{C})$.

And there is a natural transformation $\mathcal{C}^{\otimes, op} \Rightarrow \text{Span}_2(\mathcal{C})^\otimes$, giving rise to a symmetric monoidal $(\infty, 2)$ functor on the natural embedding $\mathcal{C}^{op} \rightarrow \text{Span}_2(\mathcal{C})$. Especially, the functor $\mathcal{C}^{\otimes, op}$ as the opposite of the functor $\mathcal{C}^\otimes$ corresponds to the opposite symmetric monoidal structure of $\mathcal{C}^\otimes$, as [remarked here](#), in our case, we consider the Cartesian symmetric monoidal structure on $\mathcal{C}$, and the opposite symmetric monoidal structure on $\mathcal{C}^{op}$ is coCartesian, coinciding with our previous constructions.

And we have the following universal property of the symmetric monoidal $(\infty, 2)$ category $\text{Span}_2(\mathcal{C})$:

Proposition 4.28. [7, theorem 4.33] *Given a symmetric monoidal category $\mathcal{C}^\otimes$ such that $\mathcal{C}$ admits all pullbacks and the symmetric monoidal structure on $\mathcal{C}$ is compatible with the pullbacks, then any lax symmetric monoidal biadjointable $(\infty, 2)$ functor $F : \mathcal{C}^{\otimes, op} \rightarrow \mathbb{D}^\otimes$ extends uniquely to a lax symmetric monoidal $(\infty, 2)$ functor $F' : \text{Span}_2(\mathcal{C})^\otimes \rightarrow \mathbb{D}^\otimes$.*

We will show the lax symmetric monoidal $(\infty, 2)$ functor $\psi : \mathcal{C}^{op, \Pi} \rightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^\otimes \rightarrow \overline{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}^\otimes$ is biadjointable. And it suffices to show the following diagram induced by a morphism $f : x' \rightarrow x$ in $\mathcal{C}$:

$$\begin{array}{ccc}
\text{Span}_1(\mathcal{C}_{/x}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y}) & \xrightarrow{f^* \otimes \text{id}} & \text{Span}_1(\mathcal{C}_{/x'}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y}) \\
\downarrow & & \downarrow \\
\text{Span}_1(\mathcal{C}_{/x \times y}) & \xrightarrow{(f \times \text{id})^*} & \text{Span}_1(\mathcal{C}_{/x' \times y})
\end{array}$$

is horizontally left adjointable and horizontally right adjointable in $\overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$:

Before proceeding, let's give an interpretation on the vertical morphisms in the above diagram induced by the lax symmetric monoidal functor:

$$(\text{Span}_1(\mathcal{C}_{/x}), \text{Span}_1(\mathcal{C}_{/y})) \xrightarrow{co} \text{Span}_1(\mathcal{C}_{/x}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y}) \xrightarrow{\sigma} \text{Span}_1(\mathcal{C}_{x \times y})$$

where co is the coCartesian lifting in $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^{\otimes}$ over the active map $\langle 2 \rangle \rightarrow \langle 1 \rangle$ in $\text{Comm}^{\otimes}$.

- Consider the coCartesian lifting over the active map $e : (x, y) \rightarrow x \times y$ in the ∞ -operad $\mathcal{C}^{op, \Pi}$.
- Then the above functor σ is induced by the image of e under the ∞ -operad map:

$$\psi : \mathcal{C}^{op, \Pi} \rightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^{\otimes} \subset \text{LMod}(\text{Cat})^{\times}$$

which admits a further decomposition inside the ∞ -operad $\text{LMod}(\text{Cat})^{\times}$:

$$\begin{array}{ccccc}
(\text{Span}_1(\mathcal{C}_{/x}), \text{Span}_1(\mathcal{C}_{/y})) & \xrightarrow{co} & \text{Span}_1(\mathcal{C}_{/x}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y}) & & \\
& \searrow co' & \nearrow \gamma & & \downarrow \sigma \\
& \text{Span}_1(\mathcal{C}_{/x}) \times \text{Span}_1(\mathcal{C}_{/y}) & \xrightarrow{\delta} & \text{Span}_1(\mathcal{C}_{x \times y}) &
\end{array}$$

- co' is the coCartesian edge in $\text{LMod}(\text{Cat})^{\times}$ over the active map $\langle 2 \rangle \rightarrow \langle 1 \rangle$, and $\text{Span}_1(\mathcal{C}_{/x}) \times \text{Span}_1(\mathcal{C}_{/y})$ is the product in $\text{LMod}(\text{Cat})$ and is a left $(\text{Span}_1(\mathcal{C}) \times \text{Span}_1(\mathcal{C}))$ -left module. γ and δ are induced by the universal property of the coCartesian edge co' .
- Moreover, consider the symmetric monoidal coCartesian fibration $\text{LMod}(\text{Cat})^{\times} \rightarrow \text{Mon}(\text{Cat})^{\times}$, then γ is the coCartesian lifting of the map $\text{Span}_1(\mathcal{C}) \times \text{Span}_1(\mathcal{C}) \xrightarrow{\otimes} \text{Span}_1(\mathcal{C})$ in $\text{Mon}(\text{Cat})$, justified by the following factorization for any A modules M, N for a symmetric monoidal ∞ category A :

$$\begin{array}{ccc}
\text{LMod}(\text{Cat})^\times : & (M, N) & \xrightarrow{\quad co \quad} M \otimes_A N \\
\Downarrow & \swarrow co' & \nearrow \gamma \\
& M \times N & \\
\text{Mon}(\text{Cat})^\times : & (A, A) & \xrightarrow{\quad \quad \quad} A \\
& \swarrow co'' & \nearrow \otimes \\
& A \times A &
\end{array}$$

where co, co', co'' are the coCartesian liftings over the active map $\langle 2 \rangle \rightarrow \langle 1 \rangle$ in $\text{LMod}_A(\text{Cat})^\otimes$, $\text{LMod}(\text{Cat})^\times$, $\text{Mon}(\text{Cat})^\times$ respectively, and we can take γ' to be the coCartesian lifting over map $\otimes$, then by the fact that the composition of coCartesian liftings is again coCartesian lifting and the universal property of coCartesian edge, we have $\gamma \cong \gamma'$.

•

Remark 4.29. Moreover, the $(\infty, 1)$ functor underlying the linear functor γ is exactly the first edge of the Bar construction: $\cdots M \times A \times N \rightarrow M \times N \rightarrow M \otimes_A N$, justified by the following diagram of ∞ -operads

$$\begin{array}{ccc}
\text{Mod}_A(\text{Cat})^\otimes & \xrightarrow{f, \cong} & \text{LMod}_A(\text{Cat})^\otimes \\
& \searrow & \swarrow \\
& \text{Cat}^\times &
\end{array}$$

where all the edges are the ‘forgetful’ functors: the left and right vertical arrows are the functor taking the underlying category of a (left) module category, and the horizontal edge is an equivalence by [22, corollary 5.4.11], thus it suffices to compare the coCartesian lifting γ' over the active map $\langle 2 \rangle \rightarrow \langle 1 \rangle$ in $\text{Mod}_A(\text{Cat})^\otimes$ with the first edge in the Bar construction.

And given the coCartesian lifting over the active map (which corresponds to the relative tensor product) $\gamma' : (M, N) \rightarrow M \otimes_A N$ in $\text{Mod}_A(\text{Cat})^\otimes$, which corresponds to a diagram $p : [1] \rightarrow \text{Mod}_A(\text{Cat})^\otimes$. By [22, construction 3.3.3.1] of $\text{Mod}_A(\text{Cat})$, the vertex $p(\{0\}) = (M, N)$ corresponds to a diagram $p : \mathcal{K}_{\text{Comm}} \times_{\text{Comm}^\otimes} \{\langle 2 \rangle\} \rightarrow \text{Cat}^\times$ where $\mathcal{K}_{\text{Comm}}$ is the full subcategory of $\text{Fun}([1], \text{Comm}^\otimes)$ consisting of semi-inert maps. And by precomposing with $\text{Comm}_{\langle 2 \rangle /}^\otimes \rightarrow \mathcal{K}_{\text{Comm}} \times_{\text{Comm}^\otimes} \{\langle 2 \rangle\}$, there is a diagram $q : \text{Comm}_{\langle 2 \rangle /}^\otimes \rightarrow \text{Cat}^\times$ derived from object (M, N) .

Back to the diagram $p : [1] = [0]^\triangleright \rightarrow \text{Mod}_A(\text{Cat})^\otimes$, since it’s the coCartesian lifting $(M, N) \rightarrow M \otimes_A N$, so it is an operad colimit diagram of $\text{Mod}_A(\text{Cat})^\otimes \rightarrow \text{Comm}^\otimes$. Thus the above diagram q in $\text{Cat}^\times$ constructed from (M, N) can be extended to an operadic colimit diagram $q^\triangleright$ using p by [22, theorem 3.4.4.3], where the final vertex is $M \otimes_A N$ and the first term is $\gamma' : (M, N) \rightarrow M \otimes_A N$ (in $\text{Cat}^\times$).

And by [22, the proof of theorem 4.5.2.1], there is a cofinal map $\Delta^{op} \rightarrow \text{Comm}_{\langle 2 \rangle /}^\otimes$, and by

composition with $q^\triangleright$, and push all the terms to the fiber over $\langle 1 \rangle$ by coCartesian lifting, there is a simplicial diagram which is exactly the Bar construction, and the first term $(M, N) \rightarrow M \otimes_A N$ (after coCartesian lifting) is exactly the edge $\gamma : M \times N \rightarrow M \otimes_A N$ (induced by the universal coCartesian edge $(M, N) \rightarrow M \times N$ in $\text{Cat}^\times$).

so $\gamma^\simeq : M^\simeq \times N^\simeq \rightarrow (M \otimes_A N)^\simeq$ is an epimorphism. And any object in the underlying category of A left module $M \otimes_A N$ can be written as $m \otimes n := \gamma(m, n)$ for $(m, n) \in M \times N$.

And to understand $\sigma : \text{Span}_1(\mathcal{C}_{/x}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y})$, it suffices for us to understand the $(\infty, 1)$ functor underlying the $\mathcal{C}$ -linear functor $\delta : \text{Span}_1(\mathcal{C}_{/x}) \times \text{Span}_1(\mathcal{C}_{/y})$: we can compose ψ with the evaluation (at $\mathfrak{m} \in \text{LM}^\otimes$) functor $ev_{\mathfrak{m}} : \text{LMod}_{\mathcal{C}}(\text{Cat})^\otimes \rightarrow \text{Cat}^\times$. And we show what $ev_{\mathfrak{m}} \circ \psi(e) = ev_{\mathfrak{m}}(\delta)$ is in the following steps:

Notice that $ev_{\mathfrak{m}} \circ \psi$ can be decomposed as the composition:

$$\mathcal{C}^{op, \Pi} \xrightarrow{\Phi: x \mapsto \mathcal{C}_{/x}^{op}} \text{Cat}^\times \xrightarrow{(-)^{op}} \text{Cat}^\times \supset \text{Cat}^{pb, \times} \xrightarrow{\text{Span}_1(-)} \text{Cat}^\times$$

Under the lax symmetric monoidal functor $\Psi = (-)^{op} \circ \Phi : \mathcal{C}^{op, \Pi} \rightarrow \text{Cat}^{pb, \times} \subset \text{Cat}^\times$, there

is a factorization of morphisms $\Phi(e) : (\mathcal{C}_{/x}, \mathcal{C}_{/y}) \rightarrow \mathcal{C}_{/x} \times \mathcal{C}_{/y} \xrightarrow{e'} \mathcal{C}_{/x \times y} \cong \mathcal{C}_x \times_{\mathcal{C}} \mathcal{C}_{/y}$, and we first give an interpretation on the induced morphism $e' : \mathcal{C}_{/x} \times \mathcal{C}_{/y} \rightarrow \mathcal{C}_{/x \times y}$:

The unstraightening of Φ is the coCartesian fibration $\text{Arr}(\mathcal{C})^{op, \Pi} \rightarrow \mathcal{C}^{op, \Pi}$, consider the fiber of $\text{Arr}(\mathcal{C})^{op, \Pi}$ over $(x, y) \in \mathcal{C}_{(2)}^{op, \Pi}$ lying over $\langle 2 \rangle \in \text{Comm}^\otimes$, which is the pullback along the morphism $* \rightarrow \mathcal{C}^{op, \Pi}$ specifying the object (x, y) , as shown in the diagram:

$$\begin{array}{ccccc} \mathcal{C}_{/x}^{op} \times \mathcal{C}_{/y}^{op} & \longrightarrow & \text{Arr}(\mathcal{C})_{(2)}^{op, \Pi} \cong \text{Arr}(\mathcal{C})^{op} \times \text{Arr}(\mathcal{C})^{op} & \longrightarrow & \text{Arr}(\mathcal{C})^{op, \Pi} \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ * & \xrightarrow{\{(x, y)\}} & \mathcal{C}_{(2)}^{op, \Pi} \cong \mathcal{C}^{op} \times \mathcal{C}^{op} & \longrightarrow & \mathcal{C}^{op, \Pi} \end{array}$$

By (un)straightening of symmetric monoidal coCartesian fibration (see [27, proof of theorem 2.1]), the coCartesian fibration $\text{Arr}(\mathcal{C})^{op, \Pi} \rightarrow \mathcal{C}^{op, \Pi}$ classifies a lax Cartesain structure: $\mathcal{C}^{op, \Pi} \rightarrow \text{Cat}$, which extends to the ∞ -operad map $\Phi : \mathcal{C}^{op, \Pi} \rightarrow \text{Cat}^\times$, from thus we

can see that the map e'^{op} corresponds to the coCartesian lifting of $e : (x, y) \xrightarrow{act} x \times y$ in $\text{Arr}(\mathcal{C})^{op, \Pi} \rightarrow \mathcal{C}^{op, \Pi}$, that is, for any point $(z \rightarrow x, w \rightarrow y)$ in the fiber $\text{Arr}(\mathcal{C})_{(x, y)}^{op, \Pi} \cong \mathcal{C}_{/x}^{op} \times \mathcal{C}_{/y}^{op}$, the coCartesian lifting is the image $e'(z \rightarrow x, w \rightarrow y)$, as exhibited in the following diagram in $\text{Arr}(\mathcal{C})^{op, \Pi}$:

$$\begin{array}{ccc} (z \rightarrow x, w \rightarrow y) & \longrightarrow & z \times w \rightarrow x \times y \\ \downarrow & & \downarrow \\ (x, y) & \xrightarrow{e} & x \times y \end{array}$$

Thus the functor e' is the opposite of the functor $e'^{op} : \mathcal{C}_{/x}^{op} \times \mathcal{C}_{/y}^{op} \rightarrow \mathcal{C}_{/x \times y}^{op}$, which is the restriction of the product functor $\times^{op} : \text{Arr}(\mathcal{C})^{op} \times \text{Arr}(\mathcal{C})^{op} \rightarrow \text{Arr}(\mathcal{C})^{op} = \text{Fun}([1], \mathcal{C})^{op}$ along the subcategory $\mathcal{C}_{/x}^{op} \times \mathcal{C}_{/y}^{op} \hookrightarrow \mathcal{C}_{/x \times y}^{op}$.

Since $\text{Span}_1(-)$ is product preserving, thus is symmetric monoidal and preserves the coCartesian lifting, so $ev_m \circ \delta = \text{Span}_1(\Psi(e))$ is:

$$(\text{Span}_1(\mathcal{C}_{/x}), \text{Span}_1(\mathcal{C}_{/y})) \xrightarrow{co} \text{Span}_1(\mathcal{C}_{/x}) \times \text{Span}_1(\mathcal{C}_{/y}) \xrightarrow{\text{Span}_1(e')=\delta} \text{Span}_1(\mathcal{C}_{/x \times y})$$

Thus $\delta = \text{Span}_1(e')$ where $e' : \mathcal{C}_{/x} \times \mathcal{C}_{/y} \xrightarrow{\times} \mathcal{C}_{/x \times y}$ which can be regarded as the functor taking product inside $\text{Arr}(\mathcal{C}) = \text{Fun}([1], \mathcal{C})$.

Lemma 4.30. *Given a morphism $f : x' \rightarrow x$ in $\mathcal{C}$, the $\text{Span}_1(\mathcal{C})$ -linear functor in $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$*

$$\bar{f}^* : \text{Span}_1(\mathcal{C}_{/x}) \rightarrow \text{Span}_1(\mathcal{C}_{/x'})$$

induces a morphism in $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$:

$$\bar{f}^* \otimes \text{id} : \text{Span}_1(\mathcal{C}_{/x}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y}) \rightarrow \text{Span}_1(\mathcal{C}_{/x'}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y})$$

which admits both left adjoint $\bar{f}_ \otimes \text{id}$ and right adjoint $\tilde{f}_* \otimes \text{id}$ in $\overline{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$.*

Proof. During the proof, we will abbreviate the left module $(\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/x}))$ as $\text{Span}_1(\mathcal{C}_{/x})$, and instead addressing a certain functor between $\text{Span}_1(-)$'s is in the category $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$.

- Recall the Cat -action on $\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$, which is induced by the symmetric monoidal free module functor $\text{LMod}_*(\text{Cat}) \rightarrow \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$, $K \mapsto K \times \text{Span}_1(\mathcal{C})$, and especially for any $M \in \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$, the action is given by $K \cdot M = (K \times \text{Span}_1(\mathcal{C})) \otimes_{\text{Span}_1(\mathcal{C})} M \cong K \times M$
- Especially, by taking $K = [1]$, we can see that a natural transformation between $\text{Span}_1(\mathcal{C})$ -linear functors $F, G : M \rightarrow N \in \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$ (that is, a 2-morphism in $\overline{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$) can be encoded into a $\text{Span}_1(\mathcal{C})$ -linear functor $[1] \times M \rightarrow N \in \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$ where the restrictions to the subcategory $\{0\} \times M$ and $\{1\} \times M$ recover the morphisms F and G .
- Especially, consider the 2-morphisms of unit and counit maps corresponding to the adjunction in $\overline{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$:

$$\bar{f}_* : \text{Span}_1(\mathcal{C}_{/x'}) \rightarrow \text{Span}_1(\mathcal{C}_{/x}) : \bar{f}^*$$

, denoted as

$$\eta : \text{id} \Rightarrow \bar{f}^* \bar{f}_*, [1] \times \text{Span}_1(\mathcal{C}_{/x'}) \rightarrow \text{Span}_1(\mathcal{C}_{/x'}), \epsilon : \bar{f}_* \bar{f}^* \Rightarrow \text{id}, [1] \times \text{Span}_1(\mathcal{C}_{/x}) \rightarrow \text{Span}_1(\mathcal{C}_{/x})$$

satisfying the triangle identities

$$\text{id}_{f_*(c)} : f_*(c) \xrightarrow{f_*(\eta_c)} f_*(f^* f_*(c)) \xrightarrow{\epsilon_{f^*(c)}} f_*(c), \text{id}_{f^*(c)} : f^*(c) \xrightarrow{\eta_{f^*(c)}} f^* f_*(f^*(c)) \xrightarrow{f^*(\epsilon_c)} f^*(c)$$

.

- Since the relative tensor product $\otimes_{\text{Span}_1(\mathcal{C})}$ can be computed using bar construction, thus we have $([1] \times M) \otimes_{\text{Span}_1(\mathcal{C})} N \cong [1] \times (M \otimes_{\text{Span}_1(\mathcal{C})} N)$ since product commutes with sifted colimits, or we can just take $[1] \times M \cong ([1] \times \text{Span}_1(\mathcal{C})) \otimes_{\text{Span}_1(\mathcal{C})} M$ and use the associativity of relative tensor product, for any $M, N \in \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$.
- Now consider the $\text{Span}_1(\mathcal{C})$ -linear functor $f_* \otimes \text{id} : \text{Span}_1(\mathcal{C}_{/x'}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y}) \rightarrow \text{Span}_1(\mathcal{C}_x) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y})$, then to show $\eta \otimes \text{id}, \epsilon \otimes \text{id}$ exhibits $f_* \otimes \text{id}$ as the left adjoint of $f^* \otimes \text{id}$, it suffices to show that the triangle identities hold.
- Recall that in Cat , the first term of the Bar construction $M \times N \rightarrow M \otimes_{\mathcal{C}} N$ is an epimorphism between the underlying spaces of objects, thus any object in the category $M \otimes_{\text{Span}_1(\mathcal{C})} N$ can be written as $m \otimes n$, which is the image of (m, n) under the first term of the Bar construction, for any $M, N \in \text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$.
- Thus if we consider the underlying categories of the left $\text{Span}_1(\mathcal{C})$ -modules, the underlying functor $\eta \otimes \text{id}$ and $\epsilon \otimes \text{id}$ satisfy the triangle identity, since we can write any object in $\text{Span}_1(\mathcal{C}_{x'}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y})$ in the form of $c \otimes d$ where $c \in \text{Span}_1(\mathcal{C}_{/x'})$ and $d \in \text{Span}_1(\mathcal{C}_{/y})$, and the image $\eta \otimes \text{id}(c, d) = \eta(c) \otimes d$, thus the triangle identities hold:

$$\begin{aligned} \text{id}_{f_*(c) \otimes d} : (f_* \otimes \text{id})(c \otimes d) &= f_*(c) \otimes d \xrightarrow{(f_* \otimes \text{id})(\eta \otimes \text{id})_{c \otimes d}} f_* f^* f_*(c) \otimes d \xrightarrow{(\epsilon \otimes \text{id})_{f_*(c) \otimes d}} f_*(c) \otimes d \\ \text{id}_{f^*(c) \otimes d} : (f^* \otimes \text{id})(c \otimes d) &= f^*(c) \otimes d \xrightarrow{(\eta \otimes \text{id})_{(f \otimes \text{id})^*(c \otimes d)}} f^* f_* f^*(c) \otimes d \xrightarrow{(f^* \otimes \text{id})(\epsilon_c \otimes \text{id})} f^*(c) \otimes d \end{aligned}$$

- Likewise, we can show $\tilde{f}_* \otimes \text{id}$ is the right adjoint.

□

Proposition 4.31. *The commutative diagram of left- $\text{Span}_1(\mathcal{C})$ modules*

$$\begin{array}{ccc} \text{Span}_1(\mathcal{C}_{/x}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y}) & \xrightarrow{\tilde{f}^* \otimes \text{id}} & \text{Span}_1(\mathcal{C}_{/x'}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y}) \\ \downarrow u & & \downarrow v \\ \text{Span}_1(\mathcal{C}_{/x \times y}) & \xrightarrow{(\tilde{f} \times \text{id})^*} & \text{Span}_1(\mathcal{C}_{/x' \times y}) \end{array}$$

is biadjointable in $\overline{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}$

Proof. • Since all the edges are $\text{Span}_1(\mathcal{C})$ linear functors, and fix an object z , then the Beck-Chevalley map at z is an equivalence iff it's fiberwise an equivalence, thus it suffices to show the diagram of the underlying categories is biadjointable in $\overline{\text{Cat}}$.

- Given an object in $\text{Span}_1(\mathcal{C}_{/x'}) \otimes_{\text{Span}_1(\mathcal{C})} \text{Span}_1(\mathcal{C}_{/y})$, written in the form of $m \otimes n$ for $m \rightarrow x', n \rightarrow y$.

- To show it's horizontally left adjointable: consider the Beck-chevalley map:

$$\overline{(f \times \text{id})}_* \circ v \Rightarrow \overline{(f \times \text{id})}_* \circ v \circ (\overline{f}^* \otimes \text{id}) \circ (\overline{f}_* \otimes \text{id}) \cong \overline{(f \times \text{id})}_* \circ (\overline{f \times \text{id}})^* \circ u \circ (\overline{f}^* \otimes \text{id}) \Rightarrow u \circ (\overline{f}^* \otimes \text{id})$$

Now consider the Beck-Chevalley map at $m \otimes n$, which is the image of the following map under the embedding of right legs: $\mathcal{C}_{/x \times y} \hookrightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$:

$$\begin{array}{ccccc} m \times n & \xrightarrow{\eta_m \times \text{id}_n} & (m \times_x x') \times n & \xrightarrow{\epsilon_m \times \text{id}_n} & m \times n \\ \downarrow & & \downarrow & & \downarrow \\ x' \times y & & x' \times y & & x \times y \\ \downarrow & & \downarrow & & \\ x \times y & & x \times y & & x \times y \end{array}$$

, which is equivalence by the triangle identity of $f_* \dashv f^*$.

- Likewise we can show it's horizontally right adjointable, with the Beck-Chavalley map at $m \otimes n$ being the image of the above map under the embedding of left legs: $\mathcal{C}_{x \times y}^{\text{op}} \rightarrow \text{Span}_1(\mathcal{C}_{/x \times y})$, and thus an equivalence. $\square$

Proposition 4.32. *There exists an extension of the lax symmetric monoidal functor $\psi : \mathcal{C}^{\text{op}, \Pi} \rightarrow \text{LMod}_{\mathcal{C}}(\text{Cat})^{\otimes} \hookrightarrow \overline{\text{LMod}_{\mathcal{C}}(\text{Cat})}^{\otimes}$ to a lax symmetric monoidal $(\infty, 2)$ functor:*

$$\psi : \text{Span}_1(\mathcal{C})^{\otimes} \rightarrow \overline{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}^{\otimes}$$

where $\mathcal{C}$ is an $(\infty, 1)$ category which is truncated and admits all pullbacks.

Proof. Since ψ is lax symmetric monoidal $(\infty, 2)$ biadjointable functor, by [22, theorem 4.33], it extends to a lax symmetric monoidal $(\infty, 2)$ functor $\text{Span}_2(\mathcal{C})^{\otimes} \rightarrow \overline{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})}^{\otimes}$ such that the precomposition with the symmetric monoidal functor $\mathcal{C}^{\text{op}, \Pi} \rightarrow \text{Span}_2(\mathcal{C})^{\otimes}$ is ψ . $\square$

5 $R : \mathcal{V}\text{-mod}(\text{Cat}) \hookrightarrow \text{Enr}_{\mathcal{P}(\mathcal{V})}$

5.1 construction of $L \dashv R$

This part follows [15, proposition 4.80]: the natural inclusion $R : \text{LMod}_{\mathcal{V}}(\text{Cat}) \hookrightarrow \text{PLMod}_{\mathcal{V}}$ admits a left adjoint L .

Consider the natural inclusion over $\text{Mon}(\text{Cat})$: $R : \text{LMod}(\text{Cat}) \rightarrow \text{PLMod}$, which gives rise to the following commutative diagram of ∞ -operad maps

$$\begin{array}{ccccc}
\mathrm{LMod}_{\mathcal{V}}(\mathrm{Cat})^{\otimes} & \xrightarrow{\quad} & \mathrm{LMod}(\mathrm{Cat})^{\times} & \xrightarrow{\quad R \quad} & \mathrm{PLMod}^{\times} \\
\downarrow & \searrow & \downarrow & & \downarrow \\
& & \mathrm{PLMod}_{\mathcal{V}}^{\otimes} & \xrightarrow{\quad} & \\
& & \downarrow & & \\
& & \{v^{\otimes}\} & & \\
& & \downarrow & & \\
\mathrm{Comm}^{\otimes} & \xrightarrow{\quad} & \mathrm{Mon}(\mathrm{Cat})^{\times} & \xrightarrow{\quad} & \mathrm{Mon}(\mathrm{Cat})^{\times} \\
\parallel & & \parallel & & \parallel \\
& & \mathrm{Comm}^{\otimes} & \xrightarrow{\quad} & \mathrm{Mon}(\mathrm{Cat})^{\times}
\end{array}$$

Especially, by pullback along the map $\mathrm{Comm}^{\otimes} \rightarrow \mathrm{Mon}(\mathrm{Cat})^{\times}$ classifying the commutative algebra object $v^{\otimes}$, we have an ∞ -operad map $\mathrm{LMod}_{\mathcal{V}}(\mathrm{Cat})^{\otimes} \rightarrow \mathrm{PLMod}_{\mathcal{V}}^{\otimes}$, which we again denote by R .

Now we construct a left adjoint of the natural inclusion $R : \mathrm{LMod}_{\mathcal{V}} \hookrightarrow \mathrm{PLMod}_{\mathcal{V}}$ without the ∞ -operad structure.

Recall the [15, notation 4.78]:

Construction 5.1. Given a category pseudo-enriched in $\mathcal{V}$: $\mathcal{M}^{\otimes} \rightarrow v^{\otimes}$, let

$$i : L(\mathcal{M})^{\otimes} \subset v^{\otimes} \times_{\mathcal{P}(\mathcal{V})^{\otimes}} \mathcal{P}'(\mathcal{M})^{\otimes}$$

be the full subcategory generated under $\mathcal{V}$ action by the images of $\mathcal{M} \rightarrow \mathcal{P}'(\mathcal{M})$, that is, the full weakly left tensored subcategory spanned by the objects in the form of $v \otimes m$ for $v \in \mathcal{V}$ and $m \in \mathcal{M}$, where $\mathcal{P}'(\mathcal{M})^{\otimes} = \mathrm{PLEnv}(\mathcal{M})_{\mathrm{Pseu}}^{\otimes}$.

Notation 5.2. There is a natural morphism by construction: $\mathcal{M}^{\otimes} \rightarrow L(\mathcal{M})^{\otimes}$, denoted as $\eta_{\mathcal{M}}$.

Here are some properties of our construction $L(\mathcal{M})$:

Lemma 5.3. The linear embedding $i : L(\mathcal{M})^{\otimes} \rightarrow \mathfrak{L}^*(\mathcal{P}'(\mathcal{M}))^{\otimes} = v^{\otimes} \times_{\mathcal{P}(\mathcal{V})^{\otimes}} \mathcal{P}'(\mathcal{M})^{\otimes}$ induces an equivalence:

$$\rho : \mathcal{P}(L(\mathcal{M}))^{\otimes} \rightarrow \mathcal{P}'(\mathcal{M})^{\otimes}$$

.

Proof. This is [15, lemma 4.79].

- By [14, corollary 9.32], the induced functor

$$\mathrm{LaxLin}(\mathcal{P}(L(\mathcal{M})), \mathcal{P}'(\mathcal{M})) \xrightarrow{\mathfrak{L}^*} \mathrm{LaxLin}(L(\mathcal{M}), \mathcal{P}'(\mathcal{M}))$$

admits a fully faithful left adjoint which image is exactly $\mathrm{Lin}^L(\mathcal{P}(L(\mathcal{M})), \mathcal{P}'(\mathcal{M}))$ by operadic left Kan extension. And by [14, proposition 9.30], we can restrict the Ass part to be identity, and gives rise to an equivalence:

$$\mathrm{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}(L(\mathcal{M})), \mathcal{P}'(\mathcal{M})) \rightarrow \mathrm{LaxLin}_{\mathcal{V}}(L(\mathcal{M}), \mathcal{P}'(\mathcal{M}))_{\mathfrak{L}}$$

where the right hand side denotes the pullback

$$\begin{array}{ccc}
\mathrm{LaxLin}_{\mathcal{V}}(L(\mathcal{M}), \mathcal{P}'(\mathcal{M}))_{\mathfrak{L}} & \xrightarrow{\quad} & \mathrm{LaxLin}(L(\mathcal{M}), \mathcal{P}'(\mathcal{M})) = \mathrm{Alg}_{L(\mathcal{M})/\mathrm{LM}}(\mathcal{P}'(\mathcal{N})) \\
\downarrow \lrcorner & & \downarrow \\
* & \xrightarrow{\{\mathfrak{L}\}} & \mathrm{Alg}_{\mathcal{V}/\mathrm{Ass}}(\mathcal{P}(\mathcal{V}))
\end{array}$$

and recall the Cartesian lifting in $\omega\mathrm{LMod}$: we have $\mathrm{LaxLin}_{\mathcal{V}}(L(\mathcal{M}), \mathcal{P}'(\mathcal{N}))_{\mathfrak{L}}^{\simeq} \cong \mathrm{LaxLin}_{\mathcal{V}}(L(\mathcal{M}), \mathfrak{L} * \mathcal{P}'(\mathcal{M})) = \mathcal{V}^{\otimes} \times_{\mathcal{P}(\mathcal{V})^{\otimes}} \mathcal{P}'(\mathcal{M})^{\otimes}$.

- Thus the inclusion $L(\mathcal{M})^{\otimes} \rightarrow \mathcal{V}^{\otimes} \times_{\mathcal{P}(\mathcal{V})^{\otimes}} \mathcal{P}'(\mathcal{M})^{\otimes}$ induces uniquely a $\mathcal{P}(\mathcal{V})$ -linear colimit preserving functor $\mathcal{P}(L(\mathcal{M}))^{\otimes} \rightarrow \mathcal{P}'(\mathcal{M})^{\otimes}$.
- The image of $\mathfrak{L}' : \mathcal{M} \hookrightarrow \mathcal{P}'(\mathcal{M})$ consists of $\mathcal{P}(\mathcal{V})$ -atomic objects by [proposition 2.58](#). And since the unit in $\mathcal{P}(\mathcal{V})^{\otimes}$ is complete compact, thus all the $\mathcal{P}(\mathcal{V})$ -atomic objects are also complete compact, and for $v \in \mathcal{V}$, and a $\mathcal{P}(\mathcal{V})$ -atomic object m , $v \otimes m$ satisfies the property that for any colimit $\mathrm{colim}_{\alpha} m_{\alpha}$ in $\mathcal{P}'(\mathcal{M})$, there are equivalences:

$$\begin{aligned}
\mathrm{Map}_{\mathcal{P}'(\mathcal{M})}(v \otimes m, \mathrm{colim}_{\alpha} m_{\alpha}) &\cong \mathrm{Map}_{\mathcal{P}(\mathcal{V})}(v, \mathrm{Mor}_{\mathcal{P}'(\mathcal{M})}(m, \mathrm{colim}_{\alpha} m_{\alpha})) \cong \\
\mathrm{Map}_{\mathcal{P}(\mathcal{V})}(v, \mathrm{colim}_{\alpha} \mathrm{Mor}_{\mathcal{P}'(\mathcal{M})}(m, m_{\alpha})) &\cong \mathrm{colim}_{\alpha} \mathrm{Map}_{\mathcal{P}(\mathcal{V})}(v, \mathrm{Mor}_{\mathcal{P}'(\mathcal{M})}(m, m_{\alpha})) \cong \mathrm{colim}_{\alpha} \mathrm{Map}_{\mathcal{P}'(\mathcal{M})}(v \otimes m, m_{\alpha})
\end{aligned}$$

where we use the fact that v is complete compact in $\mathcal{P}(\mathcal{V})$. Thus the objects in $L(\mathcal{M})$ in $\mathcal{P}'(\mathcal{M})$ are complete compact.

- And if we look at the $(\infty, 1)$ functor underlying the above linear functor, then it's an equivalence by the fact that the image of $L(\mathcal{M})$ in $\mathcal{P}'(\mathcal{M})$ consists of complete compact objects and then apply [\[21, corollary 5.1.6.11\]](#). And this implies the $\mathcal{V}$ -linear functor $\mathcal{P}(L(\mathcal{M}))^{\otimes} \rightarrow \mathcal{P}'(\mathcal{M})^{\otimes}$ is fiberwise equivalence, and thus is an equivalence.
- Here is another approach using the universal properties of $\mathcal{P}(\mathcal{M})$ and $\mathcal{P}'(\mathcal{M})$: consider the following diagram in PLMod :

$$\begin{array}{ccc}
\mathcal{M}^{\otimes} & & \\
\downarrow & \searrow \mathfrak{L}' & \\
L(\mathcal{M})^{\otimes} & \hookrightarrow & \mathcal{P}'(\mathcal{M})^{\otimes} \\
\downarrow \mathfrak{L} & \dashrightarrow \psi & \\
\mathcal{P}(L(\mathcal{M}))^{\otimes} & \xleftarrow{\phi} &
\end{array}$$

where ψ and ϕ are induced by the universal properties of $\mathcal{P}(\mathcal{M})$ (as shown above) and $\mathcal{P}'(\mathcal{M})$ (as a localization of presheaf category)

- To show $\psi \circ \phi$ is id, it suffices to show it after precomposition with $\mathfrak{L}$, which is true. And to show $\phi \circ \psi$ is id, it suffices to show it after precomposition with $\mathfrak{L}'$, which is also true.

□

To show $R : \mathbf{LMod}_{\mathcal{V}}(\mathbf{Cat}) \rightarrow \mathbf{PLMod}_{\mathcal{V}}$ admits a left adjoint, it suffices to show for each object $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes} \in \mathbf{PLMod}_{\mathcal{V}}$, there exists a map $\eta_{\mathcal{M}} : \mathcal{M}^{\otimes} \rightarrow L(\mathcal{M})^{\otimes}$ such that the induced mapping spaces:

$$\mathbf{Map}_{\mathbf{LMod}(\mathbf{Cat})}(L(\mathcal{M}), \mathcal{N}) \xrightarrow{R(-)} \mathbf{Map}_{\mathbf{PLMod}_{\mathcal{V}}}(RL(\mathcal{M}), R(\mathcal{N})) \xrightarrow{\eta_{\mathcal{M}}^*} \mathbf{Map}_{\mathbf{PLMod}_{\mathcal{V}}}(\mathcal{N}, R(\mathcal{N}))$$

is an equivalence for any $\mathcal{N} \in \mathbf{LMod}_{\mathcal{V}}(\mathbf{Cat})$ (This implies the coCartesian lifting classifying the functor R is also a Cartesian fibration).

Proposition 5.4. *The inclusion $\eta_{\mathcal{M}} : \mathcal{M}^{\otimes} \rightarrow L(\mathcal{M})^{\otimes}$ induces an equivalence:*

$$\mathbf{Lin}_{\mathcal{V}}(L(\mathcal{M}), \mathcal{N})^{\simeq} \rightarrow \mathbf{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})^{\simeq}$$

for any $\mathcal{N} \in \mathbf{LMod}_{\mathcal{V}}(\mathbf{Cat})$.

Proof. The proof is basically the same as [15, proposition 4.78]:

- $\mathbf{Lin}_{\mathcal{V}}(L(\mathcal{M}), \mathcal{N}) \rightarrow \mathbf{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})$ is essential surjective: given a lax linear functor $F : \mathcal{M}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$, the composition $\mathfrak{L} \circ F : \mathcal{M}^{\otimes} \rightarrow \mathcal{P}(\mathcal{N})^{\otimes}$ induces a unique colimit preserving $\mathcal{P}(\mathcal{V})$ -linear functor $\mathcal{P}'(\mathcal{M})^{\otimes} \rightarrow \mathcal{P}(\mathcal{N})^{\otimes}$, by the following commutative diagram:

$$\begin{array}{ccccc} \mathcal{M}^{\otimes} & \xrightarrow{F} & \mathcal{N}^{\otimes} & \xrightarrow{\mathfrak{L}} & \mathcal{P}(\mathcal{N})^{\otimes} \\ \eta_{\mathcal{M}} \downarrow & \nearrow & & \nearrow & \text{induced} \\ L(\mathcal{M})^{\otimes} & \longrightarrow & \mathcal{P}'(\mathcal{M})^{\otimes} & & \end{array}$$

there is a factorization from $L(\mathcal{M}) \rightarrow \mathcal{N}$ since the image of composition of the bottom edges lands in the full subcategory $\mathcal{N} \subset \mathcal{P}(\mathcal{N})$.

- Thus by the following commutative diagram:

$$\begin{array}{ccc} \mathbf{Lin}_{\mathcal{V}}(L(\mathcal{M}), \mathcal{N})^{\simeq} & \xrightarrow{\eta_{\mathcal{M}}^*} & \mathbf{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})^{\simeq} \\ \mathfrak{L} \downarrow & & \downarrow \mathfrak{L} \\ \mathbf{Lin}_{\mathcal{V}}(L(\mathcal{M}), \mathfrak{L}^* \mathcal{P}(\mathcal{N}))^{\simeq} & \longrightarrow & \mathbf{LaxLin}_{\mathcal{V}}(L(\mathcal{M}), \mathfrak{L}^* \mathcal{P}(\mathcal{N}))^{\simeq} \end{array}$$

where all the vertical edges are fully faithful, so if the bottom edge is an equivalence, then the horizontal edge is also full faithful, and thus is equivalence.

- So we replace $\mathcal{N}^{\otimes}$ by $\mathcal{P}(\mathcal{N})^{\otimes}$, and the equivalence:

$$\eta_{\mathcal{M}}^* : \mathbf{Lin}_{\mathcal{V}}(L(\mathcal{M}), \mathfrak{L}^* \mathcal{P}(\mathcal{N}))^{\simeq} \rightarrow \mathbf{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathfrak{L}^* \mathcal{P}(\mathcal{N}))^{\simeq}$$

is due to the following commutative diagram:

$$\begin{array}{ccc}
\mathrm{Lin}_{\mathcal{P}(\mathcal{V})}(\mathcal{P}'(\mathcal{M}), \mathcal{P}(\mathcal{N}))^{\simeq} & & \\
\uparrow \wr^* & \searrow \wr'^* & \\
& & \mathrm{LaxLin}_{\mathcal{V}}(\mathcal{M}, \wr^* \mathcal{P}(\mathcal{N}))^{\simeq} \\
& \nearrow \eta_{\mathcal{M}}^* & \\
\mathrm{Lin}_{\mathcal{V}}(L(\mathcal{M}), \wr^* \mathcal{P}(\mathcal{N}))^{\simeq} & &
\end{array}$$

where $\wr'^*$ and $\wr^*$ are both equivalences by the universal properties of $\mathcal{P}'(-)$ and $\mathcal{P}(-)$ and the equivalence $\mathcal{P}'(\mathcal{M})^{\otimes} \cong \mathcal{P}(L(\mathcal{M}))^{\otimes}$ by the lemma above. Thus $\eta_{\mathcal{M}}^*$ is an equivalence. $\square$

Thus we have an adjunction:

$$L : \mathrm{PLMod}_{\mathcal{V}} \rightleftarrows \mathrm{LMod}_{\mathcal{V}}(\mathrm{Cat}) : R$$

where the left adjoint L assign each $\mathcal{M} \in \mathrm{PLMod}_{\mathcal{V}}$ with the left $\mathcal{V}$ tensored ∞ -category $L(\mathcal{M})$ constructed as above.

5.2 the left adjoint L is symmetric monoidal $(\infty, 2)$ functor

In this subsection, we enhance the left adjoint $L : \mathrm{PLMod}_{\mathcal{V}} \rightarrow \mathrm{LMod}_{\mathcal{V}}$ to be a symmetric monoidal functor.

We already have a map between operads: $R : \mathrm{LMod}_{\mathcal{V}}^{\otimes} \rightarrow \mathrm{PLMod}_{\mathcal{V}}^{\otimes}$ over $\mathrm{Comm}^{\otimes}$, and fiber-wise $R_{\langle n \rangle} : (\mathrm{LMod}_{\mathcal{V}}^{\otimes})_{\langle n \rangle} \cong \mathrm{LMod}_{\mathcal{V}}^{\times n} \rightarrow \mathrm{PLMod}_{\mathcal{V}}^{\times n} \cong (\mathrm{PLMod}_{\mathcal{V}}^{\otimes})_{\langle n \rangle}$ admits a left adjoint $L^{\times n} : \mathrm{PLMod}_{\mathcal{V}}^{\times n} \rightarrow \mathrm{LMod}_{\mathcal{V}}^{\times n}$. Thus the left adjoint L can be equipped with an oplax symmetric monoidal structure by [13, proposition A]. And we have the following sufficient condition for it to be symmetric monoidal:

Proposition 5.5. *Suppose we are given a commutative diagram of ∞ operads:*

$$\begin{array}{ccc}
C^{\otimes} & \xrightarrow{G} & D^{\otimes} \\
& \searrow p \quad \swarrow q & \\
& O^{\otimes} &
\end{array}$$

where p and q are coCartesian fibrations, assume that:

- (1) For each object $X \in O$, the functor $G_X : C_X \rightarrow D_X$ admits a left adjoint L_X .
- (2) For each operation $\phi \in \mathrm{Mul}_O(\{X_i\}_i, Y)$ in $O^{\otimes}$, if we let $\phi_!^C$ and $\phi_!^D$ denote the associated functors $\prod_i C_{X_i} \rightarrow C_Y$ and $\prod_i D_{X_i} \rightarrow D_Y$, then the natural transformation $F_Y \circ \phi_!^C \rightarrow \phi_!^D \circ \prod_i F_{X_i}$ is an equivalence of functors.

Proof. By [22, 7.3.2.12]. $\square$

Especially, consider the following lax square induced by taking pullback of the above diagram along $[1] \rightarrow O^\otimes$ corresponding to the morphism ϕ :

$$\begin{array}{ccc} \prod_i C_{X_i} & \xrightarrow{\prod_i G_{X_i}} & \prod_i D_{X_i} \\ \phi_!^C \downarrow & \swarrow & \downarrow \phi_!^D \\ C_Y & \xrightarrow{G_Y} & D_Y \end{array}$$

By taking mate correspondence (Beck-Chevalley transformation), we have the natural transformation mentioned in the above proposition:

$$\begin{array}{ccc} \prod_i D_{X_i} & \xrightarrow{\prod_i F_{X_i}} & \prod_i C_{X_i} \\ \phi_!^D \downarrow & \searrow & \downarrow \phi_!^C \\ D_Y & \xrightarrow{F_Y} & C_Y \end{array}$$

which is the following composition of 2-morphisms:

$$F_Y \circ \phi_!^D \rightarrow F_Y \circ \phi_!^D \circ (\prod_i G_{X_i} \circ \prod_i F_{X_i}) \rightarrow F_Y \circ (\phi_!^D \circ \prod_i G_{X_i}) \circ \prod_i F_{X_i} \rightarrow F_Y \circ (G_Y \circ \phi_!^C) \circ \prod_i F_{X_i} \rightarrow (F_Y \circ G_Y) \circ \phi_!^C \circ \prod_i F_{X_i} \rightarrow \phi_!^C \circ \prod_i F_{X_i}.$$

Example 5.6. For the case when $O^\otimes = \text{Comm}^\otimes$ and G is an operad morphism, consider the following lax square corresponding to an inert morphism $\phi \in \text{Mul}_{\text{Comm}}(\langle n \rangle, \langle m \rangle)$, we can regard ϕ as picking the sub-interval $\langle m \rangle^\circ = J \subset I = \langle n \rangle^\circ$:

$$\begin{array}{ccc} \prod_I C & \xrightarrow{\prod_I G} & \prod_I D \\ \phi_!^C \downarrow & \swarrow & \downarrow \phi_!^D \\ \prod_J C & \xrightarrow{\prod_J G} & \prod_J D \end{array}$$

Since G is an operad morphism, the above lax square is strict, and by passing to the left adjoint of the horizontal arrows, we acquire the mate:

$$\begin{array}{ccc} \prod_I D & \xrightarrow{\prod_I F} & \prod_I C \\ \phi_!^D \downarrow & \searrow & \downarrow \phi_!^C \\ \prod_J D & \xrightarrow{\prod_J F} & \prod_J C \end{array}$$

It turns out that the above lax square is also strict:

Claim: The above mate lax square is strict.

Proof. Since both $\phi_!^C$ and $\phi_!^D$ are projection from I -product to J -product for $J \subset I$, we can regard them as $\prod_I D \rightarrow \prod_I D' = \prod_J D \times \prod_{I \setminus J} *$ and $\prod_I C \rightarrow \prod_I C' = \prod_J C \times \prod_{I \setminus J} *$.

And to see a natural transformation $\eta : [1] \times \prod_I X_i \rightarrow \prod_I Y_i$ between functors in the form

of $\prod_I f_i, \prod_I g_i : \prod_I X \rightarrow \prod_I Y$ is natural isomorphism, it suffices to check componentwise projection $\eta_i : [1] \times X_i \rightarrow Y_i$ is a natural isomorphism. In the above case, the componentwise transformation is just the triangle identity $F \rightarrow F(GF) \rightarrow ((FG)F \rightarrow F$, thus is a natural isomorphism. $\square$

In combination of the above proposition and example, to see whether the left adjoint of a lax symmetric monoidal functor is symmetric monoidal, it suffices to check the conditions (2) in [22, corollary 7.3.2.12] for the active map, and we have the following corollary:

Corollary 5.7. *Suppose we are given a commutative diagram of symmetric monoidal categories:*

$$\begin{array}{ccc} C^\otimes & \xrightarrow{G} & D^\otimes \\ & \searrow p \quad \swarrow q & \\ & \text{Comm}^\otimes & \end{array}$$

where p and q are coCartesian fibrations, assume that:

- (1) *The underlying functor $G : C \rightarrow D$ admits a left adjoint L .*
- (2) *For each active morphism $\phi \in \text{Mul}_{\text{Comm}}(\langle n \rangle, \langle 1 \rangle)$ in $O^\otimes$, if we let $\phi_!^C$ and $\phi_!^D$ denote the associated functors $\prod_i C_{X_i} \rightarrow C_Y$ and $\prod_i D_{X_i} \rightarrow D_Y$ (which corresponds to the tensor product operations), then the natural transformation $F_Y \circ \phi_!^C \rightarrow \phi_!^D \circ \prod_i F_{X_i}$ is an equivalence of functors.*

Proof. Given any morphism $\phi : \langle n \rangle \rightarrow \langle 1 \rangle$, it factors uniquely as $\phi : \phi_1 \circ \phi_0$ where ϕ_0 is inert and ϕ_1 is active.

Consider the lax square induced by ϕ , which can be factorized as:

$$\begin{array}{ccc} \prod_I C & \xrightarrow{\prod_i G} & \prod_I D \\ (\phi_0^C)_! \downarrow & \cong \nearrow & \downarrow (\phi_0^D)_! \\ \prod_J C & \xrightarrow{\prod_j G} & \prod_J D \\ (\phi_1^C)_! \downarrow & \nearrow & \downarrow (\phi_1^D)_! \\ C & \xrightarrow{G} & D \end{array}$$

By the fact that mate (Beck-Chevalley) transformation is compatible with composition (both horizontally and vertically), by the condition (2) and the above example, we see that the mate of the lax square induced by ϕ is the composition of two strict squares, which is again strict, satisfying the second condition in [22, 7.3.2.12]. $\square$

Proposition 5.8. *Under the given symmetric monoidal structures on PLMod_V and LMod_V as above, $L : \text{PLMod}_V \rightarrow \text{LMod}_V$ satisfies the following property:*

$$k : L(\boxtimes_V \mathcal{M}_i) \cong \otimes_V L(\mathcal{M}_i)$$

is an equivalence where k is induced in the following way:

$$L(\boxtimes_V \mathcal{M}_i) \xrightarrow{L(\boxtimes_{\text{unit}})} L(\boxtimes_V RL(\mathcal{M}_i)) \rightarrow LR(\otimes_V (L(\mathcal{M}_i))) \xrightarrow{\text{counit}} \otimes_V L(\mathcal{M}_i)$$

where the middle edge is the natural map induced by the lax symmetric monoidal structure on R .

Notation 5.9. We denote the morphism $\boxtimes_{\mathcal{V}} RL\mathcal{M}_i \rightarrow R(\otimes_{\mathcal{V}} L\mathcal{M}_i)$ induced by the lax symmetric monoidal structure on R by ψ .

Remark 5.10. Equivalently, k is also induced by the morphism $l = \psi \circ \boxtimes \text{unit}_i : \boxtimes_{\mathcal{V}} \mathcal{M}_i \xrightarrow{\boxtimes \text{unit}_i} \boxtimes_{\mathcal{V}} RL\mathcal{M}_i \xrightarrow{\psi} R(\otimes_{\mathcal{V}} L\mathcal{M}_i)$ under the adjunction $\text{Map}_{\text{PLMod}_{\mathcal{V}}}(L(\boxtimes_{\mathcal{V}} \mathcal{M}_i), \otimes_{\mathcal{V}} L\mathcal{M}_i) \cong \text{Map}_{\text{PLMod}_{\mathcal{V}}}(\boxtimes_{\mathcal{V}} \mathcal{M}_i, R(\otimes_{\mathcal{V}} L\mathcal{M}_i))$.

Remark 5.11. Without abuse of notation, we omit the (non-full) subcategory inclusion R when we regard a left $\mathcal{V}$ module as a category pseudo-enriched in $\mathcal{V}$.

Remark 5.12. Recall how we construct the left adjoint L , given a morphism in PLMod $l : \boxtimes_{\mathcal{V}} \mathcal{M}_i \rightarrow \otimes_{\mathcal{V}} L\mathcal{M}_i$, the corresponding morphism under adjunction, denoted as $k : L(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \rightarrow \otimes_{\mathcal{V}} L\mathcal{M}_i$ is induced from the following factorization:

$$\begin{array}{ccc}
 L(\boxtimes_{\mathcal{V}} \mathcal{M}_i) & \xhookrightarrow{i} & \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \\
 \uparrow & \nearrow \wr' & \downarrow \phi \\
 \boxtimes_{\mathcal{V}} \mathcal{M}_i & & \mathcal{P}(\otimes_{\mathcal{V}} L\mathcal{M}_i) \\
 \downarrow l & \nwarrow \wr & \\
 \otimes_{\mathcal{V}} L\mathcal{M}_i & &
 \end{array}$$

Especially, the composition $\phi \circ i : L(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \rightarrow \mathcal{P}(\otimes_{\mathcal{V}} L\mathcal{M}_i)$ factors through $\otimes_{\mathcal{V}} L\mathcal{M}_i$ by our construction, this gives us the corresponding morphism $k : L(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \rightarrow \otimes_{\mathcal{V}} L\mathcal{M}_i$

And it's essential to unpack the morphism ϕ above. It turns out that ϕ is an equivalence of presentably $\mathcal{P}(\mathcal{V})$ -left modules:

Consider the following commutative diagram :

$$\begin{array}{ccc}
 L(\boxtimes_{\mathcal{V}} \mathcal{M}_i) & \xhookrightarrow{i} & \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i) \\
 \uparrow & \nearrow \wr' & \downarrow \phi'' \\
 \boxtimes_{\mathcal{V}} \mathcal{M}_i & & \mathcal{P}'(\boxtimes_{\mathcal{V}} L\mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(L\mathcal{M}_i) \\
 \boxtimes \text{unit}_i \downarrow & \nearrow \wr' & \downarrow \phi' \\
 \boxtimes_{\mathcal{V}} L\mathcal{M}_i & & \\
 \psi \downarrow & \xrightarrow{\wr} & \mathcal{P}(\otimes_{\mathcal{V}} L\mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}(L\mathcal{M}_i) \\
 \otimes_{\mathcal{V}} L\mathcal{M}_i & &
 \end{array}$$

where

- ϕ' and ϕ'' are the colimit preserving linear functors induced by the universal property of $\mathcal{P}'$, and $\phi = \phi' \circ \phi''$ by the universal property.

- i is the inclusion in the [construction of \$L\$](#) .
- and ψ is induced by the lax symmetric monoidal structure on R , and $l = \psi \circ \boxtimes \text{unit}_i$.

And in the following subsections, we shall show

- (1) $\mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$ by [corollary 5.24](#).
- (2) $\psi : \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i) \rightarrow \mathcal{P}(\otimes_{\mathcal{V}} L \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}(L \mathcal{M}_i)$ is induced component-wise by the [equivalence](#) $\rho_i : \mathcal{P}'(\mathcal{M}_i) \cong \mathcal{P}(L \mathcal{M}_i)$, thus ψ is an equivalence, by [proposition 5.25](#).

And under the above two conditions (1), (2), then we can show L is symmetric monoidal $(\infty, 1)$ functor:

proof of proposition 5.8. First we can extract a subdiagram from the diagram above:

$$\begin{array}{ccc}
 \otimes_{\mathcal{V}} L \mathcal{M}_i & \xrightarrow{\quad \mathfrak{z} \quad} & \mathcal{P}(\otimes_{\mathcal{V}} L \mathcal{M}_i) \\
 \uparrow \scriptstyle \psi \circ (\prod \text{unit}_i) & & \uparrow \scriptstyle \cong \\
 \boxtimes_{\mathcal{V}} \mathcal{M}_i & \xrightarrow{\quad \mathfrak{z}' \quad} & \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i)
 \end{array}$$

And recall how we produce k : since $L(\boxtimes_{\mathcal{V}} \mathcal{M}_i)$ is spanned by the image of $\boxtimes_{\mathcal{V}} \mathcal{M}_i \xrightarrow{\mathfrak{z}'} \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \xrightarrow{\phi, \cong} \mathcal{P}(\otimes_{\mathcal{V}} L \mathcal{M}_i)$, we have a factorization through $L(\boxtimes_{\mathcal{V}} \mathcal{M}_i)$:

$$\begin{array}{ccccc}
 \otimes_{\mathcal{V}} L \mathcal{M}_i & \xrightarrow{\quad \mathfrak{z} \quad} & & \mathcal{P}(\otimes_{\mathcal{V}} L \mathcal{M}_i) & \\
 \uparrow \scriptstyle \psi \circ (\prod \text{unit}_i) & \searrow & \otimes_{\mathcal{V}} L \mathcal{M}_i & \nearrow & \uparrow \scriptstyle \cong \\
 \boxtimes_{\mathcal{V}} \mathcal{M}_i & \xrightarrow{\quad \mathfrak{z}' \quad} & & \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) & \\
 & \searrow & \uparrow \scriptstyle k & \nearrow \scriptstyle i & \\
 & & L(\boxtimes_{\mathcal{V}} \mathcal{M}_i) & &
 \end{array}$$

Thus $k : L(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \rightarrow \otimes_{\mathcal{V}} L \mathcal{M}_i$ is fully faithful since both $i : L(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \rightarrow \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i)$ and $\mathfrak{z} : \otimes_{\mathcal{V}} L \mathcal{M}_i \rightarrow \mathcal{P}(\otimes_{\mathcal{V}} L \mathcal{M}_i)$ are fully faithful.

And it remains to show k is essential surjective, recall we actually have the following diagram:

$$\begin{array}{ccc}
& & \otimes_{\mathcal{V}} L\mathcal{M}_i \\
& \nearrow \otimes & \uparrow \psi \\
\prod L\mathcal{M}_i & \longrightarrow & \boxtimes_{\mathcal{V}} L\mathcal{M}_i \\
\uparrow \prod \text{unit}_i & & \uparrow \boxtimes \text{unit}_i \\
\prod \mathcal{M}_i & \longrightarrow & \boxtimes_{\mathcal{V}} \mathcal{M}_i
\end{array}$$

Since the coCartesian lifting $\otimes : \prod L\mathcal{M}_i \rightarrow \otimes_{\mathcal{V}} L\mathcal{M}_i$ is the first term in the Bar construction by [remark 4.29](#), thus is essential surjective. So $\otimes_{\mathcal{V}} L\mathcal{M}_i$ is generated under $\mathcal{V}$ -linear action by the image of $\prod L\mathcal{M}_i$, and $\prod L\mathcal{M}_i$ is generated under $\prod_{i=1}^n \mathcal{V}$ -linear action by the image of $\prod \mathcal{M}_i$, thus $\otimes_{\mathcal{V}} L\mathcal{M}_i$ is generated under the $\mathcal{V}$ -linear action by the image of $\prod \mathcal{M}_i$, thus is also generated by the image of $k := \psi \circ (\boxtimes \text{unit}_i)$. $\square$

5.3 $\mathcal{P}'(\boxtimes_{\mathcal{V}}(\mathcal{M}_i)) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$

In this subsection, we show the equivalence $\mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}(\mathcal{M}_i)$. To do this, we

- First recall the suboperad $\text{PrLMod}^{\otimes} \subset \widehat{\text{LMod}}^{\times}$.
- Then show that $\mathcal{P}'(\prod_{i=1}^n \mathcal{M}_i) \cong \otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$ for $\mathcal{M}_i^{\otimes} \rightarrow \mathcal{V}_i^{\otimes} \in \text{PLMod}$.
- Then we show $\mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$

Recall in [22, 4.8.5.14], the operad $\text{PrLMod}^{\otimes}$ is a suboperad of $\text{PrLMod}^{\times} \subset \widehat{\text{LMod}}^{\times}$ consisting of:

- Object: $\{(C_i, M_i)\}_i$ where (C_i, M_i) is a presentably left tensored category, that is, an object in $\text{Alg}_{\text{LM}}(\text{Pr}^L)$.
- Morphism: a multimorphism over $\alpha : \langle n \rangle \rightarrow \langle m \rangle$ in $\widehat{\text{LMod}}^{\times}$, $F : \{C_i, M_i\}_{i \in \langle n \rangle^{\circ}} \rightarrow \{(D_j, N_j)\}_{j \in \langle m \rangle^{\circ}}$ is a multimorphism in $\text{PrLMod}^{\otimes}$ iff for each $j \in \langle m \rangle^{\circ}$ $\prod_{i: \alpha(i)=j} C_i \rightarrow C_j$ preserves colimits separately in each variable, and $\prod_{i: \alpha(i)=j} M_i \rightarrow M_j$ preserves colimits separately in each variable.
- We can identify $\text{PrLMod}^{\otimes}$ as $\text{Alg}_{\text{LM}}(\text{Pr}^L)^{\otimes}$ by [22, 4.8.5.16].

Moreover, $\text{PrLMod}^{\otimes}$ is coCartesian fibration over $\text{Comm}^{\otimes}$, and given an active map $\alpha : \langle n \rangle \rightarrow \langle 1 \rangle$, and an object $\{(C_i, M_i)\}_{i \in \langle n \rangle^{\circ}}$ lying over $\langle n \rangle$, the coCartesian lifting is $(\otimes_{i=1}^n C_i, \otimes_{i=1}^n M_i)$, since fiberwise the tensor product operation in $\text{Alg}_{\text{LM}}(\text{Pr}^L)^{\otimes}$ is just tensor product operation in $\text{Pr}^{L, \otimes}$ by [22, 3.2.4.4].

Construction 5.13. Consider the coCartesian edge over the active map α in $\text{PrLMod}^\otimes$ and its image under the inclusion $\text{PrLMod}^\otimes \hookrightarrow \widehat{\text{LMod}}^\times$, we have the following factorization in $\widehat{\text{LMod}}^\times$:

$$\begin{array}{ccccc}
 & & d & & \\
 & \nearrow d_1 & & \searrow d_2 & \\
 \{(C_i, M_i)\}_i & \xrightarrow{\quad} & (\prod C_i, \prod M_i) & \xrightarrow{\quad} & (\otimes C_i, \otimes M_i) \\
 \downarrow & & \downarrow & & \downarrow \\
 \{C_i\}_i & \xrightarrow{\quad} & \prod C_i & \xrightarrow{q} & \otimes C_i
 \end{array}$$

where d_1 is the coCartesian lifting in $\widehat{\text{LMod}}^\times$, and fiberwise d_2 is the universal componentwise colimit preserving functor $\prod C_i \rightarrow \otimes C_i, \prod M_i \rightarrow \otimes M_i$ developed in [22, 4.8.1].

And given $(D, N) \in \text{PrLMod}$, there are equivalences of mapping spaces:

$$\text{Lin}^L((\otimes C_i, \otimes M_i), (D, N))^{\simeq} \rightarrow \text{Lin}^{LL}((\prod C_i, \prod M_i), (D, N))^{\simeq} \rightarrow \text{Mul}_{\text{PrLMod}^\times}(\{(C_i, M_i)\}, (D, N))$$

And especially for $N \in \text{LMod}_{\otimes C_i}(\text{Pr}^L)$, there is an equivalence of mapping spaces induced by the universal componentwise colimit preserving functor $\prod C_i \rightarrow \otimes C_i$:

$$\text{Lin}_{\otimes C_i}(\otimes M_i, N)^{\simeq} \rightarrow \text{Lin}_{\prod C_i}(\prod M_i, N')^{\simeq}$$

is an equivalence where N' is the pullback of N along $\prod C_i \rightarrow \otimes C_i$.

Notation 5.14. Given n objects $\{\mathcal{M}_i^\otimes\}_{i=1}^n$ in PrLMod , we denote their tensor product as $\otimes \mathcal{M}_i^\otimes$.

By [22, 4.8.5.16], $\text{PrLMod}^\otimes \rightarrow \text{Alg}(\text{Pr}^L)^\otimes$ is coCartesian fibration of symmetric monoidal categories, and especially, we have a lax symmetric monoidal inclusion:

$$\begin{array}{ccc}
 \text{PrLMod}^\otimes & \hookrightarrow & \widehat{\text{LMod}}^\times \\
 \downarrow & & \downarrow \\
 \text{Alg}(\text{Pr}^L)^\otimes & \hookrightarrow & \text{Alg}(\widehat{\text{Cat}})^\times
 \end{array}$$

Recall the construction of symmetric monoidal structure on $\text{LMod}_{\mathcal{P}(\mathcal{V})}(\text{Pr}^L)^\otimes$, which is just taking fiber product along $\text{Comm}^\otimes \rightarrow \text{Alg}(\text{Pr}^L)^\otimes$ specifying the object $\mathcal{P}(\mathcal{V})$ equipped with the symmetric monoidal structure induced from $\mathcal{V}$.

Especially, given n objects $\{\mathcal{M}_i^\otimes \rightarrow \mathcal{V}^\otimes\}_{i=1}^n \in \text{PLMod}_{\mathcal{V}}$, consider the coCartesian lifting over the active map $\langle n \rangle \rightarrow \langle 1 \rangle$: $\{(\mathcal{P} \mathcal{V}, \mathcal{P}' \mathcal{M}_i)\}_{i \in \langle n \rangle^\circ} \rightarrow \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}' \mathcal{M}_i$ which corresponds to the tensor product operation, there is factorization after embedding into $\widehat{\text{LMod}}^\times$:

$$\{(\mathcal{P} \mathcal{V}, \mathcal{P}' \mathcal{M}_i)\}_{i \in \langle n \rangle^\circ} \longrightarrow (\prod_{i=1}^n \mathcal{P} \mathcal{V}, \prod_{i=1}^n \mathcal{P}' \mathcal{M}_i) \longrightarrow (\mathcal{P} \mathcal{V}, \otimes_{\mathcal{P} \mathcal{V}} \mathcal{P}' \mathcal{M}_i)$$

We notice the last morphism is induced by the universal property of coCartesian edge in $\widehat{\text{LMod}}^\times$, and we denote it by q .

Now consider the morphism under the inclusion $\text{PrLMod}^\otimes \hookrightarrow \widehat{\text{LMod}}^\times : \prod \text{loc} : \{(\mathcal{P}\text{Env } \mathcal{V}_i, \mathcal{P}\text{LEnv}(\mathcal{M}_i))\}_i \rightarrow \{(\mathcal{P} \mathcal{V}_i, \mathcal{P}'(\mathcal{M}_i))\}_i$ where each component is the localization functor (with respect to the small localization pair $\text{Pseu}(\mathcal{P}\text{Env } \mathcal{V}_i, \mathcal{P}\text{LEnv}(\mathcal{M}_i)) \rightarrow (\mathcal{P} \mathcal{V}_i, \mathcal{P}'(\mathcal{M}_i))$) as developed in [lemma 2.28](#):

$$\begin{array}{ccccc}
\prod_{i=1}^n \mathcal{M}_i^\otimes & \xleftarrow{i} & \prod_{i=1}^n \mathcal{P}\text{LEnv}(\mathcal{M}_i)^\otimes & \xrightarrow{p} & \otimes_{i=1}^n \mathcal{P}\text{LEnv}(\mathcal{M}_i)^\otimes \\
& & \downarrow & & \downarrow \\
& & \prod_{i=1}^n \mathcal{P}\text{Env}(\mathcal{V}_i)^\otimes & \xrightarrow{p'} & \otimes_{i=1}^n \mathcal{P}\text{Env}(\mathcal{V}_i)^\otimes \\
& \swarrow \text{loc} & & \searrow \gamma & \\
\prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^\otimes & \xleftarrow{\text{loc}} & \prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i) & \xrightarrow{q} & \otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i) \\
& \downarrow & & \downarrow \gamma' & \\
\prod_{i=1}^n \mathcal{P} \mathcal{V}_i^\otimes & \xrightarrow{q'} & \otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)^\otimes & \cong & \mathcal{P}(\prod_{i=1}^n \mathcal{V}_i)^\otimes
\end{array}$$

Proposition 5.15. *The composition of the red arrows above: $q \circ \prod \text{loc} \circ i : \prod_{i=1}^n \mathcal{M}_i^\otimes \rightarrow \otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$ induces an equivalence between $\otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^\otimes$ and $\mathcal{P}'(\prod_{i=1}^n \mathcal{M}_i)^\otimes$.*

To prove the claim, we shall use the following lemma, which is [proposition 2.60](#).

Lemma 5.16. *Given an absolute small pseudo-enriched category $\mathcal{M}^\otimes \rightarrow \mathcal{V}^\otimes$, and a locally small left tensored category $\mathcal{N}^\otimes \rightarrow \mathcal{P}(\mathcal{V})^\otimes$, if there exists an embedding $\psi : \mathcal{M}^\otimes \rightarrow \mathcal{N}^\otimes$, and for every $X \in \mathcal{M}$, the image $\psi(X)$ is atomic, there exists an embedding $\psi' : \mathcal{P}'(\mathcal{M}) \rightarrow \mathcal{N}$ by the universal property of $\mathcal{P}'$. Moreover, when $\mathcal{N}$ is generated under small colimits and left tensors by the essential image of $\mathcal{M}$, ψ' is an equivalence.*

Thus to prove the above claim, we shall show the following two properties of $q \circ \prod \text{loc} \circ i$:

- (1) The image consists of $\otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)$ -atomic objects.
- (2) The functor $q \circ \prod \text{loc} \circ i$ is an embedding (a fully faithful functor).

Since by [\[15, lemma 4.64\]](#), $\otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$ is generated by the essential image of $\prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$ under small colimits, thus any full subcategory of $\otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$ containing the essential image of $\prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$ and closed under colimits is equivalent to $\otimes_{i=1}^n \mathcal{P}(\mathcal{M}_i)$ itself. Thus it remains to show any full subcategory of $\otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$ containing the essential image of $\prod_{i=1}^n \mathcal{M}_i$ and closed under left tensors and colimits will contain the essential image of $\prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$, which is true by [proposition 2.58](#): the image of the embedding $\hookrightarrow' : \mathcal{M}_i^\otimes \hookrightarrow \mathcal{P}'(\mathcal{M}_i)^\otimes$ atomically generates presentably left tensored ∞ -category $\mathcal{P}'(\mathcal{M}_i)$.

Let's show the property (1) first.

Lemma 5.17. *Given $M \in \text{LMod}_{\mathcal{V}}(\text{Pr}^L)$, and $N \in \text{LMod}_{\mathcal{W}}(\text{Pr}^L)$, $m \in M$ is an $\mathcal{V}$ -atomic object, $n \in N$ is an $\mathcal{W}$ -atomic object, then $m \otimes n \in M \otimes N$ is an $\mathcal{V} \otimes \mathcal{W}$ -atomic object.*

Proof. By the definition of atomic object, $m \in M \in \mathbf{LMod}_{\mathcal{V}}(\mathbf{Pr}^L)$ is $\mathcal{V}$ -atomic iff the unique morphism $\mathcal{V} \rightarrow M$ in $\mathbf{LMod}_{\mathcal{V}}(\mathbf{Pr}^L)$ sending $1_{\mathcal{V}} \mapsto m$ is internally left adjoint in $\mathbf{LMod}_{\mathcal{V}}(\mathbf{Pr}^L)$ ([29, 3.30]).

Now we have $\mathcal{V}$ -internal left adjoint $m : \mathcal{V} \rightarrow M$ and $\mathcal{W}$ -internal left adjoint $n : \mathcal{W} \rightarrow N$. By [29, 3.40], their tensor product $m \otimes n : \mathcal{V} \otimes \mathcal{W} \rightarrow N \otimes M$ sending $1_{\mathcal{V}} \otimes 1_{\mathcal{W}}$ to $m \otimes n$ is an internal left adjoint in $\mathbf{LMod}_{\mathcal{V} \otimes \mathcal{W}}(\mathbf{Pr}^L)$. Thus $m \otimes n$ is $\mathcal{V} \otimes \mathcal{W}$ -atomic object. $\square$

Lemma 5.18. *Given a pseudo-enriched category $\mathcal{M}_i^{\otimes} \rightarrow \mathcal{V}_i^{\otimes}$, the essential image of the natural embedding $\mathcal{M}_i \rightarrow \mathcal{P}'(\mathcal{M}_i)$ consists of $\mathcal{V}_i$ -atomic object.*

Proof. This is [15, 4.50]. $\square$

Corollary 5.19. *The image of $q \circ \prod \text{loc} \circ i : \prod_{i=1}^n \mathcal{M}_i \rightarrow \prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i) \rightarrow \otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$ consists of $\otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)$ -atomic objects.*

Proof. Given an object $\{m_i\}_{i=1}^n \in \prod_{i=1}^n \mathcal{M}_i$, for each $m_i \in \mathcal{M}_i$, we have $\mathfrak{z}'(m_i) \in \mathcal{P}'(\mathcal{M}_i)$ is $\mathcal{V}_i$ -atomic object, thus $\otimes_{i=1}^n \mathfrak{z}'(i)$ is $\otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)$ -atomic object by the above lemmas. $\square$

Recall that a lax linear functor (also called an enriched functor)

$$\begin{array}{ccc} M^{\otimes} & \xrightarrow{f} & N^{\otimes} \\ \downarrow & & \downarrow \\ V^{\otimes} & \xrightarrow{f'} & W^{\otimes} \end{array} \text{ is an}$$

embedding if both f and f' are fully faithful functors.

Lemma 5.20. *Given a linear functor between left tensored categories (that is, a lax linear functor preserving left tensors)*

$$\begin{array}{ccc} M^{\otimes} & \xrightarrow{f} & N^{\otimes} \\ \downarrow & & \downarrow \\ V^{\otimes} & \xrightarrow{f'} & W^{\otimes} \end{array}$$

, if f' is a fully faithful functor and the restriction

of f to the fiber over $[0]$ $f_{[0]} : M \rightarrow N$ is fully faithful, then $f : M^{\otimes} \rightarrow N^{\otimes}$ is an embedding.

Proof. We can regard them as coCartesian fibrations over LM, then fiberwise fully faithfulness implies fully faithfulness as a whole $\square$

Now let's show the property (2): the morphism $q \circ \prod \text{loc} \circ i : \prod_{i=1}^n \mathcal{M}_i^{\otimes} \rightarrow \otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes}$ is fully faithful.

Claim 5.21. *The morphism $\gamma : \otimes_{i=1}^n \mathcal{P} \mathbf{LEnv}(\mathcal{M}_i)^{\otimes} \rightarrow \otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes}$ admits a right adjoint $\beta : \otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i) \hookrightarrow \otimes_{i=1}^n \mathcal{P} \mathbf{LEnv}(\mathcal{M}_i)$ which is a fully faithful embedding as a **lax linear functor** between left tensored ∞ -categories.*

Proof. By construction, the Ass-part of γ is a monoidal ∞ functor $\gamma' : \bigotimes_{i=1}^n \mathcal{P} \text{Env}(\mathcal{V}_i)^{\otimes} \rightarrow \bigotimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)^{\otimes}$, and is a fiberwise colimit preserving functor, where fiberwise admits a fully faithful right adjoint by [15, 4.12].

And over the fiber $[0]$, $\gamma_{[0]} : \bigotimes_{i=1}^n \mathcal{P} \text{LEnv}(\mathcal{M}_i) \rightarrow \bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$ admits a fully faithful right adjoint by [15, lemma 4.12].

Thus by [15, lemma 2.74], γ admits a right adjoint $\beta : \bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes} \rightarrow \bigotimes_{i=1}^n \mathcal{P} \text{LEnv}(\mathcal{M}_i)^{\otimes}$.

Consider the counit map $\gamma \circ \beta \xrightarrow{\eta} \text{id} : \bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes} \rightarrow \bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes}$, which is a natural transformation over Ass. Since fiberwise γ admits a fully faithful right adjoint, η is a fiberwise natural isomorphism, thus it's a natural isomorphism over Ass. This implies that the right adjoint is fully faithful as a lax linear functor $\beta : \bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes} \rightarrow \bigotimes_{i=1}^n \mathcal{P} \text{LEnv}(\mathcal{M}_i)^{\otimes}$. $\square$

Claim 5.22. *The composition of morphisms $q \circ \prod \text{loc} \circ i : \prod_{i=1}^n \mathcal{M}_i^{\otimes} \rightarrow \bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes}$ is an embedding.*

Proof. To show a lax linear functor is an embedding, by definition, it suffices to show that:

- (1) the functor itself $q \circ \prod \text{loc} \circ i$ is fully faithful.
- (2) the functor $q' \circ \prod \text{loc} \circ i : \prod_{i=1}^n \mathcal{V}_i^{\otimes} \rightarrow \prod_{i=1}^n \mathcal{P} \text{Env}(\mathcal{V}_i)^{\otimes} \rightarrow \prod_{i=1}^n \mathcal{P} \mathcal{V}_i^{\otimes} \rightarrow \bigotimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)^{\otimes}$ is fully faithful.

For (1), noticing that we have the following factorization:

$$\begin{array}{ccc}
 \prod_{i=1}^n \mathcal{M}_i^{\otimes} & \xrightarrow{\quad \tau \quad} & \bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i) \\
 & \searrow & \downarrow \beta \\
 \prod_{i=1}^n \mathcal{M}_i^{\otimes} & \xrightarrow{\quad \hookrightarrow \quad} \prod_{i=1}^n \text{LEnv}(\mathcal{M}_i)^{\otimes} \xrightarrow{\quad \hookrightarrow \quad} \mathcal{P}(\prod_{i=1}^n \text{LEnv}(\mathcal{M}_i))^{\otimes} \cong \bigotimes_{i=1}^n \mathcal{P} \text{LEnv}(\mathcal{M}_i)^{\otimes} & \\
 & \searrow q \circ \prod \text{loc} \circ i & \downarrow \gamma \\
 & & \bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes}
 \end{array}$$

where β is the fully faithful right adjoint of γ . And the factorization τ through β is due to the fact that the image of $\prod_{i=1}^n \mathcal{M}_i^{\otimes} \rightarrow \prod_{i=1}^n \text{LEnv}(\mathcal{M}_i)^{\otimes} \xrightarrow{\quad \hookrightarrow \quad} \bigotimes_{i=1}^n \mathcal{P} \text{LEnv}(\mathcal{M}_i)^{\otimes}$ lands in the full subcategory of local objects $\bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes} \subset \bigotimes_{i=1}^n \mathcal{P} \text{LEnv}(\mathcal{M}_i)^{\otimes}$ by the following facts:

- It suffices to show that the underlying functor $\prod_{i=1}^n \mathcal{M}_i \rightarrow \prod_{i=1}^n \text{LEnv}(\mathcal{M}_i) \xrightarrow{\quad \hookrightarrow \quad} \bigotimes_{i=1}^n \mathcal{P} \text{LEnv}(\mathcal{M}_i)$ lands in $\bigotimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)$. Thus we show the relation between localizations and Lurie's tensor product of presentable ∞ -categories.

- By [22, proof of proposition 4.8.1.15], given two presentable categories C, D , let $D' = \{d_\alpha\}$ be a small set of generators of D under colimits, and S is a small set of morphisms in C , then denote the image of $\{s \times \text{id}_d\}_{s \in S, d \in D'}$ under the universal functor $C \times D \rightarrow C \otimes D$ as S' , then the composition

$$S^{-1}C \times D \subset C \times D \rightarrow C \otimes D \xrightarrow{\text{localization}} S'^{-1}(C \otimes D)$$

exhibits $S'^{-1}(C \otimes D)$ as the (Lurie's) tensor product of presentable categories C and D .

- Especially, this means: induced by the colimit preserving localization functor $L : C \rightarrow S^{-1}C$, the colimit preserving functor $C \otimes D \rightarrow (S^{-1}C \otimes D)$ admits a fully faithful right adjoint, which image consists of S' -local objects, and for $c \in S^{-1}C, d \in D'$, the image of (c, d) under the universal functor $C \times D \rightarrow C \otimes D$, denoted as $c \otimes d$, lands in this full subcategory $S'^{-1}C \otimes D \subset C \otimes D$, as shown in the following commutative diagram:

$$\begin{array}{ccc} C \times D & \xrightarrow{\quad} & C \otimes D \\ L \downarrow & \hat{i}_i \downarrow & \downarrow L \\ (c, d) \in S^{-1}C \times D & \xrightarrow{\quad} & S'^{-1}(C \otimes D) \cong S^{-1}C \otimes D \end{array}$$

- Now we put an additional condition: let T be a small set of morphisms in D , and $C' = \{c_\beta\}$ be a small set of generators of C under colimits (thus $\{L(c_\beta)\}$ is also a set of generators of $S^{-1}C$), then we have the following claim: Induced by componentwise localization functors, the colimit preserving functor $C \otimes D \xrightarrow{L \otimes L} (S^{-1}C) \otimes (T^{-1}D)$ admits a fully faithful right adjoint. And for $c \in C' \cap S^{-1}C, d \in D' \cap T^{-1}D$, the image of (c, d) under the functor $C \times D \rightarrow C \otimes D$ lands in the image of the fully faithful right adjoint $(S^{-1}C) \otimes (T^{-1}D) \hookrightarrow C \otimes D$.
- The above claim can be justified by the following commutative diagram:

$$\begin{array}{ccc} C \times D & \xrightarrow{\quad} & C \otimes D \\ L \times \text{id} \downarrow & \hat{i}_i \downarrow & \downarrow L \\ S^{-1}C \times D & \xrightarrow{\quad} & S'^{-1}(C \otimes D) \cong S^{-1}C \otimes D \\ \cong \downarrow & & \downarrow \cong \\ D \times S^{-1}C & \xrightarrow{\quad} & D \otimes S^{-1}C \\ \downarrow L \times \text{id} & & \downarrow L \\ T^{-1}D \times S^{-1}C & \xrightarrow{\quad} & T'^{-1}(D \otimes S^{-1}C) \cong T^{-1}D \otimes S^{-1}C \\ \cong \downarrow & & \downarrow \cong \\ (c, d) \in S^{-1}C \times T^{-1}D & \xrightarrow{\quad} & S^{-1}C \otimes T^{-1}D \end{array}$$

where all the horizontal edges are the universal functors (preserving colimits separately), and all the L s denote the localizations, and their fully faithful right adjoints are denoted as i . And T' denotes the set of morphisms in $D \times S^{-1}C$ in the form of $\{t \times \text{id}_{L(c)}\}_{t \in T, c \in C'}$. And the compositions of the left horizontal edges are just $(L, L) : C \times D \rightarrow S^{-1}C \times T^{-1}D$, and thus the composition of the right vertical edges is $L \otimes L$, which right adjoint is a composition of fully faithful right adjoints, thus is again fully faithful.

- Especially, take $S = T = \text{Pseu}$, $C = \mathcal{P}\text{LEnv}(\mathcal{M})$, $D = \mathcal{P}\text{LEnv}(\mathcal{N})$, $C' = \text{LEnv}(\mathcal{M})$, $D' = \text{LEnv}(\mathcal{N})$, then for $(c, d) \in \mathcal{M} \times \mathcal{N} \rightarrow \text{LEnv}(\mathcal{M}) \times \text{LEnv}(\mathcal{N}) \rightarrow \mathcal{P}\text{LEnv}(\mathcal{M}) \times \mathcal{P}\text{LEnv}(\mathcal{N})$, $c \otimes d$ lands in the image of the fully faithful right adjoint of the localization functor $\mathcal{P}\text{LEnv}(\mathcal{M}) \otimes \mathcal{P}\text{LEnv}(\mathcal{N}) \rightarrow \mathcal{P}'(\mathcal{M}) \otimes \mathcal{P}'(\mathcal{N})$.

.

And (2) is similar. □

Thus we have shown the equivalence $\otimes \mathcal{P}'(\mathcal{M}_i)^{\otimes} \cong \mathcal{P}'(\prod_{i=1}^n \mathcal{M}_i)$

Corollary 5.23. Consider the commutative diagram

$$\begin{array}{ccc} \prod_{i=1}^n \mathcal{M}_i^{\otimes} & \xrightarrow{\prod \mathfrak{f}'} & \prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes} \\ & \searrow \mathfrak{f}'_{\prod} & \downarrow \gamma \\ & \mathcal{P}'(\prod_{i=1}^n \mathcal{M}_i)^{\otimes} & \xrightarrow{\cong} \otimes_{i=1}^n \mathcal{P}'(\mathcal{M}_i)^{\otimes} \end{array},$$

for any $\mathcal{N}^{\otimes} \rightarrow \otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)^{\otimes}$, the induced morphisms between mapping spaces:

$$\text{Lin}_{\otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)}^L(\mathcal{P}'(\prod_{i=1}^n \mathcal{M}_i), \mathcal{N})^{\simeq} \rightarrow \text{Lin}_{\prod_{i=1}^n \mathcal{P}(\mathcal{V}_i)}^{LL}(\prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i), \mathcal{N}')^{\simeq} \rightarrow \text{LaxLin}_{\prod_{i=1}^n \mathcal{V}_i}(\prod_{i=1}^n \mathcal{M}_i, \mathcal{N}'')^{\simeq}$$

are all equivalences, where $\mathcal{N}'$ is the corresponding pullback of $\mathcal{N}$ along the universal componentwise colimit preserving functor $\prod_{i=1}^n \mathcal{P}(\mathcal{V}_i) \rightarrow \otimes_{i=1}^n \mathcal{P}(\mathcal{V}_i)$, and $\mathcal{N}''$ is the further pullback along the product of Yoneda embeddings $\prod \mathfrak{f}_i : \prod \mathcal{V}_i \rightarrow \prod \mathcal{P}(\mathcal{V}_i)$.

Proof. The first arrow is an equivalence is by the fact that $\mathcal{P}'(\prod_{i=1}^n \mathcal{M}_i) \cong \otimes_{i=1}^n \mathcal{P}(\mathcal{M}_i)$ and the [corollary 1.74](#), and the composition of the two arrows are equivalences by the universal property of $\mathcal{P}'$. □

Corollary 5.24. Given n pseudo-enriched categories $\{\mathcal{M}_i^{\otimes} \rightarrow \mathcal{V}^{\otimes}_{i=1}^n\}$, we have:

$$\mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$$

Proof. consider the following commutative diagram in $\widehat{\text{PLMod}}$:

$$\begin{array}{ccccc} & \prod_i \mathcal{P}'(\mathcal{M}_i) & \xrightarrow{c_2} & \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i) & \\ & \downarrow & & \downarrow & \\ & \prod_{i=1}^n \mathcal{P}(\mathcal{V}) & \xrightarrow{\otimes'} & \mathcal{P}(\mathcal{V}) & \\ \uparrow \Pi \mathfrak{f}' & & & & \\ \prod_i \mathcal{M}_i & \xrightarrow{\psi} & \boxtimes_{\mathcal{V}} \mathcal{M}_i & \xrightarrow{\mathfrak{f}} & \mathcal{P}(\mathcal{V}) \\ \downarrow & \nearrow \Pi \mathfrak{f} & \downarrow & \nearrow & \\ \prod_{i=1}^n \mathcal{V} & \xrightarrow{\otimes} & \mathcal{V} & & \end{array}$$

(Note: A red arrow labeled $\tilde{c}$ connects $\boxtimes_{\mathcal{V}} \mathcal{M}_i$ to $\mathcal{P}(\mathcal{V})$.)

where

- $\otimes' : \prod \mathcal{P}(\mathcal{V}) \rightarrow \mathcal{P}(\mathcal{V})$ is the tensor product functor induced by the image under inclusion $\text{PrMon}^{\otimes} \rightarrow \widehat{\text{Mon}}^{\times}$ of the coCartesian edge $\otimes' : (\mathcal{P}(\mathcal{V}))_i \rightarrow \mathcal{P}(\mathcal{V})$ in $\text{PrMon}^{\otimes}$ over the active map $\langle n \rangle \rightarrow \langle 1 \rangle$, where $\text{PrMon}^{\otimes}$ is the suboperad of presentable monoidal ∞ -categories with colimit-preserving functors.
- $c_2 : \prod \mathcal{P}'(\mathcal{M}_i) \rightarrow \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$ is induced by the image under the inclusion $\text{PrLMod}^{\otimes} \rightarrow \widehat{\text{LMod}}^{\times}$ of the coCartesian lifting $(\mathcal{P}'(\mathcal{M}_i))_i \rightarrow \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$ over the tensor operation $\otimes'$.
- $\tilde{c}$ is induced by $c_2 \circ \prod \mathfrak{z}'$: since $\otimes' \circ \prod \cong \mathfrak{z} \circ \otimes$, by the universal property of coCartesian lifting, there is an induced morphism $\tilde{c} : \boxtimes_{\mathcal{V}} \mathcal{M}_i \rightarrow \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$.

And given any $\mathcal{N}^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$ which is presentably left tensored, the mapping space induced by $\tilde{c}$:

$$\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i), \mathcal{N})^{\simeq} \rightarrow \text{LaxLin}_{\mathcal{V}}(\boxtimes_{\mathcal{V}} \mathcal{M}_i, \mathfrak{z}^* \mathcal{N})^{\simeq}$$

is an equivalence, which is justified by the following commutative diagram:

$$\begin{array}{ccc}
 & & \text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}' \mathcal{M}_i, \mathcal{N})^{\simeq} \\
 & \nearrow c_2^*, \cong & \downarrow \tilde{c}^* \\
 \text{Lin}_{\prod_i \mathcal{P}(\mathcal{V})}^{LL}(\prod_i \mathcal{P}' \mathcal{M}_i, \otimes'^* \mathcal{N})^{\simeq} & & \text{LaxLin}(\boxtimes_{\mathcal{V}} \mathcal{M}_i, \mathcal{N})_{\mathfrak{z}}^{\simeq} \\
 \uparrow \prod \mathfrak{z}'^*, \cong & & \uparrow \cong \\
 \text{LaxLin}_{\prod_i \mathcal{V}}(\prod_i \mathcal{M}_i, \otimes^* \mathfrak{z}^* \mathcal{N} \cong \prod_i \mathfrak{z}^* \otimes'^* \mathcal{N})^{\simeq} & \xrightarrow{\psi^*, \cong} & \text{LaxLin}_{\mathcal{V}}(\boxtimes_{\mathcal{V}} \mathcal{M}_i, \mathfrak{z}^* \mathcal{N})^{\simeq}
 \end{array}$$

where

- c_2^* is an equivalence by [corollary 1.74](#).
- $\prod \mathfrak{z}'^*$ is an equivalence by the [corollary 5.23 above](#).
- ψ^* is an equivalence by the coCartesian lifting.

Recall that $\mathfrak{z}' : \boxtimes_{\mathcal{V}} \mathcal{M}_i \hookrightarrow \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i)$ satisfies the universal property:

$$\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i), \mathcal{N})^{\simeq} \xrightarrow{\mathfrak{z}'^*} \text{LaxLin}_{\mathcal{V}}(\boxtimes_{\mathcal{V}} \mathcal{M}_i, \mathfrak{z}^* \mathcal{N})$$

is an equivalence, thus by the above two univsersal properties, we have the following commutative diagram:

$$\begin{array}{ccc}
\mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) & \xrightarrow{\cong} & \otimes_{\mathcal{P}(\mathcal{V})}(\mathcal{P}' \mathcal{M}_i) \\
& \swarrow \mathfrak{z}' & \searrow c \\
& \boxtimes_{\mathcal{V}} \mathcal{M}_i &
\end{array}$$

□

5.4 $\phi : \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \mathcal{P}(\otimes_{\mathcal{V}} L \mathcal{M}_i)$

By the equivalence $\mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$, it is obvious that the morphism $\psi'' : \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \rightarrow \mathcal{P}'(\boxtimes_{\mathcal{V}} L \mathcal{M}_i)$ is induced by $\prod \mathcal{P}'(\text{unit}_i) : \prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i) \rightarrow \prod_{i=1}^n \mathcal{P}'(L \mathcal{M}_i)$, justified by the following diagram in $\widehat{\text{PLMod}}$:

$$\begin{array}{ccccc}
\prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i) & & & & \\
\uparrow \prod \mathfrak{z}' & & c_2 \searrow & & \\
\prod_{i=1}^n \mathcal{M}_i & \xrightarrow{c_1} & \boxtimes_{\mathcal{V}} \mathcal{M}_i & \xrightarrow{\mathfrak{z}'} & \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i) \\
\downarrow \prod \text{unit}_i & & \downarrow \boxtimes \text{unit}_i & & \downarrow \phi'' = \otimes \mathcal{P}'(\text{unit}_i) \\
\prod_{i=1}^n L \mathcal{M}_i & \xrightarrow{c_1} & \boxtimes_{\mathcal{V}} L \mathcal{M}_i & \xrightarrow{\mathfrak{z}'} & \mathcal{P}'(\boxtimes_{\mathcal{V}} L \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(L \mathcal{M}_i) \\
\downarrow \prod \mathfrak{z}' & & & & \\
\prod_{i=1}^n \mathcal{P}'(L \mathcal{M}_i) & & c_2 \nearrow & &
\end{array}$$

- All the c_1 s are the coCartesian liftings in PLMod over the tensor product morphism $\mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ in $\text{Mon}(\text{Cat})$.
- Consider the coCartesian lifting $(\mathcal{P}'(\mathcal{M}_i))_{i=1}^n \rightarrow \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$ in $\text{PrLMod}^{\otimes}$ over the tensor product morphism $(\mathcal{P}(\mathcal{V}))_{i=1}^n \rightarrow \mathcal{P}(\mathcal{V})$ in PrMon of presentable monoidal ∞ -categories. And under the embedding $\text{PrLMod}^{\otimes} \subset \text{LMod}^{\times}$, it admits a factorization through the coCartesian lifting in $\text{LMod}^{\times}$, which gives rise to $c_2 : \prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i) \rightarrow \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i)$. And likewise for $c_2 : \prod_{i=1}^n \mathcal{P}'(L \mathcal{M}_i) \rightarrow \mathcal{P}'(\boxtimes_{\mathcal{V}} L \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(L \mathcal{M}_i)$.
- By the universal property of $\mathfrak{z}'$, there is a unique map ϕ'' making the diagram commutes, thus we let ϕ'' be the colimit preserving $\mathcal{P}(\mathcal{V})$ -linear functor induced componentwise by $\mathcal{P}'(\text{unit}_i)$, and it remains to show this choice of ϕ'' makes the diagram commute, which is true since c_1 is the coCartesian lifting, thus it suffices to precompose with c_1 , then by design it commutes.
- And let $\mathcal{P}'(\text{unit}_i) : \mathcal{P}'(\mathcal{M}_i) \rightarrow \mathcal{P}'(L \mathcal{M}_i)$ be the colimit-preserving $\mathcal{P}(\mathcal{V})$ -linear functor induced by $\mathcal{M}_i \xrightarrow{\text{unit}_i} L \mathcal{M}_i \xrightarrow{\mathfrak{z}'} \mathcal{P}'(L \mathcal{M}_i)$ by applying the universal property of $\mathfrak{z}' : \mathcal{M}_i \rightarrow \mathcal{P}'(\mathcal{M}_i)$.

Now let's interpret $\psi' : \mathcal{P}'(\boxtimes_{\mathcal{V}} L \mathcal{M}_i) \rightarrow \mathcal{P}(\otimes_{\mathcal{V}} L \mathcal{M}_i)$ as the colimit preserving $\mathcal{P}(\mathcal{V})$ -linear

functor induced by $\prod_{i=1}^n \mathcal{P}'(L\mathcal{M}_i) \rightarrow \prod_{i=1}^n \mathcal{P}(L\mathcal{M}_i)$ where each component is the unique colimit preserving linear functor induced by the universal property of $\mathcal{P}'$, denoted as δ_i .

$$\begin{array}{ccccc}
\prod_{i=1}^n \mathcal{P}(L\mathcal{M}_i) & & & & \\
\uparrow \Pi \mathfrak{L} & \nearrow c_2 & & & \\
& \otimes_{\mathcal{V}} L\mathcal{M}_i & \xrightarrow{\mathfrak{L}} & \mathcal{P}(\otimes_{\mathcal{V}} L\mathcal{M}_i) & \\
& \uparrow c_3 & \nwarrow \psi & & \nwarrow \phi' \\
\prod_{i=1}^n L\mathcal{M}_i & \xrightarrow{c_1} & \boxtimes_{\mathcal{V}} L\mathcal{M}_i & \xrightarrow{\mathfrak{L}'} & \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \\
\downarrow \Pi \mathfrak{L}' & & & \nearrow c_2 & \\
\prod_{i=1}^n \mathcal{P}'(L\mathcal{M}_i) & & & &
\end{array}$$

where c_3 is the coCartesian lifting in $\mathbf{LMod}$ over the tensor product morphism $\prod_{i=1}^n \mathcal{V} \rightarrow \mathcal{V}$, and c_1 and c_2 are defined as in the previous case. And using the same strategy, we can show that there is a unique choice of ϕ' making the diagram commutes, and we can let ϕ' be the colimit preserving linear functor induced by $\prod \delta_i$ and show it commute with the diagram by using the fact that c_1 is the coCaartesian lifting, thus ψ' is the one induced by $\prod \delta_i$.

And to show that $\psi' \circ \psi''$ is an equivalence, it suffices to show componentwise $\delta_i \circ \mathcal{P}'(\text{unit}_i)$ is an equivalence, which is true by the following diagram:

$$\begin{array}{ccc}
\mathcal{M}_i & \xrightarrow{\mathfrak{L}'} & \mathcal{P}'(\mathcal{M}_i) \\
\text{unit} \downarrow & & \downarrow \mathcal{P}'(\text{unit}_i) \\
L\mathcal{M}_i & \xrightarrow{\mathfrak{L}'} & \mathcal{P}'(L\mathcal{M}_i) \\
\parallel & & \downarrow \delta_i \\
L\mathcal{M}_i & \xrightarrow{\mathfrak{L}} & \mathcal{P}(L\mathcal{M}_i)
\end{array}$$

where by the proof of the [blue equivalence](#) $\rho_i : \mathcal{P}'(\mathcal{M}_i) \cong \mathcal{P}(L\mathcal{M}_i)$, there is only a unique colimit preserving linear functor as the right vertical edge in the above diagram making the diagram commute, thus $\delta_i \circ \mathcal{P}'(\text{unit}_i) \cong \rho_i$ is an equivalence.

Thus we have reached the purpose:

Proposition 5.25. *The morphism $\phi = \phi' \circ \phi'' : \mathcal{P}(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \rightarrow \mathcal{P}(\otimes_{\mathcal{V}} L\mathcal{M}_i)$ in the diagram below is an equivalence.*

$$\begin{array}{ccc}
L(\boxtimes_{\mathcal{V}} \mathcal{M}_i) & \xhookrightarrow{i} & \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(\mathcal{M}_i) \\
\uparrow & \nearrow \mathfrak{y}' & \downarrow \phi'' \\
\boxtimes_{\mathcal{V}} \mathcal{M}_i & & \mathcal{P}'(\boxtimes_{\mathcal{V}} L\mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}'(L\mathcal{M}_i) \\
\boxtimes \text{unit}_i \downarrow & \nearrow \mathfrak{y}' & \downarrow \phi' \\
\boxtimes_{\mathcal{V}} L\mathcal{M}_i & & \\
\psi \downarrow & \searrow \mathfrak{y} & \\
\otimes_{\mathcal{V}} L\mathcal{M}_i & \xrightarrow{\quad} & \mathcal{P}(\otimes_{\mathcal{V}} L\mathcal{M}_i) \cong \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}(L\mathcal{M}_i)
\end{array}$$

And we can summarize the results so far in the following diagram:

$$\begin{array}{ccccc}
\prod_{i=1}^n L\mathcal{M}_i & \xrightarrow{\quad \prod \mathfrak{y} \quad} & \prod_{i=1}^n \mathcal{P}(L\mathcal{M}_i) & & \\
\downarrow & \searrow & \downarrow c_2 & \nwarrow \prod \delta_i & \\
& \prod_{i=1}^n L\mathcal{M}_i & \xrightarrow{\quad} & \prod_{i=1}^n \mathcal{P}'(L\mathcal{M}_i) & \\
\downarrow & \swarrow c_3 & \downarrow & \downarrow c_2 & \nwarrow \\
\otimes_{\mathcal{V}} L\mathcal{M}_i & \xrightarrow{\quad \mathfrak{y} \quad} & \mathcal{P}(\otimes_{\mathcal{V}} L\mathcal{M}_i) & & \prod_{i=1}^n \mathcal{P}'(\mathcal{M}_i) \\
\downarrow \psi & \swarrow c_1 & \downarrow \prod \text{unit}_i & \nwarrow \phi' & \downarrow c_2 \\
& \boxtimes_{\mathcal{V}} L\mathcal{M}_i & \xrightarrow{\quad \mathfrak{y}' \quad} & \mathcal{P}'(\boxtimes_{\mathcal{V}} L\mathcal{M}_i) & \\
& \downarrow \boxtimes \text{unit}_i & \downarrow & \nwarrow \phi'' & \\
& \boxtimes_{\mathcal{V}} \mathcal{M}_i & \xrightarrow{\quad \mathfrak{y}' \quad} & \mathcal{P}'(\boxtimes_{\mathcal{V}} \mathcal{M}_i) &
\end{array}$$

5.5 $\mathfrak{y}^* : \text{PLMod}_{\mathcal{V}} \cong \text{Enr}_{\mathcal{P}(\mathcal{V})}$

This subsection follows [15, section 4.2], where there is an equivalence by pullback along $\mathfrak{y} : \mathcal{V} \rightarrow \mathcal{P}(\mathcal{V})$

$$\mathfrak{y}^* : \text{Enr}_{\mathcal{P}(\mathcal{V})} \rightarrow \text{PLMod}_{\mathcal{V}}$$

where the right hand side is the category of absolute small $\mathcal{V}$ pseudo-enriched categories, and the left hand side is the category of $\mathcal{P}(\mathcal{V})$ enriched categories with small underlying categories. And we enhance this equivalence to be a symmetric monoidal equivalence. More precisely:

- We show that for a small symmetric monoidal ∞ -category $\mathcal{V}$, there is a full symmetric monoidal subcategory embedding $\text{PLMod}_{\mathcal{V}} \hookrightarrow \widehat{\text{PLMod}}_{\mathcal{V}}$.

- Given a presentably symmetric monoidal ∞ -category, for example, $\mathcal{P}(\mathcal{V})$, we define $\text{Enr}_{\mathcal{P}(\mathcal{V})}$ as the full symmetric monoidal subcategory of $\widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}$.
- As in the small case: there is a symmetric monoidal embedding $\widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})} \hookrightarrow \widehat{\text{PLMod}}_{\mathcal{V}}$.
- And we show that by pullback along $\mathfrak{y} : \mathcal{V} \hookrightarrow \mathcal{P}(\mathcal{V})$, the lax symmetric monoidal functor $\widehat{\text{PLMod}}_{\mathcal{P}(\mathcal{V})} \rightarrow \widehat{\text{PLMod}}_{\mathcal{V}}$ restricts to a symmetric monoidal equivalence

$$\mathfrak{y}^* : \text{Enr}_{\mathcal{P}(\mathcal{V})} \rightarrow \text{PLMod}_{\mathcal{V}}$$

, and we start with showing the underlying $(\infty, 1)$ functor admits a left adjoint $\overline{(-)}$, serving as the inverse of $\mathfrak{y}^*$, and then we show the lax symmetric monoidal structure on $\mathfrak{y}^*$ is strict.

To consider the enrichment theory of a large presetably monoidal category, we need to enlarge the universe and consider $\widehat{\text{Enr}} \subset \widehat{\text{PLMod}} \rightarrow \widehat{\text{Mon}}$, but for a small symmetric monoidal category $\mathcal{V} \in \text{Mon} \subset \widehat{\text{Mon}}$, we can identify $\text{PLMod}_{\mathcal{V}}$ as a symmetric monoidal full subcategory of $\widehat{\text{PLMod}}_{\mathcal{V}}$ by the following lemmas.

Lemma 5.26. *Given a monoidal functor between small monoidal ∞ -category $\alpha : \mathcal{V}^{\otimes} \rightarrow \mathcal{W}^{\otimes}$, and $\mathcal{M}^{\otimes} \in \widehat{\text{PLMod}}_{\mathcal{V}}$, if $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ is absolute small, then the coCartesian lifting of $\mathcal{M}^{\otimes}$ in $\widehat{\text{PLMod}}$ over α is again an absolute small pseudo-enriched ∞ -category.*

Proof. • If $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ is absolute small, then $\text{Mul}_{\mathcal{M}}(v, m_1; m_2) \cong \text{Map}_{\widehat{\mathcal{P}(\mathcal{V})}}(v, \text{Mor}_{\widehat{\mathcal{P}(\text{LEnv}(\mathcal{M}))}}(m_1, m_2))$ is small for all $v \in \mathcal{V}, m_1, m_2 \in \mathcal{M}$. Thus $\text{Mor}_{\widehat{\mathcal{P}(\text{LEnv}(\mathcal{M}))}}(m_2, m_2)(v) \in \text{Spc}$ for all $v \in \mathcal{V}$, so $\text{Mor}_{\widehat{\mathcal{P}(\text{LEnv}(\mathcal{M}))}}(m_1, m_2)$ lands in $\mathcal{P}(\mathcal{V}) \subset \widehat{\mathcal{P}(\mathcal{V})}$.

- The coCartesian lifting $f : \mathcal{M}^{\otimes} \rightarrow \alpha_!(\mathcal{M})^{\otimes}$ satisfies that f is essential surjective and for any $m_1, m_2 \in \mathcal{M}$, there is an equivalence:

$$\widehat{\mathcal{P}(\text{Env}(\alpha))}(\text{Mor}_{\widehat{\mathcal{P}(\text{LEnv}(\mathcal{M}))}}(m_1, m_2)) \cong \text{Mor}_{\alpha_!(\widehat{\mathcal{P}(\text{LEnv}(\mathcal{M}))})}(f(m_1), f(m_2))$$

, and since $\mathcal{M}^{\otimes}$ is pseudo-enriched, so is $\alpha_!(\mathcal{M})$, and the morphism objects $\text{Mor}_{\widehat{\mathcal{P}(\text{LEnv}(\mathcal{M}))}}(m_1, m_2)$ lands in $\widehat{\mathcal{P}}(\mathcal{V})$, and the above equivalence can be restricted to:

$$\widehat{\mathcal{P}}(\alpha)(\text{Mor}_{\widehat{\mathcal{P}(\text{LEnv}(\mathcal{M}))}}(m_1, m_2)) \cong \text{Mor}_{\alpha_!(\widehat{\mathcal{P}(\text{LEnv}(\mathcal{M}))})}(f(m_1), f(m_2))$$

- Since $\widehat{\mathcal{P}}(\mathcal{V}) = \text{Ind}_{\sigma}(\mathcal{P}(\mathcal{V}))$, and given a monoidal functor α , there is an induced commutative diagram:

$$\begin{array}{ccc} \mathcal{V} & \xrightarrow{\alpha} & \mathcal{W} \\ \downarrow \mathfrak{y} & & \downarrow \mathfrak{y} \\ \mathcal{P}(\mathcal{V}) & \xrightarrow{\mathcal{P}(\alpha)} & \mathcal{P}(\mathcal{W}) \\ \downarrow & & \downarrow \\ \text{Ind}_{\sigma}(\mathcal{P}(\mathcal{V})) & \xrightarrow{\widehat{\mathcal{P}}(\alpha)} & \text{Ind}_{\sigma}(\mathcal{P}(\mathcal{W})) \end{array}$$

Thus $\text{Mor}_{\alpha_i \widehat{\mathcal{P}\text{LEnv}(\mathcal{M})}}(f(m_1), f(m_2))$ lands in $\mathcal{P}(\mathcal{W}) \subset \widehat{\mathcal{P}}(\mathcal{W})$, and since f is essential surjective, thus for any $n_1, n_2 \in \mathcal{N}$, and there are:

$$\text{Mul}_{\mathcal{N}}(v, n_1; n_2) \cong \text{Mul}_{\widehat{\mathcal{P}\text{LEnv}(\mathcal{N})}}(v, n_1; n_2) \cong \text{Map}_{\widehat{\mathcal{P}}(\mathcal{W})}(v, \text{Mor}_{\widehat{\mathcal{P}\text{LEnv}(\mathcal{N})}}(n_1, n_2))$$

, by the equivalences of spaces above, we can see $\mathcal{N}^{\otimes}$ is locally small and thus is small, and $\mathcal{N}^{\otimes} \rightarrow \mathcal{W}$ is an absolute small pseudo-enriched ∞ -category. $\square$

And in the context of the above lemma, we further have:

Lemma 5.27. *The fully faithful embedding $i : \text{PLMod} \rightarrow \widehat{\text{PLMod}}$ induces a symmetric monoidal fully faithful embedding*

$$j : \text{PLMod}_{\mathcal{V}}^{\otimes} \rightarrow \widehat{\text{PLMod}}_{\mathcal{V}}^{\otimes}$$

by the diagram:

$$\begin{array}{ccc} \text{PLMod}_{\mathcal{V}}^{\otimes} & \xrightarrow{j} & \widehat{\text{PLMod}}_{\mathcal{V}}^{\otimes} \\ \downarrow \checkmark & & \downarrow \checkmark \\ \text{Comm}^{\otimes} & \xrightarrow{\text{id}} & \text{Comm}^{\otimes} \\ \downarrow \{v\} & \searrow & \downarrow \\ \text{PLMod}^{\times} & \xrightarrow{i} & \widehat{\text{PLMod}}^{\times} \\ \downarrow & & \downarrow \\ \text{Mon}^{\times} & \xrightarrow{\quad} & \widehat{\text{Mon}}^{\times} \end{array}$$

Proof. • The above lemma implies the embedding $\text{PLMod} \hookrightarrow \widehat{\text{PLMod}}$ preserves coCartesian liftings over the morphism in $\text{Mon} \hookrightarrow \widehat{\text{Mon}}$.

- And it remains to show for every morphism $f : \{A_i\} \rightarrow A$ in $\text{Mon}^{\times} \hookrightarrow \widehat{\text{Mon}}^{\times}$, the coCartesian lifting in $\text{PLMod}^{\times}$ over f is preserved under the embedding $\text{PLMod}^{\times} \hookrightarrow \widehat{\text{PLMod}}^{\times}$.
- Since $\text{Mon}^{\times}$ is coCartesian over $\text{Comm}^{\otimes}$, thus it can be factored as $f = f_1 \circ f_2$ where f_2 is the coCartesian lifting in $\text{Mon}^{\times}$ over some map in $\text{Comm}^{\otimes}$ and f_1 is an induced morphism in Mon . And since we have already know $i : \text{PLMod} \rightarrow \widehat{\text{PLMod}}$ preserves coCartesian liftings over morphisms in $\text{Mon} \subset \widehat{\text{Mon}}$, thus f_1 is preserved, and it remains to show that the coCartesian liftings over f_2 in $\text{PLMod}^{\times}$ is preserved under $i : \text{PLMod}^{\times} \hookrightarrow \widehat{\text{PLMod}}^{\times}$.
- If f_2 is inert, it's obvious true, and if f_2 is coCartesian over an active map $\alpha \in \text{Comm}^{\otimes}$, then by the dual version of [Tag 01V9](#), the coCartesian lifting in $\text{PLMod}^{\times}$ over f_2 is exactly the coCartesian lifting of $\text{PLMod}^{\times} \rightarrow \text{Comm}^{\otimes}$ over the same active map α . And since $i : \text{PLMod}^{\times} \rightarrow \widehat{\text{PLMod}}^{\times}$ is symmetric monoidal ∞ -functor, thus preserves coCartesian edges over the active map in $\text{Comm}^{\otimes}$, so i preserves the coCartesian liftings over all the morphisms in $\text{Mon}^{\times} \subset \widehat{\text{Mon}}^{\times}$.

And especially the morphism induced by i under pullback $\text{Comm}^\otimes \rightarrow \text{Mon}^\times \subset \widehat{\text{Mon}}^\times$ also preserves all the coCartesian liftings, thus $j : \text{PLMod}_\mathcal{V}^\otimes \rightarrow \widehat{\text{PLMod}}_\mathcal{V}^\otimes$ is symmetric monoidal. And it's obvious that j is also fully faithful. $\square$

Remark 5.28. By the above lemmas, when $\mathcal{V}$ is small, we can identify the small version $\text{PLMod}_\mathcal{V}^\otimes$ as a symmetric monoidal subcategory of $\widehat{\text{PLMod}}_\mathcal{V}^\otimes$ consisting of absolute small enriched ∞ -categories.

For $\mathcal{M}^\otimes \in \widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}$, if the underlying category $\mathcal{M} = \mathcal{M}_{[0]}^\otimes$ is small, then by the equivalence:

$$\text{Map}_{\mathcal{P}(\mathcal{V})}(v, \text{Mor}_\mathcal{M}(x, y)) \cong \text{Mul}_\mathcal{M}(v, x; y)$$

, since $\mathcal{P}(\mathcal{V})$ is locally small, so the mutimorphism space of $\mathcal{M}$ is locally small, and this implies $\mathcal{M}^\otimes$ is small, and $\mathcal{M}^\otimes \rightarrow \mathcal{P}(\mathcal{V})^\otimes$ is a small enriched ∞ -category (although not absolute small).

Moreover, if we take $v = \mathbf{1}_{\mathcal{P}(\mathcal{V})}$, then $\text{Map}_{\mathcal{P}(\mathcal{V})}(\mathbf{1}_{\mathcal{P}(\mathcal{V})}, \text{Mor}_\mathcal{M}(x, y)) \cong \text{Map}_\mathcal{M}(x, y)$ is small, thus a $\mathcal{P}(\mathcal{V})$ -enriched ∞ -category has small underlying category iff its underlying space is small iff it's a small enriched ∞ -category, although not absolute small.

Recall the construction of $\widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}^\otimes$, given any two $\mathcal{P}(\mathcal{V})$ -enriched ∞ -categories $\mathcal{M}, \mathcal{N}$, then the tensor product operation in $\widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}^\otimes : (\mathcal{M}, \mathcal{N}) \rightarrow \mathcal{M} \boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{N}$ factors through the tensor product operation in $\widehat{\text{Enr}}^\times(\mathcal{M}, \mathcal{N}) \rightarrow \mathcal{M} \times \mathcal{N} \rightarrow \mathcal{M} \boxtimes_\mathcal{V} \mathcal{N}$, and $\mathcal{M} \boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{N}$ is the coCartesian lifting in $\widehat{\text{Enr}}$ over the tensor product morphism $\boxtimes : \mathcal{P}(\mathcal{V}) \times \mathcal{P}(\mathcal{V}) \rightarrow \mathcal{P}(\mathcal{V})$. Especially, the coCartesian lifting is essential surjective, thus if $\mathcal{M}, \mathcal{N}$ are with small underlying spaces, so is $\mathcal{M} \times \mathcal{N}$ and $\mathcal{M} \boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{N}$.

This justifies the following notation:

Notation 5.29. Let $\mathcal{V}^\otimes$ be a small symmetric monoidal ∞ -category, and $\mathcal{P}(\mathcal{V})^\otimes$ be its presheaf category equipped with the Day convolution symmetric monoidal structure. Denote the full symmetric monoidal subcategory of $\widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}^\otimes$ consisting of enriched ∞ -categories with small underlying categories as $\text{Enr}_{\mathcal{P}(\mathcal{V})}^\otimes$.

And the above argument works for any presentabl symmetric monoidal ∞ -category $\mathcal{O}$, thus we have:

Notation 5.30. Let $\mathcal{O}^\otimes$ be a presentably symmetric monoidal ∞ -category. Denote the full symmetric monoidal subcategory of $\widehat{\text{Enr}}_{\mathcal{O}}^\otimes$ consisting of enriched ∞ -categories with small underlying categories as $\text{Enr}_{\mathcal{O}}^\otimes$.

Consider a map $[1] \rightarrow \text{CAlg}(\widehat{\text{Mon}}) \subset \text{Fun}(\text{Comm}^\otimes, \widehat{\text{Mon}}^\times)$ classifying the symmetric monoidal functor $\flat : \mathcal{V}^\otimes \rightarrow \mathcal{P}(\mathcal{V})^\otimes$, which corresponds to a functor $[1] \times \text{Comm}^\otimes \rightarrow \widehat{\text{Mon}}^\times$.

Construction 5.31. Consider the pullback along the map classifying the symmetric monoidal Yoneda embedding:

$$\begin{array}{ccc}
M & \xrightarrow{\quad} & \widehat{\text{PLMod}}^{\times} \\
\downarrow & \lrcorner & \downarrow \\
[1] \times \text{Comm}^{\otimes} & \xrightarrow{\quad} & \widehat{\text{Mon}}^{\times}
\end{array}$$

Here is some information on the above construction:

- M is Cartesian over $[1] \times (\text{Comm}^{\otimes})^{\sim}$ and coCartesian over $\text{Comm}^{\otimes}$, the fiber over $\{0\} \times \text{Comm}^{\otimes}$ is $\widehat{\text{PLMod}}_{\mathcal{V}}^{\otimes}$ and the fiber over $\{1\} \times \text{Comm}^{\otimes}$ is $\widehat{\text{PLMod}}_{\mathcal{P}(\mathcal{V})}^{\otimes}$.
- By [13, corollary 2.2.2], after projection $[1] \times \text{Comm}^{\otimes} \rightarrow [1]$, $M \rightarrow [1]$ is Cartesian.
- By [13, 2.2.4], via further projection $[1] \times \text{Comm}^{\otimes} \rightarrow [1]$, the coCartesian fibration $M \rightarrow [1]$ after straightening gives an ∞ -operad map:

$$\mathfrak{y}^* : \widehat{\text{PLMod}}_{\mathcal{P}(\mathcal{V})}^{\otimes} \rightarrow \widehat{\text{PLMod}}_{\mathcal{V}}^{\otimes}$$

- Recall that there is a symmetric monoidal functor $\widehat{\text{Enr}}^{\times} \rightarrow \widehat{\text{PLMod}}^{\times}$ induced by the product-preserving functor $\widehat{\text{Enr}} \rightarrow \widehat{\text{PLMod}}$, and this gives rise to a symmetric monoidal embedding $\widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}^{\otimes} \rightarrow \widehat{\text{PLMod}}_{\mathcal{P}(\mathcal{V})}^{\otimes}$. Given a $\mathcal{P}(\mathcal{V})$ -enriched ∞ -category $\mathcal{M}$ with small underlying category, then pullback $\mathfrak{y}^* \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ lands in $\text{PLMod}_{\mathcal{V}} \subset \widehat{\text{PLMod}}_{\mathcal{V}}$:

$$\begin{array}{ccc}
\mathfrak{y}^* \mathcal{M}^{\otimes} & \xrightarrow{\quad} & \mathcal{M}^{\otimes} \\
\downarrow & \lrcorner & \downarrow \\
\mathcal{V}^{\otimes} & \xrightarrow{\quad \mathfrak{y} \quad} & \mathcal{P}(\mathcal{V})^{\otimes}
\end{array}$$

since $\mathcal{M}^{\otimes}$ is locally small, $\mathfrak{y}^* \mathcal{M}^{\otimes}$ is local small and so is absolute small. By restricting to the full subcategory $\text{Enr}_{\mathcal{P}(\mathcal{V})}^{\otimes} \hookrightarrow \widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}^{\otimes}$, there is a factorization of lax symmetric monoidal functors

$$\begin{array}{ccc}
\text{Enr}_{\mathcal{P}(\mathcal{V})}^{\otimes} & \xrightarrow{\quad \mathfrak{y}^* \quad} & \text{PLMod}_{\mathcal{V}}^{\otimes} \\
\downarrow & & \downarrow \\
\widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}^{\otimes} & & \\
\downarrow & & \\
\widehat{\text{PLMod}}_{\mathcal{P}(\mathcal{V})}^{\otimes} & \xrightarrow{\quad \mathfrak{y}^* \quad} & \widehat{\text{PLMod}}_{\mathcal{V}}^{\otimes}
\end{array}$$

- And we denote the induced lax symmetric monoidal functor again by $\mathfrak{y}^*$:

$$\mathfrak{y}^* : \text{Enr}_{\mathcal{P}(\mathcal{V})}^{\otimes} \rightarrow \text{PLMod}_{\mathcal{V}}^{\otimes}$$

.

Construction 5.32. Consider $\mathcal{M}^{\otimes} \in \text{PLMod}_{\mathcal{V}}$, consider the embedding $\mathfrak{y}' : \mathcal{M}^{\otimes} \rightarrow \mathcal{P}'(\mathcal{M})^{\otimes} = \mathcal{P}\text{LEnv}(\mathcal{M})_{\text{Pseu}}^{\otimes}$, denote the full weakly left tensored ∞ -subcategory of $\mathcal{P}'(\mathcal{M})^{\otimes}$ generated by the image of $\mathfrak{y}'(\mathcal{M})$ as $\overline{\mathcal{M}}^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$. And obviously $\overline{\mathcal{M}}^{\otimes}$ is enriched in $\mathcal{P}(\mathcal{V})$, and we denote the induced embedding as $\iota : \mathcal{M}^{\otimes} \rightarrow \overline{\mathcal{M}}^{\otimes}$.

Lemma 5.33. *The embedding $\iota : \mathcal{M}^{\otimes} \rightarrow \overline{\mathcal{M}}^{\otimes}$ is a Cartesian edge over the Yoneda embedding $\mathfrak{y} : \mathcal{V}^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$.*

Proof. There is an induced commutative diagram:

$$\begin{array}{ccc}
 \mathcal{M}^{\otimes} & \xrightarrow{\quad \iota \quad} & \overline{\mathcal{M}}^{\otimes} \\
 \searrow \omega & & \downarrow \\
 \mathfrak{y}^* \overline{\mathcal{M}}^{\otimes} & \xrightarrow{\quad c \quad} & \overline{\mathcal{M}}^{\otimes} \\
 \downarrow \lrcorner & & \downarrow \\
 \mathcal{V}^{\otimes} & \xrightarrow{\quad \mathfrak{y} \quad} & \mathcal{P}(\mathcal{V})^{\otimes}
 \end{array}$$

And it remains to show ω is an equivalence.

ω is fully faithful since both ι and the Cartesian lifting c are fully faithful, and ω is essential surjective since the underlying category of $\mathfrak{y}^* \overline{\mathcal{M}}^{\otimes}$ is equivalent to $\overline{\mathcal{M}}_{[0]}^{\otimes} \cong \mathcal{M}$ by construction. $\square$

Lemma 5.34. *Given $\mathcal{N}^{\otimes} \in \text{LMod}_{\mathcal{P}(\mathcal{V})}(\text{Pr}^L)$, there is an equivalence induced by the embedding $\iota : \mathcal{M}^{\otimes} \hookrightarrow \overline{\mathcal{M}}^{\otimes}$:*

$$\iota^* : \text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}}, \mathcal{N}) \cong \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathfrak{y}^* \mathcal{N})$$

.

Proof. • ι^* is conservative: given a natural transformation relative to $\mathcal{P}(\mathcal{V})^{\otimes}$: $\alpha : [1] \times \overline{\mathcal{M}}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$, to show it's a natural isomorphism it suffices to show for each object $m \in \mathcal{M}^{\otimes}$, the induced map α_m is an equivalence in $\mathcal{N}^{\otimes}$, since the Ass-part is fixed, it suffices to show the case when $m \in \mathcal{M}_{[0]}^{\otimes}$.

Consider the precomposition: $\alpha \circ ([1] \times \iota) : [1] \times \mathcal{M}^{\otimes} \hookrightarrow [1] \times \overline{\mathcal{M}}^{\otimes} \rightarrow \mathcal{N}^{\otimes}$, and since $\mathcal{M}_{[0]}^{\otimes} \cong \overline{\mathcal{M}}_{[0]}^{\otimes}$, so α_m is an isomorphism iff $(\alpha \circ ([1] \times \iota))_m$ is an isomorphism, which is true when $\alpha \circ ([1] \times \iota)$ is a natural isomorphism.

- And it remains to show ι^* admits a fully faithful left adjoint, which is [15, proposition 2.65]. $\square$

Now consider the enriched Yoneda embedding $\mathfrak{y}_{\mathcal{P}(\mathcal{V})} : \overline{\mathcal{M}}^{\otimes} \hookrightarrow \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}})^{\otimes}$, we have the following corollary:

Corollary 5.35. *The composition $\mathfrak{L}_{\mathcal{P}(\mathcal{V})} : \mathcal{M}^{\otimes} \xrightarrow{\iota} \overline{\mathcal{M}}^{\otimes} \xrightarrow{\mathfrak{L}_{\mathcal{P}(\mathcal{V})}} \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}})^{\otimes}$ induces an equivalence in $\text{LMod}_{\mathcal{P}(\mathcal{V})}(\text{Pr}^L)$:*

$$\mathcal{P}'(\mathcal{M}) \cong \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}})$$

Proof. By the above lemma, the composition $\mathfrak{L}_{\mathcal{P}(\mathcal{V})} : \mathcal{M} \hookrightarrow \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}})$ induces the equivalences:

$$\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{P}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}}), \mathcal{N}) \xrightarrow{\mathfrak{L}_{\mathcal{P}(\mathcal{V})}^* \cong} \text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}}, \mathcal{N}) \xrightarrow{\iota^* \cong} \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathfrak{L}^* \mathcal{N})$$

where the first edge is the universal property of enriched presheaf. Thus we can see $\mathcal{P}_{\mathcal{P}(\mathcal{V})} \circ \iota$ and $\mathfrak{L}'$ share the same universal properties. Thus there exists a $\mathcal{P}(\mathcal{V})$ -linear equivalence between $\mathcal{P}'(\mathcal{M})$ and $\mathcal{P}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}})$:

$$\begin{array}{ccc} \mathcal{P}'(\mathcal{M}) & \xrightarrow{\cong} & \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}}) \\ \mathfrak{L}' \uparrow & & \uparrow \mathfrak{L}_{\mathcal{P}(\mathcal{V})} \\ \mathcal{M} & \xrightarrow{\iota} & \overline{\mathcal{M}} \end{array}$$

□

The following is [15, proposition 4.21]:

Corollary 5.36. *The functor $\mathfrak{L}^* : \text{Enr}_{\mathcal{P}(\mathcal{V})} \rightarrow \text{PLMod}_{\mathcal{V}}$ admits a fully faithful left adjoint $\overline{(-)}$ assigning $\mathcal{N} \in \text{PLMod}_{\mathcal{V}}$ with $\overline{\mathcal{N}} \in \text{Enr}_{\mathcal{P}(\mathcal{V})}$ constructed as above. Moreover, $\mathfrak{L}^* : \text{Enr}_{\mathcal{P}(\mathcal{V})} \rightarrow \text{PLMod}_{\mathcal{V}}$ is an equivalence.*

Proof. • There exists a commutative diagram for any small $\mathcal{P}(\mathcal{V})$ -enriched ∞ -category $\mathcal{N}$:

$$\begin{array}{ccc} \text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}}, \mathcal{N}) & \xrightarrow{\iota^*} & \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathfrak{L}^* \mathcal{N}) \\ \mathfrak{L}_{\mathcal{P}(\mathcal{V})} \downarrow & & \downarrow \\ \text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\overline{\mathcal{M}}, \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\mathcal{N})) & \xrightarrow{\cong} & \text{LaxLin}_{\mathcal{P}(\mathcal{V})}(\mathcal{M}, \mathfrak{L}^* \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\mathcal{N})) \end{array}$$

where the bottom horizontal arrow is an equivalence by the above lemma, and all the arrows except the horizontal top arrow are known to be fully faithful, thus ι^* is fully faithful, and it's essential surjective by the construction of $\overline{\mathcal{M}}$.

- Thus the embedding $\iota : \mathcal{M} \rightarrow \overline{\mathcal{M}}$ induces an equivalence of mapping space:

$$\text{Map}_{\text{Enr}_{\mathcal{P}(\mathcal{V})}}(\overline{\mathcal{M}}, \mathcal{N}) \cong \text{Map}_{\text{PLMod}_{\mathcal{V}}}(\mathcal{M}, \mathfrak{L}^* \mathcal{N})$$

Thus the construction $\overline{\mathcal{M}}$ can be assembled into an adjunction where the equivalence $\mathcal{M} \cong \mathfrak{L}^* \iota(\mathcal{M})$ is the unit map, and the left adjoint sends $\mathcal{M}$ to $\overline{\mathcal{M}}$ and is fully faithful.

- $\mathfrak{y}^*$ is conservative: Given a morphism $F : \mathcal{M}^\otimes \rightarrow \mathcal{N}^\otimes \in \text{Enr}_{\mathcal{P}(\mathcal{V})}$, if $\mathfrak{y}^*F$ is an equivalence in $\text{PLMod}_{\mathcal{V}}$, then $\mathfrak{y}^*F_{[0]} \cong F_{[0]}$, and it suffices to show $\mathfrak{y}^*F$ induces an equivalence between multimorphism spaces: For any $v \in \mathcal{V}$, and $m_1, m_2 \in \mathfrak{y}^* \mathcal{M}_{[0]}^\otimes$, there is a natural equivalence of mapping spaces:

$$\text{Mul}_{\mathfrak{y}^* \mathcal{M}}(v, m_1; m_2) \cong \text{Mul}_{\mathcal{M}}(\mathfrak{y}(v), m_1; m_2) \cong \text{Map}_{\mathcal{P}(\mathcal{V})}(\mathfrak{y}(v), \text{Mor}_{\mathcal{M}}(m_1, m_2))$$

and for $v' \in \mathcal{P}(\mathcal{V})$, v' can be written as colimit $v' = \text{colim}_{\alpha} \mathfrak{y}(v_{\alpha})$, and the equivalence

$$\mathfrak{y}^*F : \text{Mul}_{\mathfrak{y}^* \mathcal{M}}(v_{\alpha}, m_1; m_2) \cong \text{Mul}_{\mathfrak{y}^* \mathcal{N}}(v_{\alpha}, F(m_1); F(m_2))$$

implies equivalence:

$$\begin{aligned} F : \text{Mul}_{\mathcal{M}}(\text{colim}_{\alpha} v_{\alpha}, m_1; m_2) &\cong \text{Mul}_{\mathfrak{y}^* \mathcal{M}}(\text{colim}_{\alpha} v_{\alpha}, m_1; m_2) \cong \\ \text{Mul}_{\mathfrak{y}^* \mathcal{N}}(\text{colim}_{\alpha} v_{\alpha}, F(m_1); F(m_2)) &\cong \text{Mul}_{\mathcal{N}}(\text{colim}_{\alpha} v_{\alpha}, F(m_1); F(m_2)) \end{aligned}$$

and thus F is an equivalence.

- $\mathfrak{y}^*$ is conservative and admits a fully faithful left adjoint, so it's an equivalence. $\square$

Remark 5.37. The above proof can be generalized to arbitrary small localization pair $(\mathcal{V}^\otimes, S)$, and there is an equivalence:

$$\text{Enr}_{S^{-1} \mathcal{P}\text{Env}(\mathcal{V})} \cong^S \text{Enr}_{\mathcal{V}}$$

by pullback along $\mathcal{V} \hookrightarrow S^{-1} \mathcal{P}\text{Env}(\mathcal{V})$. Moreover, consider the following compositions of morphisms:

$$\text{Enr}_{\mathcal{P}\text{Env}(\mathcal{V})} \xrightarrow{\mathfrak{y}^*} \text{PLMod}_{\text{Env}(\mathcal{V})} \xrightarrow{i^*} \omega\text{LMod}_{\mathcal{V}}$$

, the first and the composition are equivalences, so is the second, and this gives us the following equivalences:

Lemma 5.38. [15, lemma 4.22(7)] *Given a small monoidal ∞ -category $\mathcal{V}$, consider the natural embedding $i : \mathcal{V} \hookrightarrow \text{Env}(\mathcal{V})$, then pullback along i induces an equivalence:*

$$\text{PLMod}_{\text{Env}(\mathcal{V})} \rightarrow \omega\text{LMod}_{\mathcal{V}}$$

.

Remark 5.39. By taking $\mathcal{V} = \emptyset^\otimes$, the above lemma gives us an equivalence $\text{PLMod}_* \cong \text{Cat}$.

Remark 5.40. Since the left adjoint is unique up to a unique natural isomorphism, there exists a natural isomorphism $\beta : \overline{(-)} \Rightarrow (\mathfrak{y}^*)^{-1}$, and especially, there is a commutative diagram by [Tag 02D8](#):

$$\begin{array}{ccc} \overline{(-)} \circ \mathfrak{y}^* & \xrightarrow{\epsilon} & \text{id} \\ \searrow \text{id} \circ \beta & & \swarrow \\ & \text{id} = \mathfrak{y}^* \circ (\mathfrak{y}^*)^{-1} & \end{array}$$

, thus ϵ is also a natural isomorphism, and especially, for any $\mathcal{M} \in \text{Enr}_{\mathcal{P}(\mathcal{V})}$, there is an equivalence $\epsilon_{\mathcal{M}} : \mathcal{M} \rightarrow \overline{\mathfrak{y}^* \mathcal{M}}$ and there is a commutative diagram in $\widehat{\text{PLMod}}$:

$$\begin{array}{ccc}
\mathfrak{L}^* \mathcal{M} & \xrightarrow{\text{Cartesian edge}} & \mathcal{M} \\
\mathfrak{L}^*(\epsilon_{\mathcal{M}}) \downarrow & & \downarrow \epsilon_{\mathcal{M}} \\
\mathfrak{L}^*(\overline{\mathfrak{L}^* \mathcal{M}}) \cong \mathfrak{L}^* \mathcal{M} & \xrightarrow{\iota} & \overline{\mathfrak{L}^* \mathcal{M}}
\end{array}$$

where both vertical edges are isomorphisms and horizontal edges are Cartesian edges. And the bottom horizontal edge can be taken as the morphism $\iota : \mathfrak{L}^* \mathcal{M} \hookrightarrow \overline{\mathfrak{L}^* \mathcal{M}}$ defined above. Thus for any $\mathcal{N} \in \text{LMod}_{\mathcal{P}(\mathcal{V})}(\text{Pr}^L)$, the Cartesian lifting $\mathfrak{L}^* \mathcal{M} \rightarrow \mathcal{M}$ also induces an equivalence between mapping spaces:

$$\text{Lin}_{\mathcal{P}(\mathcal{V})}^L(\mathcal{M}, \mathcal{N}) \cong \text{LaxLin}_{\mathcal{V}}(\mathfrak{L}^* \mathcal{M}, \mathfrak{L}^* \mathcal{N})$$

And there is a $\mathcal{P}(\mathcal{V})$ -linear equivalence between $\mathcal{P}'(\mathfrak{L}^* \mathcal{M})$ and $\mathcal{P}_{\mathcal{P}(\mathcal{V})}(\mathcal{M})$ making the following diagram commute:

$$\begin{array}{ccc}
\mathfrak{L}^* \mathcal{M} & \xrightarrow{\text{Cartesian edge}} & \mathcal{M} \\
\mathfrak{L}' \downarrow & & \downarrow \mathfrak{L}_{\mathcal{P}(\mathcal{V})} \\
\mathcal{P}'(\mathfrak{L}^* \mathcal{M}) & \xrightarrow{\cong} & \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\mathcal{M})
\end{array}$$

And in the remaining part, we show the lax symmetric monoidal functor $\mathfrak{L}^*$ is **(strong)** symmetric monoidal.

Given n small enriched categories $\{\mathcal{M}_i^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}\}$, consider the tensor product operation in $\text{Enr}_{\mathcal{P}(\mathcal{V})}^{\otimes} \subset \widehat{\text{PLMod}}^{\times}$:

$$\boxtimes_{\mathcal{P}(\mathcal{V})} : (\mathcal{M}_i) \rightarrow \boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i$$

, then the natural morphism $r : \boxtimes_{\mathcal{V}} \mathfrak{L}^* \mathcal{M}_i \rightarrow \mathfrak{L}^*(\boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i)$ induced by the lax symmetric monoidal structure on $\mathfrak{L}^*$ is encoded in the following commutative diagram:

$$\begin{array}{ccc}
(\mathcal{M}_i)_{i=1}^n & \xrightarrow{\boxtimes_{\mathcal{P}(\mathcal{V})}} & \boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i \\
\uparrow (c_i) & & \uparrow c \\
(\mathfrak{L}^* \mathcal{M}_i)_{i=1}^n & \xrightarrow{\mathfrak{L}^*(\boxtimes_{\mathcal{P}(\mathcal{V})})} & \mathfrak{L}^*(\boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i) \\
& \searrow \boxtimes_{\mathcal{V}} & \uparrow r \\
& & \boxtimes_{\mathcal{V}} \mathfrak{L}^* \mathcal{M}_i
\end{array}$$

where c_i and c are the Cartesian liftings in $\widehat{\text{PLMod}}$

Lemma 5.41. *The morphism r in the above diagram is an isomorphism.*

Proof. • Noticing that there exists an equivalence $\mathcal{P}'(\mathfrak{L}^* \mathcal{M}_i) \cong \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\mathcal{M}_i)$, and $\mathcal{P}_{\mathcal{P}(\mathcal{V})}(-)$ is symmetric monoidal by [16, corollary 5.1], and $\mathcal{P}'(-)$ is ‘symmetric monoidal’ in the sense of corollary 5.24, thus we have a commutative diagram in $\widehat{\text{PLMod}}^{\times}$:

$$\begin{array}{ccc}
(\mathcal{P}_{\mathcal{P}(\mathcal{V})}(\mathcal{M}_i) \cong \mathcal{P}'(\mathcal{J}^* \mathcal{M}_i)) & \xrightarrow{\otimes_{\mathcal{P}(\mathcal{V})}} & \otimes_{\mathcal{P}(\mathcal{V})} \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\mathcal{M}_i) \cong \mathcal{P}_{\mathcal{P}(\mathcal{V})}(\boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i) \\
\uparrow (\mathcal{J}'_i) & \xrightarrow{\boxtimes_{\mathcal{P}(\mathcal{V})}} & \uparrow \mathcal{J}_{\mathcal{P}(\mathcal{V})} \\
(\mathcal{M}_i)_{i=1}^n & \xrightarrow{\boxtimes_{\mathcal{P}(\mathcal{V})}} & \boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i \\
\uparrow (c_i) & & \uparrow c \\
(\mathcal{J}^* \mathcal{M}_i)_{i=1}^n & \xrightarrow{\mathcal{J}^*(\boxtimes_{\mathcal{P}(\mathcal{V})})} & \mathcal{J}^*(\boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i) \\
& \searrow \boxtimes_{\mathcal{V}} & \swarrow r \\
& & \boxtimes_{\mathcal{V}} \mathcal{J}^* \mathcal{M}_i
\end{array}$$

where $\mathcal{J}^*(\boxtimes_{\mathcal{P}(\mathcal{V})})$ is induced by the universal property of cartesian edge $c : \boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i \rightarrow \mathcal{J}^* \boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i$.

- Since both $\mathcal{J}'$ and $\mathcal{J}_{\mathcal{P}(\mathcal{V})} \circ c$ are fully faithful embedding, thus r is also fully faithful, and it remains to show r is essential surjective. Recall that both $\boxtimes_{\mathcal{P}(\mathcal{V})}$ and $\boxtimes_{\mathcal{V}}$ can be computed using coCartesian liftings co, co'' as shown in the following diagram:

$$\begin{array}{ccccc}
& & \boxtimes_{\mathcal{P}(\mathcal{V})} & & \\
& \searrow & & \searrow & \\
(\mathcal{M}_i)_{i=1}^n & \xrightarrow{\quad} & \prod_{i=1}^n \mathcal{M}_i & \xrightarrow{co} & \boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i \\
\uparrow (c_i) & & \uparrow \prod_{i=1}^n c_i & & \uparrow c \\
(\mathcal{J}^* \mathcal{M}_i)_{i=1}^n & \xrightarrow{\quad} & \prod_{i=1}^n \mathcal{J}^* \mathcal{M}_i & & \mathcal{J}^*(\boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i) \\
& \searrow \boxtimes_{\mathcal{V}} & \searrow co' & & \uparrow r \\
& & & & \boxtimes_{\mathcal{V}} \mathcal{J}^* \mathcal{M}_i
\end{array}$$

- Especially, $co, co'', \prod_{i=1}^n c_i, c$ are all essential surjective, Moreover, c is an equivalence between the underlying categories, thus for any $x \in (\boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i)_{[0]}^{\otimes} \cong \mathcal{J}^*(\boxtimes_{\mathcal{P}(\mathcal{V})} \mathcal{M}_i)_{[0]}^{\otimes}$, there exists $x' \in \prod_{i=1}^n \mathcal{J}^* \mathcal{M}_i$, and $co \circ \prod_{i=1}^n c_i(x') \cong x$, and we can take $co'(x') \in \boxtimes_{\mathcal{V}} \mathcal{J}^* \mathcal{M}_i$, then $r(co'(x')) \cong x$.

□

And finally we reach the following corollary by [corollary 5.7](#):

Corollary 5.42. *For a symmetric monoidal small category $\mathcal{V}$, the functor*

$$\mathcal{J}^* : \text{Enr}_{\mathcal{P}(\mathcal{V})}^{\otimes} \rightarrow \text{PLMod}_{\mathcal{V}}^{\otimes}$$

induced by pullback along $\downarrow : \mathcal{V} \hookrightarrow \mathcal{P}(\mathcal{V})$ is a symmetric monoidal equivalence.

5.6 $(\infty, 2)$ enhancement of $\mathbf{Enr}$

In this subsection,

- We enhance $\mathbf{Enr}_{\mathcal{O}}$ into a symmetric monoidal $(\infty, 2)$ category for a presentably symmetric monoidal category $\mathcal{O}$.
- And further we enhance the symmetric monoidal equivalence $\downarrow^* : \mathbf{Enr}_{\mathcal{P}(\mathcal{V})} \cong \mathbf{PLMod}_{\mathcal{V}}$ into a symmetric monoidal $(\infty, 2)$ equivalence.

And we frequently use the following facts:

- By [22, corollary 3.4.1.5, proposition 4.5.1.4], there is an equivalence $\mathbf{Fun}([1], \mathbf{CAlg}(\mathcal{C})) \cong \mathbf{Alg}_{\mathbf{LM} \otimes \mathbf{Comm}}(\mathcal{C}) \cong \mathbf{Alg}_{\mathbf{LM}}(\mathbf{CAlg}(\mathcal{C}))$: the left action $(C_a, C_m) \rightarrow C_m$ is determined by $C_a \amalg C_m \rightarrow C_m$, thus determined by a morphism between commutative algebra objects $C_a \rightarrow C_m$.
- Through the lax symmetric monoidal functor $\theta : \mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L) \rightarrow \widehat{\mathbf{Cat}}_{(\infty, 2)}$:

$$\downarrow^* : \theta(\mathbf{Enr}_{\mathcal{P}(\mathcal{V})}) \rightarrow \theta(\mathbf{PLMod}_{\mathcal{V}})$$

we transfer from $\mathbf{CAlg}(\mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L)) \subset \mathbf{CAlg}(\mathbf{LMod}(\mathbf{Pr}^L))$ to $\mathbf{CAlg}(\widehat{\mathbf{Cat}}_{(\infty, 2)})$.

Lemma 5.43. *When $\mathcal{V} = *$, there are symmetric monoidal equivalences:*

$$\mathbf{LMod}_{\mathcal{V}}(\mathbf{Cat})^{\otimes} \cong \mathbf{PLMod}_{\mathcal{V}}^{\otimes} \cong \mathbf{Enr}_{\mathcal{P}(\mathcal{V})}^{\otimes} \cong \mathbf{Cat}^{\times}$$

.

Proof. The embeddings $\mathbf{LMod}^{\times} \hookrightarrow \mathbf{PLMod}^{\times}$ over $\mathbf{Mon}^{\times}$ induce lax symmetric monoidal inclusions $R : \mathbf{LMod}_{\mathcal{V}}(\mathbf{Cat})^{\otimes} \hookrightarrow \mathbf{PLMod}_{\mathcal{V}}^{\otimes}$ by pullback along $\mathcal{V} : \mathbf{Comm}^{\otimes} \rightarrow \mathbf{Mon}^{\times}$.

When $\mathcal{V} = *$, which is the unit object in $\mathbf{Mon}$, then the underlying functor of the above embedding is just an equivalence: consider the initial operad $\emptyset$ in $\mathbf{Op}$, which consist only one object over $[0]$ and all the other fibers are emptyset. Then we have $\mathbf{Env}(\emptyset) \cong *$, thus by lemma 5.38, there is an equivalence by pullback along $\emptyset \rightarrow *^{\otimes} = \mathbf{Ass}$:

$$\mathbf{PLMod}_* \cong \omega \mathbf{LMod}_{\emptyset} = \mathbf{Cat}$$

. Moreover, if we restricts to the full subcategory $\mathbf{LMod}_*(\mathbf{Cat}) \subset \mathbf{PLMod}_*$, then the pullback along $\emptyset \rightarrow *$ (which is just taking the fiber over $[0]$) induces another equivalence $\mathbf{LMod}_*(\mathbf{Cat}) \rightarrow \mathbf{Cat}$, thus we have a composition of morphisms:

$$\mathbf{LMod}_*(\mathbf{Cat}) \xrightarrow{R} \mathbf{PLMod}_* \rightarrow \omega \mathbf{LMod}_{\emptyset} = \mathbf{Cat}$$

where the second and the composition morphisms are equivalence, so is the first morphism R .

And especially, the symmetric monoidal left adjoint L is also an equivalence, thus L is also a symmetric monoidal equivalence. And it's obvious that $\mathbf{LMod}_*(\mathbf{Cat})^{\otimes}$ is Cartesian symmetric monoidal, thus all of them are symmetric monoidal equivalent to $\mathbf{Cat}^{\times}$. $\square$

Remark 5.44. Recall that any object in $\text{LMod}_*(\text{Cat})$ can be written as a coCartesian fibration $K^\otimes$ over Ass where each fiber is equivalent to the same $(\infty, 1)$ category K , moreover, it's a property to check that all the coCartesian liftings over Ass correspond to the identity map, thus under straightening, $K^\otimes$ corresponds to a constant functor $\text{Ass} \rightarrow \text{Cat}$, so $K^\otimes$ can be written in the form of $K^\otimes \cong K \times \text{Ass}$. So is any object in PLMod_* .

Lemma 5.45. *Given a morphism $f : \mathcal{V} \rightarrow \mathcal{W} \in \text{Alg}(\text{Pr}^L)$, there exists an adjunction:*

$$f_! : \text{Enr}_{\mathcal{V}} \rightleftarrows \text{Enr}_{\mathcal{W}} : f_!^R$$

.

Proof. By [15, corollary 4.71], there exists an adjunction:

$$f_! : \widehat{\text{Enr}}_{\mathcal{V}} \rightleftarrows \widehat{\text{Enr}}_{\mathcal{W}} : f_!^R$$

And recall that for a category enriched in a locally small category, being small is equivalent to the underlying space of objects being small, and the coCartesian lifting (transfer of enrichment) is essential surjective, thus the above adjunction restricts to an adjunction:

$$f_! : \text{Enr}_{\mathcal{V}} \rightleftarrows \text{Enr}_{\mathcal{W}} : f_!^R$$

□

Remark 5.46. Especially, $f_! : \text{Enr}_{\mathcal{V}} \rightarrow \text{Enr}_{\mathcal{W}}$ preserves colimits.

The following lemma is a combination of [15, corollary 4.62 and theorem 1.3]:

Lemma 5.47. *Given a presentably symmetric monoidal ∞ -category $\mathcal{V}$, $\text{Enr}_{\mathcal{V}}$ is also presentably symmetric monoidal.*

Proof. • By [15, proposition 2.60, corollary 4.62], $\text{Enr}_{\mathcal{V}}$ is presentable if $\mathcal{V}$ is presentable.

- And it remains to show the symmetric monoidal structure on $\text{Enr}_{\mathcal{V}}^\otimes$ is compatible with colimits: By [15, theorem 1.3]: there exists an internal Hom $\text{LaxLin}(-, -)^\otimes \rightarrow \mathcal{V}^\otimes$ for the tensor product $\boxtimes_{\mathcal{V}}$ of $\mathcal{V}$ -enriched categories where Heine's notion of tensor product coincides ours (see [15, notation 5.66]), thus the functor $(-) \otimes \mathcal{M}$ admits a right adjoint $\text{LaxLin}(\mathcal{M}, -)$ for $\mathcal{M} \in \text{Enr}(\mathcal{V})$, and thus preserves all colimits.

□

For any presentably symmetric monoidal ∞ -category $\mathcal{V}$, consider the canonical symmetric monoidal colimit preserving functor $\text{Spc} = \mathcal{P}(*) \rightarrow \mathcal{V}$, especially, consider the morphism $[1] \times \text{Comm}^\otimes \rightarrow \text{Mon}^\times$ specifying this map, then by pullback along the coCartesian fibration $\text{Enr}^\times \rightarrow \text{Mon}^\times$, it induces a symmetric monoidal functor:

$$\mu : \text{Cat}^\times \cong \text{Enr}_{\text{Spc}}^\otimes \rightarrow \text{Enr}_{\mathcal{V}}^\otimes$$

which is colimit preserving, thus is a morphism in $\text{CAlg}(\text{Pr}^L)$ by the above lemmas.

Construction 5.48. Especially, the morphism $\mu \in \text{Fun}([1], \text{CAlg}(\text{Pr}^L)) \cong \text{CAlg}(\text{LMod}(\text{Pr}^L))$ gives rise to an object $(\text{Cat}, \text{Enr}_{\mathcal{V}}) \in \text{LMod}_{\text{Cat}}(\text{Pr}^L)$, and after applying functor $\theta : \text{LMod}_{\text{Cat}}(\text{Pr}^L) \rightarrow \widehat{\text{Cat}}_{(\infty, 2)}$, we denote the corresponding $(\infty, 2)$ category as $\overline{\text{Enr}}_{\mathcal{V}}$.

Construction 5.49. Moreover, for any map $f : \mathcal{V} \rightarrow \mathcal{W}$ in $\text{CAlg}(\text{Pr}^L)$, it induces commutative diagrams in $\text{CAlg}(\text{Pr}^L)$:

$$\begin{array}{ccc} \mathcal{V} & \xrightarrow{f} & \mathcal{W} \\ & \nwarrow \quad \nearrow & \\ & \text{Spc} & \end{array} \quad \xRightarrow{\quad} \quad \begin{array}{ccc} \text{Enr}_{\mathcal{V}} & \xrightarrow{f_!} & \text{Enr}_{\mathcal{W}} \\ & \nwarrow \quad \nearrow & \\ & \text{Cat} & \end{array}$$

where by the above lemma, thus $f_!$ is colimit preserving and also symmetric monoidal, thus gives rise to a morphism in $\text{CAlg}(\text{Pr}^L)$.

And the second diagram gives a morphism $[1] \rightarrow \text{Fun}([1], \text{CAlg}(\text{Pr}^L)) \cong \text{CAlg}(\text{LMod}(\text{Pr}^L))$ which lands in $\text{CAlg}(\text{LMod}_{\text{Cat}}(\text{Pr}^L))$, after applying θ functor, there is a symmetric monoidal $(\infty, 2)$ functor, denoted as:

$$\overline{f} : \overline{\text{Enr}}_{\mathcal{V}} \rightarrow \overline{\text{Enr}}_{\mathcal{W}}$$

Now consider the map $[1] \times [1] \rightarrow \text{CAlg}(\text{Cat})$ corresponding to the following commutative square:

$$\begin{array}{ccc} \mathcal{V} & \xrightarrow{\mathfrak{d}} & \mathcal{P}(\mathcal{V}) \\ \mathbf{1}_{\mathcal{V}} \uparrow & & \uparrow \mathcal{P}(\mathbf{1}_{\mathcal{V}}) \\ * & \xrightarrow{\mathfrak{d}} & \mathcal{P}(*) = \text{Spc} \end{array}$$

Consider the functor $h : [1] \times [1] \times \text{Comm}^{\otimes} \rightarrow \text{Mon}^{\times}$ specifying the above diagram, then pullback of h along the coCartesian fibration $\widehat{\text{PLMod}}^{\times} \rightarrow \text{Mon}^{\times}$ gives rise to a commutative diagram:

$$\begin{array}{ccc} \widehat{\text{PLMod}}_{\mathcal{V}}^{\otimes} & \xrightarrow{\mathfrak{d}_!} & \widehat{\text{PLMod}}_{\mathcal{P}(\mathcal{V})}^{\otimes} \\ \mathbf{1}_{\mathcal{V},!} \uparrow & & \uparrow \mathcal{P}(\mathbf{1}_{\mathcal{V}})_! \\ \widehat{\text{PLMod}}_*^{\otimes} & \xrightarrow{\mathfrak{d}_!} & \widehat{\text{PLMod}}_{\text{Spc}}^{\otimes} \end{array}$$

where all the edges are symmetric monoidal functors.

Recall that $\text{PLMod} \rightarrow \text{Mon}$ is biCartesian, thus all the edges above admit right adjoint at the level of underlying categories, thus admit adjoint as ∞ -operads, and by passing to the right adjoints, there is a commutative diagram:

$$\begin{array}{ccc} \widehat{\text{PLMod}}_{\mathcal{V}}^{\otimes} & \xleftarrow{\mathfrak{d}^*} & \widehat{\text{PLMod}}_{\mathcal{P}(\mathcal{V})}^{\otimes} \\ \downarrow \mathbf{1}_{\mathcal{V}}^* & & \downarrow \mathcal{P}(\mathbf{1}_{\mathcal{V}})^* \\ \widehat{\text{PLMod}}_*^{\otimes} & \xleftarrow{\mathfrak{d}^*} & \widehat{\text{PLMod}}_{\text{Spc}}^{\otimes} \end{array}$$

By [15, corollary 4.70] and Remark 2.75, the adjunction $\mathcal{P}(\mathbf{1}_V)_! : \widehat{\text{PLMod}}_{\text{Spc}} \rightleftarrows \widehat{\text{PLMod}}_{\mathcal{P}(V)} : \mathcal{P}(\mathbf{1}_V)^* \cong \mathcal{P}(\mathbf{1}_V)_!^R$ restricts to a pair of adjunction

$$\mathcal{P}(\mathbf{1}_V)_! : \widehat{\text{Enr}}_{\text{Spc}} \rightleftarrows \widehat{\text{Enr}}_{\mathcal{P}(V)} : \mathcal{P}(\mathbf{1}_V)^*$$

, thus there is a commutative diagram:

$$\begin{array}{ccccc} \text{Enr}_{\mathcal{P}(V)}^{\otimes} & \hookrightarrow & \widehat{\text{Enr}}_{\mathcal{P}(V)}^{\otimes} & \hookrightarrow & \widehat{\text{PLMod}}_{\mathcal{P}(V)}^{\otimes} \\ \uparrow \downarrow & & \uparrow \downarrow & & \mathcal{P}(\mathbf{1}_V)_! \uparrow \downarrow \mathcal{P}(\mathbf{1}_V)^* \\ \text{Enr}_{\text{Spc}}^{\otimes} & \hookrightarrow & \widehat{\text{Enr}}_{\text{Spc}}^{\otimes} & \hookrightarrow & \widehat{\text{Enr}}_{\text{Spc}}^{\otimes} \end{array}$$

where we implicitly impose a lax symmetric monoidal structure on the right adjoints. Thus by composition, there is a commutative square:

$$\begin{array}{ccc} \widehat{\text{PLMod}}_V^{\otimes} & \xleftarrow{\mathfrak{J}^*} & \text{Enr}_{\mathcal{P}(V)}^{\otimes} \\ \mathbf{1}_V^* \downarrow & & \downarrow \mathcal{P}(\mathbf{1}_V)^* \\ \widehat{\text{PLMod}}_*^{\otimes} & \xleftarrow{\mathfrak{J}^*} & \text{Enr}_{\text{Spc}}^{\otimes} \end{array}$$

, which factors through the small version of the left vertical map:

$$\begin{array}{ccc} \text{PLMod}_V^{\otimes} & \xleftarrow{\mathfrak{J}^*} & \text{Enr}_{\mathcal{P}(V)}^{\otimes} \\ \mathbf{1}_V^* \downarrow & & \downarrow \mathcal{P}(\mathbf{1}_V)^* \\ \text{PLMod}_*^{\otimes} & \xleftarrow{\mathfrak{J}^*} & \text{Enr}_{\text{Spc}}^{\otimes} \end{array}$$

We have already seen there is an adjunction:

$$\overline{(-)} : \text{PLMod}_V^{\otimes} \rightleftarrows \text{Enr}_{\mathcal{P}(V)}^{\otimes} : \mathfrak{J}^*$$

sending an absolute small pseudo-enriched category $\mathcal{M}$ to the full weakly left tensored subcategory $\overline{\mathcal{M}}$ of $\mathcal{P}'(\mathcal{M})$ spanned by the image of $\mathfrak{J}' : \mathcal{M} \hookrightarrow \mathcal{P}'(\mathcal{M}) = \mathcal{P}\text{LEnv}(\mathcal{M})_{\text{Pseu}}$.

By passing to the left adjoints, there is a commutative diagram:

$$\begin{array}{ccc} \text{PLMod}_V^{\otimes} & \xrightarrow{\overline{(-)}_V} & \text{Enr}_{\mathcal{P}(V)}^{\otimes} \\ \mathbf{1}_{V,!} \uparrow & & \uparrow \mathcal{P}(\mathbf{1}_V)_! \\ \text{PLMod}_*^{\otimes} \cong \text{Cat}^{\times} & \xrightarrow{\overline{(-)}_{*,\cong}} & \text{Enr}_{\text{Spc}}^{\otimes} \cong \text{Cat}^{\times} \end{array}$$

By passing to the horizontal right adjoints, there is a lax square:

$$\begin{array}{ccc}
\text{PLMod}_{\mathcal{V}}^{\otimes} & \xleftarrow{\mathfrak{L}_{\mathcal{V}}^*} & \text{Enr}_{\mathcal{P}(\mathcal{V})}^{\otimes} \\
\uparrow \mathbf{1}_{\mathcal{V},!} & \nearrow & \uparrow \mathcal{P}(\mathbf{1}_{\mathcal{V}})_! \\
\text{PLMod}_{*}^{\otimes} \cong \text{Cat}^{\times} & \xleftarrow{\mathfrak{L}_{*}^*} & \text{Enr}_{\text{Spc}}^{\otimes} \cong \text{Cat}^{\times}
\end{array}$$

where the 2-morphism is induced by the Beck-Chevalley transformation:

$$\mathbf{1}_{\mathcal{V},!} \circ \mathfrak{L}_{*}^* \Rightarrow \mathfrak{L}_{\mathcal{V}}^* \circ \overline{(-)}_{\mathcal{V}} \circ \mathbf{1}_{\mathcal{V},!} \circ \mathfrak{L}_{*}^* \Rightarrow \mathfrak{L}_{\mathcal{V}}^* \circ \mathcal{P}(\mathbf{1}_{\mathcal{V}})_! \circ \overline{(-)}_{*} \circ \mathfrak{L}_{*}^* \Rightarrow \mathfrak{L}_{\mathcal{V}}^* \circ \mathcal{P}(\mathbf{1}_{\mathcal{V}})_!$$

Lemma 5.50. *The above lax square is strict.*

Proof. Recall that the unit and counit map of the equivalence adjunction $\overline{(-)} \dashv \mathfrak{L}^*$ are all natural isomorphisms, thus all the natural transformations in the above Beck-Chevalley transformation are natural isomorphisms. $\square$

Especially, there is a commutative diagram in Pr^{L} :

$$\begin{array}{ccc}
\text{PLMod}_{\mathcal{V}} & \xleftarrow{\mathfrak{L}_{\mathcal{V}}^*} & \text{Enr}_{\mathcal{P}(\mathcal{V})} \\
\swarrow \mathbf{1}_{\mathcal{V},!} & & \nearrow \mathcal{P}(\mathbf{1}_{\mathcal{V}})_! \\
& \text{Cat} &
\end{array}$$

, and this gives us a symmetric monoidal $(\infty, 2)$ equivalence $\overline{\mathfrak{L}_{\mathcal{V}}^*} : \overline{\text{Enr}_{\mathcal{P}(\mathcal{V})}} \rightarrow \overline{\text{PLMod}_{\mathcal{V}}}$.

Thus if we regard the symmetric monoidal colimit preserving functor $\mathcal{P}(\mathbf{1}_{\mathcal{V}})_! : \text{Cat} \rightarrow \text{Enr}_{\mathcal{P}(\mathcal{V})}$ as a way to enhance the $\text{Enr}_{\mathcal{P}(\mathcal{V})}$ as a symmetric monoidal $(\infty, 2)$ category $\overline{\text{Enr}_{\mathcal{P}(\mathcal{V})}}$, and $\mathbf{1}_{\mathcal{V},!} : \text{Cat} \rightarrow \text{PLMod}_{\mathcal{V}}$ as a way to enhance the $\text{PLMod}_{\mathcal{V}}$ into a symmetric monoidal $(\infty, 2)$ category, then the above commutative diagram means $\mathfrak{L}_{\mathcal{V}}^* : \text{Enr}_{\mathcal{P}(\mathcal{V})} \rightarrow \text{PLMod}_{\mathcal{V}}$ can be enhanced into a symmetric monoidal $(\infty, 2)$ equivalence.

5.7 L is Cat-linear

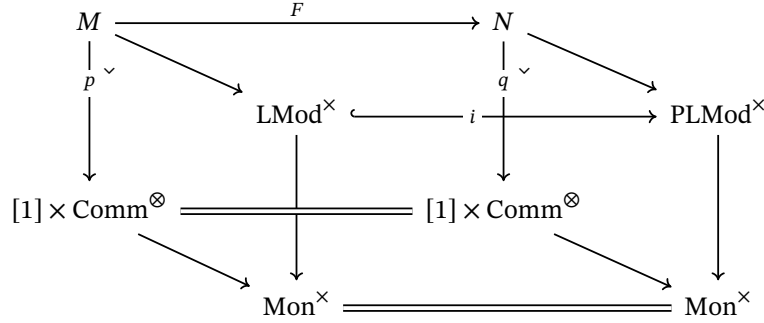
By 2.64, LMod and PLMod are Cartesian fibrations over Mon , and the natural inclusion $i : \text{LMod} \hookrightarrow \text{PLMod}$ preserves the Cartesian edges.

Now consider the Cartesian symmetric monoidal structures on LMod , PLMod and Mon , we have the following diagram of operads:

$$\begin{array}{ccc}
\text{LMod}^{\times} & \hookrightarrow & \text{PLMod}^{\times} \\
& \searrow & \swarrow \\
& \text{Mon}^{\times} &
\end{array}$$

Remark 5.51. All the arrows above are symmetric monoidal, since the underlying functors preserve products.

Now consider the pullback along the morphism $[1] \times \text{Comm}^\otimes \rightarrow \text{Mon}^\times$ (to pick such a morphism is equivalent to pick a morphism $[1] \rightarrow \text{Alg}_{\text{Comm}}(\text{Mon}) \cong \text{CAlg}(\text{Alg}(\text{Cat})) \cong \text{CAlg}(\text{Cat})$) specifying the unit map $* \rightarrow \mathcal{V}$ for the symmetric monoidal category $\mathcal{V}^\otimes$, and this gives rise to a morphism F between bifibrations over $[1] \times \text{Comm}^\otimes$ (Cartesian over $[1]$ and coCartesian over $\text{Comm}^\otimes$), as illustrated in the following diagram:



Remark 5.52. Here is some information encoded in the above diagram:

- The restriction of F to the fibers over $\{0\} \times \text{Comm}^\otimes \hookrightarrow [1] \times \text{Comm}^\otimes$ is the operad map $\text{LMod}_*^\otimes \rightarrow \text{PMod}_*^\otimes$, which is just the identity functor between $\text{Cat} \cong \text{LMod}_* \rightarrow \text{PMod}_* \cong \text{Cat}$.
- The restriction of F to the fibers over $\{1\} \times \text{Comm}^\otimes \hookrightarrow [1] \times \text{Comm}^\otimes$ is the lax symmetric monoidal functor $R : \text{LMod}_{\mathcal{V}}(\text{Cat})^\otimes \hookrightarrow \text{PMod}_{\mathcal{V}}^\otimes$.
- M and N are **biCartesian fibrations** over $[1]$ and **coCartesian fibrations** over $\text{Comm}^\otimes$ since pullback preserves Cartesian and coCartesian fibrations.
- $F : M \rightarrow N$ is a functor preserving Cartesian edges over $[1]$ and preserves the coCartesian edges over the inert maps in $\text{Comm}^\otimes$.
- Consider the fibration $M \rightarrow [1] \times \text{Comm}^\otimes$ which is biCartesian fibration over $[1]$ and coCartesian fibration over $\text{Comm}^\otimes$, by projection to $[1]$, we get an adjunction relative to $\text{Comm}^\otimes$ by [13, 2.2.4]:

$$f^L : \text{LMod}_{\mathcal{V}}^\otimes \rightleftarrows \text{LMod}_*^\otimes \cong \text{Cat}^\times : f$$

, and f^L is a symmetric monoidal ∞ -functor since a M is coCartesian over the whole $[1] \times \text{Comm}^\otimes$, and as its right adjoint, f is a lax symmetric monoidal functor.

- Consider the fibration $N \rightarrow [1] \times \text{Comm}^\otimes$ which is biCartesian fibration over $[1]$ and coCartesian fibration over $\text{Comm}^\otimes$, by projection to $[1]$, we get an adjunction relative to $\text{Comm}^\otimes$ by [13, 2.2.4]:

$$g^L : \text{PMod}_{\mathcal{V}}^\otimes \rightleftarrows \text{PMod}_*^\otimes \cong \text{Cat}^\times : g$$

, where g^L is a symmetric monoidal ∞ -functor and g is a lax symmetric monoidal functor.

- Since F preserves the Cartesian edges over $[1]$, by [13, 2.2.4], the straightening gives us a morphism $\eta : [1] \rightarrow \text{Fun}([1]^{op}, \text{Cat}_{/ \text{Comm}^\otimes})$, this is equivalent to a functor $[1] \times [1]^{op} \rightarrow \text{Cat}_{/ \text{Comm}^\otimes}$, as illustrated in the following diagram in $\text{Cat}_{/ \text{Comm}^\otimes}$:

$$\begin{array}{ccc} \text{LMod}_V^\otimes & \xrightarrow{R} & \text{PLMod}_V^\otimes \\ \downarrow f & & \downarrow g \\ \text{LMod}_*^\otimes & \xrightarrow{\cong} & \text{PLMod}_*^\otimes \end{array}$$

Since the composition of right adjoints is again right adjoint, with its left adjoint being the composition of left adjoints, thus we have the following commutative diagram:

$$\begin{array}{ccc} \text{LMod}_V^\otimes & \xleftarrow{L} & \text{PLMod}_V^\otimes \\ \uparrow f^L & & \uparrow g^L \\ \text{LMod}_*^\otimes & \xleftarrow{\cong} & \text{PLMod}_*^\otimes \end{array}$$

By composition with the commutative square in [here](#):

$$\begin{array}{ccccc} \text{LMod}_V(\text{Cat})^\otimes & \xleftarrow{L} & \text{PLMod}_V^\otimes & \xleftarrow{\wr^*, \cong} & \text{Enr}_{\mathcal{P}(V)}^\otimes \\ \uparrow f^L & & \uparrow g^L = \mathbf{1}_{V,!} & & \uparrow \mathcal{P}(\mathbf{1}_V)_! \\ \text{LMod}_*(\text{Cat})^\times & \xleftarrow{\cong} & \text{PLMod}_*^\times & \xleftarrow{\cong} & \text{Enr}_{\text{Spc}}^\times \end{array}$$

Notation 5.53. On the left side, the symmetric monoidal colimit preserving functor $f^L : \text{LMod}_*(\text{Cat}) \cong \text{Cat} \rightarrow \text{LMod}_V(\text{Cat})$ equips $\text{LMod}_V(\text{Cat})$ with a symmetric monoidal left Cat-module structure, and under the functor $\theta : \text{LMod}_{\text{Cat}}(\text{Pr}^L) \rightarrow \widehat{\text{Cat}}_{(\infty, 2)}$, we denote the corresponding symmetric monoidal $(\infty, 2)$ category as $\overline{\text{LMod}}_V(\text{Cat})$.

By 3.33, the above diagram gives us a symmetric monoidal $(\infty, 2)$ functor, denoted as:

$$L : \overline{\text{Enr}_{\mathcal{P}(V)}} \rightarrow \overline{\text{LMod}}_V(\text{Cat})$$

where we abbreviate the equivalence $\wr^*$.

And also enhance the right adjoint $R : \text{LMod}_V(\text{Cat}) \rightarrow \text{PLMod}_V \cong \text{Enr}_{\mathcal{P}(V)}$ with a lax symmetric monoidal $(\infty, 2)$ structure. Thus we have:

Corollary 5.54. *The lax symmetric monoidal functor $R : \text{LMod}_V(\text{Cat}) \rightarrow \text{PLMod}_V \cong \text{Enr}_{\mathcal{P}(V)}$ can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor:*

$$R : \overline{\text{LMod}}_V(\text{Cat}) \rightarrow \overline{\text{Enr}}_{\mathcal{P}(V)}$$

Notation 5.55. By [15, corollary 4.74], the fully faithful embedding $\mathfrak{y} : \mathcal{V} \hookrightarrow \mathcal{P}(\mathcal{V})$ induces a fully faithful embedding:

$$\mathfrak{y}_! : \widehat{\text{Enr}}_{\mathcal{V}} \rightarrow \widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}$$

which restricts to the small version:

$$\mathfrak{y}_! : \text{Enr}_{\mathcal{V}} \rightarrow \text{Enr}_{\mathcal{P}(\mathcal{V})}$$

. Moreover, $\mathfrak{y}_! : \text{Enr}_{\mathcal{V}} \rightarrow \text{Enr}_{\mathcal{P}(\mathcal{V})}$ is symmetric monoidal since $\mathfrak{y}$ is, thus we denote the full symmetric monoidal $(\infty, 2)$ subcategory of $\overline{\text{Enr}}_{\mathcal{P}(\mathcal{V})}^{\otimes}$ spanned by $\text{Enr}_{\mathcal{V}} \hookrightarrow \text{Enr}_{\mathcal{P}(\mathcal{V})}$ as $\overline{\text{Enr}}_{\mathcal{V}}^{\otimes}$, and the underlying $(\infty, 1)$ symmetric monoidal category of $\overline{\text{Enr}}_{\mathcal{V}}^{\otimes}$ is just $\text{Enr}_{\mathcal{V}}^{\otimes}$.

Take $\mathcal{V}^{\otimes} = \text{Span}_1(\mathcal{C})^{\otimes}$ where all the morphisms in $\mathcal{C}$ are truncated, then:

Corollary 5.56. *There is a lax symmetric monoidal $(\infty, 2)$ functor:*

$$R : \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\text{Cat}) \rightarrow \overline{\text{Enr}}_{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

. Moreover, by precomposition with the lax symmetric monoidal $(\infty, 2)$ functor $\psi : \text{Span}_2(\mathcal{C})^{\otimes} \rightarrow \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}^{\otimes}(\text{Cat})$, there is a lax symmetric monoidal $(\infty, 2)$ functor:

$$R \circ \psi : \text{Span}_2(\mathcal{C})^{\otimes} \rightarrow \overline{\text{Enr}}_{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$$

and it admits a factorization:

$$\text{Span}_2(\mathcal{C})^{\otimes} \rightarrow \overline{\text{Enr}}_{\text{Span}_1(\mathcal{C})}^{\otimes}$$

.

Proof. It suffices to show the left module $(\text{Span}_1(\mathcal{C}), \text{Span}_1(\mathcal{C}_{/x}))$ exhibits $\text{Span}_1(\mathcal{C}_{/x})$ as a $\text{Span}_1(\mathcal{C})$ -enriched category for all $x \in \mathcal{C}$, which is true since by [17, proposition 2.4.1], $\text{Span}_1(\mathcal{C}_{/x})$ is self enriched, especially, for any $v \in \text{Span}_1(\mathcal{C})$, and $m, n \in \text{Span}_1(\mathcal{C}_{/x})$, there exists a morphism object $\text{Mor}'_{\text{Span}_1(\mathcal{C}_{/x})}(m, n) \in \text{Span}_1(\mathcal{C}_{/x})$, such that there is an equivalence of mapping spaces:

$$\text{Map}_{\text{Span}_1(\mathcal{C}_{/x})}(j(v), \text{Mor}'_{\text{Span}_1(\mathcal{C}_{/x})}(m, n)) \cong \text{Map}_{\text{Span}_1(\mathcal{C}_{/x})}(j(v) \otimes m, n)$$

where $j : \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_1(\mathcal{C}_{/x})$ is the symmetric monoidal functor by pullback along $x \rightarrow *$, which gives $\text{Span}_1(\mathcal{C}_{/x})$ a left $\text{Span}_1(\mathcal{C})$ left module structure.

And recall that there is an adjunction:

$$j : \text{Span}_1(\mathcal{C}) \rightleftarrows \text{Span}_1(\mathcal{C}_{/x}) : G$$

where G is the forgetful functor, thus we have an equivalence of mapping spaces:

$$\text{Map}_{\text{Span}_1(\mathcal{C})}(v, G(\text{Mor}'_{\text{Span}_1(\mathcal{C}_{/x})}(m, n))) \cong \text{Map}_{\text{Span}_1(\mathcal{C}_{/x})}(j(v), \text{Mor}'_{\text{Span}_1(\mathcal{C}_{/x})}(m, n))$$

Thus $G(\text{Mor}'_{\text{Span}_1(\mathcal{C}_{/x})}(m, n))$ is the morphism object of m, n under the left $\text{Span}_1(\mathcal{C})$ -action. $\square$

$$6 \quad \overline{\text{Enr}}_{\text{Span}_1(\mathcal{C})} \rightarrow \overline{\text{Enr}}_{\text{Pr}^L}$$

6.1 Enr_{Pr^L}

In this part, we consider the category of Pr^L -enriched categories, which can be equipped with a symmetric monoidal $(\infty, 2)$ structure.

Notation 6.1. Denote σ as the smallest large cardinal.

And now we enlarge the universe, and consider the very large version of coCartesian fibration $\widehat{\widehat{\text{Enr}}} \rightarrow \widehat{\widehat{\text{Mon}}}$, then consider the following pullback diagram:

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{\quad} & \widehat{\widehat{\text{Enr}}} \\ p \downarrow & \lrcorner & \downarrow \\ [1] \times \text{Comm}^\otimes & \xrightarrow{\quad \zeta \quad} & \widehat{\widehat{\text{Mon}}} \end{array}$$

where ζ specifies the symmetric monoidal fully faithful functor $\text{Pr}^{L,\otimes} \rightarrow \widehat{\widehat{\text{Cat}}}_{\text{cc}}^\otimes$, and $\mathcal{M}$ after straightening is a symmetric monoidal functor

$$\alpha : \widehat{\widehat{\text{Enr}}}_{\text{Pr}^L} \rightarrow \widehat{\widehat{\text{Enr}}}_{\widehat{\widehat{\text{Cat}}}_{\text{cc}}}$$

And the above symmetric monoidal functor is again fully faithful by the very large version of [15, corollary 4.74], and restricts to a symmetric monoidal fully faithful functor between the respective full subcategories consisting of those enriched categories with the underlying spaces being large.

$$\alpha : \widehat{\widehat{\text{Enr}}}_{\text{Pr}^L} \rightarrow \widehat{\widehat{\text{Enr}}}_{\widehat{\widehat{\text{Cat}}}_{\text{cc}}}$$

Recall that we have the following lemma on the size:

Lemma 6.2. $\widehat{\widehat{\text{Cat}}}_{\text{cc}}$ is large presentable and σ -compactly generated, especially, the full subcategory consisting of σ -compact objects are Pr^L .

Proof. This is [33, proposition 5.1.4]. □

And by the large version of lemma 5.47, we have

Lemma 6.3. $\widehat{\widehat{\text{Enr}}}_{\widehat{\widehat{\text{Cat}}}_{\text{cc}}}$ is large presentably symmetric monoidal.

Now let's enhance it into an $(\infty, 2)$ category:

Consider the unique symmetric monoidal large colimit preserving functor: $\widehat{\widehat{\text{Spc}}}^\times \rightarrow \widehat{\widehat{\text{Cat}}}_{\text{cc}}$, which gives rise to the following symmetric monoidal functor

$$\widehat{\widehat{\text{Cat}}}^\times \cong \widehat{\widehat{\text{Enr}}}_{\widehat{\widehat{\text{Spc}}}}^\times \rightarrow \widehat{\widehat{\text{Enr}}}_{\widehat{\widehat{\text{Cat}}}_{\text{cc}}}^\otimes$$

And this gives rise to an object in $\text{CAlg}(\widehat{\text{LMod}}_{\widehat{\text{Cat}}}(\widehat{\text{Pr}}^L))$, and through the large version of the lax symmetric monoidal functor

$$\widehat{\theta} : \widehat{\text{LMod}}_{\widehat{\text{Cat}}}(\widehat{\text{Pr}}^L) \rightarrow \widehat{\text{Cat}}_{(\infty, 2)}$$

We acquire a large symmetric monoidal $(\infty, 2)$ category $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{cc}}$.

And consider the symmetric monoidal fully faithful functor $\alpha : \widehat{\text{Enr}}_{\text{Pr}^L} \rightarrow \widehat{\text{Enr}}_{\widehat{\text{Cat}}_{cc}}$, we have the following definition of $\widehat{\text{Enr}}_{\text{Pr}^L}$ and $\widehat{\text{Enr}}_{\text{Pr}^L}$:

Definition 6.4. Let $\widehat{\text{Enr}}_{\text{Pr}^L}^{\otimes}$ be the full symmetric monoidal $(\infty, 2)$ subcategory of $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{cc}}^{\otimes}$ consisting of the image of $\alpha : \widehat{\text{Enr}}_{\text{Pr}^L} \rightarrow \widehat{\text{Enr}}_{\widehat{\text{Cat}}_{cc}}$, and let $\widehat{\text{Enr}}_{\text{Pr}^L}^{\otimes} \subset \widehat{\text{Enr}}_{\text{Pr}^L}^{\otimes}$ be the corresponding symmetric monoidal $(\infty, 2)$ subcategory consisting of those enriched ∞ -categories which underlying spaces are small.

6.2 Mapping Category of $\widehat{\text{Enr}}_{\text{Pr}^L}$

In the above subsection, we have defined the symmetric monoidal $(\infty, 2)$ structure on $\widehat{\text{Enr}}_{\text{Pr}^L}$ regarded as full $(\infty, 2)$ subcategory of the symmetric monoidal $(\infty, 2)$ category $\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{cc}}$. In this subsection, we show under the above $(\infty, 2)$ structure on $\widehat{\text{Enr}}_{\text{Pr}^L}$, the mapping category of $\widehat{\text{Enr}}_{\text{Pr}^L}$ is the $(\infty, 1)$ category $\text{LaxLin}_{\widehat{\text{Pr}^L}}(-, -)$. More precisely:

- First we show that under the left-Cat module structure on $\text{PLMod}_{\mathcal{V}}$ induced by the symmetric monoidal functor $\mathbf{1}_{\mathcal{V}, !} : \text{PLMod}_* \rightarrow \text{PLMod}_{\mathcal{V}}$, the mapping category of $\text{PLMod}_{\mathcal{V}}$ is $\text{LaxLin}_{\mathcal{V}}(-, -)$.
- Since $\mathfrak{y}^* : \widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})} \rightarrow \widehat{\text{PLMod}}_{\mathcal{V}}$ is an $(\infty, 2)$ equivalence, thus the mapping category of $\widehat{\text{Enr}}_{\mathcal{P}(\mathcal{V})}$ is also equivalent to $\text{LaxLin}_{\mathcal{V}}(-, -)$.
- If $\mathcal{V}$ is a presentably symmetric monoidal ∞ -category, then we show that there is a fully faithful (lax linear) embedding $(\text{Cat}, \text{Enr}_{\mathcal{V}}) \hookrightarrow (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})})$ induced by the transfer of enrichment $\mathfrak{y}_! : \text{Enr}_{\mathcal{V}} \rightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}$, thus there is an equivalence between the morphism objects

$$\text{Fun}_{\widehat{\text{Enr}}(\mathcal{V})}(\mathcal{M}, \mathcal{N}) \cong \text{Mor}_{\widehat{\text{Enr}}_{\mathcal{V}}}(\mathcal{M}, \mathcal{N}) \cong \text{Mor}_{\widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}}(\mathfrak{y}_! \mathcal{M}, \mathfrak{y}_! \mathcal{N})$$

for any $\mathcal{M}, \mathcal{N} \in \widehat{\text{Enr}}_{\mathcal{V}}$.

- After enlarging the universe, we show that the $(\infty, 2)$ equivalence $\mathfrak{y}^* : \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})} \rightarrow \widehat{\text{PLMod}}_{\mathcal{V}}$ restricts to a functor $\widehat{\text{Enr}}_{\mathcal{V}} \hookrightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})} \xrightarrow{\mathfrak{y}^*} \widehat{\text{PLMod}}_{\mathcal{V}} \supset \widehat{\text{Enr}}_{\mathcal{V}}$, thus for any $\mathcal{M}, \mathcal{N} \in \widehat{\text{Enr}}_{\mathcal{V}}$, the mapping category between them in the $(\infty, 2)$ category $\widehat{\text{Enr}}_{\mathcal{V}}$ is equivalent to the corresponding mapping category inside $\widehat{\text{PLMod}}_{\mathcal{V}}$, which is $\text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})$.
- And we take $\mathcal{V} = \widehat{\text{Cat}}_{cc}$, and further we can show $\text{LaxLin}_{\widehat{\text{Pr}}^L}(\mathcal{M}, \mathcal{N}) \cong \text{LaxLin}_{\widehat{\text{Cat}}_{cc}}(\mathcal{M}, \mathcal{N})$ for $\mathcal{M}, \mathcal{N} \in \widehat{\text{Enr}}_{\widehat{\text{Pr}}^L} \subset \widehat{\text{Enr}}_{\widehat{\text{Cat}}_{cc}}$.

First we give an interpretation on the left action functor $act : \text{Cat} \times \text{PLMod}_{\mathcal{V}} \rightarrow \text{PLMod}_{\mathcal{V}}$ for the left Cat-module $\text{PLMod}_{\mathcal{V}}$:

Notation 6.5. By [remark 5.44](#), any object $K^{\otimes} \rightarrow \text{Ass}$ in $\text{PLMod}_* \cong \text{Cat}$ can be written in the form of a projection map $K \times \text{Ass} \rightarrow \text{Ass}$. And thus consider the product in the ∞ -category PLMod : $K^{\otimes} \times_{\text{Ass}} \mathcal{M}^{\otimes} \rightarrow \text{Ass} \times_{\text{Ass}} \mathcal{V}^{\otimes}$ for any $\mathcal{M} \in \text{PLMod}_{\mathcal{V}}$, and it can be written as:

$$K^{\otimes} \times_{\text{Ass}} \mathcal{M}^{\otimes} \cong (K \times \text{Ass}) \times_{\text{Ass}} \mathcal{M}^{\otimes} \cong K \times \mathcal{M}^{\otimes} \xrightarrow{\text{projection}} \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$$

Thus for any $K^{\otimes} = K \times \text{Ass} \in \text{PLMod}_*$ and $\mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, we denote their product in PLMod as $K \times \mathcal{M}^{\otimes} \rightarrow \mathcal{V}^{\otimes}$, and sometimes use its underlying category $K \times \mathcal{M}$ to represent it. Also we use K represent the object $K^{\otimes} = K \times \text{Ass} \rightarrow \text{Ass}$.

Given an $(\infty, 1)$ category K , and $\mathcal{M} \in \text{PLMod}_{\mathcal{V}}$, and consider the symmetric monoidal functor $\mathbf{1}_! : \text{PLMod}_* \cong \text{Cat} \rightarrow \text{PLMod}_{\mathcal{V}}$, then $\mathbf{1}_!$ gives $\text{PLMod}_{\mathcal{V}}$ the left Cat-linear structure, especially, $K \otimes \mathcal{M}$ can be computed through the functor $(K, \mathcal{M}) \in \text{PLMod}_* \times \text{PLMod}_{\mathcal{V}} \xrightarrow{\mathbf{1}_! \times \text{id}} \text{PLMod}_{\mathcal{V}} \times \text{PLMod}_{\mathcal{V}} \xrightarrow{\boxtimes_{\mathcal{V}}} \text{PLMod}_{\mathcal{V}} \ni K \otimes \mathcal{M}$, and can be computed using the following coCartesian liftings inside PLMod :

$$\begin{array}{ccccc} K \times \mathcal{M} & \xrightarrow{co_1} & \mathbf{1}_!(K) \times \mathcal{M} & \xrightarrow{co_2} & K \otimes \mathcal{M} \\ \downarrow & & \downarrow & & \downarrow \\ * \times \mathcal{V} & \xrightarrow{\mathbf{1}_{\mathcal{V}} \times \text{id}} & \mathcal{V} \times \mathcal{V} & \xrightarrow{\otimes_{\mathcal{V}}} & \mathcal{V} \end{array}$$

where co_1, co_2 are the coCartesian liftings, and especially, the composition $co_2 \circ co_1$ is the coCartesian lifting over $\text{id} \cong \otimes_{\mathcal{V}} \circ (\mathbf{1} \times \text{id}) : * \times \mathcal{V} \rightarrow \mathcal{V}$, thus $(K \otimes \mathcal{M})^{\otimes} \cong K \times \mathcal{M}^{\otimes} \in \text{PLMod}_{\mathcal{V}}$.

Construction 6.6. And we can make the above process functorial: consider the coCartesian fibration $\text{PLMod}^{\times} \rightarrow \text{Mon}^{\times}$, and the morphism

$$act : (*, \mathcal{V}) \xrightarrow{(\mathbf{1}, \text{id})} (\mathcal{V}, \mathcal{V}) \xrightarrow{\otimes_{\mathcal{V}}} \mathcal{V}$$

, which is isomorphic to the morphism:

$$act : (*, \mathcal{V}) \xrightarrow{\times} * \times \mathcal{V} \xrightarrow{\mathbf{1} \times \text{id}} \mathcal{V} \times \mathcal{V} \xrightarrow{\otimes_{\mathcal{V}}} \mathcal{V}$$

then consider the functor $[1] \rightarrow \text{Mon}^{\times}$ classifying the morphism act , and also the pullback of $\text{PLMod}^{\times} \rightarrow \text{Mon}^{\times}$ along it, then we have the following functors:

$$(act)_! : \text{PLMod}_* \times \text{PLMod}_{\mathcal{V}} \xrightarrow{\times} \text{PLMod}_{* \times \mathcal{V} = \mathcal{V}} \xrightarrow{\mathbf{1}_! \times \text{id}} \text{PLMod}_{\mathcal{V} \times \mathcal{V}} \xrightarrow{\otimes_{\mathcal{V}, !} = \boxtimes_{\mathcal{V}}} \text{PLMod}_{\mathcal{V}}$$

and noticing that the last two functors are the coCartesian liftings over $\text{id} \cong \otimes_{\mathcal{V}} \circ (\mathbf{1} \times \text{id})$, thus the left action functor $act : \text{PLMod}_* \times \text{PLMod}_{\mathcal{V}} \rightarrow \text{PLMod}_{\mathcal{V}}$ is equivalent to the functor $\text{PLMod}_* \times \text{PLMod}_{\mathcal{V}} \xrightarrow{\times} \text{PLMod}_{* \times \mathcal{V} = \mathcal{V}}$ where $\times$ is inherited from the Cartesian symmetric

monoidal structure on $\text{PLMod}^\times$, where we have a functor $\times : \text{PLMod} \times \text{PLMod} \rightarrow \text{PLMod}$ sending two pseudo-enriched categories to their product.

And recall in [14, remark 3.68], there is an equivalence between $(\infty, 1)$ categories:

$$\text{Fun}(K, \text{LaxLin}_V(\mathcal{M}, \mathcal{N})) \cong \text{LaxLin}_V(K \times \mathcal{M}, \mathcal{N})$$

inherited from the equivalence:

$$\text{Fun}(K, \text{Fun}_V(\mathcal{M}, \mathcal{N})) \cong \text{Fun}_V(K \times \mathcal{M}, \mathcal{N})$$

inherited further from the equivalence:

$$\text{Fun}(K, \text{Fun}(\mathcal{M}, \mathcal{N})) \cong \text{Fun}(K \times \mathcal{M}, \mathcal{N})$$

from the closed symmetric monoidal structure on Cat .

Thus we have the following lemma:

Lemma 6.7. *Under the left Cat -linear action on PLMod_V induced from the symmetric monoidal colimit preserving functor $\text{PLMod}_* \cong \text{Cat} \rightarrow \text{PLMod}_V$, for any $\mathcal{M}, \mathcal{N} \in \text{PLMod}_V$, $\text{LaxLin}(\mathcal{M}, \mathcal{N})$ is the morphism object $\text{Mor}_{\text{PLMod}_V}(\mathcal{M}, \mathcal{N})$, that is, the functor $(-) \times \mathcal{M}^\otimes : \text{Cat} \cong \text{PLMod}_* \xrightarrow{\text{id} \times \{\mathcal{M}\}} \text{PLMod}_* \times \text{PLMod}_V \xrightarrow{\text{act}} \text{PLMod}_V$ admits a right adjoint sending $\mathcal{N} \in \text{PLMod}_V$ to $\text{LaxLin}(\mathcal{M}, \mathcal{N}) \in \text{Cat}$.*

Proof.

It suffices to show there exists a morphism $s'' : \text{LaxLin}_V(\mathcal{M}, \mathcal{N}) \times \mathcal{M} \rightarrow \mathcal{N}$ in PLMod_V such that the induced morphism:

$$\text{Map}_{\text{Cat}}(K, \text{LaxLin}_V(\mathcal{M}, \mathcal{N})) \xrightarrow{(-) \times \mathcal{M}} \text{Map}_{\text{PLMod}_V}(K \times \mathcal{M}, \text{LaxLin}_V(\mathcal{M}, \mathcal{N}) \times \mathcal{M}) \xrightarrow{s''} \text{Map}_{\text{PLMod}_V}(K \times \mathcal{M}, \mathcal{N})$$

is an equivalence.

Consider the closed symmetric monoidal structure on Cat , then there exists a functor $s : \text{Fun}(\mathcal{M}^\otimes, \mathcal{N}^\otimes) \times \mathcal{M}^\otimes \rightarrow \mathcal{N}^\otimes$ such that:

$$\text{Fun}(K, \text{Fun}(\mathcal{M}^\otimes, \mathcal{N}^\otimes)) \xrightarrow{(-) \times \mathcal{M}^\otimes} \text{Fun}(K \times \mathcal{M}^\otimes, K \times \text{Fun}(\mathcal{M}^\otimes, \mathcal{N}^\otimes)) \xrightarrow{s} \text{Fun}(K \times \mathcal{M}^\otimes, \mathcal{N}^\otimes)$$

is an equivalence. Now consider the full subcategory $\text{Fun}_{/V^\otimes}(\mathcal{M}^\otimes, \mathcal{N}^\otimes) \subset \text{Fun}(\mathcal{M}^\otimes, \mathcal{N}^\otimes)$ consisting of functors over $V^\otimes$, and there is an induced morphism $s' : \text{Fun}_{/V^\otimes}(\mathcal{M}^\otimes, \mathcal{N}^\otimes) \times \mathcal{M}^\otimes \rightarrow \mathcal{N}^\otimes$, then the above equivalence restricts to equivalences:

$$\begin{array}{ccccc} \cong : \text{Fun}(K, \text{Fun}(\mathcal{M}^\otimes, \mathcal{N}^\otimes)) & \xrightarrow{(-) \times \mathcal{M}^\otimes} & \text{Fun}(K \times \mathcal{M}^\otimes, K \times \text{Fun}(\mathcal{M}^\otimes, \mathcal{N}^\otimes)) & \xrightarrow{s^*} & \text{Fun}(K \times \mathcal{M}^\otimes, \mathcal{N}^\otimes) \\ \uparrow & & \uparrow & & \uparrow \\ \cong : \text{Fun}(K, \text{Fun}_{/V^\otimes}(\mathcal{M}^\otimes, \mathcal{N}^\otimes)) & \longrightarrow & \text{Fun}_{/V^\otimes}(K \times \mathcal{M}^\otimes, \text{Fun}_{/V^\otimes}(\mathcal{M}^\otimes, \mathcal{N}^\otimes) \times \mathcal{M}^\otimes) & \xrightarrow{s'^*} & \text{Fun}_{/V^\otimes}(K \times \mathcal{M}^\otimes, \mathcal{N}^\otimes) \end{array}$$

And likewise, consider the full subcategory $\text{LaxLin}_V(\mathcal{M}, \mathcal{N}) \subset \text{Fun}_{/V^\otimes}(\mathcal{M}^\otimes, \mathcal{N}^\otimes)$ consisting of

$\mathcal{V}$ lax linear functors, and there is an induced morphism $s'' : \text{LaxLin}(\mathcal{M}, \mathcal{N}) \times \mathcal{M} \rightarrow \mathcal{N}$, and the above equivalence restricts to an equivalence:

$$\text{Fun}(K, \text{LaxLin}(\mathcal{M}, \mathcal{N})) \xrightarrow{(-) \times \mathcal{M}^\otimes} \text{LaxLin}_{\mathcal{V}}(K \times \mathcal{M}, \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N}) \times \mathcal{M}) \xrightarrow{s''^*} \text{LaxLin}_{\mathcal{V}}(K \times \mathcal{M}, \mathcal{N})$$

□

Thus under the symmetric monoidal functor $\text{Cat} \cong \text{PLMod}_* \rightarrow \text{PLMod}_{\mathcal{V}}$, the mapping category of the enhanced $(\infty, 2)$ category $\overline{\text{PLMod}}_{\mathcal{V}}$ is in the form of $\text{LaxLin}_{\mathcal{V}}(-, -)$.

Lemma 6.8. *Given a fully faithful monoidal functor $\alpha : \mathcal{V}^\otimes \rightarrow \mathcal{W}^\otimes$, there is an induced equivalence:*

$$\text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N}) \rightarrow \text{LaxLin}_{\mathcal{W}}(\alpha_! \mathcal{M}, \alpha_! \mathcal{N})$$

for any $\mathcal{M}, \mathcal{N} \in \text{PLMod}_{\mathcal{V}}$.

Proof. • First of all, we show that for any $K \in \text{Cat}$, there is an equivalence $\alpha_!(K \otimes \mathcal{M}) \cong K \otimes \alpha_!(\mathcal{M})$ where $\otimes$ refers to the left-Cat-action defined in the above lemma.

- By lemma 2.76, the coCartesian lifting of product of morphisms is the product of coCartesian liftings, thus the coCartesian lifting of $K \times \mathcal{M} \rightarrow * \times \mathcal{V}$ over $\text{id} \times \alpha \cong \alpha : * \times \mathcal{V} \cong \mathcal{V} \rightarrow * \times \mathcal{W} \cong \mathcal{W}$ can be computed componentwise, thus $\alpha_!(K \times \mathcal{M}) \cong \text{id}_!(K) \times \alpha_!(\mathcal{M}) \rightarrow * \times \mathcal{W} \cong \mathcal{W}$.
- Thus for any $K \in \text{Cat} \cong \text{PLMod}_*$, there are natural equivalences:

$$\begin{aligned} \text{Map}_{\text{Cat}}(K, \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})) &\cong \text{Map}_{\text{PLMod}_{\mathcal{V}}}(K \otimes \mathcal{M}, \mathcal{N}) \xrightarrow{\alpha_! \cong} \text{Map}_{\text{PLMod}_{\mathcal{W}}}(\alpha_!(K \otimes \mathcal{M}), \alpha_! \mathcal{N}) \\ &\cong \text{Map}_{\text{PLMod}_{\mathcal{W}}}(K \otimes \alpha_! \mathcal{M}, \mathcal{N}) \cong \text{Map}_{\text{Cat}}(K, \text{LaxLin}_{\mathcal{W}}(\alpha_! \mathcal{M}, \alpha_! \mathcal{N})) \end{aligned}$$

- Thus $\text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N}) \cong \text{LaxLin}_{\mathcal{W}}(\alpha_!(\mathcal{M}), \alpha_!(\mathcal{N}))$

□

Given a presentably symmetric monoidal ∞ -category $\mathcal{V}$, and there is an symmetric monoidal adjunction:

$$\text{loc} : \widehat{\mathcal{P}}(\mathcal{V}) \rightleftarrows \widehat{\text{Ind}}(\mathcal{V}) : i$$

Especially the left adjoint loc can be assembled into the following commutative diagram in $\text{CAlg}(\widehat{\text{Pr}}^{\text{L}})$:

$$\begin{array}{ccc} \widehat{\mathcal{P}}(\mathcal{V}) & \xrightarrow{\text{loc}, \otimes} & \widehat{\text{Ind}}(\mathcal{V}) \\ & \nwarrow \scriptstyle 1_! \quad \nearrow \scriptstyle 1_! & \\ & \widehat{\text{Spc}} & \end{array}$$

And by the large version of lemma 5.45 and lemma 5.47, the above diagram gives rise to a diagram in $\text{CAlg}(\widehat{\text{Pr}}^{\text{L}})$:

$$\begin{array}{ccc}
\widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})} & \xrightarrow{\text{loc}_!, \otimes} & \widehat{\text{Enr}}_{\widehat{\text{Ind}}(\mathcal{V})} \\
& \nwarrow \mathbf{1}_! \quad \nearrow \mathbf{1}_! & \\
& \widehat{\text{Enr}}_{\widehat{\text{Spc}}} \cong \widehat{\text{Cat}} &
\end{array}$$

thus loc can be enhanced into a large presentably left $\widehat{\text{Cat}}$ -linear morphism:

$$\text{loc}_! : (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}) \rightarrow (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\text{Ind}}(\mathcal{V})})$$

and the fully faithful right adjoint $i_! : \widehat{\text{Enr}}_{\widehat{\text{Ind}}(\mathcal{V})} \hookrightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}$ is enhanced into a fully faithful lax linear functor between left tensored ∞ -categories:

$$i : (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\text{Ind}}(\mathcal{V})}) \hookrightarrow (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})})$$

And on the hand, induced by the natural embedding $\mathcal{V} \rightarrow \widehat{\text{Ind}}(\mathcal{V})$, there is a commutative diagram in $\text{CAlg}(\widehat{\text{Cat}})$ where all the arrows are symmetric monoidal and preserve **small** colimits:

$$\begin{array}{ccc}
\mathcal{V} & \xrightarrow{j} & \widehat{\text{Ind}}(\mathcal{V}) \\
\mathbf{1}_! \uparrow & & \uparrow \mathbf{1}_! \\
\text{Spc} & \hookrightarrow & \widehat{\text{Spc}} = \widehat{\text{Ind}}(\text{Spc})
\end{array}$$

where $\mathbf{1}_!$ s are the small colimit preserving symmetric monoidal functors induced by the unit maps $\mathbf{1} : * \rightarrow \mathcal{V}$ and $\mathbf{1} : * \rightarrow \widehat{\text{Ind}}(\mathcal{V})$. And the above diagram gives rise to a commutative diagram in $\text{CAlg}(\widehat{\text{Cat}})$:

$$\begin{array}{ccccc}
\text{Enr}_{\mathcal{V}} & \hookrightarrow & \widehat{\text{Enr}}_{\mathcal{V}} & \xrightarrow{j_!} & \widehat{\text{Enr}}_{\widehat{\text{Ind}}(\mathcal{V})} \\
\mathbf{1}_! \uparrow & & \mathbf{1}_! \uparrow & & \uparrow \mathbf{1}_! \\
\text{Cat} \cong \text{Enr}_{\text{Spc}} & \hookrightarrow & \widehat{\text{Enr}}_{\text{Spc}} & \hookrightarrow & \widehat{\text{Enr}}_{\widehat{\text{Spc}}} \cong \widehat{\text{Cat}}
\end{array}$$

Especially, the outer side square gives us a fully faithful linear functor:

$$j_! : (\text{Cat} \cong \text{Enr}_{\text{Spc}}, \text{Enr}_{\mathcal{V}}) \hookrightarrow (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\text{Ind}}(\mathcal{V})})$$

through the equivalence $\text{Fun}([1], \text{CAlg}(\widehat{\text{Cat}})) \cong \text{CAlg}(\text{LMod}(\widehat{\text{Cat}}))$. Thus we have:

Proposition 6.9. *The fully faithful embedding $\mathfrak{J}_! : \text{Enr}_{\mathcal{V}} \hookrightarrow \widehat{\text{Enr}}_{\mathcal{V}} \hookrightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}$ can be enhanced into a fully faithful lax linear embedding:*

$$(\text{inclusion}, \widehat{\mathfrak{J}}_!) : (\text{Cat} \cong \text{Enr}_{\text{Spc}}, \text{Enr}_{\mathcal{V}}) \hookrightarrow (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})})$$

.

Proof. There is a factorization of $\widehat{\mathfrak{J}} : \mathcal{V} \hookrightarrow \widehat{\mathcal{P}}(\mathcal{V})$ through $\mathcal{V} \xrightarrow{j} \widehat{\text{Ind}}(\mathcal{V}) \xrightarrow{i} \widehat{\mathcal{P}}(\mathcal{V})$, and we have seen there is a fully faithful linear embedding:

$$j : (\text{Cat}, \text{Enr}_{\mathcal{V}}) \hookrightarrow (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\text{Ind}}(\mathcal{V})})$$

and a fully faithful lax linear embedding:

$$i : (\widehat{\text{Cat}} \cong \widehat{\text{Enr}}_{\widehat{\text{SpC}}}, \widehat{\text{Enr}}_{\widehat{\text{Ind}}(\mathcal{V})}) \hookrightarrow (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})})$$

and by composition, we have the lax fully faithful embedding as required. $\square$

Remark 6.10. Noticing that the natural embedding $j : \mathcal{V} \hookrightarrow \widehat{\text{Ind}}(\mathcal{V})$ is symmetric monoidal and preserves small colimits, while $\widehat{\mathfrak{L}} : \mathcal{V} \hookrightarrow \widehat{\mathcal{P}}(\mathcal{V})$ doesn't preserve small colimits, thus we do the above factorization.

Corollary 6.11. *For a presentably monoidal ∞ -category, the natural embedding $\widehat{\mathfrak{L}}_! : \text{Enr}_{\mathcal{V}} \rightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}$ induces an equivalence between morphism objects:*

$$\text{Mor}_{\text{Enr}_{\mathcal{V}}}(\mathcal{M}, \mathcal{N}) \cong \text{Mor}_{\widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}}(\widehat{\mathfrak{L}}_!(\mathcal{M}), \widehat{\mathfrak{L}}_!(\mathcal{N}))$$

, thus there is an equivalence between the mapping categories:

$$\text{Fun}_{\text{Enr}_{\mathcal{V}}}(\mathcal{M}, \mathcal{N}) \cong \text{Fun}_{\widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}}(\widehat{\mathfrak{L}}_! \mathcal{M}, \widehat{\mathfrak{L}}_! \mathcal{N})$$

.

Proof. By the above proposition and the [fact that fully faithfulness of lax linear functor implies enriched fully faithfulness](#). $\square$

Now we show there exists a commutative diagram for a large presentably monoidal $(\infty, 1)$ category $\mathcal{V}$:

$$\begin{array}{ccc} \widehat{\text{PLMod}}_{\mathcal{V}} & \xleftarrow{\widehat{\mathfrak{L}}^*} & \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})} \\ \uparrow & & \uparrow \widehat{\mathfrak{L}}_{\mathcal{V},!} \\ \text{Enr}_{\mathcal{V}} & \xlongequal{\text{id}} & \text{Enr}_{\mathcal{V}} \end{array}$$

where for $\mathcal{M} \in \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}$, $\widehat{\mathfrak{L}}^*(\mathcal{M}) := \mathcal{V}^{\otimes} \times_{\widehat{\mathcal{P}}(\mathcal{V})^{\otimes}} \mathcal{M}^{\otimes}$. And for simplicity, we go back to the small universe and show that:

Lemma 6.12. *Given a small monoidal ∞ -category $\mathcal{V}$, for any $\mathcal{M} \in \text{Enr}_{\mathcal{V}}$, then $\widehat{\mathfrak{L}}^* \widehat{\mathfrak{L}}_! \mathcal{M} \cong \mathcal{M} \in \text{Enr}_{\mathcal{V}}$, where $\widehat{\mathfrak{L}}^* : \text{Enr}_{\mathcal{V}} \rightarrow \widehat{\text{PLMod}}_{\mathcal{V}}$ is the pullback functor along $\widehat{\mathfrak{L}} : \mathcal{V} \hookrightarrow \widehat{\mathcal{P}}(\mathcal{V})$ and $\widehat{\mathfrak{L}}_!$ is the transfer of enrichment.*

Proof. • Let $\alpha : \mathcal{M}^{\otimes} \rightarrow \widehat{\mathfrak{L}}_!(\mathcal{M})^{\otimes}$ be the coCartesian lifting over the map $\widehat{\mathfrak{L}} : \mathcal{V} \hookrightarrow \widehat{\mathcal{P}}(\mathcal{V})$, then there is a commutative diagram in $\widehat{\text{PLMod}}$:

$$\begin{array}{ccc} \widehat{\mathfrak{L}}^* \widehat{\mathfrak{L}}_! \mathcal{M}^{\otimes} & & \\ \uparrow \text{induced} & \searrow \text{Cartesian lifting} & \\ \mathcal{M}^{\otimes} & \xrightarrow{\alpha} & \widehat{\mathfrak{L}}_! \mathcal{M}^{\otimes} \end{array}$$

and we denote the induced map by $\beta : \mathcal{M}^{\otimes} \rightarrow \widehat{\mathfrak{L}}^* \widehat{\mathfrak{L}}_! \mathcal{M}^{\otimes}$.

- β is obviously essential surjective since coCartesian lifting is essential surjective and the Cartesian lifting induces equivalence between the underlying categories.
- And it remains to show β is fully faithful embedding, and since the Cartesian lifting is a fully faithful embedding between weakly left tensored ∞ -categories, thus it suffices to show $\alpha : \mathcal{M}^{\otimes} \hookrightarrow \mathfrak{L}_! \mathcal{M}^{\otimes}$ is a fully faithful embedding.
- And it suffices to show the induced morphism between multimorphism spaces is an equivalence for any $v \in \mathcal{V}, x, y \in \mathcal{M}$:

$$\text{Mul}_{\mathcal{M}}(v, x; y) \rightarrow \text{Mul}_{\mathfrak{L}_! \mathcal{M}}(\mathfrak{L}_!(v), \alpha(x); \alpha(y))$$

which is true by the following commutative diagram:

$$\begin{array}{ccc} \text{Mul}_{\mathcal{M}}(v, x; y) & \xrightarrow{\quad\quad\quad} & \text{Mul}_{\mathfrak{L}_! \mathcal{M}}(\mathfrak{L}_!(v), \alpha(x); \alpha(y)) \\ \cong \uparrow & & \uparrow \cong \\ \text{Map}_{\mathcal{V}}(v, \text{Mor}_{\mathcal{M}}(x, y)) & \xrightarrow{\quad \cong \quad} & \text{Map}_{\mathcal{P}(\mathcal{V})}(\mathfrak{L}_!(v), \mathfrak{L}_!(\text{Mor}_{\mathcal{M}}(x, y)) \cong \text{Mor}_{\mathfrak{L}_! \mathcal{M}}(\alpha(x), \alpha(y))) \end{array}$$

where the bottom morphism is an equivalence by the fully faithfulness of $\mathfrak{L}_! : \mathcal{V} \hookrightarrow \mathcal{P}(\mathcal{V})$. And we also use the fact that [transfer of enrichment preserves morphism objects \(see description \(4\) of coCartesian lifting\)](#), thus $\mathfrak{L}_!(\text{Mor}_{\mathcal{M}}(x, y)) \cong \text{Mor}_{\mathfrak{L}_! \mathcal{M}}(\alpha(x), \alpha(y))$. $\square$

Thus by the above lemmas, we have the following equivalences:

$$\begin{aligned} \text{Fun}_{\overline{\text{Enr}}_{\mathcal{V}}}(\mathcal{M}, \mathcal{N}) &\cong \text{Mor}_{\text{Enr}_{\mathcal{V}}}(\mathcal{M}, \mathcal{N}) \cong \text{Mor}_{\widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}}(\mathfrak{L}_! \mathcal{M}, \mathfrak{L}_! \mathcal{N}) \cong \text{Fun}_{\overline{\widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}}}(\mathfrak{L}_! \mathcal{M}, \mathfrak{L}_! \mathcal{N}) \\ &\cong \text{Fun}_{\overline{\text{PLMod}_{\mathcal{V}}}}(\mathfrak{L}_!^* \mathfrak{L}_! \mathcal{M} = \mathcal{M}, \mathfrak{L}_!^* \mathfrak{L}_! \mathcal{N} = \mathcal{N}) \cong \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N}) \end{aligned}$$

where

- The first line of equivalences is due to the fully faithful lax linear embedding $(\text{Cat}, \text{Enr}_{\mathcal{V}}) \hookrightarrow (\widehat{\text{Cat}}, \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})})$.
- The first equivalence in the second line is by the $(\infty, 2)$ equivalence $\mathfrak{L}_!^* : \overline{\widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}} \cong \overline{\text{PLMod}_{\mathcal{V}}}$, and the second equivalence in the second line is by the [description of mapping category of PLMod \$_{\mathcal{V}}\$](#) .

Corollary 6.13. *Given a presentably symmetric monoidal ∞ -category $\mathcal{V}$, under the symmetric monoidal functor $\mathbf{1}_{\mathcal{V},!} : \text{Enr}_{\text{Spc}} \rightarrow \text{Enr}_{\mathcal{V}}$, the mapping category of the enhanced $(\infty, 2)$ category $\overline{\text{Enr}}_{\mathcal{V}}$ can be described as:*

$$\text{Fun}_{\overline{\text{Enr}}_{\mathcal{V}}}(\mathcal{M}, \mathcal{N}) \cong \text{LaxLin}_{\mathcal{V}}(\mathcal{M}, \mathcal{N})$$

for any $\mathcal{M}, \mathcal{N} \in \text{Enr}_{\mathcal{V}}$.

Now we enlarge the universe and consider the large presentably symmetric monoidal ∞ -category $\widehat{\text{Cat}}_{cc}$, then for any $\mathcal{M}, \mathcal{N} \in \widehat{\text{Enr}}_{\widehat{\text{Cat}}_{cc}}$, there is an equivalence:

$$\text{Fun}_{\widehat{\text{Enr}}_{\widehat{\text{Cat}}_{cc}}}(\mathcal{M}, \mathcal{N}) \cong \text{LaxLin}_{\widehat{\text{Cat}}_{cc}}(\mathcal{M}, \mathcal{N})$$

and if $\mathcal{M}, \mathcal{N}$ lands in the image of the (transfer of enrichment) functor $\text{Enr}_{\text{PrL}} \hookrightarrow \widehat{\text{Enr}}_{\text{PrL}} \hookrightarrow \widehat{\text{Enr}}_{\widehat{\text{Cat}}_{cc}}$, and by [lemma 6.8](#), we have:

Corollary 6.14. *For $\mathcal{M}, \mathcal{N} \in \text{Enr}_{\text{PrL}}$, there is an equivalence:*

$$\text{Fun}_{\widehat{\text{Enr}}_{\text{PrL}}}(\mathcal{M}, \mathcal{N}) \cong \text{LaxLin}_{\text{PrL}}(\mathcal{M}, \mathcal{N})$$

where $\widehat{\text{Enr}}_{\text{PrL}}$ is [defined as above](#).

6.3 $\text{Span}_2(\mathcal{C})^\otimes \rightarrow \widehat{\text{Enr}}_{\text{PrL}}^\otimes$

Fix a category $\mathcal{C}$ which admits all pullbacks and every morphism in $\mathcal{C}$ is truncated.

Recall that we have already got two lax symmetric monoidal $(\infty, 2)$ functor:

$$\begin{aligned} \psi : \text{Span}_2(\mathcal{C})^\otimes &\rightarrow \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^\otimes \\ R : \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^\otimes &\rightarrow \widehat{\text{Enr}}_{\mathcal{P}(\text{Span}_1(\mathcal{C}))}^\otimes \end{aligned}$$

where

- ψ is defined in [proposition 4.32](#) and R is defined in [corollary 5.56](#).
- $\overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\text{Cat})$ is a symmetric monoidal $(\infty, 2)$ category constructed in [construction 4.15](#) through a symmetric monoidal colimit preserving functor $\text{Cat}^\times \cong \text{LMod}_*(\text{Cat})^\otimes \rightarrow \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\text{Cat})^\otimes$.
- $\widehat{\text{Enr}}_{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ is a symmetric monoidal $(\infty, 2)$ category constructed in [construction 5.48](#) through a symmetric monoidal colimit preserving functor $\text{Cat}^\times \cong \text{Enr}_{\text{Spc}}^\otimes \rightarrow \widehat{\text{Enr}}_{\mathcal{P}(\text{Span}_1(\mathcal{C}))}^\otimes$ induced by the unique symmetric monoidal colimit preserving functor $\mathcal{P}(\ast) \cong \text{Spc} \rightarrow \mathcal{P}(\text{Span}_1(\mathcal{C}))$.

Now we enlarge the universe, there are again two lax symmetric monoidal $(\infty, 2)$ functors of large version:

$$\begin{aligned} \psi : \text{Span}_2(\mathcal{C})^\otimes &\rightarrow \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\widehat{\text{Cat}})^\otimes \\ R : \overline{\text{LMod}}_{\text{Span}_1(\mathcal{C})}(\widehat{\text{Cat}})^\otimes &\rightarrow \widehat{\widehat{\text{Enr}}}_{\widehat{\mathcal{P}}(\text{Span}_1(\mathcal{C}))}^\otimes \end{aligned}$$

Now consider the following adjunction for any small symmetric monoidal category $\mathcal{V}$:

$$L : \text{Env}(\mathcal{C}) \rightleftarrows \mathcal{C} : i$$

where L is symmetric monoidal and i is lax symmetric monoidal. And it induces an adjunction after applying the large presheaf functor $\widehat{\mathcal{P}}(-)$:

$$\widehat{\mathcal{P}}(L) : \widehat{\mathcal{P}} \text{Env}(\mathcal{V}) \rightleftarrows \widehat{\mathcal{P}}(\mathcal{V}) : \widehat{\mathcal{P}}(i)$$

where $\widehat{\mathcal{P}}(L)$ is symmetric monoidal and $\widehat{\mathcal{P}}(i)$ is lax symmetric monoidal. And the left adjoint gives rise to the following morphism in $\text{CAlg}(\widehat{\text{Pr}}^{\text{L}})$ (as in [construction 5.49](#):

$$\begin{array}{ccc} \widehat{\text{Enr}}_{\widehat{\mathcal{P}} \text{Env}(\mathcal{V})}^{\otimes} & \xrightarrow{\widehat{\mathcal{P}}(L)_!} & \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}^{\otimes} \\ & \nwarrow \quad \nearrow & \\ & \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(*)}^{\otimes} \cong \widehat{\text{Cat}}^{\times} & \end{array}$$

and by applying the large version of the lax symmetric monoidal functor $\theta: \widehat{\theta} : \text{LMod}_{\widehat{\text{Cat}}}(\widehat{\text{Pr}}^{\text{L}}) \rightarrow \widehat{\text{Cat}}_{(\infty, 2)}$, we have:

Lemma 6.15. $\widehat{\mathcal{P}}(L)_! : \widehat{\text{Enr}}_{\widehat{\mathcal{P}} \text{Env}(\mathcal{V})} \rightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}$ can be enhanced into a symmetric monoidal $(\infty, 2)$ functor:

$$\widehat{\mathcal{P}}(L)_! : \widehat{\text{Enr}}_{\widehat{\mathcal{P}} \text{Env}(\mathcal{V})} \rightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})}$$

and the right adjoint $\widehat{\mathcal{P}}(i)_!$ thus can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor:

$$\widehat{\mathcal{P}}(i)_! : \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})} \rightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{P}} \text{Env}(\mathcal{V})}$$

.

Given a 3-functor formalism, which is a lax symmetric monoidal functor:

$$D : \text{Span}_1(\mathcal{C})^{\otimes} \rightarrow \text{Pr}^{\text{L}}{}^{\otimes}$$

, and by composition with the symmetric monoidal embedding

$$\text{Pr}^{\text{L}}{}^{\otimes} \hookrightarrow \widehat{\text{Cat}}_{cc}^{\otimes}$$

, we have a lax symmetric monoidal functor:

$$D' : \text{Span}_1(\mathcal{C})^{\otimes} \rightarrow \widehat{\text{Cat}}_{cc}^{\otimes}$$

, and since $\widehat{\text{Cat}}_{cc}^{\otimes}$ is large presentable, thus there is an induced functor:

$$\begin{array}{ccc}
\mathrm{Span}_1(\mathcal{C})^\otimes & \xrightarrow{D'} & \widehat{\mathrm{Cat}}_{cc}^\otimes \\
\downarrow i & \nearrow & \\
\mathrm{Env}(\mathrm{Span}_1(\mathcal{C}))^\otimes & \xrightarrow{D''} & \\
\downarrow \widehat{\mathfrak{L}} & \nearrow & \\
\widehat{\mathcal{P}} \mathrm{Env}(\mathrm{Span}_1(\mathcal{C}))^\otimes & &
\end{array}$$

where D'' is symmetric monoidal and colimit preserving.

By applying the large version of [construction 5.49](#), there is a symmetric monoidal $(\infty, 2)$ functor,

$$D'_! : \widehat{\mathrm{Enr}}_{\widehat{\mathcal{P}}(\mathrm{Env}(\mathcal{V}))}^\otimes \rightarrow \widehat{\mathrm{Enr}}_{\widehat{\mathrm{Cat}}_{cc}}^\otimes$$

And by composition of the functor $\widehat{\mathcal{P}}(i)$ and $D'_!$, we have:

Proposition 6.16. *There is a lax symmetric monoidal $(\infty, 2)$ functor:*

$$D'_! \circ \widehat{\mathcal{P}}(i)_! : \overline{\mathrm{Enr}}_{\widehat{\mathcal{P}}(\mathcal{V})} \rightarrow \overline{\mathrm{Enr}}_{\widehat{\mathrm{Cat}}_{cc}}$$

.

If we take a look at the underlying $(\infty, 1)$ categories, there is a commutative diagram:

$$\begin{array}{ccc}
\mathrm{Span}_1(\mathcal{C}) & \xrightarrow{D} & \mathrm{Pr}^L \\
\downarrow \widehat{\mathfrak{L}} & \searrow & \downarrow \\
\widehat{\mathcal{P}}(\mathrm{Span}_1(\mathcal{C})) & \xrightarrow{D'} & \widehat{\mathrm{Cat}}_{cc} \\
\downarrow \widehat{\mathcal{P}}(i) & \searrow & \downarrow \\
\widehat{\mathcal{P}} \mathrm{Env}(\mathrm{Span}_1(\mathcal{C})) & \xrightarrow{D''} & \widehat{\mathrm{Cat}}_{cc}
\end{array}$$

By applying the large version of the [transfer of enrichment](#), there is a commutative diagram of $(\infty, 1)$ categories:

$$\begin{array}{ccc}
\widehat{\mathrm{Enr}}_{\mathrm{Span}_1(\mathcal{C})} & \xrightarrow{D_!} & \widehat{\mathrm{Enr}}_{\mathrm{Pr}^L} \\
\downarrow \widehat{\mathfrak{L}}_! & \searrow & \downarrow \\
\widehat{\mathrm{Enr}}_{\widehat{\mathcal{P}}(\mathrm{Span}_1(\mathcal{C}))} & \xrightarrow{D'_!} & \widehat{\mathrm{Enr}}_{\widehat{\mathrm{Cat}}_{cc}} \\
\downarrow \widehat{\mathcal{P}}(i)_! & \searrow & \downarrow \\
\widehat{\mathrm{Enr}}_{\widehat{\mathcal{P}} \mathrm{Env}(\mathrm{Span}_1(\mathcal{C}))} & \xrightarrow{D''_!} & \widehat{\mathrm{Enr}}_{\widehat{\mathrm{Cat}}_{cc}}
\end{array}$$

Thus the image of the composition of $(\infty, 2)$ functors $\mathrm{Span}_2(\mathcal{C}) \xrightarrow{\widehat{\psi}} \mathrm{LMod}_{\mathrm{Span}_1(\mathcal{C})}(\widehat{\mathrm{Cat}}) \xrightarrow{\widehat{R}}$

$\widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\text{Span}_1(\mathcal{C}))} \xrightarrow{D'_!} \widehat{\text{Enr}}_{\widehat{\mathcal{C}}_{\text{atcc}}}$ lands in the $(\infty, 2)$ full subcategory $\widehat{\text{Enr}}_{\text{Pr}^L} \hookrightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{C}}_{\text{atcc}}}$, since by [corollary 5.56](#), the functor $R\circ\widehat{\psi} : \text{Span}_2(\mathcal{C}) \rightarrow \widehat{\text{Enr}}_{\widehat{\mathcal{P}}(\text{Span}_1(\mathcal{C}))}$ actually lands in $\widehat{\text{Enr}}_{\text{Span}_1(\mathcal{C})}$.

Thus we have:

Corollary 6.17. *There is a lax symmetric monoidal $(\infty, 2)$ functor:*

$$\text{Span}_2(\mathcal{C}) \rightarrow \widehat{\text{Enr}}_{\text{Pr}^L}$$

where $\mathcal{C}$ admits all pullbacks and every morphism in $\mathcal{C}$ is truncated. And the symmetric monoidal $(\infty, 2)$ structure on $\widehat{\text{Enr}}_{\text{Pr}^L}$ is defined in [definition 6.4](#).

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